



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

February/March 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **20** pages. Any blank pages are indicated.



1 Which row about a change of state is correct?

	change of state	energy change	process
A	solid → liquid	heat given out	melting
B	gas → liquid	heat taken in	evaporation
C	solid → gas	heat taken in	sublimation
D	liquid → solid	heat given out	condensing

2 Gases are separated from liquid air by fractional distillation.

The boiling points of four gases are shown.

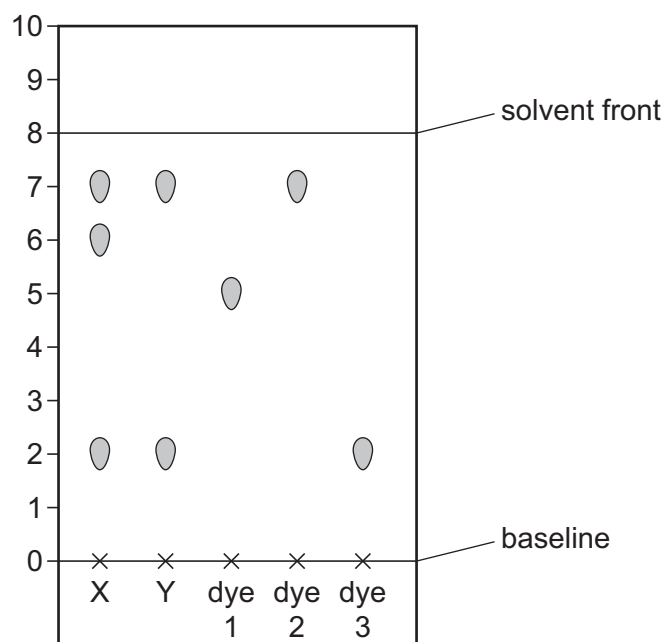
Which gas is both monoatomic and a liquid at $-200\text{ }^{\circ}\text{C}$?

	gas	boiling point/ $^{\circ}\text{C}$
A	argon	-186
B	helium	-269
C	neon	-246
D	nitrogen	-196

3 Two different food colourings, X and Y, are tested using chromatography.

Three pure dyes, 1, 2 and 3, are also tested.

The chromatogram is shown.



Which statements are correct?

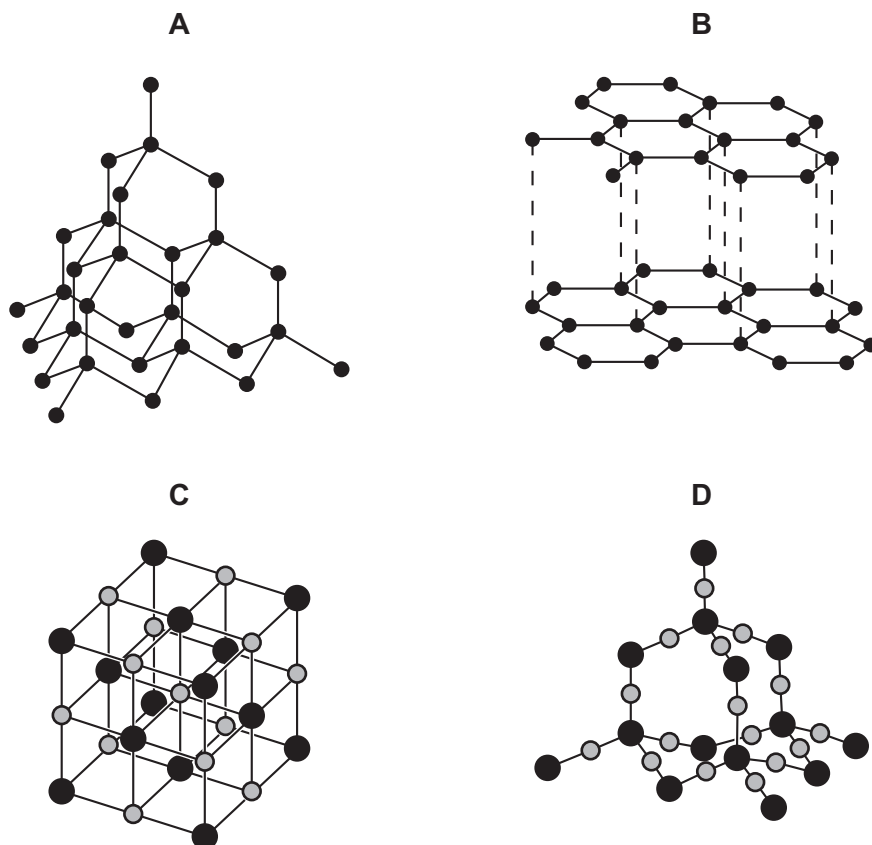
- 1 X and Y both contain two or more dyes.
- 2 Dyes 2 and 3 are present in both X and Y.
- 3 The R_f of dye 1 is 0.625.

A 1 and 2 only **B** 1 and 3 only **C** 1, 2 and 3 **D** 2 and 3 only

4 Which statement about the atoms of all the isotopes of carbon is correct?

- A** They are all radioactive.
- B** They have the same mass.
- C** They have the same number of neutrons.
- D** They have the same number of electrons in the outer shell.

5 Which diagram represents the structure of silicon(IV) oxide?



6 Lithium and fluorine react to form lithium fluoride.

A student writes three statements about the reaction.

- 1 Lithium atoms lose an electron when they react.
- 2 Each fluoride ion has one more electron than a fluorine atom.
- 3 Lithium fluoride is a mixture of elements.

Which statements are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

7 How many electrons are used to form covalent bonds in a molecule of methanol, CH_3OH ?

- A** 5 **B** 6 **C** 8 **D** 10

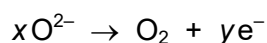
- 8 Magnesium oxide has a high melting point.

Carbon dioxide has a low melting point.

Which row identifies the attractive forces that are broken when these compounds are melted?

	magnesium oxide	carbon dioxide
A	strong attractions between molecules	weak attractions between atoms
B	strong attractions between molecules	weak attractions between molecules
C	strong attractions between ions	weak attractions between atoms
D	strong attractions between ions	weak attractions between molecules

- 9 The ionic half-equation for the formation of oxygen during the electrolysis of aluminium oxide is shown.



What are the values of x and y ?

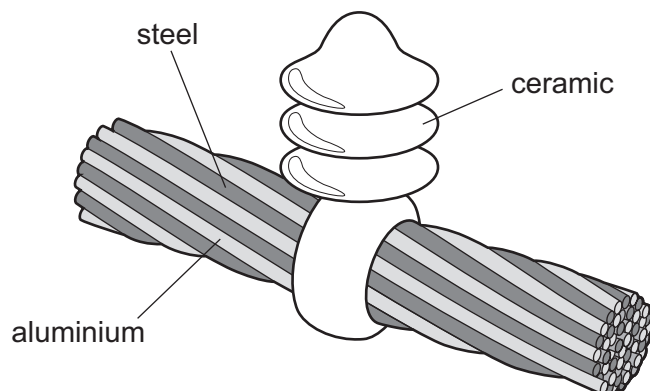
	x	y
A	1	2
B	1	4
C	2	2
D	2	4

- 10 A compound has the formula XF_2 and has a relative mass of 70.

What is element X?

- A** gallium
- B** germanium
- C** sulfur
- D** ytterbium

11 The diagram shows a section of an overhead power cable.



Which statement explains why a particular substance is used?

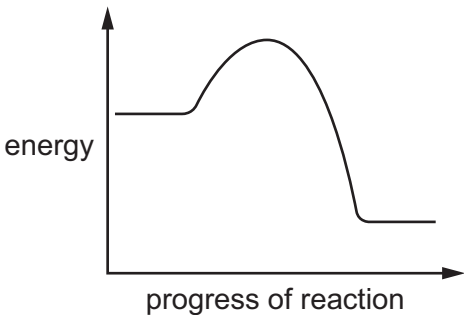
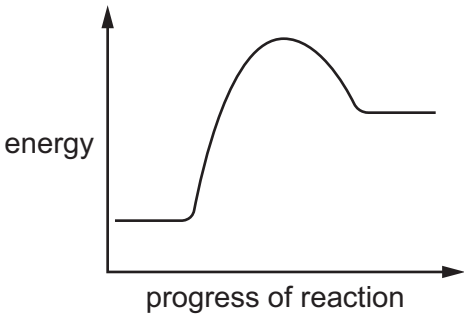
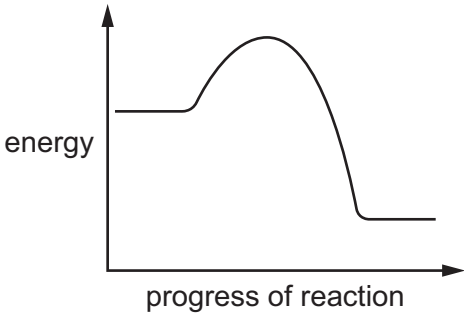
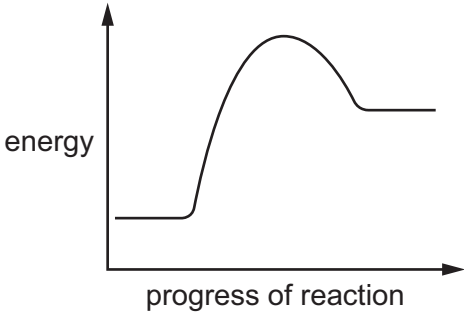
- A Aluminium has a low density and is a good conductor of electricity.
- B Ceramic is a good conductor of electricity.
- C Steel can rust in damp air.
- D Steel is more dense than aluminium.

12 During the electrolysis of dilute sulfuric acid, hydrogen is collected at the cathode.

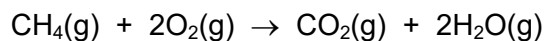
What is the ionic half-equation for this reaction?

- A $\text{H}^+ + \text{e}^- \rightarrow \text{H}$
- B $\text{H}^+ \rightarrow \text{H} + \text{e}^-$
- C $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- D $2\text{H}^+ \rightarrow \text{H}_2 + 2\text{e}^-$

13 Which row describes an endothermic reaction?

	energy level diagram	energy transfer
A		energy is transferred from the surroundings to the reaction
B		energy is transferred from the surroundings to the reaction
C		energy is transferred from the reaction to the surroundings
D		energy is transferred from the reaction to the surroundings

14 The equation for the complete combustion of methane is shown.



The bond energies are shown in the table.

bond	bond energy in kJ/mol
C–H	+410
C=O	+805
O–H	+460
O=O	+496

What is the energy change for the reaction?

- A** –818 kJ/mol **B** –359 kJ/mol **C** –323 kJ/mol **D** +102 kJ/mol

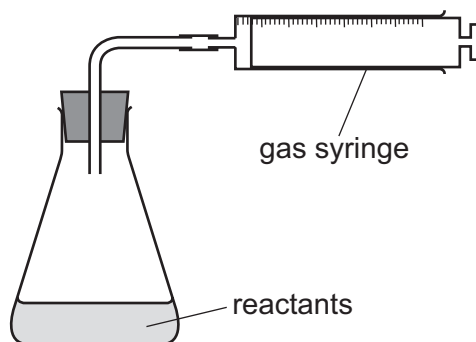
15 Hydrogen fuel cells can be used to power cars.

Which statements about a fuel cell are correct?

- 1 The balanced equation for the reaction is $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$.
- 2 The fuel cell generates electricity.
- 3 In the fuel cell hydrogen is reduced.
- 4 The reactants are gases at room temperature.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

- 16 The apparatus shown is used to measure the rate of a reaction.



Which equation represents a reaction where the rate can be measured using this apparatus?

- A** $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$
- B** $\text{HCl(aq)} + \text{NaOH(aq)} \rightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}$
- C** $\text{Fe(s)} + \text{CuSO}_4\text{(aq)} \rightarrow \text{Cu(s)} + \text{FeSO}_4\text{(aq)}$
- D** $2\text{Na(s)} + \text{Br}_2\text{(l)} \rightarrow 2\text{NaBr(s)}$
- 17 P is a hydrated metal salt with a blue colour. When P is heated, water is given off, leaving solid Q.

R is a hydrated metal salt with a pink colour. When R is heated, water is given off, leaving solid S.

Which row gives the name of P and the colour of S?

	name of P	colour of S
A	hydrated cobalt(II) chloride	blue
B	hydrated cobalt(II) chloride	white
C	hydrated copper(II) sulfate	blue
D	hydrated copper(II) sulfate	white

- 18 Magnesium reacts with copper(II) oxide to give magnesium oxide and copper.

Which substance is the oxidising agent in this reaction?

- A** copper
- B** copper(II) oxide
- C** magnesium
- D** magnesium oxide

19 Part of the Periodic Table is shown.

Which element forms an acidic oxide?

A simplified periodic table diagram with 18 columns and 4 rows. The first two columns are labeled 'A' and 'C' in the top row. The next six columns are empty. The next six columns are labeled 'B' and 'D' in the top row. The last two columns are empty. A small square box is located above the 10th column.

20 When aqueous sodium hydroxide is added to a solution of a metal ion, a grey-green precipitate forms, which dissolves in excess to form a dark green solution.

What is the identity of the metal ion?

- A** chromium(III)
B iron(II)
C iron(III)
D copper(II)

21 Which statements about strong acids are correct?

- 1 They have a high concentration of OH^- ions.
- 2 They have a pH value of 1.
- 3 They completely ionise in water.
- 4 They turn red litmus blue.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

22 Metal X reacts with non-metal Y to form an ionic compound with the formula X_2Y .

Which statements are correct?

- 1 X is in Group I of the Periodic Table.
- 2 X is in Group II of the Periodic Table.
- 3 Y is in Group VI of the Periodic Table.
- 4 Y is in Group VII of the Periodic Table.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

23 The table gives some properties of Group IV elements.

element	density g/cm ³	boiling point /°C
carbon	2.2	4827
silicon		
germanium	5.3	2830
tin	5.8	2270
lead	11.3	1755

Which row describes the properties of silicon?

	density g/cm ³	boiling point /°C
A	2.3	3 265
B	3.1	1 997
C	6.2	2 920
D	24.6	11 682

24 The metal beryllium does not react with cold water.

It reacts with hydrochloric acid but cannot be extracted from its ore by using carbon.

Where is beryllium placed in the reactivity series?

magnesium

A

zinc

B

iron

C

copper

D

25 Why is cryolite used in the extraction of aluminium from bauxite?

A as a catalyst for the process

B as a solvent for aluminium oxide

C it stops the carbon anodes burning away

D it reduces aluminium ions in aluminium oxide

26 Which statements about the uses of metals are correct?

- 1 Iron is used to make aircraft because iron has a low density.
- 2 Copper is used to make electric cables because copper is a good conductor of electricity.
- 3 Aluminium is used to make brass because aluminium is strong and hard.
- 4 Iron is mixed with additives to make an alloy used in chemical plant.

A 1 and 2 **B** 3 and 4 **C** 1 and 3 **D** 2 and 4

27 Which row describes the reactions of magnesium hydroxide and magnesium oxide?

	effect of heat on hydroxide	effect of heating oxide with carbon
A	forms magnesium oxide	magnesium and carbon dioxide formed
B	forms magnesium oxide	no reaction
C	no reaction	magnesium and carbon dioxide formed
D	no reaction	no reaction

28 The properties of an element are listed.

Its melting point is 3414 °C.

Some of its compounds are catalysts.

It has variable oxidation states.

Where is the element found in the Periodic Table?

- A** alkali metals
B halogens
C noble gases
D transition elements

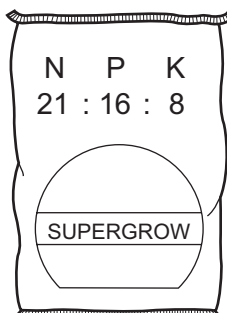
29 Petrol burns in a car engine to produce waste gases which leave through the car exhaust.

One of these waste gases is an oxide of nitrogen.

Which statement describes how this oxide of nitrogen is formed?

- A** Carbon dioxide reacts with nitrogen in the catalytic converter.
B Nitrogen reacts with oxygen in the car engine.
C Nitrogen reacts with oxygen in the catalytic converter.
D Petrol combines with nitrogen in the car engine.

30 Which combination of chemical compounds can be used to produce the fertiliser shown?



- A $(\text{NH}_4)_3\text{PO}_4$, KCl
- B NH_4NO_3 , $\text{Ca}_3(\text{PO}_4)_2$
- C NH_4NO_3 , $\text{CO}(\text{NH}_2)_2$
- D NH_4NO_3 , K_2SO_4 , $(\text{NH}_4)_2\text{SO}_4$

31 Which process does **not** produce carbon dioxide?

- A combustion of a hydrocarbon
- B photosynthesis
- C reaction between an acid and a metal carbonate
- D respiration

32 Which substance is used as a bleach in the manufacture of paper?

- A carbon dioxide
- B nitrogen dioxide
- C silicon dioxide
- D sulfur dioxide

33 What is an industrial use of calcium carbonate?

- A cracking of hydrocarbons
- B manufacture of aluminium
- C manufacture of cement
- D purification of water

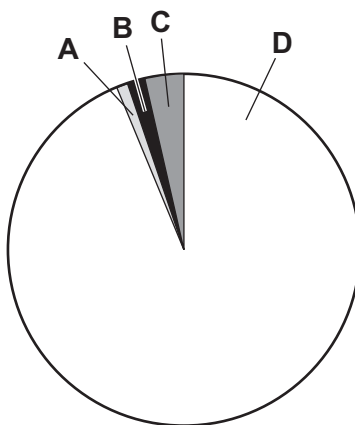
34 Propane reacts with chlorine.

Which row shows a condition required for this reaction and identifies the type of reaction?

	condition	type of reaction
A	phosphoric acid catalyst	addition
B	phosphoric acid catalyst	substitution
C	ultraviolet light	addition
D	ultraviolet light	substitution

35 The pie chart represents the composition of natural gas.

Which sector represents methane?

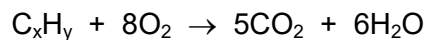


36 Which statement describes the reaction between ethene and steam?

- A** a cracking reaction which produces ethane and hydrogen gas as products
- B** an addition reaction which produces ethanol as the only product
- C** an oxidation reaction which produces ethanoic acid as the only product
- D** a slow reaction producing ethanol and carbon dioxide

- 37 The formula of a hydrocarbon is C_xH_y .

The equation for its complete combustion is shown.



What are the values of x and y?

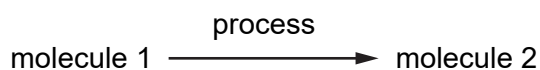
	x	y
A	5	6
B	5	12
C	6	5
D	12	5

- 38 The formula of an ester is $CH_3CH_2CH_2COOCH_2CH_2CH_3$.

Which acid and alcohol react together to make the ester?

	acid	alcohol
A	butanoic acid	butanol
B	butanoic acid	propanol
C	propanoic acid	butanol
D	propanoic acid	propanol

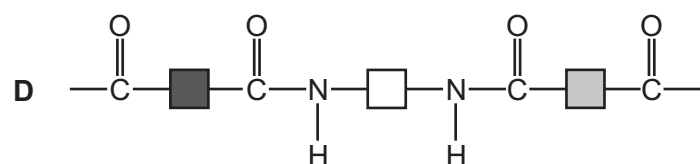
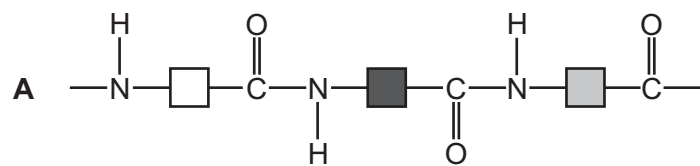
- 39 Molecule 1 undergoes a process to make molecule 2.



Which row describes the molecules and the process?

	molecule 1	process	molecule 2
A	monomer	cracking	polymer
B	monomer	polymerisation	polymer
C	small molecule	polymerisation	monomer
D	small molecule	cracking	monomer

40 Which structure represents a protein?



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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23											12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	119 Uue unbinilium —	120 Uuh ununilium —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



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1 Which gas has the fastest rate of diffusion?

- A** H_2 **B** CH_4 **C** CO_2 **D** SO_2

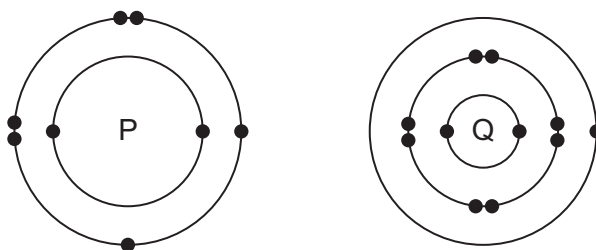
2 In which state does 1 dm^3 of methane contain the most particles?

- A** gas at 100°C
B gas at room temperature
C liquid
D solid

3 Which statement explains why isotopes of the same element have the same chemical properties?

- A** They have the same electronic structure.
B They have the same relative mass.
C They have the same nucleon number.
D They have the same proton number.

4 The electronic structures of atoms P and Q are shown.



P and Q form an ionic compound.

What is the formula of the compound?

- A** PQ **B** P_2Q **C** P_2Q_3 **D** PQ_2

5 Fermentation of sugar produces a mixture of ethanol solution and solid yeast.

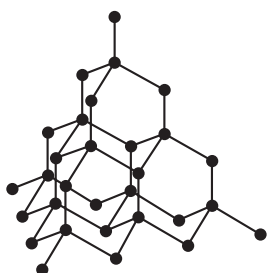
How is the solid yeast removed from the mixture?

- A** crystallisation
B distillation
C filtration
D fractional distillation

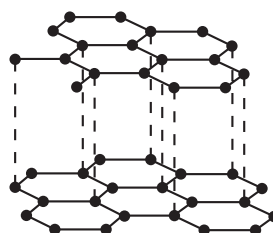
- 6 Which row explains why copper is a good conductor of electricity at room temperature?

	copper ions move freely	electrons move freely
A	no	no
B	no	yes
C	yes	no
D	yes	yes

- 7 Which pair of statements about diamond and graphite is correct?



diamond



graphite

- A** Diamond and graphite are both pure carbon. They are both macromolecules.
B Diamond and graphite can both be used as electrodes. Graphite is also used as a lubricant.
C Diamond has covalent bonds. Graphite has ionic bonds.
D Diamond is hard with a high melting point. Graphite is soft with a low melting point.
- 8 Sodium nitride contains the nitride ion, N^{3-} .

Sodium nitride is unstable and decomposes into its elements.

What is the equation for the decomposition of sodium nitride?

- A** $2\text{NaN}_3 \rightarrow 2\text{Na} + 3\text{N}_2$
B $2\text{Na}_3\text{N} \rightarrow 6\text{Na} + \text{N}_2$
C $2\text{NaN}_3 \rightarrow \text{Na}_2 + 3\text{N}_2$
D $2\text{Na}_3\text{N} \rightarrow 6\text{Na} + 2\text{N}$
- 9 Compound X contains carbon, hydrogen and oxygen only.

By mass, it contains 26.7% carbon and 2.2% hydrogen.

What is the empirical formula of X?

- A** CHO **B** C_2HO **C** CH_2O **D** CHO_2

10 Caesium fluoride is an ionic compound.

Which statements about caesium fluoride are correct?

- 1 It conducts electricity when solid.
- 2 It has a high melting point.
- 3 It is soluble in water.
- 4 It is highly volatile.

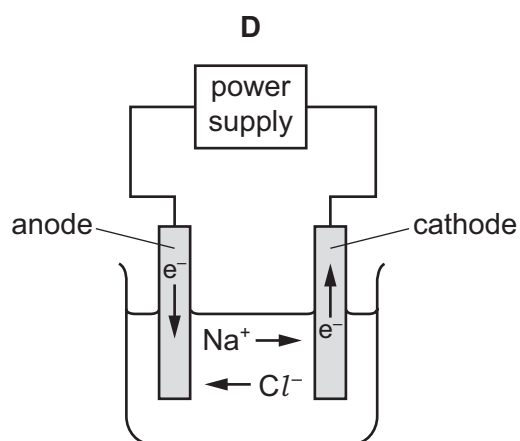
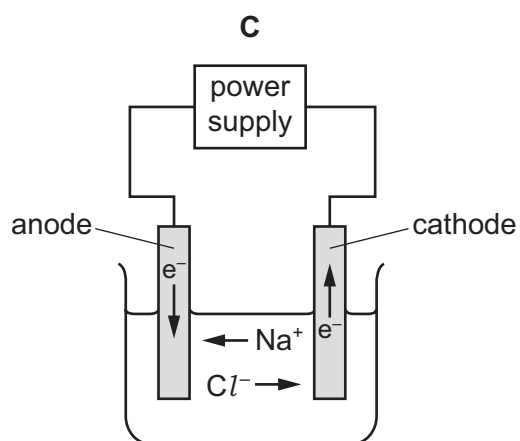
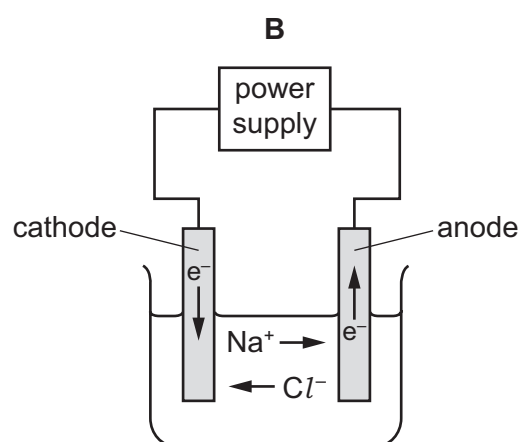
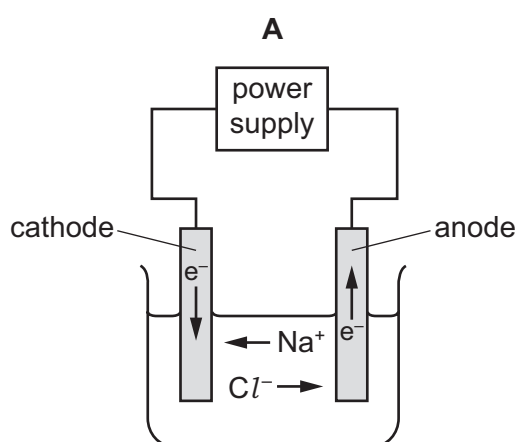
A 1 and 2

B 1 and 4

C 2 and 3

D 3 and 4

11 Which diagram shows the direction of movement of ions and electrons during the electrolysis of molten sodium chloride?



- 12** Calcium carbonate, CaCO_3 , reacts with dilute hydrochloric acid to produce carbon dioxide.

The equation for the reaction is shown. The relative formula mass of calcium carbonate is 100.



10 g of calcium carbonate is reacted with an excess of dilute hydrochloric acid.

Which mass of carbon dioxide is produced?

- A** 2.2 g **B** 2.8 g **C** 4.4 g **D** 44 g

- 13** Molten sodium chloride and concentrated aqueous sodium chloride are electrolysed using platinum electrodes.

What are the products at the negative electrode (cathode) in each electrolysis?

	molten sodium chloride	concentrated aqueous sodium chloride
A	hydrogen	hydrogen
B	hydrogen	sodium
C	sodium	hydrogen
D	sodium	sodium

- 14** An object is electroplated with silver using an aqueous silver salt as the electrolyte.

Which row is correct?

	the object to be electroplated is the	the other electrode is made from
A	anode	carbon
B	anode	silver
C	cathode	carbon
D	cathode	silver

15 Which row describes the changes that occur in an endothermic reaction?

	energy change	temperature
A	energy given out to the surroundings	decreases
B	energy given out to the surroundings	increases
C	energy taken in from the surroundings	decreases
D	energy taken in from the surroundings	increases

16 Which statement about fuels is correct?

- A** Heat energy is only produced by burning fuels.
- B** Hydrogen is used as a fuel although it is difficult to store.
- C** Methane is a good fuel because it produces only water when burned.
- D** Uranium is burned in air to produce energy.

17 Which statement about endothermic and exothermic reactions is correct?

- A** In an endothermic reaction, less energy is absorbed in bond breaking than is released in bond forming.
- B** In an endothermic reaction, the activation energy is always higher than in an exothermic reaction.
- C** In an exothermic reaction, more energy is absorbed in bond breaking than is released in bond forming.
- D** In an exothermic reaction, the reactants are higher on an energy level diagram than the products.

- 18** The reaction used to manufacture ammonia from nitrogen and hydrogen is reversible.

An equilibrium is established between ammonia, nitrogen and hydrogen.

Which statement describes the equilibrium?

- A** Both the forward reaction and the backward reaction have the same rate.
- B** The rate of the backward reaction is greater than the rate of the forward reaction.
- C** The rate of the forward reaction is greater than the rate of the backward reaction.
- D** The forward and backward reactions have both stopped.

- 19** How does increasing the concentration affect the reacting particles in a chemical reaction?

	increases the collision rate	increases the proportion of particles with the activation energy
A	✓	✗
B	✓	✓
C	✗	✗
D	✗	✓

- 20** Methyl orange is added to dilute hydrochloric acid and to aqueous sodium hydroxide.

What is the colour of the methyl orange in each solution?

	colour in dilute hydrochloric acid	colour in aqueous sodium hydroxide
A	orange	red
B	red	yellow
C	red	orange
D	yellow	red

21 Zinc oxide is an amphoteric oxide.

Which types of substances will react with zinc oxide?

- A acids and bases
- B acids only
- C bases only
- D neither acids nor bases

22 Information about some silver compounds is shown.

compound	formula	solubility in water
silver carbonate	Ag_2CO_3	insoluble
silver chloride	AgCl	insoluble
silver nitrate	AgNO_3	soluble
silver oxide	Ag_2O	insoluble

Which equation shows a reaction which **cannot** be used to make a silver salt?

- A $\text{AgNO}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{HNO}_3(\text{aq})$
- B $\text{Ag}_2\text{O}(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- C $\text{Ag}_2\text{CO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow 2\text{AgNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{CO}_2(\text{g})$
- D $2\text{Ag}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{AgCl}(\text{s}) + \text{H}_2(\text{g})$

23 Aqueous ethanoic acid is a weak acid.

Aqueous sodium hydroxide is a strong base.

Aqueous ethanoic acid is neutralised by aqueous sodium hydroxide.

Which statements are correct?

- 1 Aqueous ethanoic acid accepts protons from hydroxide ions.
- 2 The aqueous ethanoic acid used is fully dissociated into ions.
- 3 The aqueous sodium hydroxide used is fully dissociated into ions.
- 4 The reaction produces a salt and water.

- A 1 and 2
- B 1 and 3
- C 2 and 4
- D 3 and 4

24 Three properties of element X are listed.

- It contains atoms with a full outer shell of electrons.
- It is monoatomic.
- It is unreactive.

In which part of the Periodic Table is the element placed?

- A** Group I
- B** Group VII
- C** Group VIII
- D** transition elements

25 Which word equation represents a reaction that occurs?

- A** sodium oxide + carbon $\rightarrow$ sodium + carbon dioxide
- B** sodium oxide + iron $\rightarrow$ sodium + iron(II) oxide
- C** iron(II) oxide + copper $\rightarrow$ iron + copper(II) oxide
- D** iron(III) oxide + carbon $\rightarrow$ iron + carbon dioxide

26 Which statement about the extraction of aluminium is correct?

- A** Aluminium is formed at the cathode during the electrolysis of aluminium oxide.
- B** Hematite is mainly aluminium oxide.
- C** Molten cryolite is used to raise the melting point of the aluminium oxide.
- D** Oxygen gains electrons at the anode during the electrolysis of aluminium oxide.

27 Metal M is mixed with copper to produce brass.

What is M?

- A** chromium
- B** nickel
- C** vanadium
- D** zinc

28 The table gives some properties of an element.

melting point in °C	3422
appearance of the element	grey
appearance of the chloride of the element	dark blue
density in g / cm ³	19.2
electrical conductivity when solid	good

Which other property does this element have?

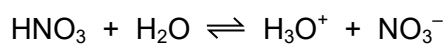
- A** acts as a catalyst
- B** brittle
- C** forms an acidic oxide
- D** highly reactive with water

29 Ammonia is produced using the Haber process.

Which row shows the source of the raw materials and the reaction conditions?

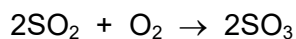
	source of nitrogen	source of hydrogen	temperature / °C	pressure / atm
A	air	hydrocarbons	200	200
B	hydrocarbons	air	450	2
C	air	hydrocarbons	450	200
D	air	hydrocarbons	450	2

30 How many species are acting as bases in this reversible reaction?



- A** 3
- B** 2
- C** 1
- D** 0

- 31 The equation for a reaction occurring in the Contact process is shown.



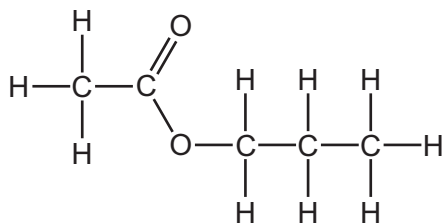
What is the catalyst used in this reaction?

- A iron
- B phosphoric(V) acid
- C sulfuric acid
- D vanadium(V) oxide

- 32 Which information about carbon dioxide and methane is correct?

		carbon dioxide	methane	key ✓ = correct ✗ = not correct
A	formed when vegetation decomposes	✓	✗	
B	greenhouse gas	✓	✓	
C	present in unpolluted air	✗	✗	
D	produced during respiration	✗	✓	

- 33 The structure of an ester is shown.



What are the names of the carboxylic acid and the alcohol that react together to form this ester?

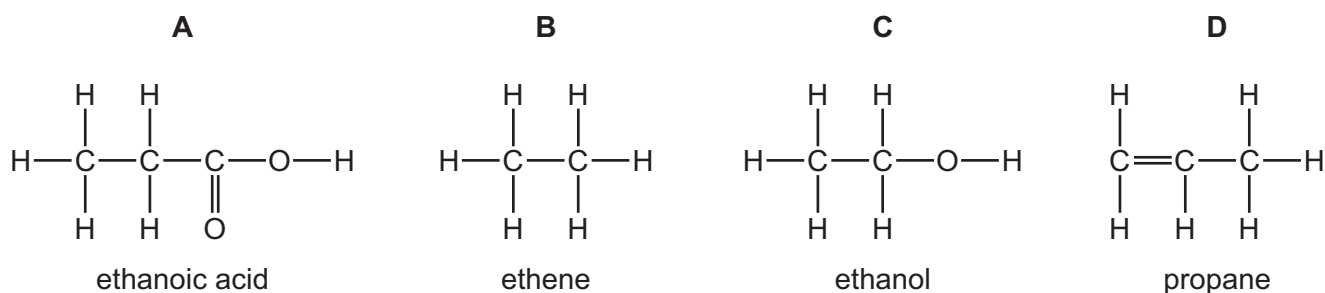
	carboxylic acid	alcohol
A	ethanoic acid	ethanol
B	ethanoic acid	propan-1-ol
C	propanoic acid	ethanol
D	propanoic acid	propan-1-ol

34 Which statements about lime are correct?

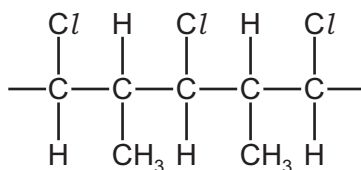
- 1 Lime is made by heating calcium carbonate (limestone).
- 2 Lime is used to desulfurise flue gases.
- 3 Lime is used to treat alkaline soil.
- 4 The chemical name for lime is calcium oxide.

A 1 and 3 **B** 1, 2 and 4 **C** 1 and 4 only **D** 2, 3 and 4

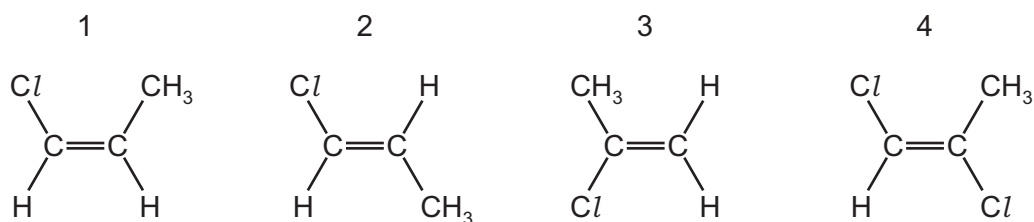
35 Which structure is correctly named?



36 The structure of part of a polymer is shown.



Which monomers can be used to make this polymer?



A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

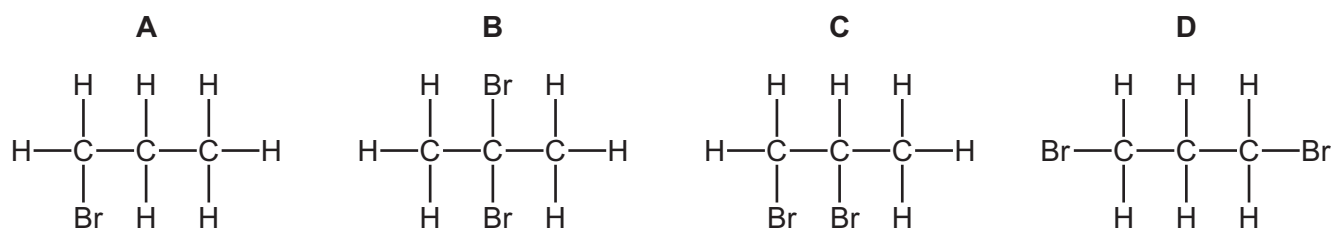
37 Carboxylic acids are made by the oxidation of alcohols.

Which carboxylic acid is produced from $\text{CH}_3\text{CH}_2\text{OH}$?

- A** butanoic acid
- B** ethanoic acid
- C** methanoic acid
- D** propanoic acid

38 Propene, C_3H_6 , reacts with bromine, Br_2 , in an addition reaction.

Which structure represents the product of this reaction?

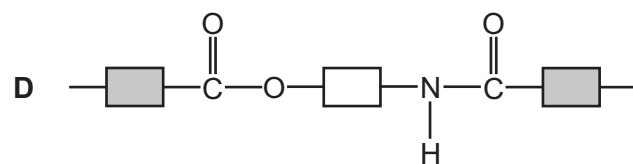
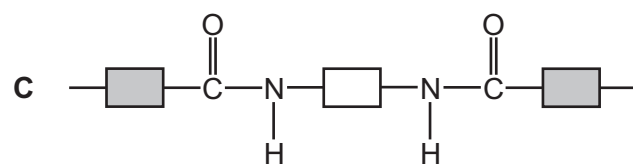
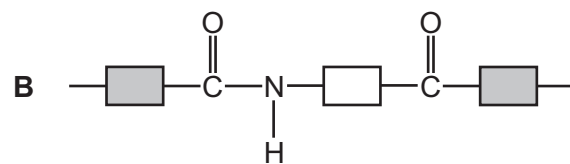
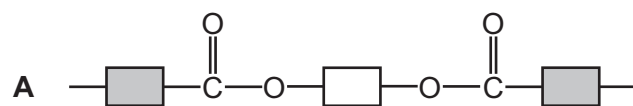


39 Which statements about ethanol are correct?

- 1 Ethanol is used as a solvent.
- 2 Ethanol can be made directly from ethane.
- 3 Ethanol is a covalent compound.

- A** 1 only
- B** 1 and 2
- C** 1 and 3
- D** 2 and 3

40 Which diagram represents the structure of nylon?



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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	118 Og oganesson —	119 Ts tennessine —	120 Nh nihonium —	121 Lr lawrencium —

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

February/March 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

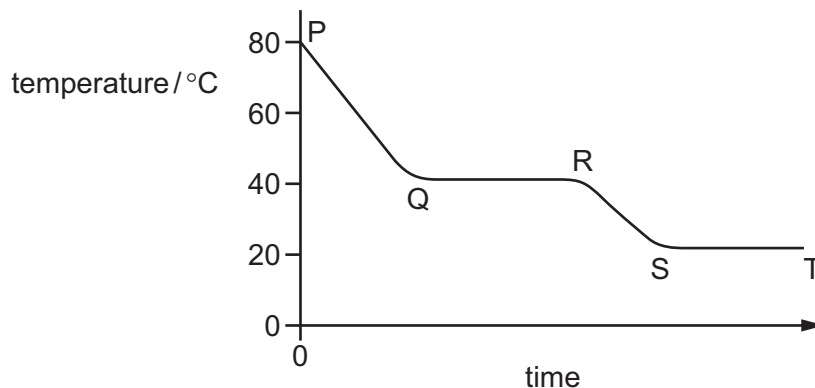
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- 1 Substance M is a solid at 30 °C.

The substance is heated to 80 °C and its temperature measured as it cools down to room temperature.

The cooling curve is shown.



Between which times is substance M freezing?

- A** P to Q **B** Q to R **C** R to S **D** S to T
- 2 Which gas has the fastest rate of diffusion?
- A** Ar **B** C₂H₆ **C** HCl **D** H₂S

- 3 There are two stable isotopes of bromine.

The mass number of isotope 1 is 79.

The mass number of isotope 2 is 81.

Which statement is correct?

- A** The isotopes have the same number of neutrons.
B The isotopes have different chemical properties.
C The isotopes have different numbers of protons.
D The isotopes have the same number of outer electrons.
- 4 Which statement about ions and ionic bonds is correct?
- A** Bromine atoms form negatively charged bromide ions.
B Ionic bonds form between elements in Group VII of the Periodic Table.
C Positive ions are formed when atoms lose protons.
D Potassium iodide contains negatively charged potassium ions.

5 Part of the Periodic Table is shown.

A blank periodic table grid is shown, consisting of a 4x16 grid of squares. The grid is divided into four main sections by a central gap. The top row has a single square in the center, above the gap. The second row has two squares on the left, labeled 'F' and 'G', and two squares on the right. The third and fourth rows are empty.

Which type of chemical bonding is present in the oxide of F and in the oxide of G?

	oxide of F	oxide of G
A	covalent	covalent
B	covalent	ionic
C	ionic	covalent
D	ionic	ionic

6 Elements X and Y react to form a compound.

Element X loses two electrons and element Y gains one electron.

What is the charge on the ions of elements X and Y and what is the formula of the compound?

	charge on X	charge on Y	formula of compound
A	2+	−	X_2Y
B	2+	−	XY_2
C	2−	+	X_2Y
D	2−	+	XY_2

7 Which statement about graphite explains why it is used as an electrode?

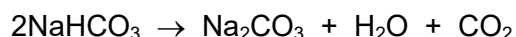
- A** It contains ions.
- B** It has a giant covalent structure.
- C** It is a metal.
- D** It has mobile electrons.

- 8 Methane, CH₄, burns in air to form carbon dioxide and water.

What is the balanced equation for this reaction?

- A CH₄(g) + O₂(g) → CO₂(g) + 2H₂O(g)
- B CH₄(g) + 2O₂(g) → CO₂(g) + 2H₂O(g)
- C CH₄(g) + 2O₂(g) → CO₂(g) + H₂O(g)
- D CH₄(g) + 3O₂(g) → CO₂(g) + 2H₂O(g)

- 9 The equation for the thermal decomposition of sodium hydrogencarbonate is shown.



The M_r of sodium hydrogencarbonate, NaHCO₃, is 84.

The M_r of sodium carbonate, Na₂CO₃, is 106.

In an experiment, 2.1 g of sodium hydrogencarbonate is heated but not all of it decomposes. All of the carbon dioxide is collected and measured at room temperature and pressure. The total volume of carbon dioxide produced is 0.21 dm³.

The volume of 1 mole of a gas at room temperature and pressure is 24 dm³.

Which statement is correct?

- A The mass of sodium carbonate produced is 0.93 g.
 - B The mass of sodium carbonate produced is 1.33 g.
 - C The percentage yield of carbon dioxide is 10%.
 - D The percentage yield of carbon dioxide is 35%.
- 10 An electrolysis experiment is done using carbon electrodes.

Hydrogen and oxygen are formed at the electrodes.

What is the electrolyte?

- A aqueous copper(II) sulfate
- B concentrated hydrochloric acid
- C dilute aqueous sodium chloride
- D molten potassium oxide

- 11 Concentrated aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which ionic half-equation describes the reaction taking place at the cathode?

- A $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$
- C $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
- D $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

- 12 When powdered sodium carbonate and aqueous ethanoic acid are mixed, the temperature of the mixture falls.

Which statement about this reaction is correct?

- A The reaction is endothermic and ΔH is negative.
- B The reaction is endothermic and ΔH is positive.
- C The reaction is exothermic and ΔH is negative.
- D The reaction is exothermic and ΔH is positive.

- 13 Magnesium powder reacts with an excess of dilute hydrochloric acid to produce hydrogen gas.

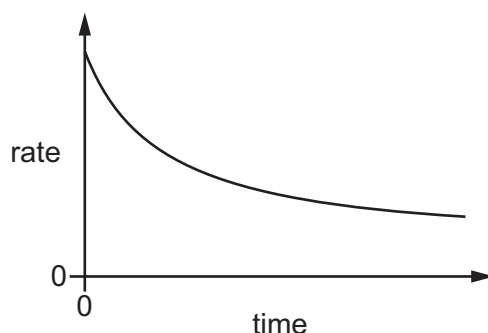
Which statements about this reaction are correct?

- 1 The smaller the particles of magnesium powder, the more slowly the hydrogen is produced.
- 2 The higher the temperature, the faster the magnesium powder disappears.
- 3 The lower the concentration of dilute hydrochloric acid, the faster the rate of reaction.
- 4 The faster the magnesium powder disappears, the faster the rate of reaction.

- A 1 and 2 B 2 and 3 C 2 and 4 D 3 and 4

- 14** The reaction between two aqueous compounds, X and Y, is slow and exothermic.

The graph shows how the rate of this reaction changes with time.



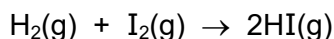
A student suggests that the rate of reaction decreases with time because:

- 1 the activation energy decreases
- 2 the speed of the molecules of X and Y decreases
- 3 the concentration of both X and Y decreases with time.

Which suggestions are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

- 15** Hydrogen reacts with iodine to form hydrogen iodide.



Which statements explain why the reaction is faster when the pressure is increased, at constant temperature?

- 1 At higher pressure, the molecules are moving faster.
- 2 At higher pressure, more of the molecules have the required activation energy.
- 3 At higher pressure, the molecules are closer together.
- 4 At higher pressure, the molecules collide more frequently.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

16 Ammonium sulfate is used as a fertiliser.

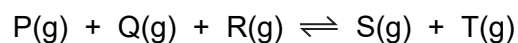
It is made from ammonia and sulfuric acid.

The1..... is made by the2..... process in which3..... is used as a catalyst.

Which words complete gaps 1, 2 and 3?

	1	2	3
A	ammonia	Contact	iron
B	ammonia	Haber	vanadium(V) oxide
C	sulfuric acid	Contact	vanadium(V) oxide
D	sulfuric acid	Haber	iron

17 The reversible reaction shown takes place in a closed system at constant temperature.



When the reaction has reached equilibrium, more T is added.

After the addition of T, which other substances increase in concentration?

- A** P, Q, R and S
- B** P and Q only
- C** P, Q and R only
- D** S only

18 In which equation is the underlined substance acting as a reducing agent?

- A** $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} + 3\text{CO}_2$
- B** $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
- C** $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$
- D** $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

- 19** An aqueous solution reacts with a solid. The products are an alkaline gas, a salt and water.

What are the aqueous solution and the solid?

	aqueous solution	solid
A	sodium hydroxide	magnesium carbonate
B	hydrochloric acid	magnesium carbonate
C	hydrochloric acid	ammonium chloride
D	sodium hydroxide	ammonium chloride

- 20** Butanoic acid partially dissociates in aqueous solution.

Which row about butanoic acid is correct?

	pH	effect on thymolphthalein
A	3	turns blue
B	5	turns colourless
C	8	turns blue
D	10	turns colourless

- 21** Copper(II) sulfate is prepared by adding excess copper(II) carbonate to sulfuric acid.

Why is an **excess** of copper(II) carbonate added?

- A** to ensure all the copper(II) carbonate has reacted
- B** to ensure all the sulfuric acid has reacted
- C** to increase the rate of reaction
- D** to increase the amount of copper(II) sulfate produced

- 22** Part of the Periodic Table is shown.

Which element has two electrons in its outer shell and three electron shells?

- 23** Elements in Group I and Group II show the same trends in their reactions with water and in their density.

Which row shows how the properties of barium compare with calcium?

	reaction with water	density
A	faster	higher
B	faster	lower
C	slower	higher
D	slower	lower

- 24** Which pair of compounds shows a transition element in two different oxidation states?

- A** Cr_2O_3 and $\text{Cr}_2(\text{SO}_4)_3$
B Cu_2O and CuCO_3
C ZnS and ZnSO_4
D NiO and $\text{Ni}(\text{NO}_3)_2$

- 25** Which description of brass is correct?

- A** a compound of copper and zinc
B a compound of copper and tin
C a mixture of copper and zinc
D a mixture of copper and tin

- 26** What is the symbol of the metal used in the manufacture of aircraft because of its low density?

- A** Al **B** Cu **C** Fe **D** Zn

- 27** Which substances react to form hydrogen gas?

- 1 calcium and water
 2 silver and dilute hydrochloric acid
 3 magnesium and steam
 4 zinc and dilute hydrochloric acid

- A** 1, 3 and 4 **B** 1 and 3 only **C** 2 and 4 **D** 4 only

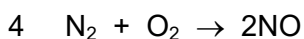
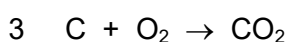
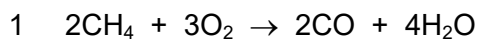
28 Coke (carbon) and limestone are two raw materials used in the extraction of iron from hematite.

Which type of reaction occurs when each substance is heated during the process?

	coke	limestone
A	redox	redox
B	redox	thermal decomposition
C	thermal decomposition	redox
D	thermal decomposition	thermal decomposition

29 Some combustion reactions produce pollutant gases.

Which reactions produce a pollutant gas that is **not** present in clean air?



A 1 and 3

B 1 and 4

C 2 and 3

D 3 and 4

30 One mole of alkane Y produces 72 dm^3 of carbon dioxide when burned in excess oxygen, measured at room temperature and pressure.

What is Y?

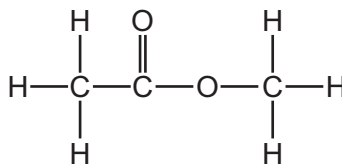
A butane

B ethane

C methane

D propane

31 The structure of organic compound X is shown.



What is X?

- A** ethyl ethanoate
B ethyl methanoate
C methyl ethanoate
D methyl methanoate
- 32 What is the structural formula of the compound formed in the addition reaction of propene with bromine?
- A** $\text{CH}_3\text{CHBrCH}_2\text{Br}$
B $\text{CH}_2\text{BrCH}_2\text{CH}_2\text{Br}$
C $\text{CHBr}_2\text{CH}_2\text{CH}_3$
D $\text{CH}_3\text{CBr}_2\text{CH}_3$
- 33 Ethanol is produced industrially by fermentation and also by a catalysed addition reaction involving steam.

Which row describes one advantage of each process?

	fermentation	catalysed addition reaction involving steam
A	the reactant used is renewable	it is a continuous process
B	the reactant used is renewable	it requires little energy
C	it is a very rapid reaction	it is a continuous process
D	it is a very rapid reaction	it requires little energy

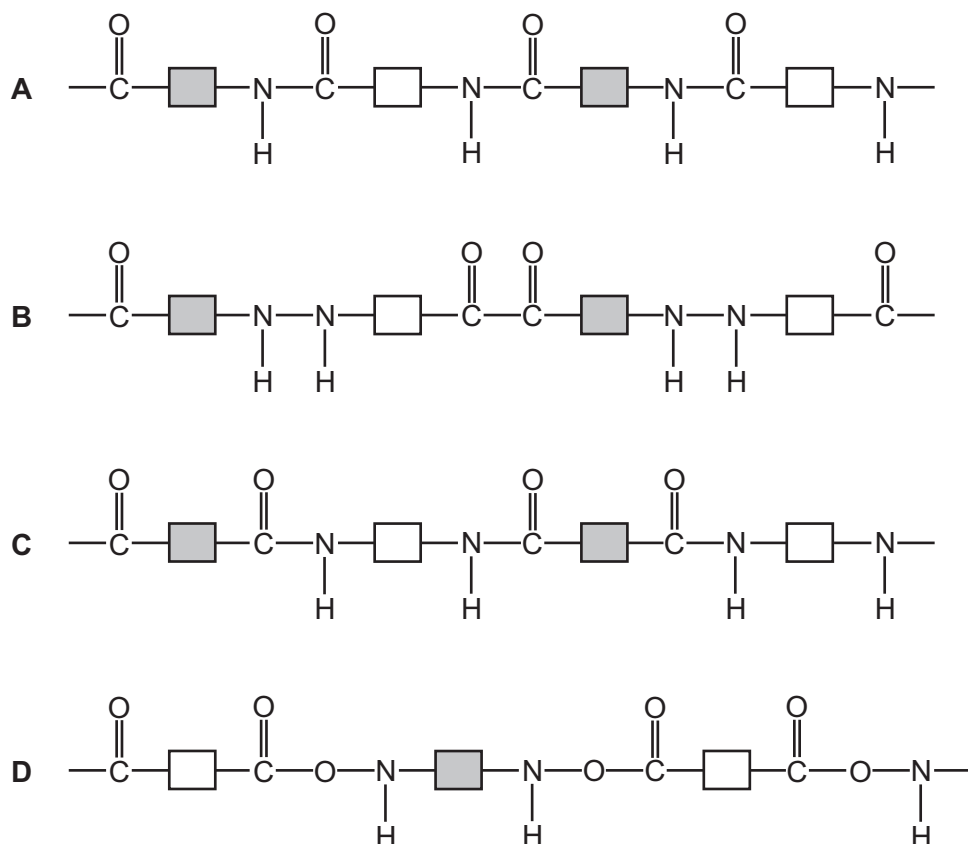
34 Carboxylic acids react with alcohols when warmed with an acid catalyst.

Which type of substance is formed in this reaction?

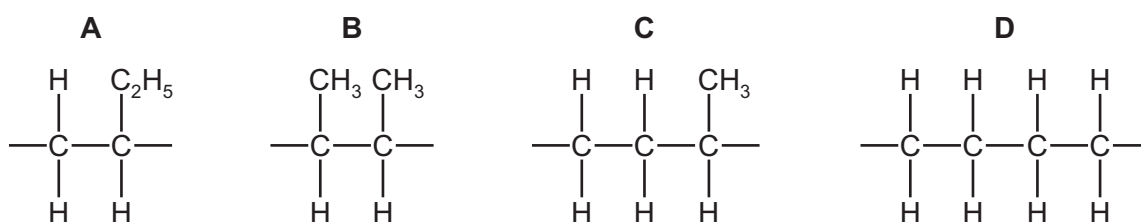
- A** an alkene
B an ester
C a salt
D a polymer

35 Nylon is formed by condensation polymerisation.

Which structure represents nylon?



36 Which structure represents the repeat unit of the addition polymer formed from but-1-ene?

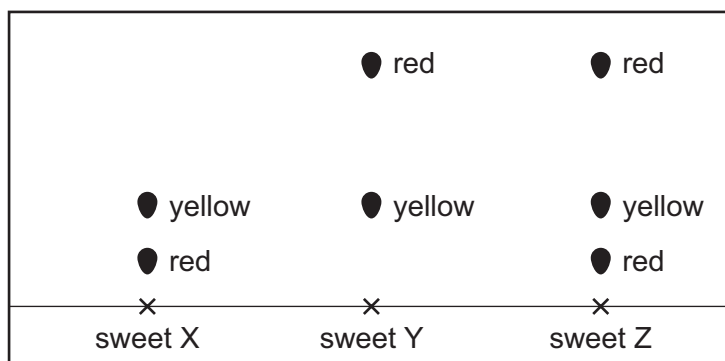


37 2.00 g of powdered calcium carbonate is added to 50.0 cm³ of hydrochloric acid.

Which apparatus is used to measure these quantities of calcium carbonate and hydrochloric acid?

	calcium carbonate	hydrochloric acid
A	balance	burette
B	balance	thermometer
C	pipette	burette
D	pipette	thermometer

- 38 The diagram shows a chromatogram obtained from the colours of three different sweets, X, Y and Z.



How many different **red** dyes are present in the sweets?

- A** 1 **B** 2 **C** 3 **D** 4
- 39 A mixture contains sand and an aqueous solution of sodium chloride.
- Which processes are used to obtain a sample of solid sand **and** a sample of solid sodium chloride from the mixture?
- A** crystallisation followed by filtration
B evaporation followed by filtration
C filtration followed by crystallisation
D simple distillation followed by crystallisation
- 40 A student tests an unknown compound M.

The compound:

- produces a lilac flame using a flame test
- produces a gas which turns limewater cloudy when dilute hydrochloric acid is added.

What is M?

- A** sodium sulfate
B sodium carbonate
C potassium sulfate
D potassium carbonate

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The Periodic Table of Elements

Group

I	II											III	IV	V	VI	VII	VIII	
												1 H hydrogen 1						
												atomic number atomic symbol name relative atomic mass						
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	
lanthanoids																		
57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175				
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —				
actinoids																		

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

February/March 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

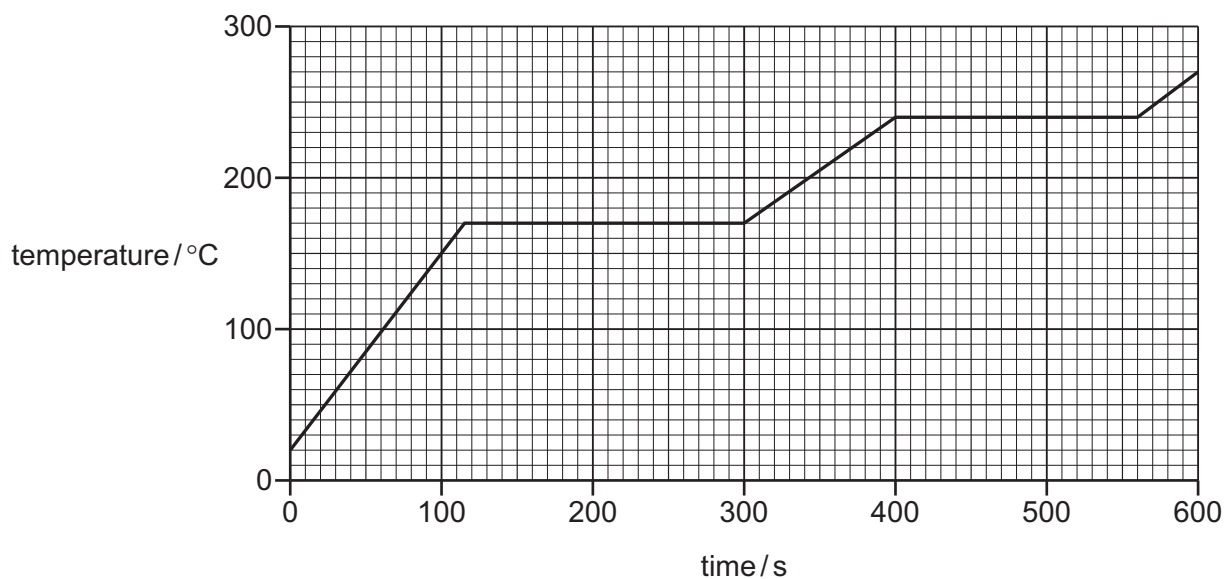
- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Solid X is heated for 600 seconds.

The graph shows the heating curve that is obtained.



What is the melting point of X?

- A** 20 °C **B** 170 °C **C** 240 °C **D** 270 °C

- 2 Which statements about diffusion are correct?

- 1 Aqueous ions cannot diffuse in water.
- 2 Diffusion is caused by the random movement of particles.
- 3 Particles spread out in all directions in diffusion.
- 4 Diffusion can only take place in solids and liquids.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 3 Which statement about an atom of fluorine, $^{19}_{9}\text{F}$, is correct?

- A** It contains a total of 28 protons, neutrons and electrons.
B It contains more protons than neutrons.
C Its isotopes contain different numbers of protons.
D Its nucleus contains 9 neutrons.

- 4 Two of the isotopes of calcium are represented as ${}^{40}_{20}\text{Ca}$ and ${}^{44}_{20}\text{Ca}$.

Which statement explains why these isotopes of calcium have identical chemical properties?

- A Both isotopes have the same number of neutrons.
- B Both isotopes have an electronic configuration of 2,8,8,2.
- C Both isotopes have a mass number of 20.
- D Both isotopes have four fully occupied electron shells.

- 5 Which statement describes a property of potassium iodide?

- A It is insoluble in water.
- B It is a volatile substance.
- C It has a low melting point.
- D It conducts electricity when molten.

- 6 Methanal, CH_2O , has a boiling point of -19°C .

At -20°C , the liquid methanal is a non-conductor of electricity.

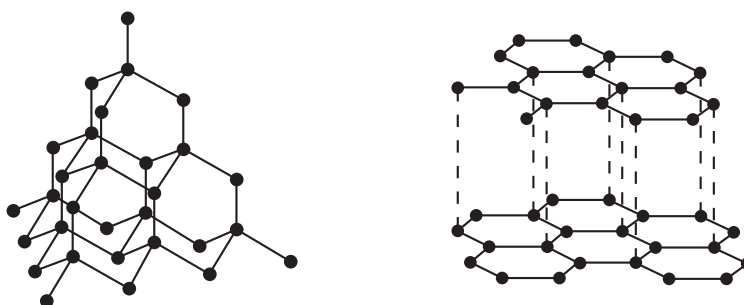
In a sample of methanal, each atom of carbon, hydrogen and oxygen has noble gas electronic configuration. Each atom has achieved this electronic configuration in one of three ways:

- gaining electrons
- losing electrons
- sharing electrons.

Which statement describes the bonding between the carbon atom and the oxygen atom in methanal?

- A The carbon atom and the oxygen atom share four electrons.
- B The carbon atom and the oxygen atom share two electrons.
- C Carbon is a negative ion and oxygen is a positive ion. These two ions attract each other.
- D Carbon is a positive ion and oxygen is a negative ion. These two ions attract each other.

- 7 The structures of diamond and graphite are shown.



Which statement about diamond and graphite is correct?

- A** Diamond and graphite contain strong covalent bonds between carbon atoms.
B Diamond and graphite have delocalised electrons.
C Diamond and graphite have layered structures.
D Diamond and graphite have low melting points.
- 8 Which row contains a description of metallic bonding and a property that is explained by reference to metallic bonding?

	description of metallic bonding	property explained by reference to metallic bonding
A	a lattice of negative ions in a sea of delocalised electrons	a metal will react with an acid, producing hydrogen
B	a lattice of negative ions in a sea of delocalised electrons	a piece of a metal can be moulded into different shapes
C	a lattice of positive ions in a sea of delocalised electrons	a metal will react with an acid, producing hydrogen
D	a lattice of positive ions in a sea of delocalised electrons	a piece of a metal can be moulded into different shapes

- 9 What is the relative molecular mass, M_r , of sulfur dioxide?

A 24 **B** 32 **C** 48 **D** 64

- 10 Magnetite is an ore of iron which contains the ions Fe^{2+} , Fe^{3+} and O^{2-} only.

What is the formula of magnetite?

A Fe_2O **B** Fe_2O_3 **C** Fe_3O_2 **D** Fe_3O_4

- 11** Concentrated aqueous sodium chloride and dilute sulfuric acid are both electrolysed using inert electrodes.

Which row identifies the product at the cathode in each electrolysis?

	aqueous sodium chloride	dilute sulfuric acid
A	hydrogen	oxygen
B	hydrogen	hydrogen
C	chlorine	oxygen
D	chlorine	hydrogen

- 12** Electrolytes can be broken down by electrolysis.

Which rows are correct for each electrolyte?

	electrolyte	reaction at cathode	product at anode
1	dilute aqueous potassium chloride	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	oxygen
2	concentrated hydrochloric acid	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	chlorine
3	molten aluminium oxide	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	aluminium
4	concentrated aqueous sodium bromide	$\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$	bromine

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

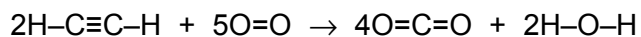
- 13** Which statement about hydrogen–oxygen fuel cells is correct?

- A** Hydrogen is extracted from clean, dry air.
B The only product is carbon dioxide.
C The reaction is endothermic.
D No toxic gases are produced.

- 14** Which statement defines the activation energy, E_a , for a reaction?

- A** It is the minimum energy that colliding particles must have to react.
B It is the minimum energy that endothermic reactions take in from their surroundings.
C It is the maximum energy that exothermic reactions transfer to their surroundings.
D It is the maximum energy released when the bonds in the products of a reaction form.

- 15** The equation for the complete combustion of ethyne, $\text{H}-\text{C}\equiv\text{C}-\text{H}$, is shown.



The bond energies are listed.

bond	bond energy in kJ/mol
$\text{C}\equiv\text{C}$	837
$\text{C}-\text{H}$	415
$\text{O}=\text{O}$	498
$\text{C}=\text{O}$	805
$\text{O}-\text{H}$	464

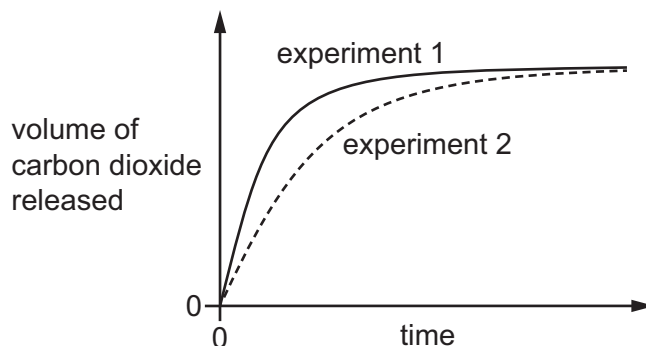
What is the enthalpy change of the reaction when **1 mol** of ethyne is completely burned?

- A** -2472 kJ/mol
- B** -1236 kJ/mol
- C** $+1236 \text{ kJ/mol}$
- D** $+2472 \text{ kJ/mol}$

- 16** In experiment 1, small lumps of limestone are added to dilute ethanoic acid at 40 °C.

The volume of carbon dioxide released is measured at regular time intervals.

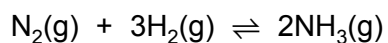
A graph of the results is shown.



Which changes give the results shown in experiment 2?

	limestone	temperature / °C
A	large lumps	40
B	powder	40
C	powder	60
D	small lumps	60

- 17** In the Haber process, nitrogen and hydrogen are reacted to make ammonia.



The forward reaction is exothermic.

Which conditions produce the maximum yield of ammonia?

	pressure	temperature
A	high	high
B	high	low
C	low	high
D	low	low

18 The Ostwald process is used to make nitric acid.

The conditions used in this process are:

- 1 a catalyst containing a transition element
- 2 a pressure of 10 atm
- 3 a temperature of 800 °C.

Which of these conditions are also used in the Contact process?

- A** 1 and 2
B 1 only
C 2 and 3
D 3 only

19 Hydrogen iodide is dissolved in water.



Which row describes the final colours seen when the solution is tested with damp red litmus paper and with acidified aqueous potassium manganate(VII)?

	damp red litmus paper	acidified aqueous potassium manganate(VII)
A	blue	brown
B	blue	colourless
C	red	brown
D	red	colourless

20 Which statements about aqueous ethanoic acid are correct?

- 1 It can be produced by oxidising ethanol with potassium iodide.
- 2 It reacts with magnesium to produce hydrogen gas.
- 3 It has an approximate pH value of 3.
- 4 It produces esters called methanoates.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

21 Which element forms an acidic oxide?

- A calcium
- B lithium
- C magnesium
- D sulfur

22 Which statement describes the properties of hydrochloric acid?

- A Carbon dioxide is produced when limestone reacts with hydrochloric acid.
- B Hydrogen is produced when sodium hydroxide reacts with hydrochloric acid.
- C Methyl orange turns yellow in strong hydrochloric acid.
- D Red litmus paper turns blue when dipped into hydrochloric acid.

23 Elements P and Q have the same number of electron shells.

An atom of Q has more electrons in its outer electron shell than an atom of P.

Which statements are correct?

- 1 P and Q are in the same group of the Periodic Table.
- 2 P and Q are in the same period of the Periodic Table.
- 3 P has a greater tendency to form positive ions than Q.
- 4 The oxide of Q is more basic than the oxide of P.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

24 Which substance reacts with dilute sulfuric acid to form a salt that can be removed from the resulting mixture by filtration?

- A aqueous barium chloride
- B aqueous sodium hydroxide
- C copper
- D copper(II) carbonate

25 Astatine is below iodine in Group VII in the Periodic Table.

Which row describes the properties of astatine?

	state at room temperature	reactivity
A	gas	displaces chlorine, bromine and iodine
B	gas	displaces iodine but does not displace chlorine or bromine
C	solid	displaces iodine but does not displace chlorine or bromine
D	solid	does not displace chlorine, bromine or iodine

26 Which property of copper explains why it is classified as a transition element?

- A** Copper can be bent into different shapes.
- B** Copper forms Cu^{2+} and Cu^+ ions.
- C** Copper is a good conductor of electricity.
- D** Copper has a low density.

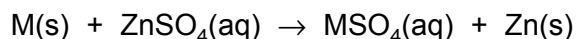
27 Brass is an alloy that is formed from copper and zinc.

Which statements are correct?

- 1 Brass, copper and zinc all conduct electricity.
- 2 Brass is a compound of copper and zinc.
- 3 Brass is harder than zinc.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

28 The equation for the reaction of metal M with aqueous zinc sulfate is shown.



Which statement explains why metal M reacts with aqueous zinc sulfate?

- A** Zinc is less reactive than M because M is able to accept electrons from zinc ions.
- B** Zinc is a more powerful reducing agent than M.
- C** Zinc is more reactive than M because it can lose electrons more easily than M.
- D** Zinc ions can remove electrons from M.

- 29 In the blast furnace, the impurity silicon(IV) oxide is removed by the formation of slag.

Which equation represents the formation of the substance which reacts with silicon(IV) oxide to form slag?

- A $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- B $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$
- C $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- D $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

- 30 Aluminium is extracted from bauxite by electrolysis.

Which statement is correct?

- A Aluminium ions are oxidised to form aluminium.
- B The cathode has to be replaced regularly because it reacts with the oxygen which is formed.
- C Carbon dioxide is produced at the anode.
- D Cryolite is added to remove impurities.

- 31 Iron rusts but aluminium does **not** easily corrode.

Which statement explains why aluminium does **not** easily corrode?

- A It is an alloy.
- B It is below iron in the reactivity series.
- C It is a transition element.
- D Its surface is protected by an oxide layer.

- 32 Which chemicals can be used as a fertiliser to provide the three elements needed for improved plant growth?

- A $(\text{NH}_2)_2\text{CO}$ and KCl
- B $(\text{NH}_4)_2\text{HPO}_4$ and K_2SO_4
- C $(\text{NH}_4)_2\text{HPO}_4$ and $(\text{NH}_2)_2\text{CO}$
- D $(\text{NH}_2)_2\text{CO}$ and K_2SO_4

33 What is the colour change when water is added to anhydrous cobalt(II) chloride?

- A** blue to white
- B** blue to pink
- C** white to blue
- D** white to pink

34 How do carbon dioxide and methane cause global warming?

- A** They emit the thermal energy they have absorbed back to the Earth.
- B** They absorb the radiation directly from the Sun.
- C** They increase thermal energy loss to space.
- D** They reduce reflection of thermal energy from the Earth's surface.

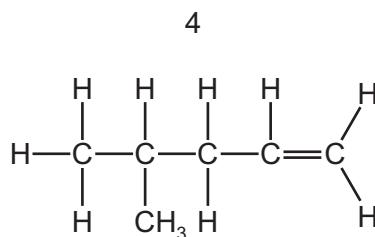
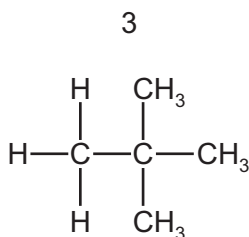
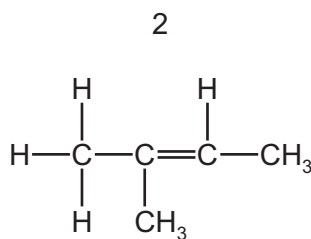
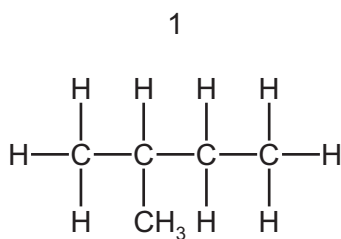
35 Four statements about photosynthesis are listed.

- 1 Chlorophyll is required for photosynthesis.
- 2 The equation for photosynthesis is $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$.
- 3 Photosynthesis requires energy from light.
- 4 Photosynthesis releases carbon dioxide, which can lead to climate change.

Which statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

36 Which molecules are structural isomers?



- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

37 Which statements about the reaction of ethene with steam are correct?

- 1 The product has a higher molecular mass than ethane.
- 2 The product reacts with aqueous bromine.
- 3 The number of electrons shared between carbon atoms decreases.
- 4 The reaction produces an alcohol and hydrogen.

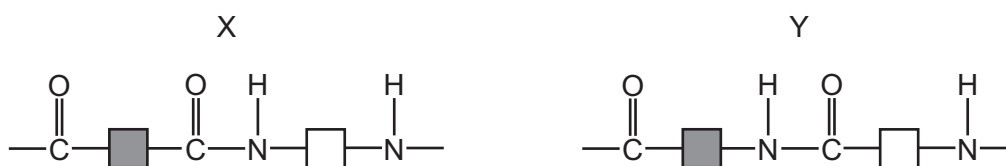
- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

38 Methane and chlorine react to form chloromethane.

Which row describes the necessary reaction condition and the type of reaction?

	reaction condition	type of reaction
A	ultraviolet light	substitution
B	nickel catalyst	substitution
C	nickel catalyst	addition
D	ultraviolet light	addition

39 Parts of the structure of two different polymers, X and Y, are shown.

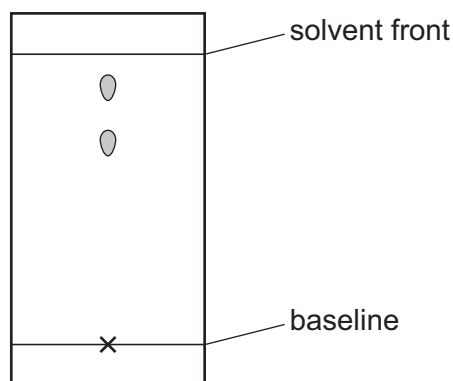


Which row about the monomers and the linkages between the monomers in polymers X and Y is correct?

	monomers in X and Y	linkages
A	different	the linkages in X are different from the linkages in Y
B	different	the linkages in X are the same as the linkages in Y
C	same	the linkages in X are different from the linkages in Y
D	same	the linkages in X are the same as the linkages in Y

40 Substance Q is tested using paper chromatography.

The resulting chromatogram is shown.



Which statement is correct?

- A** Q is a pure substance.
- B** The R_f value of the lower spot is 0.25.
- C** Q is a mixture of at least two different substances.
- D** Q is a compound of two elements.

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The Periodic Table of Elements

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3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
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lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



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0620/22

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- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

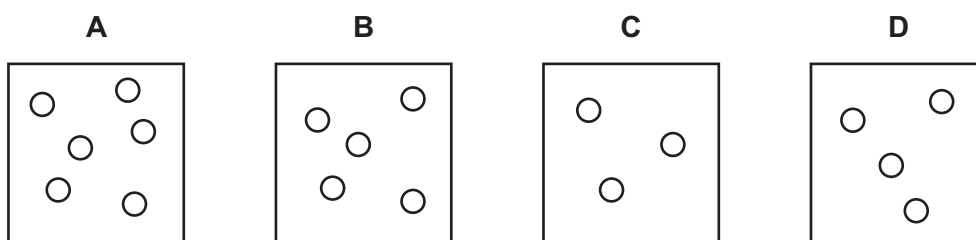
- The total mark for this paper is 40.
- Each correct answer will score one mark.
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- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

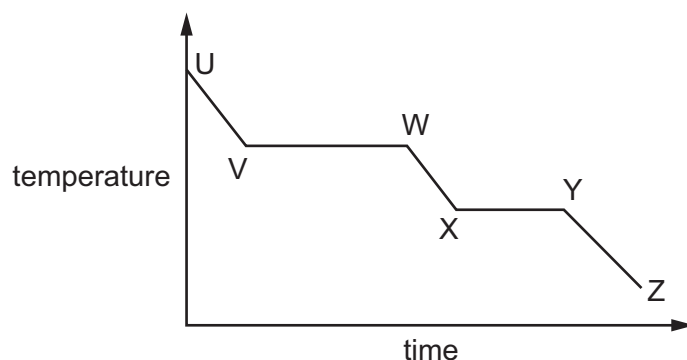


- 1 The diagrams show containers of gas at the same temperature. All containers have the same size.

Which container contains gas at the highest pressure?



- 2 A cooling curve for a substance is shown.



Which statement is correct?

- A Between U and V, the substance is condensing.
 - B Between V and W, heat is being absorbed from the surroundings.
 - C Between W and X, the particles are close together and randomly arranged.
 - D Between Y and Z, the substance is changing from a liquid to a solid.
- 3 Samples of four gases are released in a room at the same time.

The gases are carbon dioxide, CO_2 , hydrogen chloride, HCl , hydrogen sulfide, H_2S , and nitrogen dioxide, NO_2 .

Which gas diffuses fastest?

- A carbon dioxide
- B hydrogen chloride
- C hydrogen sulfide
- D nitrogen dioxide

- 4 Sulfur atoms can form the negative ion S^{2-} .

Three other atoms or ions are listed.

- argon, Ar
- calcium, Ca
- oxide, O^{2-}

How many of these atoms or ions have the same electronic configuration as S^{2-} ?

- A** 0 **B** 1 **C** 2 **D** 3

- 5 Element T has two isotopes, $^{12}_6\text{T}$ and $^{14}_6\text{T}$.

Which statement about these isotopes is correct?

- A** They have different chemical properties because they have different numbers of neutrons.
- B** They have the same chemical properties because they have the same number of outer shell electrons.
- C** They have the same nucleon number because the sum of the number of protons and electrons is the same.
- D** They have different positions in the Periodic Table because they have different numbers of neutrons.

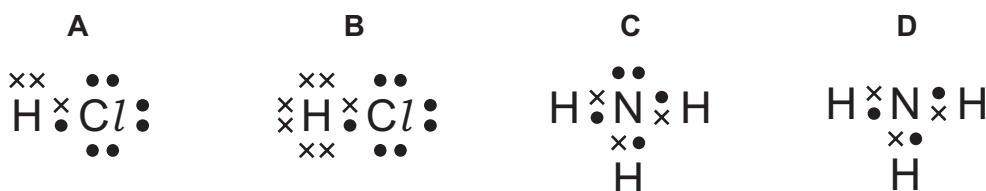
- 6 Lithium is in Group I of the Periodic Table. Nitrogen is in Group V of the Periodic Table.

Lithium reacts with nitrogen to form the ionic compound lithium nitride, Li_3N .

What happens to the electrons when lithium atoms and nitrogen atoms form ions?

	lithium	nitrogen
A	each lithium atom loses one electron to form an Li^+ ion	each nitrogen atom gains three electrons to form an N^{3-} ion
B	each lithium atom loses one electron to form an Li^+ ion	each nitrogen atom gains five electrons to form an N^{5-} ion
C	each lithium atom gains one electron to form an Li^- ion	each nitrogen atom loses three electrons to form an N^{3+} ion
D	each lithium atom gains one electron to form an Li^- ion	each nitrogen atom loses five electrons to form an N^{5+} ion

- 7 For which covalent compound does the dot-and-cross diagram correctly show the outer shell electrons?



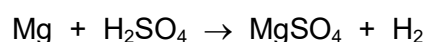
- 8 Which row identifies the positive and the negative particles present in a giant metallic lattice?

	positive particles	negative particles
A	anions	cations
B	anions	delocalised electrons
C	cations	anions
D	cations	delocalised electrons

- 9 Which formula for the named compound is correct?

- A** calcium oxide, CaO
B cobalt(II) chloride, Co_2Cl
C sulfur dioxide, S_2O_2
D anhydrous copper(II) sulfate, $\text{Cu}(\text{SO}_4)_2$

- 10 The equation for the reaction of magnesium with dilute sulfuric acid is shown.



[M_r : MgSO_4 , 120]

Which mass of magnesium sulfate is formed when 12 g of magnesium completely reacts with dilute sulfuric acid?

- A** 5g **B** 10g **C** 60g **D** 120g

- 11 An organic compound, Q, contains carbon, hydrogen and oxygen only.

Q contains 40.0% carbon and 6.7% hydrogen by mass.

What is the empirical formula of Q?

- A** CHO **B** CH_2O **C** C_2HO_2 **D** $\text{C}_3\text{H}_6\text{O}_3$

- 12 The value of the Avogadro constant is 6.02×10^{23} .

What is the total number of atoms in 2.00 mol of ammonia gas?

- A** 1.20×10^{24} **B** 2.41×10^{24} **C** 4.82×10^{24} **D** 2.89×10^{25}

- 13 Three aqueous solutions, L, M and N, are electrolysed using inert electrodes.

L is concentrated hydrochloric acid.

M is concentrated aqueous sodium chloride.

N is dilute aqueous sodium chloride.

Which solutions produce a pale yellow-green gas at the anode?

- A** L and M **B** L only **C** M and N **D** N only

- 14 Dilute sulfuric acid is electrolysed using inert electrodes.

What are the ionic half-equations for the reactions that take place at each electrode?

	positive electrode	negative electrode
A	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
B	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$
C	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
D	$4\text{OH}^- + 4\text{H}^+ \rightarrow 4\text{H}_2\text{O}$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

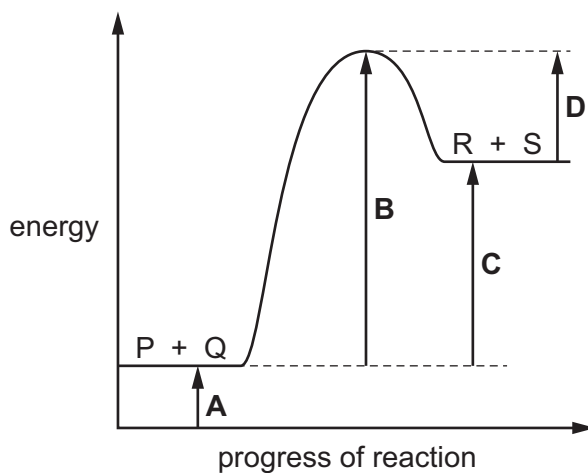
- 15 Which statements about hydrogen–oxygen fuel cells are correct?

- 1 They convert chemical energy into electrical energy.
- 2 Hydrogen is reduced in the fuel cells.
- 3 They do **not** produce any atmospheric pollutants.

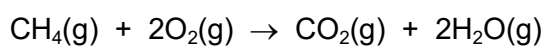
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 16 The reaction pathway diagram for the reaction between P and Q to form R and S is shown.

Which letter represents the enthalpy change for the reaction?



- 17 The equation for the complete combustion of methane is shown.



The table shows some bond energies.

bond	bond energy in kJ/mol
C–H	410
C=O	805
O=O	496
O–H	460

What is the enthalpy change for this reaction?

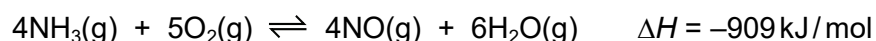
- A** –1458 kJ/mol
B –818 kJ/mol
C –359 kJ/mol
D +102 kJ/mol
- 18 Which change is a physical change?
- A** cracking an alkane
B evaporating ethanol
C fermenting glucose
D neutralising an acid

19 Which statements explain why increasing the temperature in a reaction involving gases increases the rate of reaction?

- 1 It increases the collision frequency between the gas particles.
- 2 It lowers the activation energy.
- 3 It increases the kinetic energy of the gas particles.
- 4 It increases the number of gas particles per unit volume.

A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

20 The equation for the reaction between ammonia and oxygen is shown.

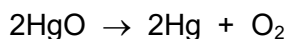


Which two changes to the reaction conditions will **both** move the position of equilibrium to the right?

	change 1	change 2
A	increasing the temperature	decreasing the pressure
B	increasing the temperature	increasing the pressure
C	decreasing the temperature	increasing the pressure
D	decreasing the temperature	decreasing the pressure

21 Mercury(II) oxide, HgO , decomposes when heated.

The equation is shown.



Why is this a reduction reaction?

- A The products weigh less than the reactants.
- B There are fewer reactants than products.
- C There is a gain of oxygen.
- D There is a loss of oxygen.

22 Some information about two dilute acids is shown.

dilute acid	acid concentration in mol/dm ³	pH
nitric acid	0.1	1.0
propanoic acid	0.4	2.6

Three statements about the acids are listed.

- 1 Nitric acid has a lower pH because it dissociates more than propanoic acid.
- 2 Propanoic acid has a lower concentration of hydrogen ions than nitric acid.
- 3 Propanoic acid has a higher pH because it has a higher concentration.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

23 Element E is a metal in Group I of the Periodic Table and element G is a non-metal in Group VII. Both of these elements form oxides.

Which statement about their oxides is correct?

- A** Both oxides are acidic.
B Both oxides are basic.
C The oxide of E is acidic and the oxide of G is basic.
D The oxide of G is acidic and the oxide of E is basic.

24 Lead(II) sulfate is an insoluble salt.

Which method is suitable for obtaining pure solid lead(II) sulfate?

- A** Mix aqueous lead(II) nitrate and aqueous potassium sulfate, heat to evaporate all of the water, collect the solid and then wash and dry it.
B Mix aqueous lead(II) nitrate and aqueous potassium sulfate, filter, collect the filtrate, crystallise, then wash and dry the crystals.
C Mix aqueous lead(II) nitrate and dilute sulfuric acid, filter, then wash and dry the residue.
D Titrate aqueous lead(II) hydroxide with dilute sulfuric acid, crystallise, then wash and dry the crystals.

- 25 The elements oxygen and sulfur are in the same group of the Periodic Table.

Which statement about oxygen and sulfur is **not** correct?

- A They are non-metals.
- B They have giant covalent structures.
- C They have six electrons in the outer electron shells of their atoms.
- D They react together to form an acidic oxide.

- 26 Tennessine, Ts, is at the bottom of Group VII of the Periodic Table.

What are the predicted properties of tennessine at room temperature?

- A a black solid, more dense than iodine
- B a black solid, more reactive than iodine
- C a colourless gas, less dense than chlorine
- D a colourless gas, less reactive than chlorine

- 27 An example of sacrificial protection is the fitting of zinc blocks to the outside of a ship's steel hull.

Which statement explains why zinc is used to protect the iron in the steel from rusting?

- A Zinc is more reactive than iron so it loses electrons more easily.
- B Zinc is less reactive than iron so it loses electrons more easily.
- C Zinc is more reactive than iron so it gains electrons more easily.
- D Zinc is less reactive than iron so it gains electrons more easily.

- 28 Which statement about alloys is correct?

- A Alloys are harder than pure metals because they contain strong intermolecular forces.
- B Brass is an alloy containing mainly copper and tin.
- C The different-sized atoms in an alloy mean that the layers **cannot** easily slide over each other.
- D There are no alloys containing carbon because carbon is a non-metal.

29 Separate pieces of aluminium foil and copper foil are heated in air.

The copper foil reacts to give a black solid.

The aluminium foil does **not** react.

Which statement explains these observations?

- A** Aluminium has an unreactive layer, but copper does **not**.
- B** Aluminium is below copper in the reactivity series.
- C** Copper reacts with moisture in the air, but aluminium does **not**.
- D** Copper reacts with nitrogen in the air, but aluminium does **not**.

30 Which row gives the symbol equation for the formation of carbon monoxide and for the reduction of iron(III) oxide in a blast furnace?

	equation for the formation of carbon monoxide	equation for the reduction of iron(III) oxide
A	$2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$	$\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
B	$\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$	$\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
C	$\text{C} + \text{O}_2 \rightarrow \text{CO}_2$	$\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
D	$\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$	$\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

31 A sample of river water contains a high concentration of nitrates from fertilisers.

Which statements about the river water are correct?

- 1 It has a boiling point of 100 °C.
- 2 Its melting point is below 0 °C.
- 3 It turns anhydrous cobalt(II) chloride from pink to blue.
- 4 It turns anhydrous copper(II) sulfate from white to blue.

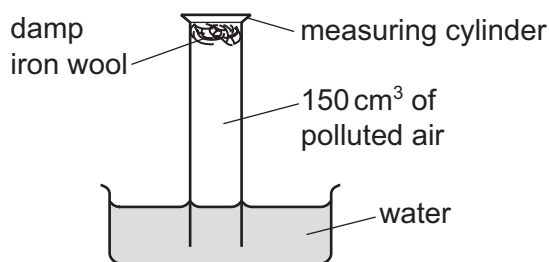
- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

32 Which statements about the treatment of domestic water supplies are correct?

- 1 The water undergoes sedimentation to remove dissolved solids.
- 2 The water is filtered to remove insoluble solids.
- 3 The water is treated with carbon to improve the taste.
- 4 The water is chlorinated to decrease the pH.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 33 An experiment to find the percentage of oxygen in 150 cm^3 of polluted air is shown.



The apparatus is left for one week.

After this time, the volume of gas in the measuring cylinder is 122 cm^3 .

What is the percentage of oxygen, to the nearest whole number, in the polluted air?

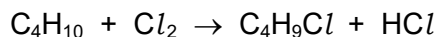
- A** 19% **B** 21% **C** 28% **D** 81%
- 34 Nitrogen monoxide, NO, and carbon monoxide, CO, are both removed from the exhaust gases of a car by a catalytic converter.
- Which statement describes how nitrogen monoxide and carbon monoxide are removed by a catalytic converter?
- A** Nitrogen monoxide and carbon monoxide are both reduced.
B Nitrogen monoxide and carbon monoxide are both oxidised.
C Nitrogen monoxide is oxidised and carbon monoxide is reduced.
D Nitrogen monoxide is reduced and carbon monoxide is oxidised.
- 35 Propan-1-ol is oxidised by acidified aqueous potassium manganate(VII) in a similar way to ethanol.

Which compound is produced by the oxidation of propan-1-ol with acidified aqueous potassium manganate(VII)?

- A** $\text{CH}_3\text{CH}_2\text{OH}$
B $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
C CH_3COOH
D $\text{CH}_3\text{CH}_2\text{COOH}$

36 The structural formula of methyl propane is $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$.

The equation represents the reaction of methyl propane with chlorine.



How many structural isomers with the molecular formula $\text{C}_4\text{H}_9\text{Cl}$ can be formed from this reaction?

- A** 1 **B** 2 **C** 3 **D** 4

37 Which statements describe disadvantages of manufacturing ethanol by fermentation?

- 1 The process uses a renewable resource.
- 2 The process produces impure ethanol.
- 3 The process requires a high temperature.
- 4 The process is slow.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

38 Nylon is made in a polymerisation reaction.

Which row describes the type of polymerisation and identifies the other product of the reaction?

	type of polymerisation	other product
A	addition	water
B	addition	none
C	condensation	water
D	condensation	none

39 Which ion forms a green precipitate with aqueous sodium hydroxide that dissolves in an excess of aqueous sodium hydroxide?

- A** Ca^{2+} **B** Cr^{3+} **C** Cu^{2+} **D** Fe^{2+}

- 40 A mixture of soluble substances can be separated by paper chromatography. Each substance can be identified using its R_f value.

Which formula shows how the R_f value is calculated?

- A $R_f = \frac{\text{distance travelled by solvent}}{\text{distance travelled by substance}}$
- B $R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$
- C $R_f = \text{distance travelled by solvent} - \text{distance travelled by substance}$
- D $R_f = \text{distance travelled by solvent} \times \text{distance travelled by substance}$

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The Periodic Table of Elements

Group																	
I	II											III	IV	V	VI	VII	VIII
1 H hydrogen 1 Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9											5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/21

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

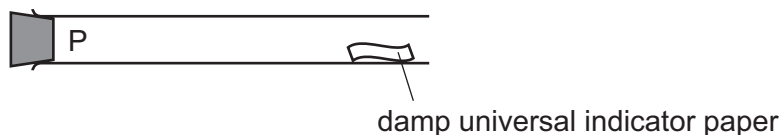
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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

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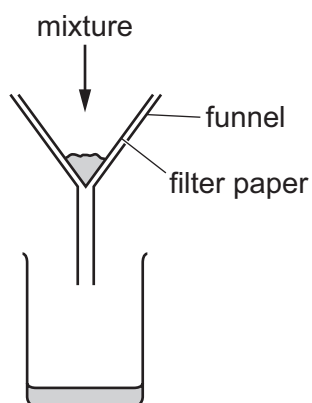


- 1 A gas is released at point P in the apparatus shown.



Which gas turns the damp universal indicator paper red most quickly?

- A ammonia, NH_3
 - B chlorine, Cl_2
 - C hydrogen chloride, HCl
 - D sulfur dioxide, SO_2
- 2 A mixture is separated using the apparatus shown.



What is the mixture?

- A aqueous copper(II) sulfate and aqueous sodium chloride
 - B aqueous copper(II) sulfate and copper
 - C copper and sulfur
 - D ethanol and ethanoic acid
- 3 Which statement about paper chromatography is correct?
- A A solvent is needed to dissolve the paper.
 - B Paper chromatography separates mixtures of solvents.
 - C The solvent should cover the baseline.
 - D The baseline should be drawn in pencil.

- 4 Element X has 7 protons.

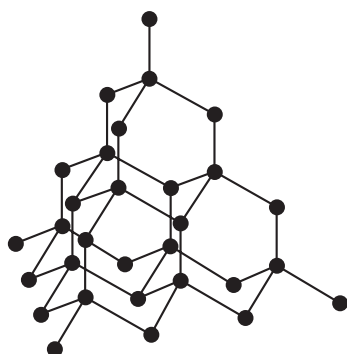
Element Y has 8 more protons than X.

Which statement about element Y is correct?

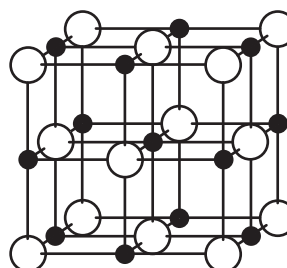
- A** Y has more electron shells than X.
- B** Y has more electrons in its outer shell than X.
- C** Y is in a different group of the Periodic Table from X.
- D** Y is in the same period of the Periodic Table as X.
- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A** ammonia, NH_3
- B** chlorine, Cl_2
- C** methane, CH_4
- D** water, H_2O
- 6 The arrangement of particles in each of two solids, S and T, are shown.



S



T

What are S and T?

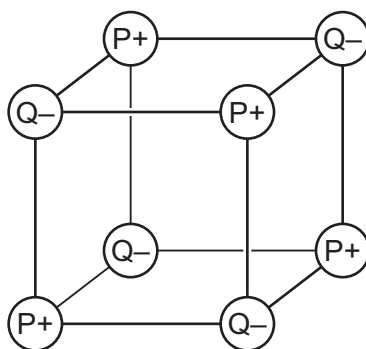
	S	T
A	diamond	silicon(IV) oxide
B	diamond	sodium chloride
C	graphite	silicon(IV) oxide
D	graphite	sodium chloride

7 Which statement about metals is correct?

- A Metals conduct electricity when molten because negative ions are free to move.
- B Metals conduct electricity when solid because positive ions are free to move.
- C Metals are malleable because the bonds between the atoms are weak.
- D Metals are malleable because the layers of ions can slide over each other.

8 Two elements, P and Q, are in the same period of the Periodic Table.

P and Q react together to form an ionic compound. Part of the lattice of this compound is shown.



Which statement is correct?

- A An ion of P has more electrons than an ion of Q.
- B Element P is non-metallic.
- C P is to the left of Q in the Periodic Table.
- D The formula of the compound is P_4Q_4 .

9 2.56 g of a metal oxide, MO_2 , is reduced to 1.92 g of the metal, M.

What is the relative atomic mass of M?

- A 48 B 96 C 128 D 192

10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.

What would happen in **both** experiments?

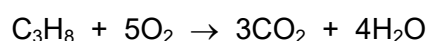
- A A halogen would be formed at the anode.
- B A metal would be formed at the cathode.
- C Hydrogen would be formed at the anode.
- D Hydrogen would be formed at the cathode.

11 What is the ionic half-equation for the reaction that occurs at the cathode when molten lead(II) bromide is electrolysed?

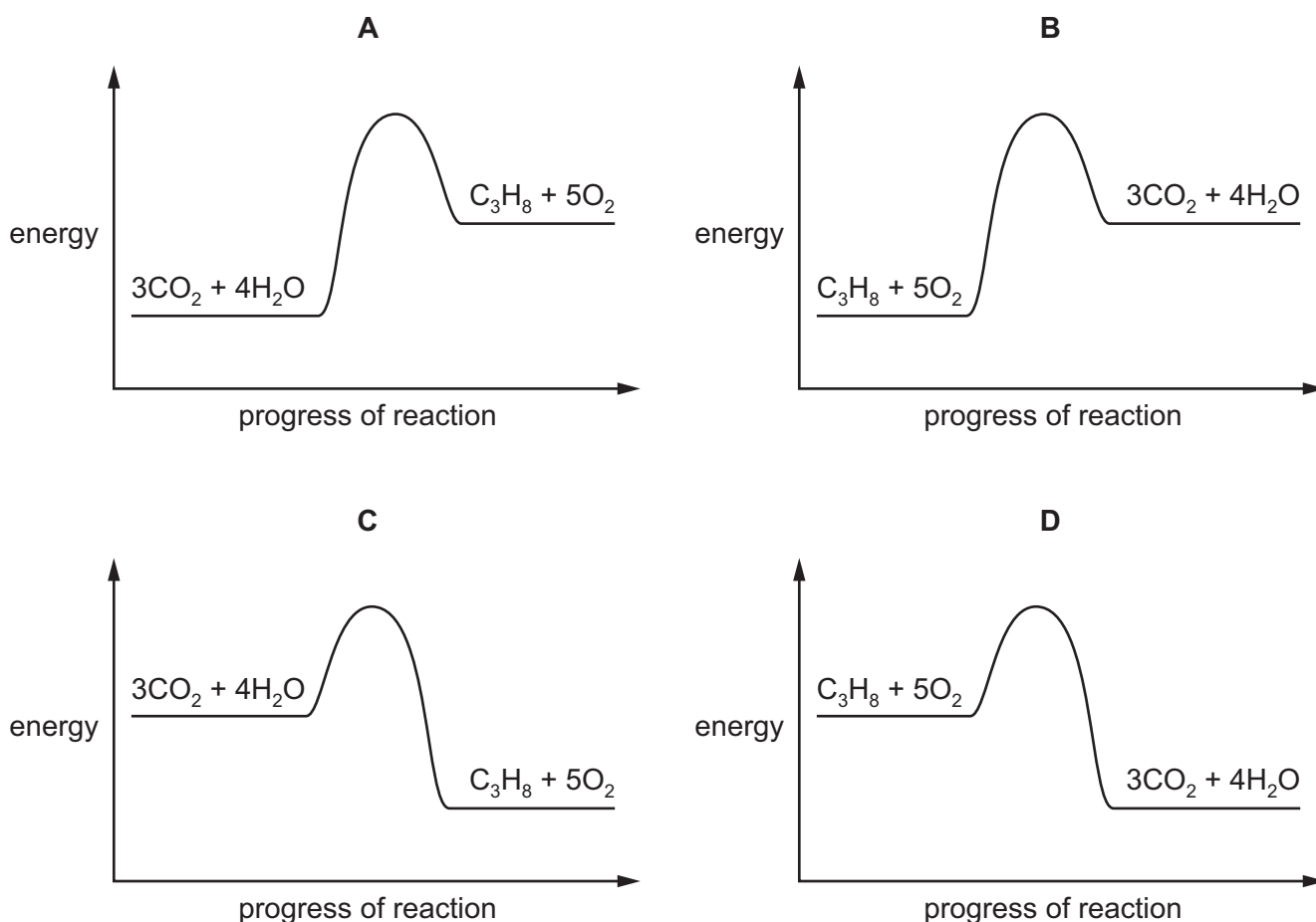
- A $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$
 B $2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$
 C $\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$
 D $\text{Pb} \rightarrow \text{Pb}^{2+} + 2\text{e}^-$

12 The complete combustion of propane is exothermic.

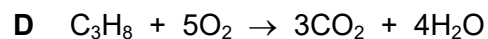
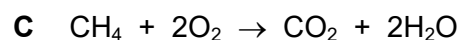
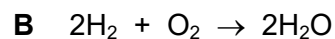
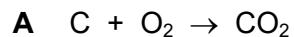
The equation for this reaction is shown.



Which energy level diagram represents the complete combustion of propane?



13 Which equation represents a reaction that takes place in a fuel cell?



14 When sulfur is heated it undergoes a1..... change as it melts.

Further heating causes the sulfur to undergo a2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

	1	2
A	chemical	chemical
B	chemical	physical
C	physical	chemical
D	physical	physical

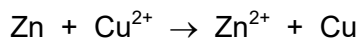
15 Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 2 and 3 only

16 An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A Zn is the oxidising agent and it oxidises Cu^{2+} .
- B Zn is the oxidising agent and it reduces Cu^{2+} .
- C Zn is the reducing agent and it oxidises Cu^{2+} .
- D Zn is the reducing agent and it reduces Cu^{2+} .

17 Which statement about a reaction in equilibrium is correct?

- A Both the forward and the backward reactions are proceeding at the same rate.
- B Neither the forward nor the backward reaction is proceeding.
- C The amount of product present is no longer affected by changes in temperature or pressure.
- D The amount of product present is only affected by a change in pressure.

18 Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

	element X	nature of oxide, XO
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

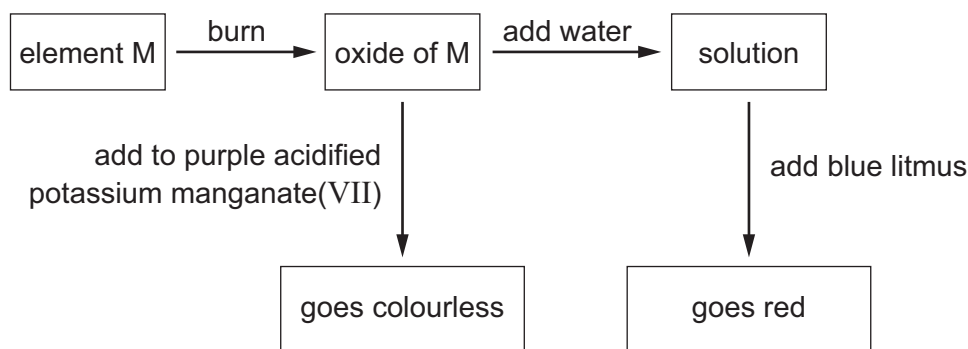
19 Copper(II) sulfate is prepared by adding excess copper(II) oxide to warm dilute sulfuric acid.

Which purification methods are used to obtain pure solid copper(II) sulfate from the reaction mixture?

- 1 crystallisation
- 2 filtration
- 3 chromatography
- 4 distillation

- A 1 and 4
- B 1 and 2
- C 2 and 3
- D 3 and 4

20 Some reactions of element M are shown.



What is element M?

- A carbon
- B iron
- C magnesium
- D sulfur

21 In which equation is the underlined reactant acting as a base?

- A $\text{CH}_3\text{COO}^- + \underline{\text{H}_3\text{O}^+} \rightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
- B $\underline{\text{NH}_4^+} + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$
- C $\text{CO}_2 + 2\underline{\text{H}_2\text{O}} \rightarrow \text{H}_3\text{O}^+ + \text{HCO}_3^-$
- D $\underline{\text{H}^+} + \text{OH}^- \rightarrow \text{H}_2\text{O}$

22 Why is helium used to fill balloons?

- A Helium is monoatomic.
- B Helium is in Group VIII of the Periodic Table.
- C Helium has a full outer electron shell.
- D Helium is less dense than air.

23 Which elements in the table are transition elements?

element	property
E	forms E^{3+} ions only
F	forms F^+ and F^{2+} ions
G	forms only white salts
H	used in catalytic converters

- A E and G
- B E and H
- C F and G
- D F and H

24 Element R forms a covalent compound R_2Si with silicon.

Which row describes R?

	metallic or non-metallic character	group number in the Periodic Table
A	metallic	II
B	metallic	VI
C	non-metallic	II
D	non-metallic	VI

25 Some properties of metal J are listed.

- J does not react with cold water.
- J reacts with dilute hydrochloric acid.
- No reaction occurs when the oxide of J is heated with carbon.

What is J?

- A** copper
- B** iron
- C** magnesium
- D** sodium

26 Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A** calcium
- B** copper
- C** sodium
- D** zinc

27 Which substances are used in the extraction of aluminium?

- A bauxite and cryolite
- B bauxite and hematite
- C cryolite and zinc blende
- D hematite and zinc blende

28 Different types of steel alloys are manufactured by changing the percentage of carbon in the alloy.

The properties of four steel alloys are shown.

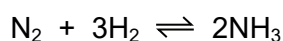
alloy mixture	percentage of carbon in the alloy	strength of the alloy	hardness of the alloy
1	0.00 to 0.20	high	low
2	0.21 to 0.30	high	medium
3	0.31 to 0.40	medium	high
4	0.41 to 1.50	low	high

What are the properties of the steel alloy containing 0.23% of carbon?

	strength	hardness
A	high	low
B	low	high
C	high	medium
D	medium	high

29 Ammonia is made by reacting nitrogen with hydrogen in the Haber process.

The equation for the process is shown.



Which changes in reaction conditions would produce a greater yield of ammonia?

- 1 adding more iron catalyst
- 2 increasing the reaction pressure
- 3 increasing the particle size of the iron catalyst

- A 1 only
- B 2 only
- C 1 and 2
- D 2 and 3

- 30 Which process removes carbon dioxide from the atmosphere?
- A combustion of fossil fuels
 - B fermentation
 - C photosynthesis
 - D respiration
- 31 Which catalyst is used in the Contact process?
- A calcium oxide
 - B iron
 - C manganese(II) oxide
 - D vanadium(V) oxide
- 32 A white solid Z reacts with dilute hydrochloric acid to produce a gas.
The same gas is produced when compound Z is heated strongly.
What is Z?
- A calcium
 - B calcium carbonate
 - C calcium hydroxide
 - D calcium oxide
- 33 What is the structure of butanoic acid?
- A $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
 - B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 - C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$
 - D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$

34 Compound Z contains carbon, hydrogen and oxygen.

Molecules of compound Z have four hydrogen atoms and two carbon atoms.

Compound Z can be made by oxidation of an alcohol.

What is compound Z?

- A** ethene
- B** ethanol
- C** ethanoic acid
- D** methyl methanoate

35 Which statement about homologous series and isomerism is correct?

- A** Butane and butene are structural isomers.
- B** Compounds in the same homologous series have the same general formula.
- C** Compounds in the same homologous series have the same molecular formula.
- D** Structural isomers have different molecular formulae.

36 Which statement about alkanes is correct?

- A** They burn in oxygen.
- B** They contain carbon, hydrogen and oxygen atoms.
- C** They contain double bonds.
- D** They contain ionic bonds.

37 What is an advantage of manufacturing ethanol by fermentation?

- A** The process is very fast.
- B** The ethanol requires no separation.
- C** The raw materials used are renewable.
- D** There are no other products formed.

38 P, Q, R and S are four organic compounds.

P is an unsaturated hydrocarbon.

Q burns but otherwise is unreactive.

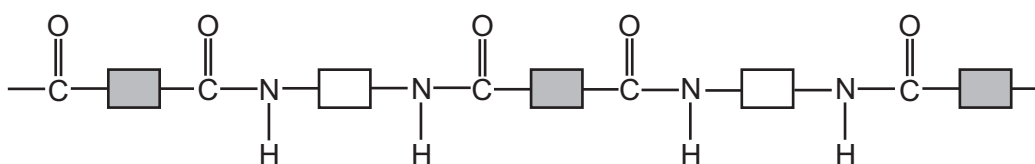
R contains a C–C single bond and a C=C double bond.

S undergoes addition polymerisation.

Which compounds are alkenes?

A P and R only **B** P, R and S **C** P, Q and S **D** Q, R and S

39 The structure of a synthetic polymer is shown.



The structure shows that it is a1..... . It is formed by2..... polymerisation.

Which words complete gaps 1 and 2?

	1	2
A	polyamide	addition
B	polyamide	condensation
C	polyester	addition
D	polyester	condensation

40 Which substance is a natural polymer?

- A** ethene
- B** *Terylene*
- C** nylon
- D** protein

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							2 He helium 4
11 Na sodium 23	12 Mg magnesium 24	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —				

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

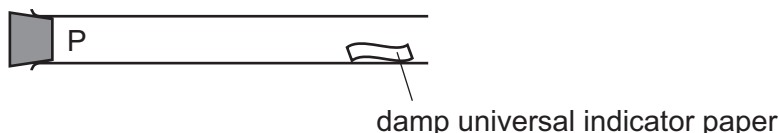
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A gas is released at point P in the apparatus shown.



Which gas turns the damp universal indicator paper red most quickly?

- A ammonia, NH_3
 - B chlorine, Cl_2
 - C hydrogen chloride, HCl
 - D sulfur dioxide, SO_2
- 2 A mixture of colourless compounds is separated using chromatography.
- Which type of reagent is used to detect these compounds after separation?
- A a dehydrating agent
 - B a locating agent
 - C an oxidising agent
 - D a reducing agent
- 3 Which statement about paper chromatography is correct?
- A A solvent is needed to dissolve the paper.
 - B Paper chromatography separates mixtures of solvents.
 - C The solvent should cover the baseline.
 - D The baseline should be drawn in pencil.
- 4 Element X has 7 protons.

Element Y has 8 more protons than X.

Which statement about element Y is correct?

- A Y has more electron shells than X.
- B Y has more electrons in its outer shell than X.
- C Y is in a different group of the Periodic Table from X.
- D Y is in the same period of the Periodic Table as X.

- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A** ammonia, NH_3
B chlorine, Cl_2
C methane, CH_4
D water, H_2O

- 6 Information about four substances E, F, G and H is shown.

	melting point / °C	electrical conductivity
E	1710	does not conduct when solid
F	3500	conducts when solid
G	120	does not conduct
H	801	conducts when molten

E, F, G and H are graphite, poly(ethene), sodium chloride and silicon(IV) oxide but not in that order.

What are E, F, G and H?

	E	F	G	H
A	graphite	poly(ethene)	silicon(IV) oxide	sodium chloride
B	sodium chloride	graphite	poly(ethene)	silicon(IV) oxide
C	poly(ethene)	sodium chloride	graphite	silicon(IV) oxide
D	silicon(IV) oxide	graphite	poly(ethene)	sodium chloride

- 7 Chemical compounds formed from a Group I element and a Group VII element contain ionic bonds.

How are the ionic bonds formed?

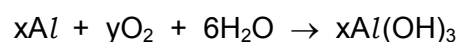
- A** Electrons are transferred from Group VII atoms to Group I atoms.
B Electrons are shared between Group I atoms and Group VII atoms.
C Electrons are lost by Group I atoms and Group VII atoms.
D Electrons are transferred from Group I atoms to Group VII atoms.

- 8 Some information about particles P, Q, R and S is shown.

	nucleon number	number of neutrons	number of electrons
P	12	6	6
Q	24	12	10
R	16	8	10
S	14	8	6

Which two particles are isotopes of the same element?

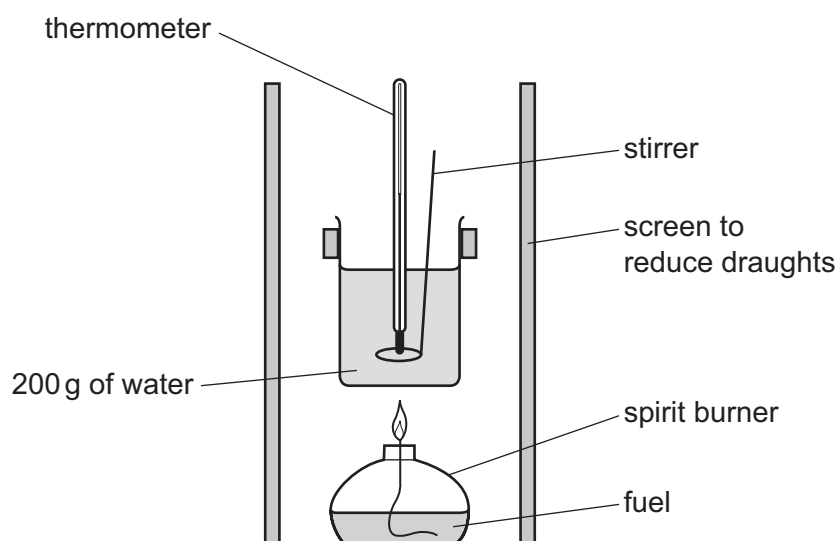
- A** P and Q **B** P and S **C** Q and R **D** R and S
- 9 Chlorine gas will react with iron metal.
- Exactly 21.3 g of chlorine reacts with 11.2 g of iron.
- How many iron atoms react with 30 molecules of chlorine?
- A** 10 **B** 15 **C** 20 **D** 30
- 10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.
- What would happen in **both** experiments?
- A** A halogen would be formed at the anode.
B A metal would be formed at the cathode.
C Hydrogen would be formed at the anode.
D Hydrogen would be formed at the cathode.
- 11 A reaction involving aluminium is shown.



Which values of x and y balance the equation?

	x	y
A	2	3
B	3	2
C	3	4
D	4	3

- 12 Four different fuels are used to heat a beaker of water, for the same amount of time, using the apparatus shown.



The initial temperature of the water and the temperature after heating by the fuel are recorded.

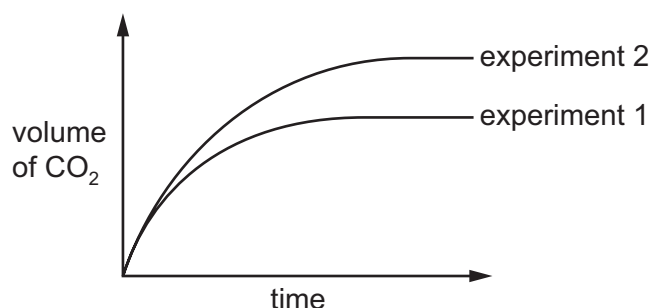
Which fuel releases the most heat energy?

	initial temperature / °C	temperature after heating / °C
A	17	46
B	24	52
C	26	61
D	30	62

- 13** An excess of calcium carbonate reacts with dilute hydrochloric acid. The volume of carbon dioxide produced is measured at regular time intervals. The results are shown as experiment 1.

The experiment is repeated with only **one** change to the reaction conditions.

The results are shown as experiment 2.



Which change is made in experiment 2?

- A** The concentration of the acid is increased.
 - B** The volume of acid is increased.
 - C** The mass of calcium carbonate is increased.
 - D** The calcium carbonate is powdered.
- 14** When sulfur is heated it undergoes a1..... change as it melts.
- Further heating causes the sulfur to undergo a2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

	1	2
A	chemical	chemical
B	chemical	physical
C	physical	chemical
D	physical	physical

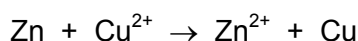
15 Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 2 and 3 only

16 An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A** Zn is the oxidising agent and it oxidises Cu^{2+} .
B Zn is the oxidising agent and it reduces Cu^{2+} .
C Zn is the reducing agent and it oxidises Cu^{2+} .
D Zn is the reducing agent and it reduces Cu^{2+} .

17 When bismuth(III) chloride, BiCl_3 , reacts with water, a white precipitate of bismuth(III) oxychloride, BiOCl , is formed. The equation for the reaction is shown.



The reaction is in equilibrium.

Which changes cause the white precipitate to dissolve?

- 1 adding acid
- 2 adding water
- 3 adding sodium chloride solution

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

18 Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

	element X	nature of oxide, XO
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

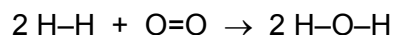
19 Information about the solubility of salts is shown.

salt	solubility
chlorides	soluble (except for lead(II) chloride and silver chloride)
nitrates	soluble
sulfates	soluble (except for barium sulfate and lead(II) sulfate)

Aqueous solutions of which two compounds would produce a precipitate when added together?

- A** $\text{Ba}(\text{NO}_3)_2$ and CaCl_2
- B** CuSO_4 and $\text{Zn}(\text{NO}_3)_2$
- C** KCl and Na_2SO_4
- D** $\text{Pb}(\text{NO}_3)_2$ and MgSO_4

- 20 The equation shows the reaction between hydrogen and oxygen.



The bond energies are shown.

	bond energy in kJ/mol
H-H	436
O=O	495
O-H	463

Which row shows the energy change and the type of reaction?

	energy change in kJ/mol	type of reaction
A	441	exothermic
B	441	endothermic
C	485	exothermic
D	485	endothermic

- 21 Burning fossil fuels releases sulfur dioxide which leads to acid rain.

Which ion in the rain water causes it to be acidic?

- A** H^+ **B** OH^- **C** O^{2-} **D** SO_4^{2-}

- 22 Which statement about the trends shown by the elements of Period 3 in the Periodic Table is **not** correct?

- A** The elements become less metallic across the period.
B The group number increases across the period.
C The number of electron shells increases across the period.
D The number of outer electrons increases across the period.

- 26** Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A** calcium
- B** copper
- C** sodium
- D** zinc

- 27** Aluminium is extracted from its ore by electrolysis.

Which equation represents the reaction that occurs at the anode during the electrolysis?

- A** $Al^{3+} + 3e^{-} \rightarrow Al$
- B** $Al^{3+} \rightarrow Al + 3e^{-}$
- C** $2O^{2-} \rightarrow O_2 + 4e^{-}$
- D** $2O^{2-} + 2e^{-} \rightarrow O_2$

- 28** Mild steel consists mostly of iron. Mild steel can be prevented from rusting by a process called galvanising.

Copper is not a very strong metal. However, if it is mixed with a suitable metal a strong alloy called brass is produced.

Which statement is correct?

- A** Copper corrodes very quickly when wet and brass does not.
- B** Copper is mixed with zinc to produce brass.
- C** Galvanising mild steel changes it from a pure metal into an alloy.
- D** When a steel object is galvanised this means it is coated with a thin layer of tin.

- 29** Water is used for the irrigation of crops and for drinking water.

For which uses must water be chlorinated?

	irrigation	drinking
A	✓	✓
B	✓	x
C	x	✓
D	x	x

30 Which natural resource **cannot** provide a raw material for the manufacture of ammonia?

- A air
- B limestone
- C petroleum
- D water

31 Ammonia is made in the Haber process.

Which conditions are used in the Haber process?

	temperature / °C	pressure / atmospheres	catalyst used
A	450	200	iron
B	450	5	vanadium(V) oxide
C	200	450	iron
D	200	5	vanadium(V) oxide

32 Which process in the carbon cycle is responsible for removing carbon dioxide from the atmosphere?

- A combustion
- B decomposition
- C photosynthesis
- D respiration

33 The equations represent two reactions, P and Q, of lime (calcium oxide).

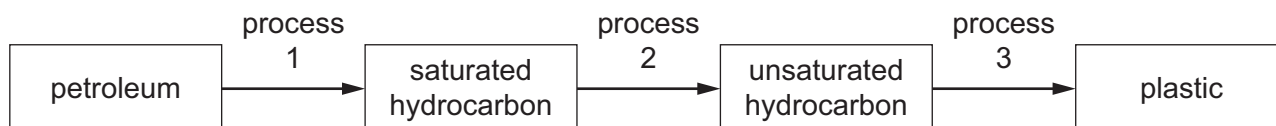


In which processes do the reactions occur?

	P	Q
A	extraction of iron	extraction of iron
B	extraction of iron	flue gas desulfurisation
C	flue gas desulfurisation	extraction of iron
D	flue gas desulfurisation	flue gas desulfurisation

- 34 Which statement about ethanol is **not** correct?
- A Ethanol can be made by fermentation.
 - B Ethanol is oxidised to make ethanoic acid.
 - C Ethanol reacts with oxygen exothermically, making it a good fuel.
 - D Ethanol reacts with propanoic acid to make propyl ethanoate.
- 35 Which pair of formulae represents two alkanes?
- A CH_4 and C_8H_{18}
 - B C_2H_6 and C_5H_8
 - C C_3H_6 and C_5H_{12}
 - D C_{10}H_8 and C_4H_8
- 36 Which statement about alkanes is correct?
- A They burn in oxygen.
 - B They contain carbon, hydrogen and oxygen atoms.
 - C They contain double bonds.
 - D They contain ionic bonds.
- 37 Which statements about ethanoic acid are correct?
- 1 It is a strong acid.
 - 2 It reacts with ethanol to form an ester.
 - 3 It has the formula CH_3COOH .
- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

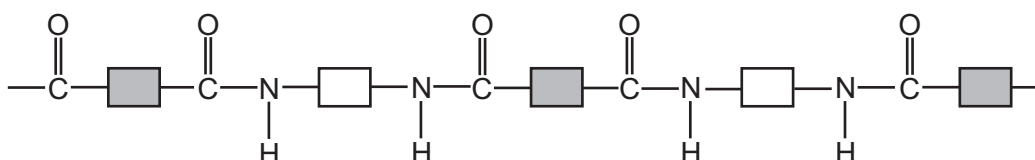
38 The flow chart shows how petroleum may be turned into a plastic.



What are processes 1, 2 and 3?

	process 1	process 2	process 3
A	cracking	fractional distillation	polymerisation
B	cracking	polymerisation	fractional distillation
C	fractional distillation	cracking	polymerisation
D	fractional distillation	polymerisation	cracking

39 The structure of a synthetic polymer is shown.



The structure shows that it is a1..... . It is formed by2..... polymerisation.

Which words complete gaps 1 and 2?

	1	2
A	polyamide	addition
B	polyamide	condensation
C	polyester	addition
D	polyester	condensation

40 Which substance is a natural polymer?

- A** ethene
- B** Terylene
- C** nylon
- D** protein

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—	—	—	—

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/23

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

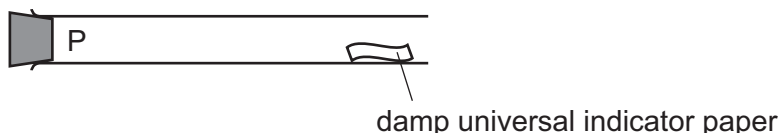
INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A gas is released at point P in the apparatus shown.



Which gas turns the damp universal indicator paper red most quickly?

- A ammonia, NH_3
 - B chlorine, Cl_2
 - C hydrogen chloride, HCl
 - D sulfur dioxide, SO_2
- 2 A 1 cm^3 sample of substance X is taken. This is sample 1.
- X is then converted to a different physical state and a 1 cm^3 sample is taken. This is sample 2.
- Sample 2 contains more particles in the 1 cm^3 than sample 1.
- Which process caused this increase in the number of particles in 1 cm^3 ?
- A boiling of liquid X
 - B condensation of gaseous X
 - C evaporation of liquid X
 - D sublimation of solid X
- 3 Which statement about paper chromatography is correct?
- A A solvent is needed to dissolve the paper.
 - B Paper chromatography separates mixtures of solvents.
 - C The solvent should cover the baseline.
 - D The baseline should be drawn in pencil.
- 4 Element X has 7 protons.
- Element Y has 8 more protons than X.
- Which statement about element Y is correct?
- A Y has more electron shells than X.
 - B Y has more electrons in its outer shell than X.
 - C Y is in a different group of the Periodic Table from X.
 - D Y is in the same period of the Periodic Table as X.

- 5 A covalent molecule Q contains only six shared electrons.

What is Q?

- A ammonia, NH_3
- B chlorine, Cl_2
- C methane, CH_4
- D water, H_2O

- 6 Which statement explains why metals are malleable?

- A The atoms release electrons to become cations.
- B The electrons are free to move.
- C The electrons and the cations are attracted to each other.
- D The layers of ions can slide over each other.

- 7 Which statement about isotopes of the same element is correct?

- A They have different numbers of electrons.
- B They have different numbers of neutrons.
- C They have different numbers of protons.
- D They have the same mass number.

- 8 The element silicon has the same structure as diamond.

Which statement about silicon is correct?

- A Every silicon atom is bonded to three other atoms only.
- B Silicon has a high melting point.
- C Silicon is a good conductor of electricity.
- D Silicon is used as a lubricant.

- 9 Three ionic compounds of vanadium have the formulae V_2O , VCl_2 and V_2O_3 .

What is the charge on the vanadium ion in each compound?

	V_2O	VCl_2	V_2O_3
A	+1	-2	+2
B	+1	+2	+3
C	+2	-2	+2
D	+2	+2	+3

- 10 In separate experiments, electricity was passed through concentrated aqueous sodium chloride and molten lead(II) bromide.

What would happen in **both** experiments?

- A A halogen would be formed at the anode.
- B A metal would be formed at the cathode.
- C Hydrogen would be formed at the anode.
- D Hydrogen would be formed at the cathode.

- 11 The equation for the decomposition of calcium carbonate is shown.



What mass of calcium oxide is produced when 10 g of calcium carbonate is heated?

- A 4.4 g B 5.0 g C 5.6 g D 10.0 g

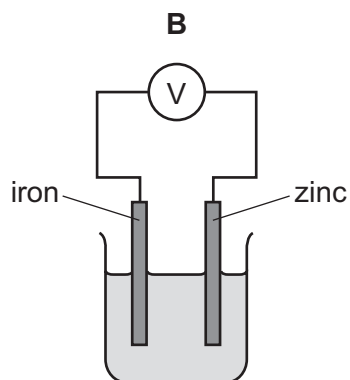
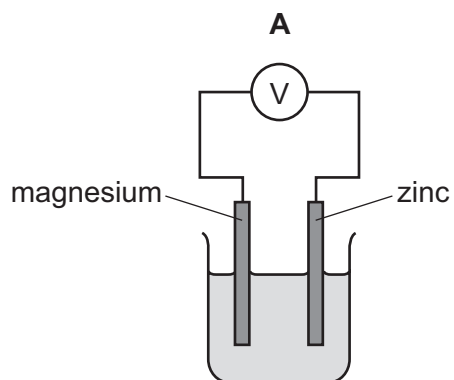
- 12 Gas syringe X contains 100 cm³ of hydrogen bromide gas, HBr.

Gas syringe Y contains 100 cm³ of carbon dioxide gas. The volume of each gas is measured at room temperature and pressure.

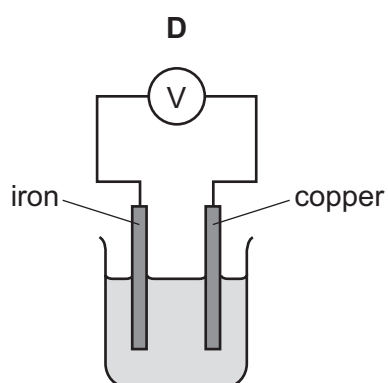
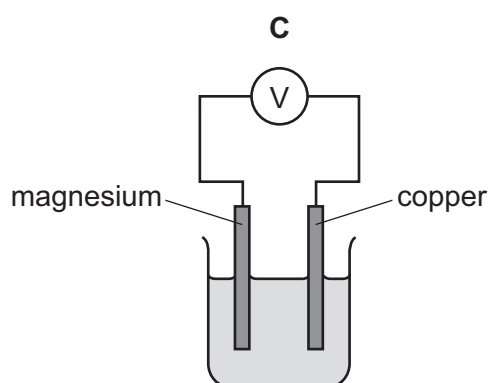
Which statement is correct?

- A The mass of HBr is less than the mass of CO₂.
- B The number of molecules of HBr equals the number of molecules of CO₂.
- C The gas in syringe X contains more atoms than the gas in syringe Y.
- D The number of moles of HBr is more than the number of moles of CO₂.

13 Which simple cell produces the most electrical energy?



key
 = voltmeter



14 When sulfur is heated it undergoes a1..... change as it melts.

Further heating causes the sulfur to undergo a2..... change and form sulfur dioxide.

Which words complete gaps 1 and 2?

	1	2
A	chemical	chemical
B	chemical	physical
C	physical	chemical
D	physical	physical

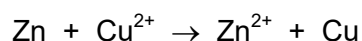
15 Four statements about the effect of increasing temperature on a reaction are shown.

- 1 The activation energy becomes lower.
- 2 The particles move faster.
- 3 There are more collisions between reacting particles per second.
- 4 There are more collisions which have energy greater than the activation energy.

Which statements are correct?

- A** 1, 2 and 3 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 2 and 3 only

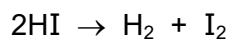
16 An example of a redox reaction is shown.



Which statement about the reaction is correct?

- A** Zn is the oxidising agent and it oxidises Cu^{2+} .
- B** Zn is the oxidising agent and it reduces Cu^{2+} .
- C** Zn is the reducing agent and it oxidises Cu^{2+} .
- D** Zn is the reducing agent and it reduces Cu^{2+} .

- 17 The equation for the decomposition of hydrogen iodide is shown.



Some bond energies are shown.

bond	bond energy in kJ/mol
H–H	440
I–I	150
H–I	300

What is the energy change for the reaction?

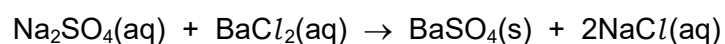
- A** –290 kJ/mol **B** –10 kJ/mol **C** +10 kJ/mol **D** +290 kJ/mol

- 18 Element X forms an oxide, XO, that neutralises sulfuric acid.

Which row describes X and XO?

	element X	nature of oxide, XO
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

- 19 Aqueous solutions of sodium sulfate and barium chloride are mixed.



Which process is used to separate a sample of barium sulfate from the reaction mixture?

- A** precipitation
B filtration
C evaporation
D distillation

20 Information about element J is shown.

- Its atoms have four electrons in their outer shell.
- It is a non-metal.
- Its oxide has a macromolecular structure.
- It has a high melting point.

What is J?

- A** beryllium
- B** carbon
- C** silicon
- D** sulfur

21 Which property is shown by transition elements?

- A** low density
- B** low melting point
- C** variable oxidation state
- D** white compounds

22 Helium and neon exist as monoatomic gases at room temperature and pressure.

statement 1 Helium and neon have eight electrons in their outer shell.

statement 2 Helium and neon are unreactive.

Which option is correct?

- A** Statement 1 and statement 2 are incorrect.
- B** Statement 1 is correct and explains statement 2.
- C** Statement 1 is correct, but does not explain statement 2.
- D** Statement 1 is incorrect, but statement 2 is correct.

23 What are possible effects of an inadequate water supply during a drought?

- 1 crop failure
- 2 wastage of water
- 3 human disease
- 4 death of farm animals

A 1, 2 and 3 **B** 1 and 2 only **C** 1, 3 and 4 **D** 3 and 4 only

24 Which statement explains why galvanising prevents iron from rusting?

- A** Zinc is more reactive than iron and corrodes in preference to iron.
- B** Zinc is more reactive than iron and loses electrons less easily than iron.
- C** Zinc is less reactive than iron and corrodes in preference to iron.
- D** Zinc is less reactive than iron and loses electrons more easily than iron.

25 Some metal nitrates and carbonates decompose when heated strongly.

Metal Q has a nitrate that decomposes to give a salt and a colourless gas only.

The carbonate of metal Q does not decompose when heated with a Bunsen burner.

What is metal Q?

- A** calcium
- B** copper
- C** sodium
- D** zinc

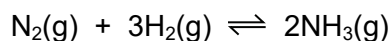
26 Which compounds are released by the extraction of zinc from zinc blende and by respiration?

	extraction of zinc	respiration
A	CO ₂ and SO ₂	CO ₂ only
B	CO ₂ and SO ₂	CO ₂ and H ₂ O
C	CO ₂ only	CO ₂ only
D	CO ₂ only	CO ₂ and H ₂ O

27 Which gas is an air pollutant that causes acid rain?

- A argon
- B carbon monoxide
- C methane
- D nitrogen dioxide

28 Ammonia is made from nitrogen and hydrogen. The equation for the reaction is shown.



The forward reaction is exothermic.

Which conditions give the greatest equilibrium yield of ammonia?

	temperature / °C	pressure / atm
A	200	15
B	200	150
C	500	15
D	500	150

29 Which reaction does **not** occur during the extraction of iron from hematite in a blast furnace?

- A $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- B $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
- C $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$
- D $4\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_2\text{O}_3$

30 Which substance is used as a catalyst in the manufacture of sulfuric acid by the Contact process?

- A iron
- B nickel
- C phosphoric acid
- D vanadium(V) oxide

- 31** Metal X is a good conductor of electricity and is used for electrical wiring.

Metal Y is used to make an alloy which is resistant to corrosion and is used to make cutlery.

Metal Z is light and strong and is used in the manufacture of aircraft.

What are X, Y and Z?

	X	Y	Z
A	aluminium	iron	copper
B	copper	iron	aluminium
C	aluminium	copper	iron
D	copper	aluminium	iron

- 32** The formulae of two compounds of manganese are MnO_2 and KMnO_4 .

In these two compounds the oxidation state of potassium is +1 and the oxidation state of oxygen is -2.

What are the oxidation states of manganese in each of these two compounds?

	MnO_2	KMnO_4
A	+2	+3
B	+2	+7
C	+4	+3
D	+4	+7

- 33** Which statement about calcium carbonate is correct?

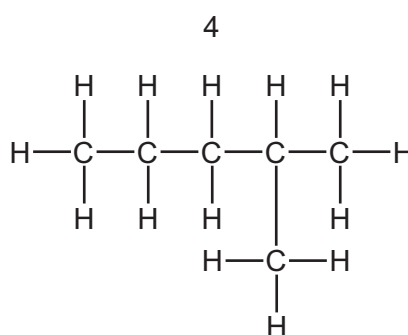
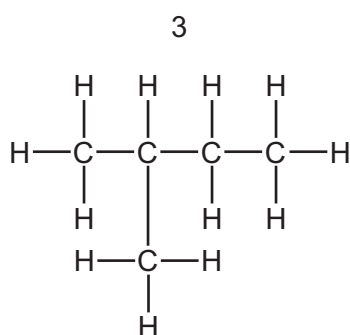
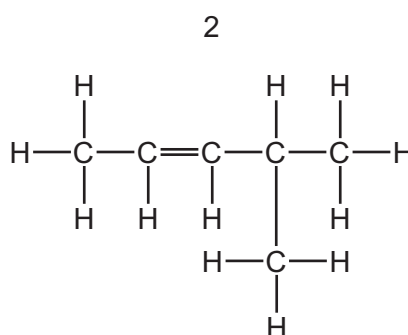
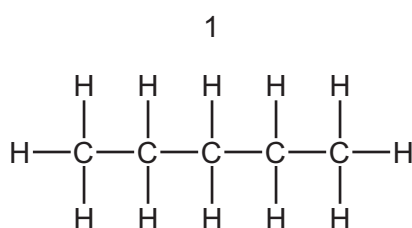
- A** It is made by the thermal decomposition of limestone.
- B** It is used to neutralise alkaline soils.
- C** It is a reactant in the test for carbon dioxide.
- D** It is used to remove impurities in iron extraction.

34 Ethanol is reacted with acidified potassium manganate(VII).

Which row describes the type of reaction and the type of organic compound formed?

	type of reaction	organic compound
A	oxidation	carboxylic acid
B	oxidation	alkene
C	dehydration	carboxylic acid
D	dehydration	alkene

35 The diagrams show the structural formulae of four compounds.



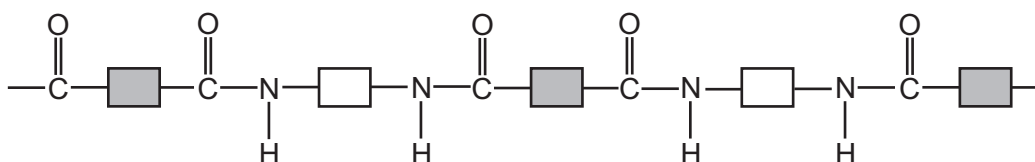
Which two compounds are structural isomers?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

36 Which statement about alkanes is correct?

- A** They burn in oxygen.
B They contain carbon, hydrogen and oxygen atoms.
C They contain double bonds.
D They contain ionic bonds.

- 37 How much hydrogen is needed to react completely with 0.02 moles of butene to make butane?
A 0.24 dm³ **B** 0.48 dm³ **C** 0.96 dm³ **D** 1.20 dm³
- 38 What is an advantage of the fermentation process for producing ethanol compared with the catalytic addition of steam to ethene?
A Fermentation requires less heat energy.
B Ethanol from fermentation needs to be distilled.
C Raw materials for fermentation are non-renewable.
D The fermentation process is carried out in batches rather than continuously.
- 39 The structure of a synthetic polymer is shown.



The structure shows that it is a1..... . It is formed by2..... polymerisation.

Which words complete gaps 1 and 2?

	1	2
A	polyamide	addition
B	polyamide	condensation
C	polyester	addition
D	polyester	condensation

- 40 Which substance is a natural polymer?
A ethene
B *Terylene*
C nylon
D protein

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lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

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0620/21

May/June 2022

45 minutes

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- 1 Which two gases will diffuse at the same rate, at the same temperature?
- A** carbon monoxide and carbon dioxide
B carbon monoxide and nitrogen
C chlorine and fluorine
D nitrogen and oxygen
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm³ of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
 2 measuring cylinder
 3 thermometer
 4 balance

- A** 1, 2 and 4 **B** 1 and 2 only **C** 1 and 4 only **D** 2, 3 and 4
- 3 The numbers of protons and neutrons and the electronic structures of four particles, W, X, Y and Z, are shown.

	number of protons	number of neutrons	electronic structure
W	8	8	2,8
X	8	10	2,6
Y	8	8	2,6
Z	10	8	2,8

Which particles have the same chemical properties?

- A** W and Y **B** W and Z **C** X and Y **D** X and Z
- 4 Which substance should be pure for the intended use?
- A** a drug for curing disease
B limestone for iron extraction
C petroleum for fractional distillation
D water for washing a car

- 5 Metals and ionic compounds have similarities and differences in their structure and properties.

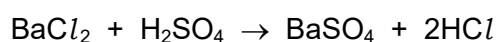
Which row about metals and ionic compounds is correct?

	similarity	difference
A	both contain positive ions	only ionic compounds contain anions
B	both contain positive ions	ionic compounds conduct using a 'sea of electrons'
C	both are malleable	only ionic compounds contain anions
D	both are malleable	ionic compounds conduct using a 'sea of electrons'

- 6 Which diagram represents the outer-shell electron arrangement in a nitrogen molecule?



- 7 The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

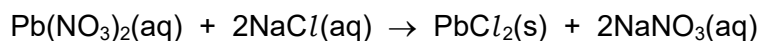
	BaCl_2	H_2SO_4	BaSO_4	2HCl
A	(aq)	(aq)	(s)	(aq)
B	(aq)	(l)	(s)	(aq)
C	(l)	(aq)	(s)	(l)
D	(aq)	(l)	(aq)	(l)

- 8 The relative atomic mass, A_r , of an element is determined by comparing the mass of one atom of the element with the mass of one atom of element Q.

What is Q?

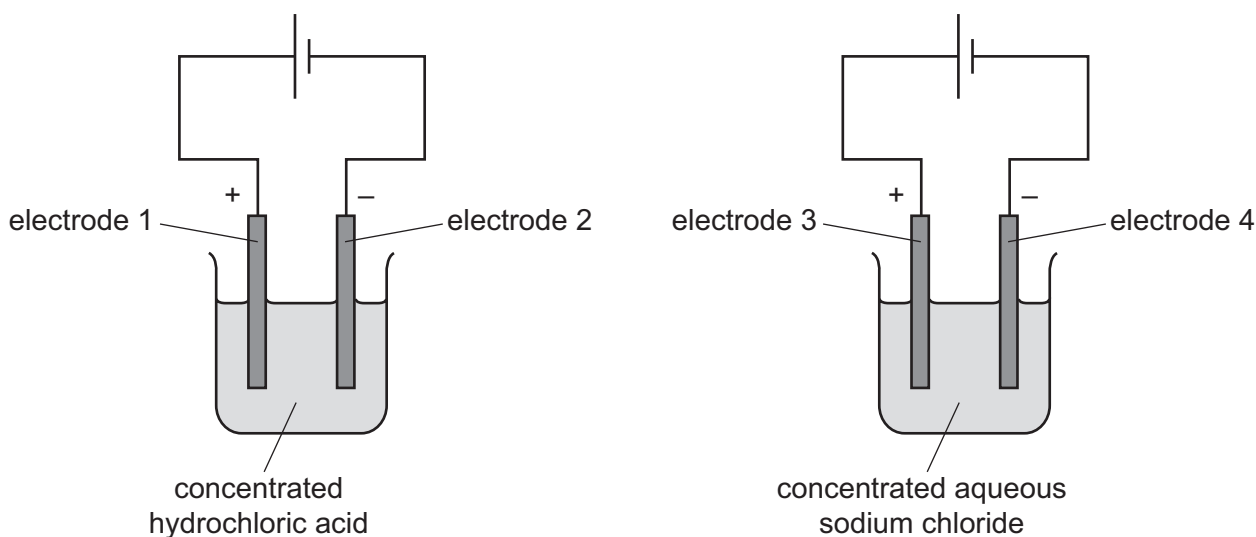
- A** carbon
- B** chlorine
- C** hydrogen
- D** oxygen

- 9 The equation for the reaction between aqueous lead(II) nitrate and aqueous sodium chloride is shown.



If 100 cm^3 of aqueous lead(II) nitrate of concentration 0.1 mol/dm^3 is reacted with an excess of aqueous sodium chloride, which mass of lead(II) chloride is obtained?

- A 1.16 g B 2.42 g C 2.78 g D 3.31 g
- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
 B electrodes 1 and 3
 C electrode 2 only
 D electrodes 2 and 4
- 11 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

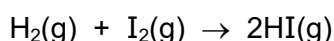
What is the ionic half-equation for the reaction at the cathode?

- A $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
 B $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
 C $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 D $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$

12 Which row identifies a chemical change and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

13 The equation for the reaction between gaseous hydrogen and gaseous iodine to form gaseous hydrogen iodide is shown.



The reaction is exothermic.

Which statement explains why the reaction is exothermic?

- A** Energy is released when H–H and I–I bonds are broken.
- B** The bond energies of the reactants are larger than the bond energies of the products.
- C** The products are at a higher energy level than the reactants.
- D** More energy is released when two HI bonds are formed than is used when the H–H and I–I bonds are broken.

14 Acidified aqueous silver nitrate is added to a test-tube containing aqueous chloride ions.

The test-tube is then left in direct sunlight.

Which row describes the observations and explains what happens to the reaction mixture?

	observation on adding aqueous silver nitrate	observation after leaving in sunlight	explanation
A	yellow precipitate	precipitate dissolves	silver chloride forms
B	yellow precipitate	precipitate turns grey	silver ions are reduced
C	white precipitate	precipitate dissolves	silver chloride forms
D	white precipitate	precipitate turns grey	silver ions are reduced

15 Water is added to anhydrous copper(II) sulfate.

What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

16 Aqueous iron(III) chloride, FeCl_3 , reacts with aqueous potassium iodide, KI .



Which statements are correct?

- 1 In the balanced equation, v , w , x and y have the same value.
- 2 Potassium iodide is an oxidising agent.
- 3 A dark brown solution is produced in the reaction.

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 2 and 3

17 Which statement about acids is correct?

- A** A strong acid has a higher pH than a weak acid of the same concentration.
- B** A strong acid is a proton acceptor.
- C** A weak acid is a proton donor.
- D** A weak acid is fully ionised in aqueous solution.

- 18** The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

oxide tested	pH of solution
X	1
Y	13

Which information about X and Y is correct?

	oxide is acidic	oxide is basic	metal	non-metal
A	X	Y	X	Y
B	X	Y	Y	X
C	Y	X	X	Y
D	Y	X	Y	X

- 19** An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration
B evaporation → crystallisation → filtration
C filtration → crystallisation → evaporation
D filtration → evaporation → crystallisation

- 20** The electronic structure of element Z is 2,8,1.

Which statements about Z are correct?

- 1 It is a metal.
 2 It has two outer-shell electrons.
 3 It is in Period 3.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 only

21 Elements in Group IV of the Periodic Table are shown.

carbon
silicon
germanium
tin
lead

What does **not** occur in Group IV as it is descended?

- A The proton number of the elements increases.
- B The elements become more metallic.
- C The elements have more electrons in their outer shell.
- D The elements have more electron shells.

22 Element M forms both M^+ and M^{2+} ions.

In which part of the Periodic Table is M placed?

- A Group I
- B Group II
- C Group III
- D transition elements

23 In the extraction of aluminium by electrolysis, cryolite is added to the bauxite ore.

Which row describes the role of cryolite and gives the ionic half-equation at the cathode?

	role of cryolite	ionic half-equation at the cathode
A	catalyst	$Al^{3+} + 3e^- \rightarrow Al$
B	catalyst	$Al^{3+} + 3e^- \rightarrow 3Al$
C	lowers melting point of electrolyte	$Al^{3+} + 3e^- \rightarrow Al$
D	lowers melting point of electrolyte	$Al^{3+} + 3e^- \rightarrow 3Al$

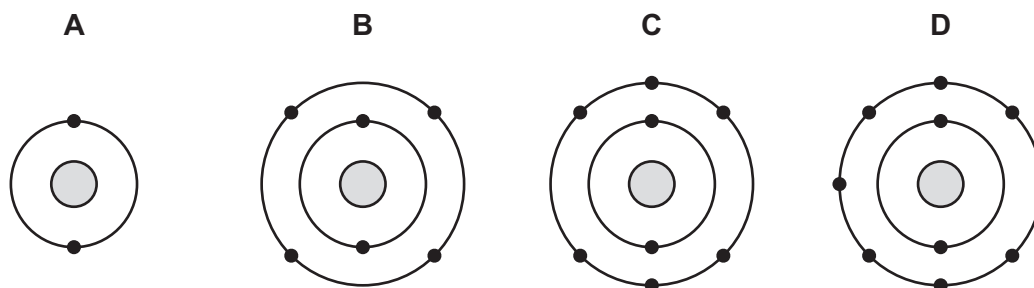
24 Mild steel is galvanised to prevent corrosion of the iron.

Which statements about galvanising are correct?

- 1 Galvanising prevents corrosion because the zinc forms an alloy.
- 2 If the coating is damaged, water and oxygen do not corrode the iron.
- 3 Zinc is a sacrificial metal and corrodes in preference to iron.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 2 and 3

25 Which diagram represents the arrangement of the outer-shell electrons of a noble gas?



26 Which statements about the general properties of metals are correct?

- 1 They are good conductors of heat and electricity.
- 2 They have low melting points.
- 3 They react with dilute acids to form a salt and water.
- 4 They react with oxygen to form basic oxides.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

27 Reactions of three metals and their oxides are shown.

metal	add dilute hydrochloric acid to metal	heat metal oxide with carbon	key ✓ = reacts ✗ = does not react
1	✓	✓	
2	✓	✗	
3	✗	✓	

What is the order of reactivity of these metals, from most reactive to least reactive?

A 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 2 → 3 → 1

28 Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

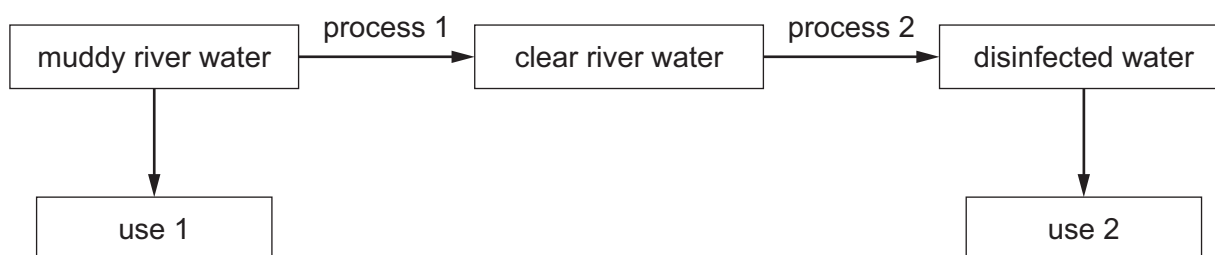
K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

	J	K	L
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

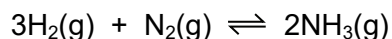
29 The diagram shows the uses and treatment processes of muddy river water.



Which row identifies uses 1 and 2 and processes 1 and 2?

	use 1	use 2	process 1	process 2
A	drinking	watering crops	chlorination	filtration
B	drinking	watering crops	filtration	chlorination
C	watering crops	drinking	chlorination	filtration
D	watering crops	drinking	filtration	chlorination

- 30 The equation for the manufacture of ammonia in the Haber process is shown.



The forward reaction is exothermic.

Which row describes the effect of the stated change on the reaction rate and the yield of ammonia?

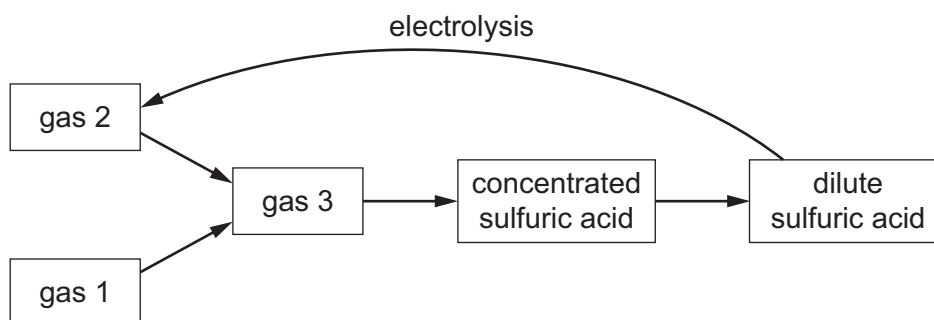
	change	effect on reaction rate	effect on yield of ammonia
A	decrease pressure	increases	decreases
B	decrease temperature	decreases	increases
C	increase pressure	increases	decreases
D	increase temperature	increases	increases

- 31 Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{SO}_4$
B $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_3\text{PO}_4$
C KNO_3 and $(\text{NH}_4)_2\text{SO}_4$
D KNO_3 and $(\text{NH}_4)_3\text{PO}_4$

- 32 The flow chart shows part of the process for the manufacture of sulfuric acid and its electrolysis.



What are gases 1, 2 and 3?

	gas 1	gas 2	gas 3
A	sulfur dioxide	hydrogen	sulfur trioxide
B	sulfur dioxide	oxygen	sulfur trioxide
C	sulfur trioxide	hydrogen	sulfur dioxide
D	sulfur trioxide	oxygen	sulfur dioxide

33 Which statements about sulfur dioxide are correct?

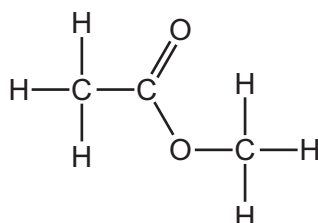
- 1 Sulfur dioxide decolourises acidified potassium manganate(VII).
- 2 Sulfur dioxide forms when acids react with carbonates.
- 3 Sulfur dioxide is used as a bleach.
- 4 Sulfur dioxide is used to treat acidic soil.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

34 What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
B calcium hydroxide and carbon monoxide
C calcium oxide and carbon dioxide
D calcium oxide and carbon monoxide

35 The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

	name of ester W	carboxylic acid	alcohol
A	ethyl methanoate	ethanoic acid	methanol
B	ethyl methanoate	methanoic acid	ethanol
C	methyl ethanoate	ethanoic acid	methanol
D	methyl ethanoate	methanoic acid	ethanol

36 Ethanol is made industrially by the fermentation of glucose or by the catalytic addition of steam to ethene.

Which statement describes an advantage of fermentation compared to catalytic addition?

- A** Ethanol is the only product of fermentation.
B Fermentation uses a batch process but catalytic addition is continuous.
C Fermentation uses a higher temperature than catalytic addition.
D Fermentation uses a renewable resource.

37 Some properties of colourless liquid L are listed.

- It boils at 65 °C.
- When added to water, two layers form which do not mix.
- It does not react with sodium carbonate.
- It has no effect on bromine water.

What is L?

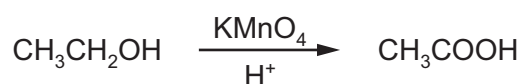
- A** ethanol
B hexane
C hexene
D ethanoic acid

38 A molecule of compound P contains two carbon atoms and four hydrogen atoms.

Which row represents P?

	name of compound	M_r	reacts with aqueous bromine
A	ethane	30	$\times$
B	ethene	16	✓
C	ethene	28	✓
D	ethene	28	$\times$

39 The reaction of ethanol with acidified potassium manganate(VII) is shown.



Which type of reaction is taking place?

- A** addition
B condensation
C hydrolysis
D oxidation

40 Which polymer is a synthetic polyamide?

- A nylon
- B poly(ethene)
- C protein
- D *Terylene*

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1							
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40							9 F fluorine 19	10 Ne neon 20				
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —						

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



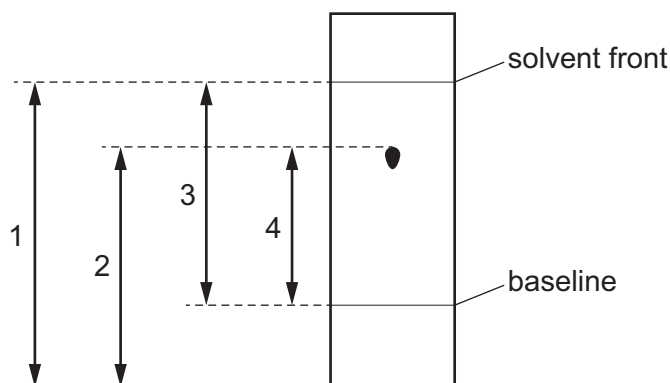
- 1 Which two gases will diffuse at the same rate, at the same temperature?
- A** carbon monoxide and carbon dioxide
B carbon monoxide and nitrogen
C chlorine and fluorine
D nitrogen and oxygen
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm³ of dilute sulfuric acid.

Which apparatus is essential to complete the experiment?

- 1 stop-clock
2 measuring cylinder
3 thermometer
4 balance

A 1, 2 and 4 **B** 1 and 2 only **C** 1 and 4 only **D** 2, 3 and 4

- 3 A chromatogram of a single substance T is shown.



Which measurements are used to find the R_f value of T?

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

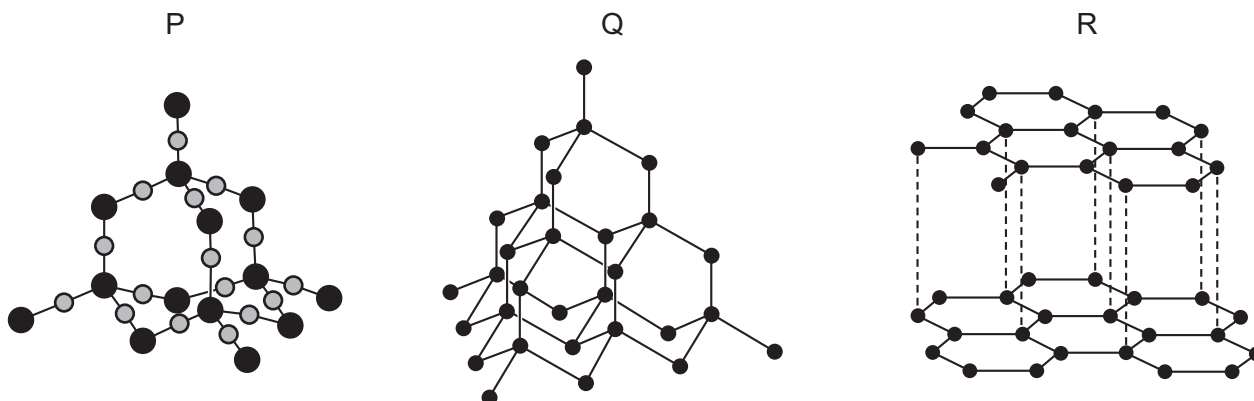
- 4 X and Y are two different elements.

X and Y have the same number of nucleons.

Which statement about X and Y is correct?

- A** They have the same physical properties.
B Their atoms have the same number of electrons.
C They are in different groups of the Periodic Table.
D They have different relative masses.

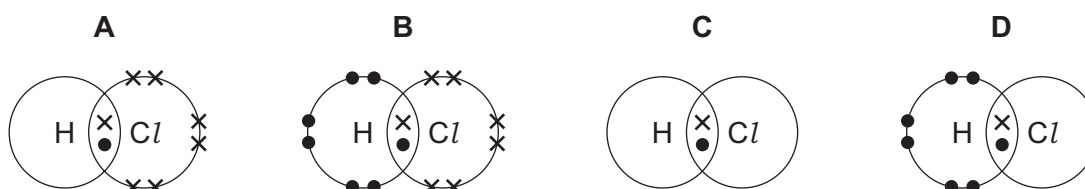
- 5 The diagrams show the structures of three macromolecules P, Q and R.



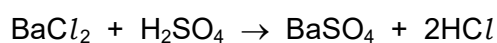
What are P, Q and R?

	P	Q	R
A	diamond	silicon(IV) oxide	graphite
B	graphite	diamond	silicon(IV) oxide
C	silicon(IV) oxide	diamond	graphite
D	silicon(IV) oxide	graphite	diamond

- 6 Which dot-and-cross diagram shows the arrangement of outer shell electrons in a molecule of hydrogen chloride?



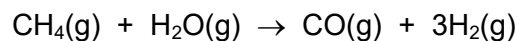
- 7 The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

	BaCl_2	H_2SO_4	BaSO_4	2HCl
A	(aq)	(aq)	(s)	(aq)
B	(aq)	(l)	(s)	(aq)
C	(l)	(aq)	(s)	(l)
D	(aq)	(l)	(aq)	(l)

- 8 Methane and steam react in the presence of a catalyst.



0.5 mol of methane reacts completely with 0.5 mol of steam.

What is the volume of carbon monoxide and hydrogen produced, measured at room temperature and pressure?

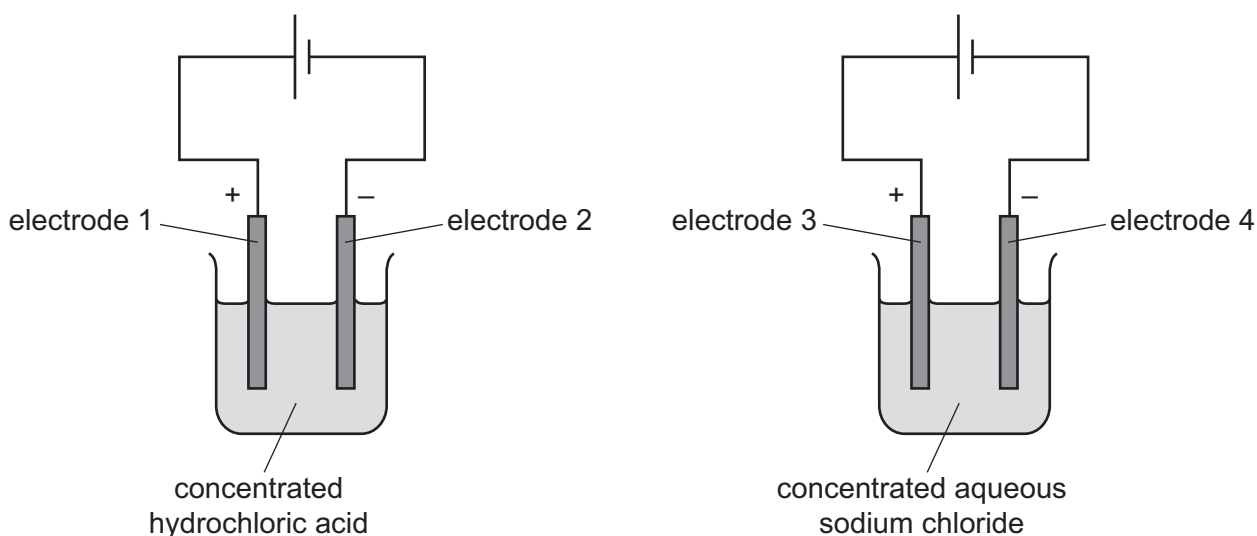
	volume of CO / dm ³	volume of H ₂ / dm ³
A	0.5	1.5
B	1.0	3.0
C	12.0	12.0
D	12.0	36.0

- 9 A compound of element X has the formula X₂O and a relative formula mass of 144.

What is element X?

- A** copper, Cu
- B** gadolinium, Gd
- C** sulfur, S
- D** tellurium, Te

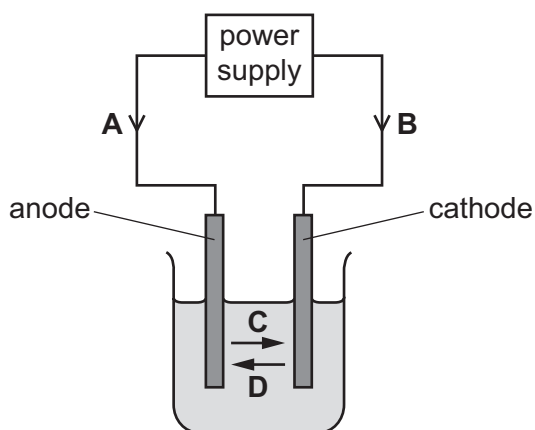
- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A electrode 1 only
 - B electrodes 1 and 3
 - C electrode 2 only
 - D electrodes 2 and 4
- 11 The diagram shows the electrolysis of aqueous copper(II) sulfate using inert electrodes.

Which arrow shows the movement of electrons in the circuit?



12 Which row identifies a chemical change and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

13 Which statements explain why increasing the concentration of a reactant increases the rate of reaction?

- 1 It increases the collision rate of particles.
- 2 It lowers the activation energy.
- 3 A greater proportion of the colliding molecules have the required activation energy.
- 4 There are more particles per unit volume.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

14 When the colourless gas N_2O_4 is heated, it forms the brown gas NO_2 .

When the reaction mixture is cooled, the brown colour fades and turns back to colourless.

Which type of reaction is described by these observations?

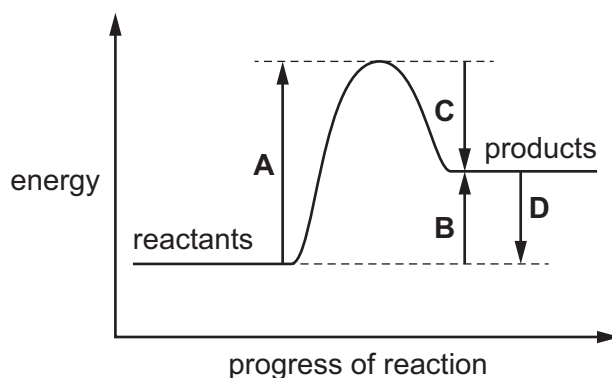
- A** decomposition
- B** displacement
- C** reduction
- D** reversible

15 Water is added to anhydrous copper(II) sulfate.

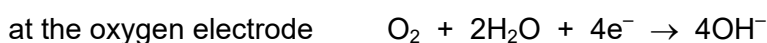
What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

- 16 Which arrow on the energy level diagram shows the overall energy change for an endothermic reaction?



- 17 When a hydrogen–oxygen fuel cell is in operation, a different reaction happens at each electrode.



The electrons that are lost at the hydrogen electrode travel through the external circuit to the oxygen electrode, where they are gained by the oxygen and water.

A hydrogen–oxygen fuel cell is operated for a period of time and four moles of oxygen molecules are consumed.

Which mass of hydrogen is consumed?

- A** 2.0g **B** 4.0g **C** 8.0g **D** 16.0g

- 18 The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

oxide tested	pH of solution
X	1
Y	13

Which information about X and Y is correct?

	oxide is acidic	oxide is basic	metal	non-metal
A	X	Y	X	Y
B	X	Y	Y	X
C	Y	X	X	Y
D	Y	X	Y	X

- 19** An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

- A** crystallisation → evaporation → filtration
- B** evaporation → crystallisation → filtration
- C** filtration → crystallisation → evaporation
- D** filtration → evaporation → crystallisation

- 20** Substance J takes part in a redox reaction.

In the reaction, J gains electrons.

Which statement is correct?

- A** J is the oxidising agent and it is oxidised in the reaction.
- B** J is the oxidising agent and it is reduced in the reaction.
- C** J is the reducing agent and it is oxidised in the reaction.
- D** J is the reducing agent and it is reduced in the reaction.

- 21** Elements in Group IV of the Periodic Table are shown.

carbon
silicon
germanium
tin
lead

What does **not** occur in Group IV as it is descended?

- A** The proton number of the elements increases.
- B** The elements become more metallic.
- C** The elements have more electrons in their outer shell.
- D** The elements have more electron shells.

22 Which statement about acids is correct?

- A Acids are proton acceptors.
- B Acids transfer electrons to bases in aqueous solution.
- C Hydrochloric acid reacts with ammonium hydroxide to produce ammonia.
- D Ethanoic acid partially ionises in aqueous solution.

23 Which elements have both a high melting point and variable oxidation states?

- A alkali metals
- B transition elements
- C halogens
- D noble gases

24 Lithium, sodium and potassium are elements in Group I of the Periodic Table.

Chlorine, bromine and iodine are elements in Group VII of the Periodic Table.

Which row identifies the **least** dense of these elements in each group?

	Group I	Group VII
A	lithium	chlorine
B	lithium	iodine
C	potassium	chlorine
D	potassium	iodine

25 The reactions of metals P, Q, R and S are shown.

metal	reaction with water	reaction with hydrochloric acid	reduction of the metal oxide with carbon
P	no reaction	no reaction	reduced
Q	slow	vigorous	no reaction
R	vigorous	vigorous	no reaction
S	very slow	vigorous	reduced

What is the order of reactivity of the metals?

	least reactive ————→ most reactive			
A	P	S	Q	R
B	P	Q	S	R
C	R	S	Q	P
D	R	Q	S	P

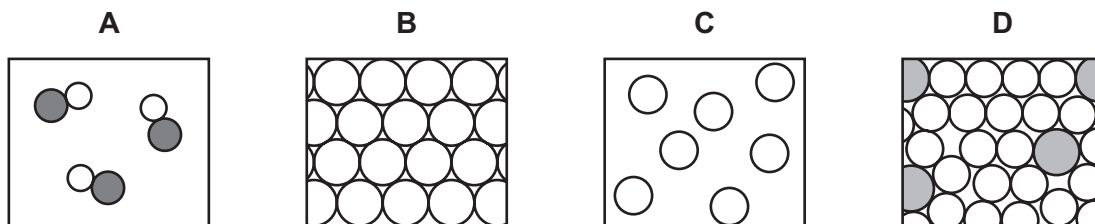
26 The number of protons and the number of neutrons in the atoms of elements X, Y and Z are shown.

	number of protons	number of neutrons
X	6	6
Y	7	6
Z	8	10

Which statement about the elements is correct?

- A** X and Y are isotopes of the same element.
- B** Z forms an ion with a +2 charge.
- C** X and Z react together to form an ionic compound.
- D** X, Y and Z are non-metals.

27 Which diagram represents the arrangement of atoms in an alloy?



28 Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

	J	K	L
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

29 Processes involved in the extraction of zinc are listed.

1 Heat zinc oxide with carbon.

2 Condense zinc vapour.

3 Vaporise the zinc.

4 Roast zinc ore in air.

In which order are the processes carried out?

A 1 → 2 → 3 → 4

B 4 → 3 → 1 → 2

C 4 → 1 → 3 → 2

D 1 → 4 → 3 → 2

30 Which process uses sacrificial protection to prevent steel from rusting?

- A galvanising
- B oiling
- C copper plating
- D painting

31 Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

- A $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{SO}_4$
- B $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_3\text{PO}_4$
- C KNO_3 and $(\text{NH}_4)_2\text{SO}_4$
- D KNO_3 and $(\text{NH}_4)_3\text{PO}_4$

32 Which processes produce carbon dioxide?

- 1 respiration
- 2 photosynthesis
- 3 fermentation
- 4 combustion of hydrogen

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

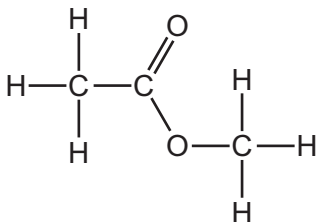
33 Which reaction in the Contact process requires the use of a catalyst?

- A $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$
- B $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$
- C $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$
- D $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$

34 What are the products when limestone (calcium carbonate) is heated strongly?

- A calcium hydroxide and carbon dioxide
- B calcium hydroxide and carbon monoxide
- C calcium oxide and carbon dioxide
- D calcium oxide and carbon monoxide

35 The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

	name of ester W	carboxylic acid	alcohol
A	ethyl methanoate	ethanoic acid	methanol
B	ethyl methanoate	methanoic acid	ethanol
C	methyl ethanoate	ethanoic acid	methanol
D	methyl ethanoate	methanoic acid	ethanol

36 Ethene reacts with substance X to form ethanol.

What is X?

- A** ethanoic acid
- B** glucose
- C** hydrogen
- D** steam

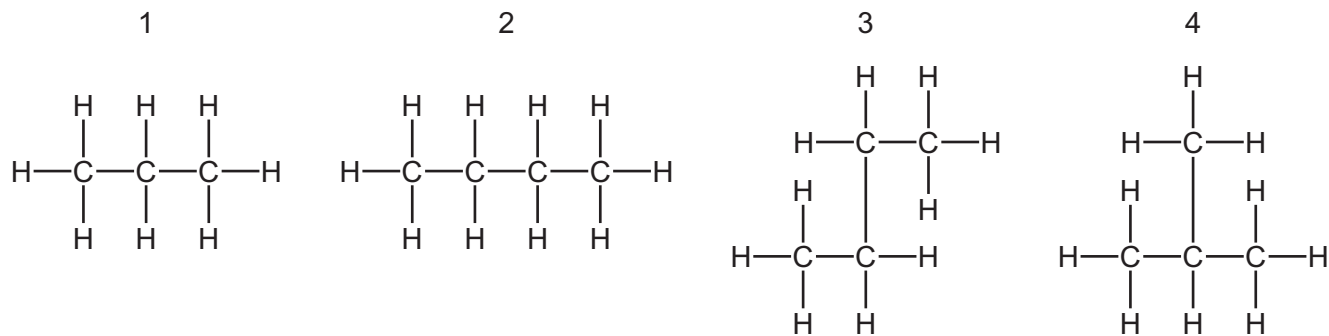
37 Alkenes can be produced by cracking large hydrocarbon molecules to form smaller hydrocarbon molecules.

Which equations represent possible reactions when tetradecane, $C_{14}H_{30}$, is cracked?

- 1 $C_{14}H_{30} \rightarrow C_2H_6 + C_3H_6 + C_4H_8 + C_5H_{10}$
- 2 $C_{14}H_{30} \rightarrow H_2 + C_2H_4 + C_3H_6 + C_4H_8 + C_5H_{10}$
- 3 $C_{14}H_{30} \rightarrow C_2H_6 + 4C_3H_6$
- 4 $C_{14}H_{30} \rightarrow C_2H_6 + C_3H_8 + C_9H_{18}$

- A** 1 only **B** 1 and 4 **C** 1, 2 and 3 **D** 3 and 4

38 The structures of some hydrocarbons are shown.

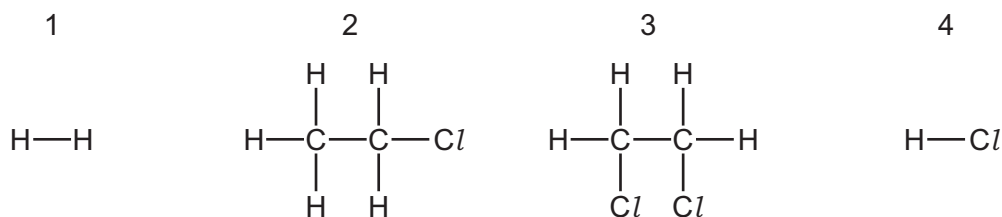


Which statement about the hydrocarbons is correct?

- A** 1 and 2 have a different general formula.
- B** 1 and 4 are in different homologous series.
- C** 2 and 3 are structural isomers.
- D** 3 and 4 have the same empirical formula.

39 Ethane reacts with chlorine in the presence of ultraviolet light.

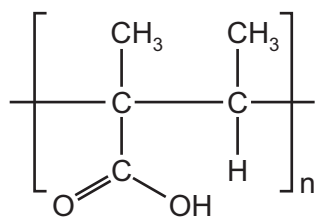
Which substances are produced in the reaction?



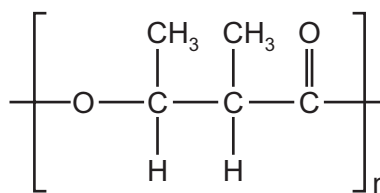
- A** 1, 2 and 3 **B** 1 and 3 only **C** 2, 3 and 4 **D** 2 and 4 only

40 Which polymer structure has the same linkages as *Terylene*?

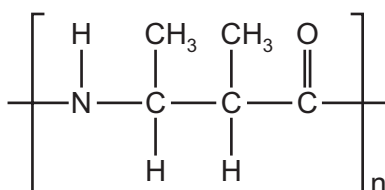
A



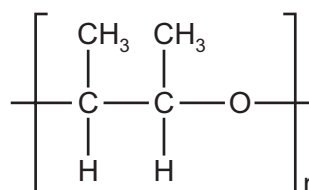
B



C



D



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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23											12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	119 Nh nihonium —	120 Fl flerovium —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/23

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Which two gases will diffuse at the same rate, at the same temperature?
- A** carbon monoxide and carbon dioxide
B carbon monoxide and nitrogen
C chlorine and fluorine
D nitrogen and oxygen
- 2 A student measures the time taken for 2.0 g of magnesium to dissolve in 50 cm³ of dilute sulfuric acid.

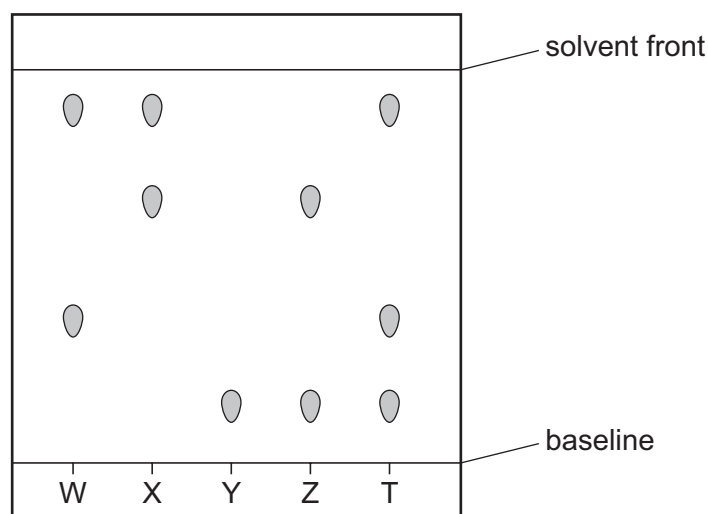
Which apparatus is essential to complete the experiment?

- 1 stop-clock
2 measuring cylinder
3 thermometer
4 balance

- A** 1, 2 and 4 **B** 1 and 2 only **C** 1 and 4 only **D** 2, 3 and 4
- 3 Which statement describes the properties of both diamond and silicon(IV) oxide?
- A** They are brittle, with a low melting point, and are insoluble in water.
B They are hard, with a high melting point, and are electrical insulators.
C They are malleable, with a high melting point, and are electrical conductors.
D They are soft, with a low melting point, and are electrical insulators.

- 4 Paper chromatography is used to separate four different coloured inks, W, X, Y and Z, and an unknown ink T.

The chromatogram is shown.



Which inks are present in ink T?

- A** W and X **B** W and Y **C** X and Z **D** Y and Z

- 5 Particle P has an atomic number of 18, a mass number of 40 and no overall charge.

Particle Q has an atomic number of 19, a mass number of 40 and a single positive charge.

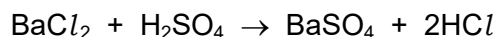
Which statement is correct?

- A** They are isotopes of the same element.
B They are both ions.
C Q has more neutrons than P.
D They have the same number of electrons in their outer shell.

- 6 Which statement about the properties of metals is correct?

- A** Metals are malleable because the layers of positive ions can slide over each other.
B Metals conduct electricity when solid because the positive ions move freely through the metal.
C Metals conduct electricity because there is a strong force of attraction between the positive ions and the delocalised electrons.
D Metals have a high melting point because the positive ions attract each other strongly.

- 7 The equation for the reaction between barium chloride and dilute sulfuric acid is shown.



Which row shows the state symbols for this equation?

	BaCl_2	H_2SO_4	BaSO_4	2HCl
A	(aq)	(aq)	(s)	(aq)
B	(aq)	(l)	(s)	(aq)
C	(l)	(aq)	(s)	(l)
D	(aq)	(l)	(aq)	(l)

- 8 A 0.5 g sample of calcium carbonate is reacted with excess dilute hydrochloric acid.



Which volume of CO_2 is produced at r.t.p.?

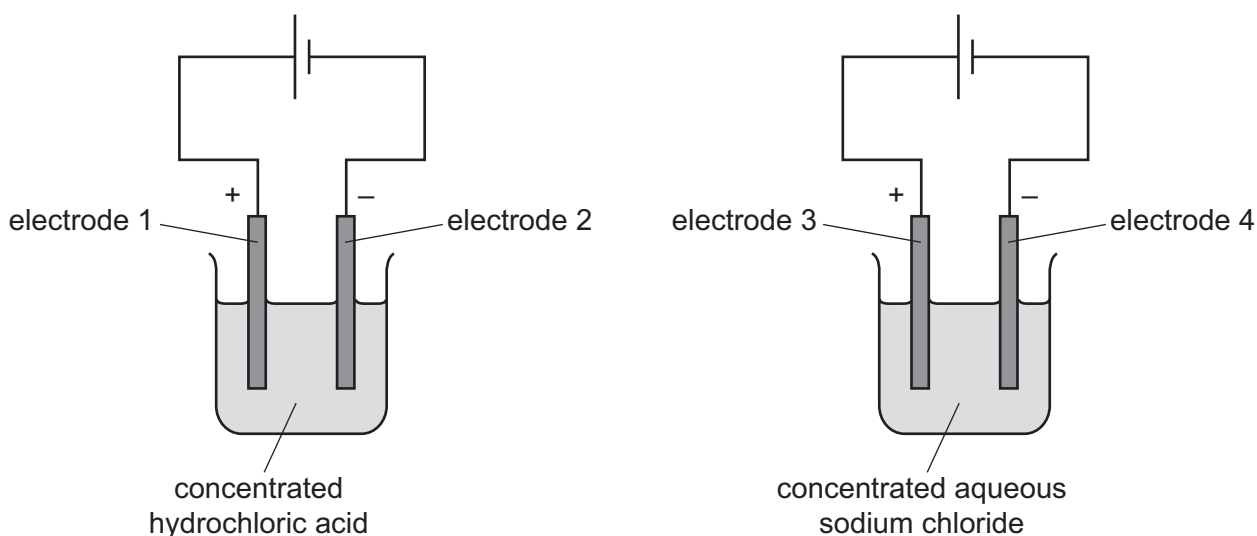
- A** 0.12 dm³ **B** 0.18 dm³ **C** 0.24 dm³ **D** 12 dm³

- 9 Aluminium is manufactured from aluminium oxide by electrolysis.

Which row shows the ionic half-equations at each electrode and describes the role of cryolite in the process?

	reaction at anode	reaction at cathode	role of cryolite
A	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow 3\text{Al}$	catalyst
B	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	solvent for aluminium oxide
C	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	solvent for aluminium oxide
D	$\text{Al}^{3+} + 3\text{e}^- \rightarrow 3\text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	catalyst

- 10 The diagram shows the electrolysis of concentrated hydrochloric acid and concentrated aqueous sodium chloride using carbon electrodes.



At which electrodes is hydrogen produced?

- A** electrode 1 only
B electrodes 1 and 3
C electrode 2 only
D electrodes 2 and 4
- 11 Which statement about fuels is correct?
- A** Coal and ethanol are examples of non-renewable energy sources.
B Hydrogen and oxygen can be reacted to produce an electric current.
C Large amounts of energy are taken in by a fuel when it burns.
D Radioactive isotopes are burned to produce heat.
- 12 Which row identifies a chemical change and a physical change?

	chemical change	physical change
A	boiling ethanol	burning ethanol
B	burning ethanol	evaporating ethanol
C	dissolving ethanol in water	burning ethanol
D	evaporating ethanol	dissolving ethanol in water

- 13** Metal M reacts with steam and produces gas G.

Which row identifies gas G and the type of reaction when metal M reacts with steam?

	gas G	type of reaction
A	hydrogen	redox
B	hydrogen	neutralisation
C	oxygen	redox
D	oxygen	neutralisation

- 14** Which statement explains why increasing the concentration of a reactant increases the rate of the reaction?

- A** A greater proportion of the particles have the activation energy so there are more successful collisions.
- B** Particles have more energy so there are more frequent collisions.
- C** There are more particles in the same volume so there are more frequent collisions.
- D** The particles move more quickly so there are more frequent collisions.

- 15** Water is added to anhydrous copper(II) sulfate.

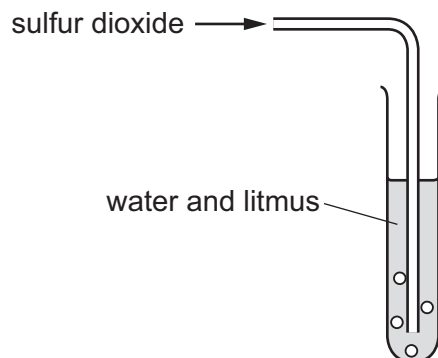
What happens during the reaction?

- A** The copper(II) sulfate turns blue and the solution formed gets colder.
- B** The copper(II) sulfate turns blue and the solution formed gets hotter.
- C** The copper(II) sulfate turns white and the solution formed gets colder.
- D** The copper(II) sulfate turns white and the solution formed gets hotter.

- 16** Which statement explains why lime is added to soil?

- A** to decrease the pH of acidic soil
- B** to decrease the pH of alkaline soil
- C** to increase the pH of acidic soil
- D** to increase the pH of alkaline soil

- 17 Sulfur dioxide is bubbled through water containing litmus.



Which row describes and explains what happens to the litmus?

	observation	explanation
A	it turns blue	sulfur dioxide is a basic oxide
B	it turns blue	sulfur dioxide is an acidic oxide
C	it turns red	sulfur dioxide is an acidic oxide
D	it turns red	sulfur dioxide is a basic oxide

- 18 The oxides of two elements, X and Y, are separately dissolved in water and the pH of each solution tested.

oxide tested	pH of solution
X	1
Y	13

Which information about X and Y is correct?

	oxide is acidic	oxide is basic	metal	non-metal
A	X	Y	X	Y
B	X	Y	Y	X
C	Y	X	X	Y
D	Y	X	Y	X

19 An acid is neutralised by adding an excess of an insoluble solid base.

A soluble salt is formed.

How is the pure salt obtained from the reaction mixture?

A crystallisation → evaporation → filtration

B evaporation → crystallisation → filtration

C filtration → crystallisation → evaporation

D filtration → evaporation → crystallisation

20 Which ion forms a precipitate that dissolves in excess with both aqueous ammonia and with aqueous sodium hydroxide?

A calcium ion, Ca^{2+}

B copper(II) ion, Cu^{2+}

C iron(III) ion, Fe^{3+}

D zinc ion, Zn^{2+}

21 Elements in Group IV of the Periodic Table are shown.

carbon

silicon

germanium

tin

lead

What does **not** occur in Group IV as it is descended?

A The proton number of the elements increases.

B The elements become more metallic.

C The elements have more electrons in their outer shell.

D The elements have more electron shells.

22 W, X, Y and Z are elements in Period 3 of the Periodic Table.

The numbers of outer-shell electrons in an atom of each element are shown.

element	number of outer-shell electrons
W	1
X	2
Y	7
Z	8

Which elements are non-metals?

- A** W, X and Y **B** W and X only **C** Y and Z **D** Z only

23 Selenium is an element in Group VI.

Group VI elements follow similar trends to Group VII elements.

Which statement about selenium is correct?

- A** It has a higher density than sulfur.
B It has a lower melting point than sulfur.
C It has six electron shells.
D It is a monoatomic element.

24 Which row describes the properties of a typical transition element?

	melting point	density	used as catalyst
A	high	high	yes
B	high	low	no
C	low	high	yes
D	low	low	no

25 Which row describes an atom of a noble gas?

	number of protons	number of neutrons	number of electrons
A	2	2	0
B	2	2	2
C	8	8	8
D	8	8	10

26 Some properties of four elements, P, Q, R and S, are shown.

Solid P reacts with dilute hydrochloric acid to give hydrogen.

Solid Q does not conduct electricity.

Solid R is used to make saucepans because it is a good conductor of heat.

Solid S reacts with oxygen to form a compound where atoms of S share electrons with atoms of oxygen.

Which elements are metals?

A P and R **B** P and S **C** Q and R **D** Q and S

27 Which substance is used to reduce zinc oxide in the manufacture of zinc?

- A** carbon
- B** carbon dioxide
- C** hydrogen
- D** sulfur dioxide

28 Three metal compounds, J, K and L, are heated using a Bunsen burner.

The results are shown.

J colourless gas produced, which relights a glowing splint

K colourless gas produced, which turns limewater milky

L no reaction

Which row identifies J, K and L?

	J	K	L
A	magnesium carbonate	potassium carbonate	potassium nitrate
B	magnesium carbonate	potassium nitrate	potassium carbonate
C	potassium nitrate	magnesium carbonate	potassium carbonate
D	potassium nitrate	potassium carbonate	magnesium carbonate

29 Nitrogen oxide, NO, is formed in the engine of petrol-powered cars.

One constituent of petrol is octane, C₈H₁₈.

Nitrogen oxide is removed from exhaust fumes by catalytic converters.

Which row identifies the reactants that produce nitrogen oxide and a reaction that removes it in a catalytic converter?

	reactants that produce NO	reaction that removes NO
A	octane + one gas found in air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
B	octane + one gas found in air	$\text{NO} + \text{CO}_2 \rightarrow \text{NO}_2 + \text{CO}$
C	two gases found in air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
D	two gases found in air	$\text{NO} + \text{CO}_2 \rightarrow \text{NO}_2 + \text{CO}$

30 A magnesium block is attached to iron to prevent it from rusting.

Which statement about this method of rust prevention is correct?

- A** Magnesium corrodes instead of iron because it is more reactive.
- B** Magnesium prevents oxygen from reaching the iron.
- C** The iron does not rust because it has a greater tendency to form ions than magnesium.
- D** This method of rust prevention is called galvanising.

31 Fertilisers are used to provide three of the elements needed for plant growth.

Which two compounds would give a fertiliser containing all three of these elements?

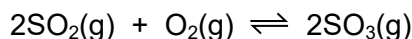
- A** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{SO}_4$
- B** $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_3\text{PO}_4$
- C** KNO_3 and $(\text{NH}_4)_2\text{SO}_4$
- D** KNO_3 and $(\text{NH}_4)_3\text{PO}_4$

32 Which processes increase the amount of carbon dioxide in the air?

- 1 combustion of hydrogen
- 2 combustion of methane
- 3 photosynthesis by plants
- 4 thermal decomposition of limestone

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

33 In the Contact process, sulfur dioxide is converted into sulfur trioxide.



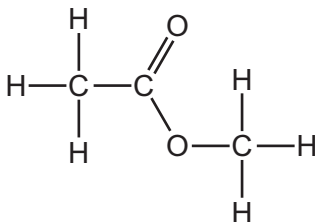
What is the effect of lowering the pressure on the rate of formation and percentage yield of sulfur trioxide at equilibrium?

	rate of formation	percentage yield
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

34 What are the products when limestone (calcium carbonate) is heated strongly?

- A** calcium hydroxide and carbon dioxide
- B** calcium hydroxide and carbon monoxide
- C** calcium oxide and carbon dioxide
- D** calcium oxide and carbon monoxide

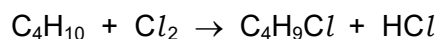
35 The structure of ester W is shown.



Which row gives the names of ester W and the carboxylic acid and alcohol from which it is made?

	name of ester W	carboxylic acid	alcohol
A	ethyl methanoate	ethanoic acid	methanol
B	ethyl methanoate	methanoic acid	ethanol
C	methyl ethanoate	ethanoic acid	methanol
D	methyl ethanoate	methanoic acid	ethanol

36 The equation for the reaction between butane, C_4H_{10} , and chlorine is shown.



Which type of reaction does butane undergo when it reacts with chlorine?

- A** addition
- B** reduction
- C** acid–base
- D** substitution

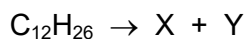
37 Butene has three structural isomers which are alkenes.

Which statements about these isomers are correct?

- 1 They have the same molecular formula.
- 2 They have different numbers of bonds in the molecule.
- 3 They have a $C=C$ bond in the structure.

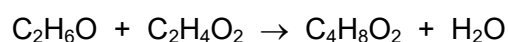
- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

- 38 The hydrocarbon $C_{12}H_{26}$ is cracked to give X and Y, as shown.



Which statement is correct?

- A** If X is C_6H_{12} then Y will react with aqueous bromine.
- B** If X is $C_{10}H_{22}$ then Y can be used to make a polymer.
- C** If X is a hydrogen molecule then Y is an alkane.
- D** X and Y could be structural isomers.
- 39 An ester, $C_4H_8O_2$, is made by reacting 0.06 mol of ethanol, C_2H_6O , and 0.05 mol of ethanoic acid, $C_2H_4O_2$.



0.0375 mol of the ester was made.

What is the percentage yield and the M_r of the ester?

	percentage yield / %	M_r
A	62.5	48
B	75.0	48
C	62.5	88
D	75.0	88

- 40 Which type of compound is made when a protein is hydrolysed?

- A** alkene
- B** amino acid
- C** carboxylic acid
- D** sugar

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	11 Na sodium 23												12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
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87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —		

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/21

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
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- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows the result of dropping a purple crystal into water.



Which processes take place in this experiment?

	chemical reaction	diffusing	dissolving
A	✓	✓	x
B	✓	x	x
C	x	x	✓
D	x	✓	✓

- 2 Which row about elements, mixtures and compounds is correct?

	metallic element	non-metallic element	mixture	compound
A	copper	methane	brass	sulfur
B	brass	sulfur	copper	methane
C	copper	sulfur	brass	methane
D	brass	methane	copper	sulfur

- 3 The atomic structures of four particles, W, X, Y and Z, are shown.

	electrons	neutrons	protons
W	2	2	2
X	2	2	3
Y	2	3	2
Z	3	2	3

Which particles are isotopes of the same element?

- A** W and X **B** W and Y **C** X and Y **D** X and Z

- 4 Which statement explains why isotopes of the same element have the same chemical properties?
- A They have the same number of outer shell electrons.
 B They have the same number of neutrons.
 C They have different numbers of protons.
 D They have different mass numbers.

- 5 Nitrogen forms a nitride ion with the formula N^{3-} .

Which particle does **not** have the same electronic configuration as the nitride ion?

- A Al^{3+} B Cl^- C Na^+ D O^{2-}

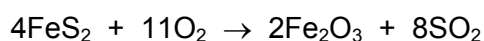
- 6 Which row describes the formation of single covalent bonds in methane?

A	atoms share a pair of electrons	both atoms gain a noble gas electronic structure
B	atoms share a pair of electrons	both atoms have the same number of electrons in their outer shell
C	electrons are transferred from one atom to another	both atoms gain a noble gas electronic structure
D	electrons are transferred from one atom to another	both atoms have the same number of electrons in their outer shell

- 7 Which formula is an empirical formula?

- A $\text{C}_2\text{H}_4\text{O}$
 B $\text{C}_4\text{H}_8\text{O}_2$
 C $\text{C}_3\text{H}_7\text{COOH}$
 D $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

- 8 Heating iron sulfide, FeS_2 , in air produces sulfur dioxide.



What is the maximum mass of sulfur dioxide produced from 120 kg of iron sulfide?

- A 64 kg B 128 kg C 240 kg D 512 kg

9 Which substance produces hydrogen and bromine when electrolysed?

- A concentrated aqueous copper(II) bromide
- B concentrated aqueous sodium bromide
- C dilute aqueous potassium bromide
- D molten lead(II) bromide

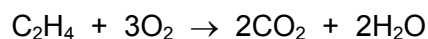
10 Which statements about hydrogen fuel cells are correct?

- 1 Water is formed as the only waste product.
- 2 Both water and carbon dioxide are formed as waste products.
- 3 The overall reaction is $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.
- 4 The overall reaction is endothermic.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

11 Ethene gas, C_2H_4 , is completely burned in excess oxygen to form carbon dioxide and water.

The equation for this exothermic reaction is shown.



The table shows the bond energies involved in the reaction.

bond	bond energy in kJ/mol
C=C	614
C-H	413
O=O	495
C=O	799
O-H	467

What is the total energy change in this reaction?

- A -954 kJ/mol
- B -1010 kJ/mol
- C -1313 kJ/mol
- D -1369 kJ/mol

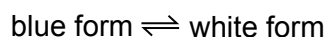
- 12 Which row describes the effect on the activation energy and the frequency of particle collisions when the temperature of a chemical reaction is increased?

	activation energy	frequency of collisions
A	increases	increases
B	no change	increases
C	increases	no change
D	no change	no change

- 13 Solid copper(II) sulfate exists in two different forms, anhydrous and hydrated.

One of these forms is blue and the other is white.

The change between these two forms is reversible.



What is the blue form and how is the change from the blue form to the white form brought about?

	blue form	change to white form
A	anhydrous	add water
B	anhydrous	heat
C	hydrated	add water
D	hydrated	heat

- 14 Sodium ions, Na^+ , and oxygen ions, O^{2-} , combine with chromium ions to form a salt.

The salt sodium dichromate has the formula $\text{Na}_2\text{Cr}_2\text{O}_7$.

What is the oxidation state of chromium in this salt?

- A** +2 **B** +3 **C** +6 **D** +12

- 15 The concentration of hydrogen ions in 100 cm^3 of 0.1 mol/dm^3 hydrochloric acid is higher than the concentration of hydrogen ions in 100 cm^3 of 0.1 mol/dm^3 ethanoic acid.

Which statement explains the difference in hydrogen ion concentration?

- A** Ethanoic acid is an organic acid.
B Ethanoic acid has a lower pH than hydrochloric acid.
C Ethanoic acid is partially dissociated.
D Ethanoic acid is a strong acid.

16 Which oxide is classified as an amphoteric oxide?

- A** aluminium oxide
- B** calcium oxide
- C** copper(II) oxide
- D** nitrogen oxide

17 Which method produces the salt copper(II) carbonate?

- A** Add copper(II) oxide to water, then add excess aqueous sodium carbonate. Filter off the precipitate.
- B** Add copper(II) oxide to dilute sulfuric acid, then add excess aqueous sodium carbonate. Filter off the precipitate.
- C** Add copper to dilute hydrochloric acid, then add aqueous sodium carbonate. Filter off the precipitate.
- D** Add copper(II) oxide to excess aqueous sodium carbonate. Filter off the precipitate.

18 Which statements about the trends across a period of the Periodic Table are correct?

- 1 Aluminium is more metallic than sodium.
- 2 Beryllium is more metallic than carbon.
- 3 Boron is more metallic than lithium.
- 4 Magnesium is more metallic than silicon.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

19 Some information about elements in Group II of the Periodic Table is shown.

element	time taken to make 10 cm ³ of hydrogen gas when 1 g of metal is added to cold water	density in g/cm ³	melting point/°C
beryllium	no reaction	1.85	1280
magnesium	>300 seconds	1.74	650
calcium	60 seconds	1.54	850
strontium	30 seconds	2.62	768
barium	10 seconds	3.51	714

Which row shows the correct trends in reactivity, density and melting point of the elements going down Group II of the Periodic Table?

	reactivity	density	melting point
A	decreases down group	increases down group	decreases down group
B	decreases down group	decreases down group	no clear trend
C	increases down group	no clear trend	increases down group
D	increases down group	no clear trend	no clear trend

20 A new element oxfordium, Ox, was discovered with the following properties.

solubility	electrical conduction	formula of element	bonding in a molecule of Ox ₂
insoluble in water	does not conduct	Ox ₂	Ox≡Ox

In which group of the Periodic Table should the new element be placed?

- A** Group III
- B** Group V
- C** Group VII
- D** Group VIII

21 Which row describes a similarity and a difference between chlorine and bromine?

	similarity	difference
A	both are gases at room temperature and pressure	chlorine and bromine have different colours
B	both exist as diatomic molecules	chlorine is more dense than bromine
C	both have atoms with seven outer-shell electrons	only bromine will react with aqueous sodium chloride
D	both react with aqueous potassium iodide	chlorine is more reactive than bromine

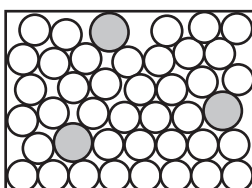
22 Which statement describes transition elements?

- A** They have high densities and high melting points.
- B** They have high densities and low melting points.
- C** They have low densities and high melting points.
- D** They have low densities and low melting points.

23 Which gas is made when powdered zinc is added to dilute hydrochloric acid?

- A** carbon dioxide
- B** chlorine
- C** hydrogen
- D** oxygen

24 The diagram represents the structure of a solid.



Which solids does the diagram represent?

	brass	graphite	sodium chloride
A	✓	✓	x
B	✓	x	x
C	x	✓	✓
D	x	x	✓

25 Steel is an alloy of iron.

Which statement explains why steel is stronger than iron?

- A** Steel contains carbon which is a very hard substance.
- B** The carbon atoms in steel bond together very strongly.
- C** The carbon atoms in steel make the iron atoms bond together very strongly.
- D** The carbon atoms prevent layers of iron atoms from sliding over each other.

26 Three students, X, Y and Z, are told that solid P reacts with dilute acids and also conducts electricity.

The table shows the students' suggestions about the identity of P.

X	Y	Z
copper	iron	graphite

Which students are correct?

- A** X, Y and Z **B** X only **C** Y only **D** Z only

27 Which statement explains why aluminium appears to be unreactive?

- A** It is coated in an oxide layer.
- B** It has a low density.
- C** It is low in the reactivity series.
- D** It is solid at room temperature.

28 During the electrolysis of aluminium oxide, the mass of the carbon anode changes.

Which row describes the change and gives a reason for this change?

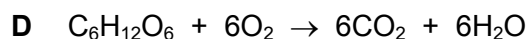
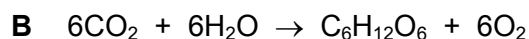
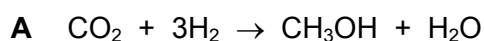
	mass change of the anode	reason
A	decreases	carbon reacts to form carbon dioxide
B	decreases	carbon dissolves in molten cryolite
C	increases	electrodes become coated with cryolite
D	increases	electrodes become coated with aluminium

29 Several processes are used to treat domestic water.

Which row identifies a reason for the given process?

	process	reason
A	chlorination	removes impurities
B	filtration	removes insoluble solids
C	sedimentation	removes soluble solids
D	use of carbon	kills bacteria

30 What is the equation for photosynthesis?



31 Which statement describes how the C–H bonds in methane gas in the atmosphere contribute to global warming?

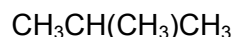
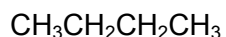
A They absorb thermal energy from the Sun and emit some of this energy into space.

B They absorb thermal energy from the Sun and emit all of this energy towards the Earth.

C They absorb thermal energy from the Earth and emit all of this energy towards the Earth.

D They absorb thermal energy from the Earth and emit some of this energy towards the Earth.

32 The structural formulae of two hydrocarbons are shown.



Which statement about the hydrocarbons is correct?

A They are both alkenes.

B They decolourise aqueous bromine.

C They are structural isomers.

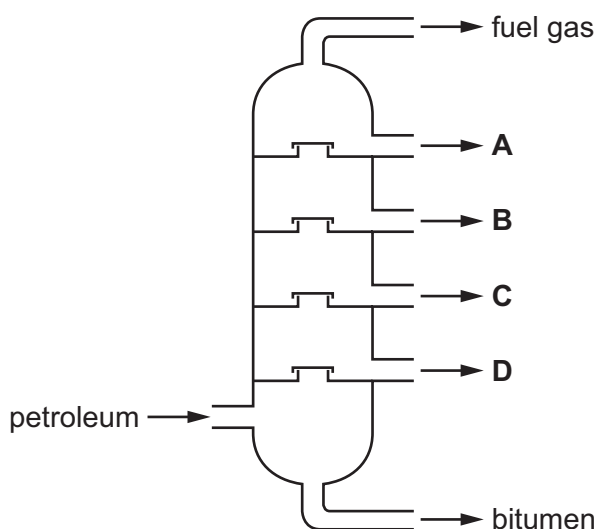
D They undergo addition reactions.

- 33 The structural formula of compound Q is given.



What is compound Q?

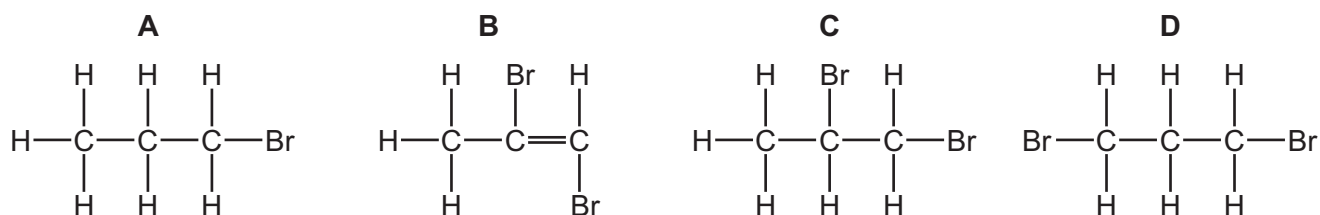
- A** butyl butanoate
B butyl propanoate
C propyl butanoate
D propyl propanoate
- 34 The fractional distillation of petroleum is shown.
- Which fraction contains hydrocarbons with the longest chain length?



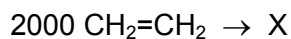
- 35 Which equation represents the cracking of an alkane?

- A** $3\text{C}_2\text{H}_4 \rightarrow \text{C}_6\text{H}_{12}$
B $\text{C}_6\text{H}_{12} + \text{H}_2 \rightarrow \text{C}_6\text{H}_{14}$
C $\text{C}_6\text{H}_{14} \rightarrow 6\text{C} + 7\text{H}_2$
D $\text{C}_6\text{H}_{14} \rightarrow \text{C}_2\text{H}_4 + \text{C}_4\text{H}_{10}$

- 36 What is the structure of the product of the reaction of propene with bromine?



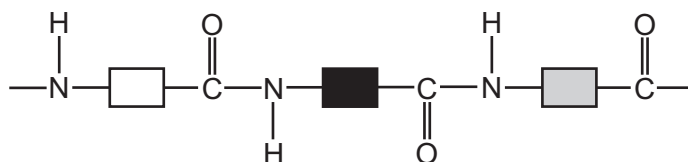
37 In reaction R, 2000 molecules of $\text{CH}_2=\text{CH}_2$ react to form a single molecule X only.



Which terms describe reaction R, $\text{CH}_2=\text{CH}_2$ and X?

	reaction R	$\text{CH}_2=\text{CH}_2$	X
A	addition	monomer	polymer
B	addition	polymer	monomer
C	substitution	monomer	polymer
D	substitution	polymer	monomer

38 Part of the structure of a polymer is shown.



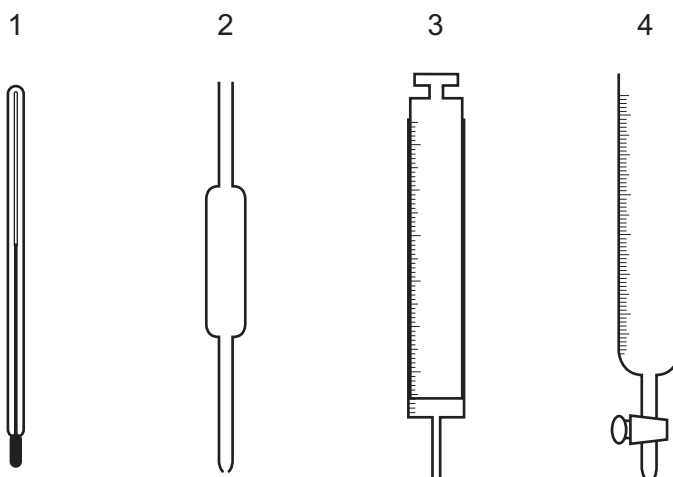
Which statements about the polymer are correct?

- 1 The polymer is nylon.
- 2 The polymer is formed by condensation polymerisation.
- 3 There are ester linkages between the monomers.

A 1 and 2 **B** 2 and 3 **C** 2 only **D** 3 only

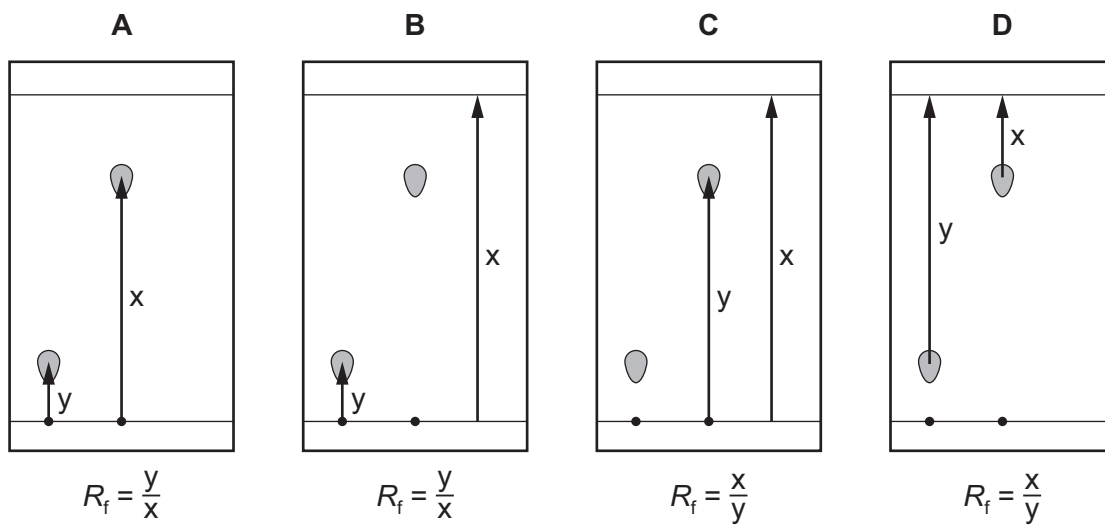
39 The concentration of acids and alkalis can be determined by titration.

Which pieces of equipment are needed to perform a titration?



A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

40 Which chromatogram shows how the R_f value of a substance is calculated?



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CHEMISTRY

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0620/22

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1 Four physical changes of ethanol are listed.

- 1 condensation
- 2 evaporation
- 3 freezing
- 4 boiling

In which changes do the particles move further apart?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

2 An atom of element X contains:

- 5 protons
- 6 neutrons
- 5 electrons.

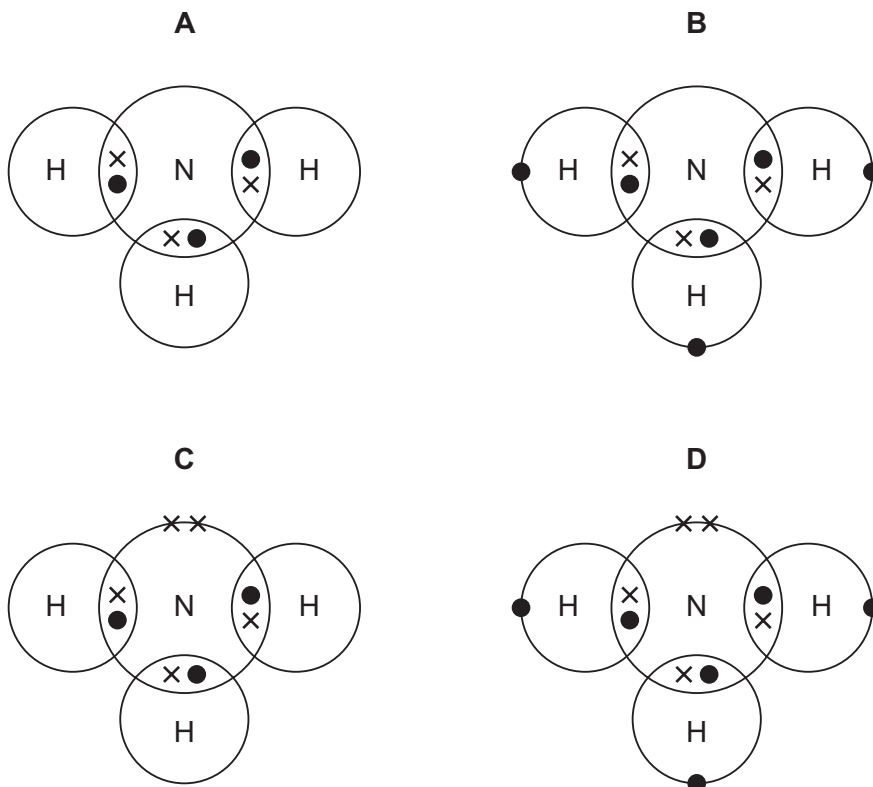
Which statements about element X are correct?

- 1 X has an atomic number of 6.
- 2 X has a nucleon number of 11.
- 3 X is in Group II of the Periodic Table.
- 4 X is in the second period of the Periodic Table.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 3 Ammonia, NH_3 , is a covalent molecule.

Which diagram shows the outer-shell electron arrangement in a molecule of ammonia?



- 4 Which structure does silicon(IV) oxide most closely resemble?

- A** carbon dioxide
- B** diamond
- C** graphite
- D** sodium chloride

- 5 Substance P conducts electricity when solid.

Which particles move in solid P so that it can conduct electricity?

- 1 anions
- 2 cations
- 3 electrons

- A** 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

6 Which equation represents a chemical change?

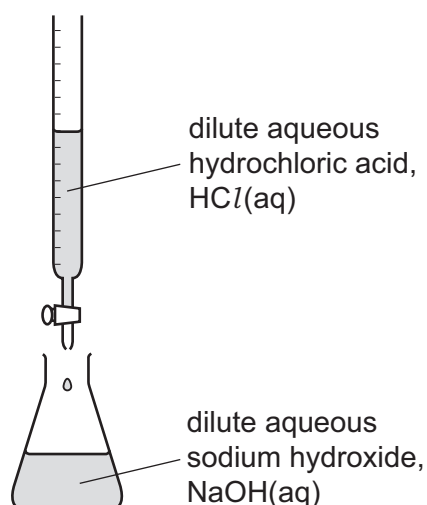
- A $\text{BaCl}_2(\text{s}) \rightarrow \text{BaCl}_2(\text{l})$
- B $\text{Ca}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{CaSO}_4(\text{s})$
- C $\text{KCl}(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{K}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
- D $\text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq}) \rightarrow \text{NaNO}_3(\text{aq})$

7 Which sample contains the largest number of molecules?

- A 16 g of methane, $\text{CH}_4(\text{g})$
- B 16 g of oxygen, $\text{O}_2(\text{g})$
- C 16 g of phosphorus, $\text{P}_4(\text{s})$
- D 16 dm^3 of methane at r.t.p., $\text{CH}_4(\text{g})$

- 8 The concentration of a sample of dilute aqueous sodium hydroxide is found by titration.

The apparatus used is shown.



Which information is needed to calculate the concentration of the dilute aqueous sodium hydroxide in mol/dm^3 ?

	concentration of HCl	volume of HCl used	molar mass of HCl	volume of NaOH used	molar mass of NaOH
A	✓	✓	✓	✓	✓
B	✓	✓	x	✓	x
C	x	✓	✓	✓	x
D	✓	x	x	x	✓

key

✓ = needed

x = not needed

- 9 In experiment 1, aqueous copper(II) sulfate is electrolysed using graphite electrodes.

In experiment 2, aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which statement identifies a half-equation for a reaction at one of the electrodes?

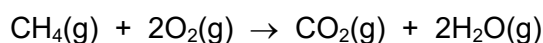
- A** In experiment 1, the half-equation for the anode reaction is $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$.
- B** In experiment 1, the half-equation for the cathode reaction is $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$.
- C** In experiment 2, the half-equation for the anode reaction is $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$.
- D** In experiment 2, the half-equation for the cathode reaction is $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$.

10 Which substance is **not** produced during the electrolysis of concentrated aqueous sodium chloride?

- A chlorine
- B hydrogen
- C sodium
- D sodium hydroxide

11 Methane burns in excess oxygen.

The equation is shown.



Bond energies are shown.

bond	bond energy in kJ/mol
C=O	805
C–H	410
O=O	496
O–H	460

What is the energy change for the reaction?

- A $(4 \times 410 + 2 \times 496) - (2 \times 805 + 4 \times 460)$
- B $(2 \times 805 + 4 \times 460) - (4 \times 410 + 2 \times 496)$
- C $(410 + 2 \times 496) - (805 + 2 \times 460)$
- D $(410 + 496) - (805 + 460)$

12 Which change increases the rate of reaction by decreasing the activation energy, E_a ?

- A addition of a catalyst
- B decrease in size of solid reactants
- C increase in concentration of solutions
- D increase in temperature

- 13** In the Contact process, sulfur dioxide is reacted with oxygen to form sulfur trioxide.

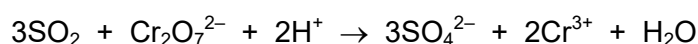
Which conditions are used in this reaction?

	temperature /°C	pressure /kPa	catalyst
A	300	200	iron
B	300	20 000	vanadium(V) oxide
C	450	200	vanadium(V) oxide
D	450	20 000	iron

- 14** Which reaction is reversible?

- A** an iron nail rusting when left in moist air
- B** limestone reacting with an acid to form carbon dioxide gas
- C** magnesium burning in air to produce a white ash
- D** white anhydrous copper(II) sulfate turning blue when water is added

- 15** The equation for the reaction of sulfur dioxide with acidified potassium dichromate(VI) is shown.



What is oxidised and what is the oxidising agent?

	oxidised	oxidising agent
A	SO_2	$\text{Cr}_2\text{O}_7^{2-}$
B	SO_2	H^+
C	$\text{Cr}_2\text{O}_7^{2-}$	H^+
D	$\text{Cr}_2\text{O}_7^{2-}$	$\text{Cr}_2\text{O}_7^{2-}$

- 16** What is the definition of a strong acid?

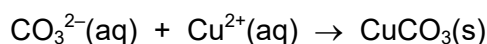
- A** a proton acceptor that is completely dissociated in aqueous solution
- B** a proton acceptor that is partially dissociated in aqueous solution
- C** a proton donor that is completely dissociated in aqueous solution
- D** a proton donor that is partially dissociated in aqueous solution

17 Which statement about amphoteric oxides is correct?

- A They are made by combining an acidic oxide with a basic oxide.
- B They react with water to give a solution of pH 7.
- C They react with both acids and bases.
- D They do not react with acids or bases.

18 Copper(II) carbonate is formed when aqueous sodium carbonate is added to aqueous copper(II) nitrate.

The ionic equation for the reaction is shown.



How is pure copper(II) carbonate obtained from the reaction mixture?

- A evaporate → filter → dry
- B evaporate → wash → crystallise
- C filter → evaporate → crystallise
- D filter → wash → dry

19 Q and R are elements in the same period of the Periodic Table.

Q has 7 electrons in its outer shell and R has 2 electrons in its outer shell.

Which statement about Q and R is correct?

- A Q is a metal and R is a non-metal.
- B Q and R have different numbers of electron shells.
- C R is found to the right of Q in the Periodic Table.
- D The proton number of R is less than the proton number of Q.

20 Lead(II) sulfate is an insoluble salt.

Which reaction produces a mixture from which lead(II) sulfate is obtained by filtration?

- A adding solid lead(II) carbonate to dilute sulfuric acid
- B adding solid lead(II) hydroxide to dilute sulfuric acid
- C adding metallic lead to dilute sulfuric acid
- D adding aqueous lead(II) nitrate to dilute sulfuric acid

21 Which statement about alkali metals is correct?

- A Lithium is more dense than sodium.
- B Sodium is more reactive than potassium.
- C Sodium has a higher melting point than potassium.
- D They are in Group II of the Periodic Table.

22 Which row describes the properties of a transition element?

	melting point	density	forms coloured compounds
A	high	low	no
B	high	high	yes
C	low	low	no
D	low	low	yes

23 Which row identifies the properties of zinc?

	thermal conductivity	reacts with dilute acid
A	good	yes
B	good	no
C	poor	yes
D	poor	no

24 Uses of metals depend on their properties.

Which property is necessary for the use given?

	use of the metal	property of the metal
A	car bodies	ductile
B	cutlery	conducts heat
C	food containers	resists corrosion
D	overhead electrical cables	high density

25 Which compounds **both** contribute to acid rain?

- A** carbon monoxide and carbon dioxide
- B** carbon monoxide and oxides of nitrogen
- C** oxides of nitrogen and sulfur dioxide
- D** sulfur dioxide and carbon dioxide

26 P, Q, R and S are metals.

P reacts with dilute hydrochloric acid, forming hydrogen.

Q reacts violently with water.

R reacts with water to give hydrogen.

S is formed by heating its oxide with carbon.

Which row identifies the metals?

	P	Q	R	S
A	copper	sodium	potassium	iron
B	zinc	magnesium	calcium	iron
C	zinc	sodium	calcium	magnesium
D	iron	potassium	sodium	zinc

27 Which compound is formed when iron rusts?

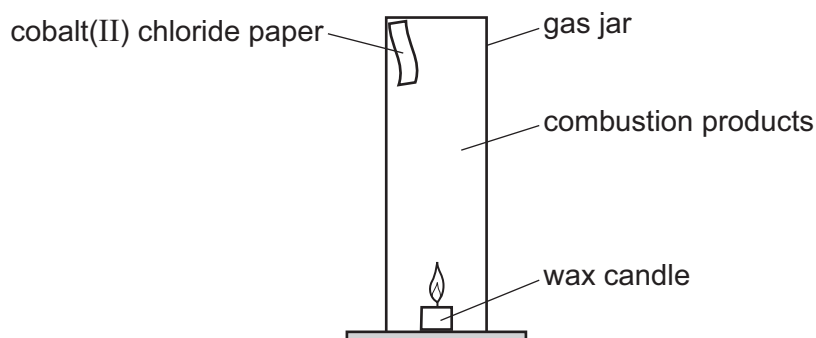
- A** anhydrous iron(II) oxide
- B** anhydrous iron(III) oxide
- C** hydrated iron(III) hydroxide
- D** hydrated iron(III) oxide

28 Why is cryolite used in the extraction of aluminium by electrolysis?

- A** It dissolves the aluminium oxide.
- B** It protects the anodes from corrosion.
- C** It changes bauxite to aluminium oxide.
- D** It decreases the melting point of the aluminium.

- 29 A wax candle is made from a mixture of hydrocarbons.

The candle is lit and placed in a gas jar along with a strip of cobalt(II) chloride test paper as shown.



After a short time, the oxygen in the jar is used up and the candle flame goes out.

Which substance does the cobalt(II) chloride paper identify?

- A carbon dioxide
 - B carbon monoxide
 - C sulfur dioxide
 - D water
- 30 The hydrocarbon C_4H_8 has two structural isomers, but-1-ene and but-2-ene.
- Which statement is correct?
- A But-2-ene has the structural formula $CH_3CH=CHCH_3$ and the same general formula as butane.
 - B But-2-ene has the structural formula $CH_3CH=CHCH_3$ and the same empirical formula as ethene.
 - C But-1-ene has the structural formula $CH_3CH_2CH=CH_2$ and the same general formula as butane.
 - D But-1-ene has the structural formula $CH_3CHCH_2=CH$ and the same empirical formula as ethene.
- 31 Which compound rapidly decolourises aqueous bromine?
- A propane
 - B propanoic acid
 - C propanol
 - D propene

32 What are the products of the addition reactions of ethene with bromine and hydrogen?

	bromine	hydrogen
A	$\text{CH}_2\text{BrCH}_2\text{Br}$	CH_3CH_3
B	$\text{CH}_2\text{BrCH}_2\text{Br}$	CH_2CH_2
C	$\text{CH}_3\text{CH}_2\text{Br}$	CH_3CH_3
D	$\text{CH}_3\text{CH}_2\text{Br}$	CH_2CH_2

33 Ethanol is manufactured by fermentation and the catalytic addition of steam to ethene.

Which row describes an advantage of both methods?

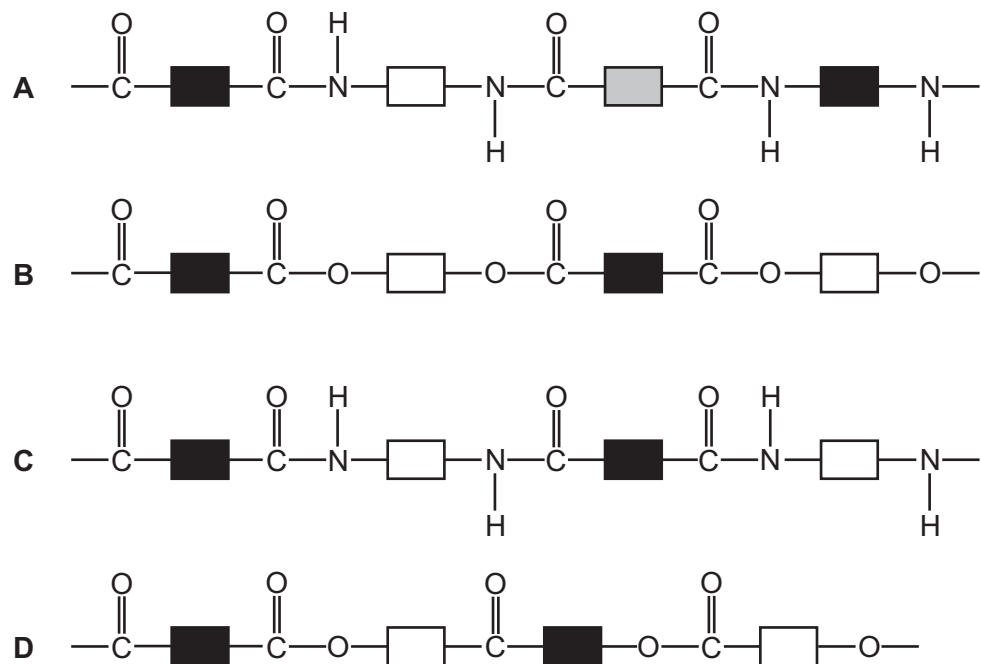
	from sugar by fermentation	from ethene and steam
A	ethanol needs to be purified	the process is continuous
B	it is a batch process	ethene comes from petroleum
C	the process is slow	the process is rapid
D	renewable resources are used	the ethanol produced is pure

34 Methanoic acid and propan-1-ol react to form an ester.

What is the structural formula of the ester?

- A** $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$
- B** $\text{CH}_3\text{CH}_2\text{COOCH}_3$
- C** $\text{CH}_3\text{COOCH}_2\text{CH}_3$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

35 What is the correct structure of PET?



36 Alkanes undergo substitution reactions in the presence of UV light.

Which equation represents a substitution reaction of ethane?

- A $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4 + 2\text{HCl}$
- B $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{HCl}$
- C $\text{C}_2\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_4\text{Cl}_2 + \text{H}_2$
- D $\text{C}_2\text{H}_6 + \text{HCl} \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{H}_2$

37 Methane reacts with chlorine in substitution reactions.

How many different products, containing a single carbon atom, can be made during the reactions?

- A 2 B 3 C 4 D 5

38 Rock salt is a mixture of salt and sand.

The method used to separate the sand from the salt is listed.

step 1 Crush the rock salt, add to warm water and stir.

step 2 Pour the mixture through a filter paper held in a funnel.

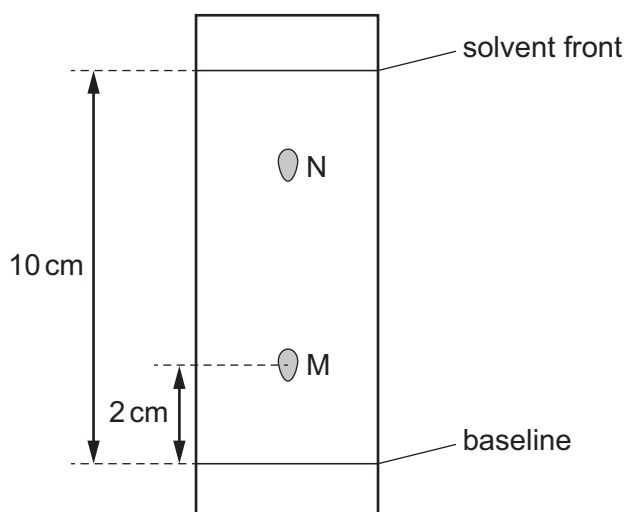
step 3 Evaporate the water to crystallise the salt.

Which statement about the method is correct?

- A The filtrate in step 2 is pure water.
- B The residue in step 2 is pure crystals of salt.
- C The solute is salt.
- D The solvent is a mixture of salt and water.

39 Two compounds, M and N, are dissolved in water and separated by chromatography.

The results are shown.



What is the R_f value of M and which compound is most soluble in water?

	R_f value of M	most soluble compound
A	0.2	M
B	0.2	N
C	5.0	M
D	5.0	N

40 When acid is added to salt X, a gas is produced which turns limewater milky.

When sodium hydroxide is added to salt X, a gas is produced which turns litmus paper blue.

What is X?

A CaCO_3

B $(\text{NH}_4)_2\text{CO}_3$

C NH_4NO_3

D ZnCO_3

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1 H hydrogen 1																	
		Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —		

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/23

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



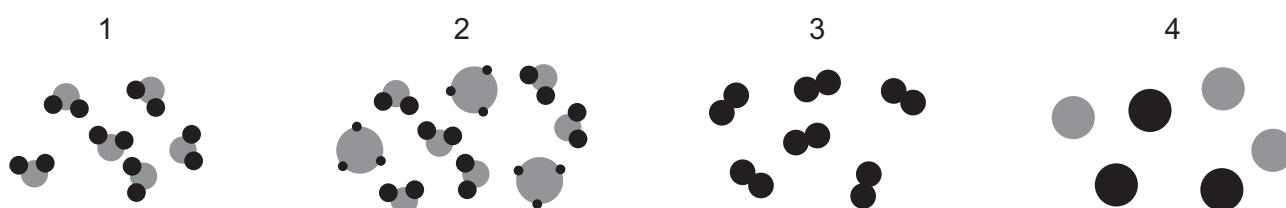
- 1 Nitrogen is heated in a balloon, which expands slightly.

Which statements about the molecules of nitrogen are correct?

- 1 They move further apart.
- 2 They move more quickly.
- 3 They remain the same distance apart.
- 4 Their speed remains unchanged.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 2 The diagrams represent some elements, compounds and mixtures.



Which row describes the numbered substances?

	1	2	3	4
A	element	mixture of compounds	compound	mixture of elements
B	compound	mixture of compounds	element	mixture of elements
C	element	mixture of elements	compound	mixture of compounds
D	compound	mixture of elements	element	mixture of compounds

- 3 Two atoms, X and Y, have the same mass number but different atomic numbers.

Which statement about X and Y is correct?

- A** They have the same number of protons.
- B** They have the same number of electrons.
- C** They are in the same group of the Periodic Table.
- D** They have different numbers of neutrons.

- 4 A sample of pure iron contains three isotopes only.

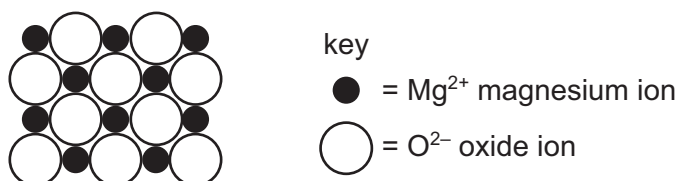
percentage abundance of isotope / %	isotope
2	${}^n\text{Fe}$
6	${}^{54}\text{Fe}$
92	${}^{56}\text{Fe}$

The iron in the sample has a relative atomic mass of 55.9.

What is the value of n ?

- A 53 B 55 C 57 D 58
- 5 Magnesium oxide is a white solid at room temperature and pressure.

Part of the structure of solid magnesium oxide is shown.



Three statements are listed.

- 1 Magnesium ions are smaller than oxide ions because they contain fewer electrons.
- 2 Magnesium oxide has good electrical conductivity when molten because the ions are mobile.
- 3 Magnesium oxide has a high melting point because of the strong electrostatic attraction between the ions and delocalised electrons in the giant lattice.

Which statements are correct?

- A 1 and 2 B 1 and 3 C 2 and 3 D 2 only
- 6 In which molecule are all the outer-shell electrons involved in covalent bonding?
- A Cl_2 B CH_4 C HCl D NH_3

7 Which row describes the properties of silicon(IV) oxide?

	giant covalent structure	melting point
A	no	high
B	no	low
C	yes	high
D	yes	low

8 Which row describes the structure of a solid metal and explains the property?

	structure of solid metal	property of solid metal
A	lattice of negative ions in a sea of electrons	conducts electricity because the electrons are free to move
B	lattice of negative ions in a sea of electrons	is malleable because the layers of ions can slide over each other
C	lattice of positive ions in a sea of electrons	conducts electricity because the ions are free to move
D	lattice of positive ions in a sea of electrons	is malleable because the layers of ions can slide over each other

9 What is the formula of potassium oxide?

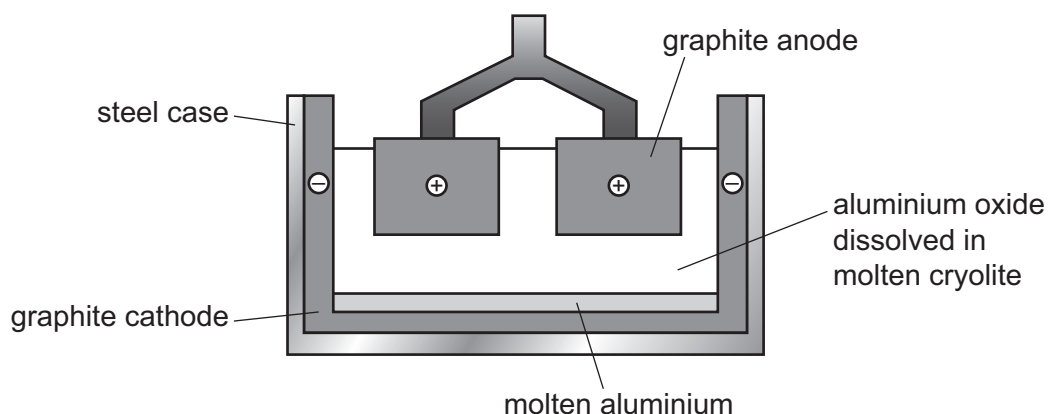
- A** P_2O **B** PO_2 **C** KO **D** K_2O

10 A dilute aqueous solution of sodium bromide is electrolysed using inert electrodes.

Which row identifies the product at the cathode and at the anode?

	cathode	anode
A	bromine	hydrogen
B	hydrogen	bromine
C	hydrogen	oxygen
D	oxygen	hydrogen

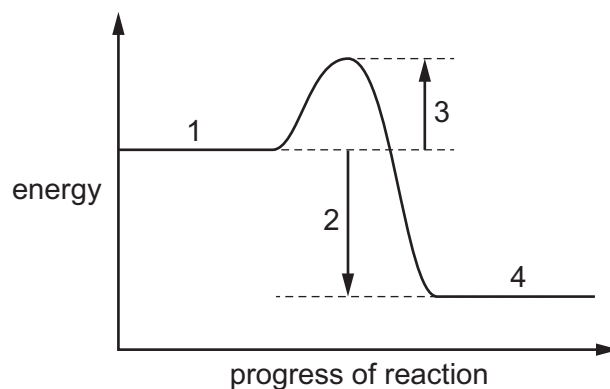
11 Aluminium is extracted by electrolysis, as shown.



Which row shows the ionic half-equations at the cathode and the anode?

	cathode	anode
A	$\text{Al}^{3+} \rightarrow \text{Al} + 3\text{e}^-$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
B	$\text{Al}^{3+} \rightarrow \text{Al} + 3\text{e}^-$	$2\text{O}^{2-} + 4\text{e}^- \rightarrow \text{O}_2$
C	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
D	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} + 4\text{e}^- \rightarrow \text{O}_2$

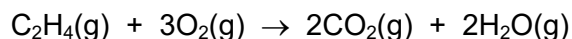
12 The reaction pathway diagram for an exothermic reaction is shown.



Which row identifies labels 1, 2, 3 and 4?

	1	2	3	4
A	reactants	ΔH	E_a	products
B	products	ΔH	E_a	reactants
C	reactants	E_a	ΔH	products
D	products	E_a	ΔH	reactants

- 13 The equation for the complete combustion of ethene is shown.



Some bond energies are listed.

bond	bond energy in kJ/mol
C–H	412
C–C	348
C=C	612
C–O	360
C=O	743
O–O	146
O=O	496
O–H	463

What is the overall energy change when one mole of ethene is completely burned?

- A** –456 **B** –1076 **C** –1340 **D** –2126

- 14 Magnesium reacts with hydrochloric acid to form magnesium chloride and hydrogen.

Why does magnesium powder react faster than magnesium ribbon?

- A** The magnesium atoms in the powder have a lower activation energy.
B The powder has a smaller surface area.
C The magnesium atoms in the powder have more frequent collisions with acid particles.
D The magnesium atoms in the powder have greater kinetic energy.

- 15 Which row shows the conditions used in the Contact process?

	catalyst	pressure / atm	temperature / °C
A	iron	2	100
B	iron	200	450
C	vanadium(V) oxide	2	450
D	vanadium(V) oxide	200	100

- 16 A student heats hydrated copper(II) sulfate. The blue crystals change to a white powder.

How can the student reverse this reaction?

- A Add anhydrous copper(II) sulfate to the white powder.
- B Add water to the white powder.
- C Cool the white powder.
- D Reheat the white powder.

- 17 Which reaction of hydrochloric acid is a redox reaction?

- A $\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O} + \text{CO}_2$
- B $\text{Mg}(\text{OH})_2 + 2\text{HCl} \rightarrow \text{MgCl}_2 + 2\text{H}_2\text{O}$
- C $\text{MgO} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$
- D $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2$

- 18 Which oxide is amphoteric?

- A Al_2O_3
- B CaO
- C Na_2O
- D SO_2

- 19 Four statements about strong acids are listed.

- 1 They react with carbonates to form carbon dioxide.
- 2 They completely dissociate in aqueous solution.
- 3 They react with ammonium salts to form ammonia.
- 4 They are proton acceptors.

Which statements are correct?

- A 1 and 2
- B 1 and 3
- C 2 and 4
- D 3 and 4

- 20 Which reaction mixture will produce a precipitate?

- A aqueous Na_2CO_3 and aqueous CuSO_4
- B dilute H_2SO_4 and aqueous NaOH
- C dilute HNO_3 and solid MgO
- D solid CuO and dilute H_2SO_4

21 Which set of elements shows the change from metallic to non-metallic character across a period of the Periodic Table?

- A** beryllium → magnesium → calcium
- B** fluorine → bromine → iodine
- C** oxygen → boron → lithium
- D** sodium → silicon → chlorine

22 A sample of ethanoic acid and a sample of hydrochloric acid have the same concentration.

How do the hydrogen ion concentration and pH of ethanoic acid compare to those of hydrochloric acid?

	ethanoic acid compared to hydrochloric acid	
	hydrogen ion concentration	pH
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

23 What is a typical property of transition elements?

- A** can act as catalysts
- B** poor electrical conductivity
- C** low melting point
- D** low density

24 Which statement about copper or aluminium is correct?

- A** Aluminium is more dense than copper.
- B** Aluminium is less reactive than copper.
- C** Copper has high ductility.
- D** Copper has poor electrical conductivity.

- 25** Water from a reservoir flows to the water works where purification process 1 takes place followed by process 2.

What are processes 1 and 2?

	process 1	process 2
A	chlorination	filtration
B	filtration	chlorination
C	fractional distillation	filtration
D	filtration	fractional distillation

- 26** Calcium reacts with cold water to produce hydrogen.

Lead reacts slowly when heated in air to form an oxide but has almost no reaction with steam.

Silver does not react with either air or water.

Zinc reacts when heated with steam to produce hydrogen.

What is the order of reactivity starting with the least reactive?

	least reactive $\longrightarrow$ most reactive			
A	calcium	lead	zinc	silver
B	calcium	zinc	lead	silver
C	silver	lead	zinc	calcium
D	silver	zinc	lead	calcium

- 27** Blocks of magnesium are attached to the bottom of a steel boat to prevent rusting.

Which equation describes a change that prevents the steel from rusting?

- A** $\text{Fe} \rightarrow \text{Fe}^{3+} + 3\text{e}^{-}$
- B** $\text{Fe}_2\text{O}_3 + 3\text{Mg} \rightarrow 2\text{Fe} + 3\text{MgO}$
- C** $3\text{Mg}^{2+} + 2\text{Fe} \rightarrow 2\text{Fe}^{3+} + 3\text{Mg}$
- D** $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^{-}$

28 Which statements about the extraction of iron in a blast furnace are correct?

- 1 The temperature inside the blast furnace is increased by burning carbon.
- 2 Iron(III) oxide is reduced to iron by carbon monoxide.
- 3 The thermal decomposition of calcium carbonate forms slag.
- 4 Slag reacts with acidic impurities.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

29 Which statements about water are correct?

- 1 Tap water has fewer impurities than distilled water.
- 2 Tap water will turn anhydrous cobalt(II) chloride pink.
- 3 The domestic water supply is treated with carbon to kill microbes.
- 4 Phosphates from fertilisers can cause deoxygenation of water.

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

30 Oxides of nitrogen form in car engines and are removed by catalytic converters.

Which equation represents a reaction that occurs in a catalytic converter?

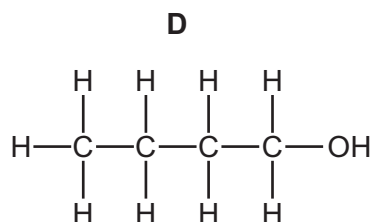
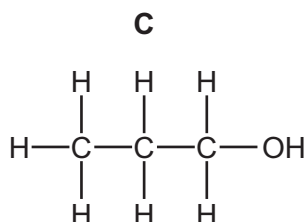
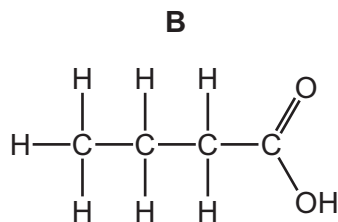
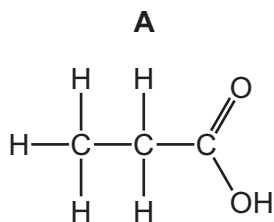
- A** $\text{CO} + \text{NO}_2 \rightarrow \text{NO} + \text{CO}_2$
- B** $2\text{CO} + 2\text{NO} \rightarrow \text{N}_2 + 2\text{CO}_2$
- C** $\text{CO}_2 + \text{NO} \rightarrow \text{NO}_2 + \text{CO}$
- D** $\text{CO}_2 + 2\text{NO}_2 \rightarrow \text{N}_2 + 3\text{O}_2 + \text{C}$

31 An alkene is represented by the formula $\text{CH}_3\text{CH}=\text{CH}_2$.

Which name is given to this type of formula?

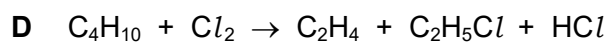
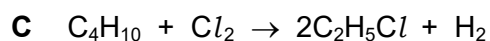
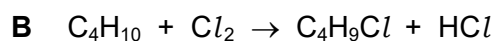
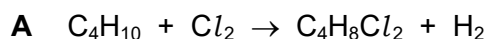
- A** displayed
- B** empirical
- C** general
- D** structural

32 What is the structure of propanoic acid?



33 Butane reacts with chlorine in the presence of ultraviolet radiation.

What is the equation for this reaction?



34 A hydrocarbon P is cracked to make compound Q and hydrogen.

Compound R is formed by the addition polymerisation of compound Q.

To which homologous series do P, Q and R belong?

	alkene	alkane
A	P only	Q and R
B	Q only	P and R
C	P and Q	R only
D	P and R	Q only

35 Which substances are structural isomers?

A but-2-ene and propene

B ethyl ethanoate and butanoic acid

C methyl methanoate and ethanol

D propan-1-ol and butan-1-ol

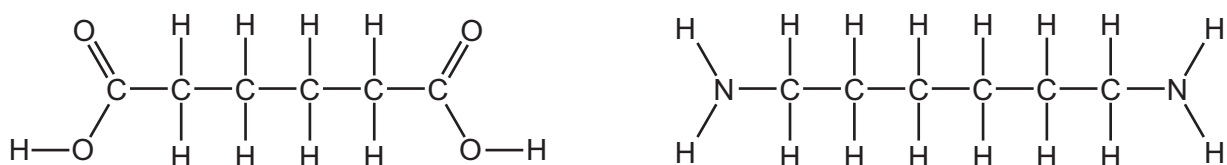
36 Ethanol is produced by:

- 1 the catalytic addition of steam to ethene
- 2 fermentation.

Which statement is correct?

- A Both processes use similar amounts of energy.
- B Both processes use a catalyst.
- C Process 1 uses a temperature of 25–35 °C.
- D Process 2 uses a pressure of 60 atm.

37 The two monomers shown can be used to form a condensation polymer.



Which small molecule is released during this reaction?

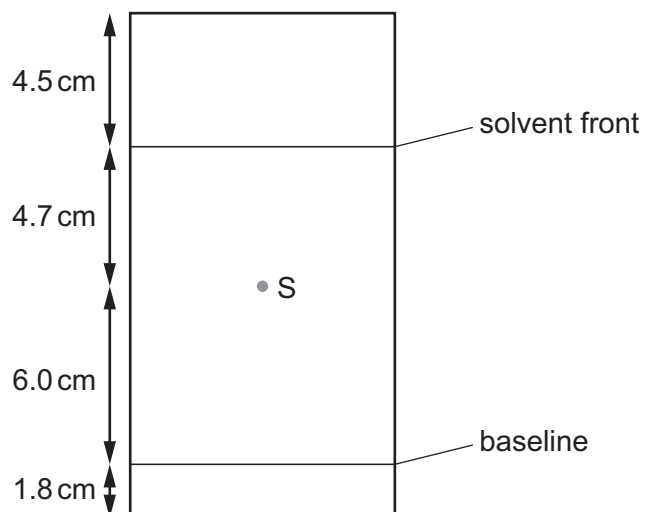
- A H_2O B NH_3 C CO_2 D CONH_2

38 Dilute hydrochloric acid is titrated into a conical flask containing sodium hydroxide solution and a few drops of methyl orange indicator.

Which piece of apparatus is used to add the hydrochloric acid?

- A beaker
- B burette
- C measuring cylinder
- D pipette

- 39 The chromatogram obtained from a chromatography experiment on substance S is shown.



What is the R_f value of S?

- A** 0.39 **B** 0.46 **C** 0.56 **D** 0.62
- 40 Element X burns in air to form an acidic gas that decolourises potassium manganate(VII).

What is X?

- A** carbon
B nitrogen
C magnesium
D sulfur

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The Periodic Table of Elements

Group																				
I	II											III	IV	V	VI	VII	VIII			
		1 H hydrogen 1																		
		Key atomic number atomic symbol name relative atomic mass																		
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19		
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84			
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131			
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —			
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —			

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/21

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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INFORMATION

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- Each correct answer will score one mark.
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This document has **16** pages. Any blank pages are indicated.



- 1 A gas is heated. The pressure is kept constant.

Which statement describes the behaviour of the particles in the gas?

- A The particles move faster and become closer together.
- B The particles move faster and become further apart.
- C The particles move more slowly and become closer together.
- D The particles move more slowly and become further apart.

- 2 A mixture of ice and water is left to stand and the ice melts.

Which row describes what happens as the ice is melting?

	temperature of mixture	energy change
A	increases	average kinetic energy of particles decreases
B	increases	energy is used to overcome attractive forces
C	stays the same	average kinetic energy of particles decreases
D	stays the same	energy is used to overcome attractive forces

- 3 Hydrogen chloride gas [M_r : HCl, 36.5] is released at P in the apparatus shown.

The universal indicator paper turns red after 38 s.



The experiment is repeated using sulfur dioxide gas [M_r : SO₂, 64].

What is the result for sulfur dioxide gas?

	universal indicator paper turns	time for universal indicator paper to change colour / s
A	blue	26
B	blue	51
C	red	26
D	red	51

4 Four statements about atoms are listed.

- 1 The centre of an atom is positively charged.
- 2 Protons and electrons are located in the nucleus.
- 3 Protons and electrons have the same mass.
- 4 Most of the mass of an atom is in the nucleus.

Which statements are correct?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

5 The electronic configurations of two elements are given.

element L: 2,8,8,1

element M: 2,8,4

Which row identifies the group number and the period number for element L and element M?

	element L		element M	
	group number	period number	group number	period number
A	I	4	IV	3
B	I	4	III	4
C	IV	1	III	4
D	IV	1	IV	3

6 Which statement explains why isotopes of the same element have the same chemical properties?

- A** They have different numbers of protons in their nucleus.
- B** They have different numbers of neutrons in their nucleus.
- C** They have the same electronic configuration.
- D** They have the same number of electrons as protons.

7 Which statements about potassium chloride are correct?

- 1 It conducts electricity when solid because its ions are free to move.
- 2 It has a high melting point because it has strong intermolecular forces.
- 3 Its structure is a giant lattice of alternating positive and negative ions.
- 4 It is soluble in water.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

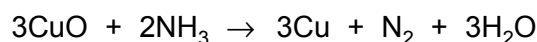
8 How many electrons are shared in **one** molecule of nitrogen and in **one** molecule of ethene?

	nitrogen	ethene
A	2	12
B	2	8
C	6	12
D	6	8

9 What is the total number of electrons in **one** molecule of ammonia, NH_3 ?

A 6 **B** 8 **C** 10 **D** 11

10 When heated, copper(II) oxide, CuO , reacts with ammonia, NH_3 .



8.5 g of ammonia reacts with an excess of copper(II) oxide to produce 26.4 g of copper.

What is the percentage yield of copper in this reaction?

A 27.5% **B** 32.2% **C** 55.0% **D** 82.5%

11 What is the empirical formula of ethanoic acid?

A CHO **B** CH_2O **C** $\text{C}_2\text{H}_2\text{O}$ **D** $\text{C}_2\text{H}_4\text{O}_2$

- 12 Magnesium chloride, MgCl_2 , contains magnesium ions and chloride ions.

How many chloride ions are present in **two** moles of magnesium chloride?

- A 6.02×10^{23}
 B 1.204×10^{24}
 C 2.408×10^{24}
 D 3.612×10^{24}

- 13 A metal object is electroplated with copper.

One electrode is the metal object and the other electrode is copper. The electrolyte is aqueous copper(II) sulfate.

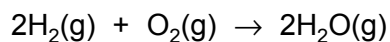
Which row shows the ionic half-equation for the reaction at the anode and the observation of the electrolyte?

	anode	electrolyte
A	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour fades
B	$\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$	blue colour does not change
C	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour fades
D	$\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$	blue colour does not change

- 14 Which statement about electrolysis is correct?

- A Chemical energy is converted to electrical energy.
 B Electrons flow through the electrolyte.
 C Ionic compounds are broken down.
 D Metals are formed at the positive electrode.

- 15 The reaction between hydrogen and oxygen releases 486 kJ/mol of energy.

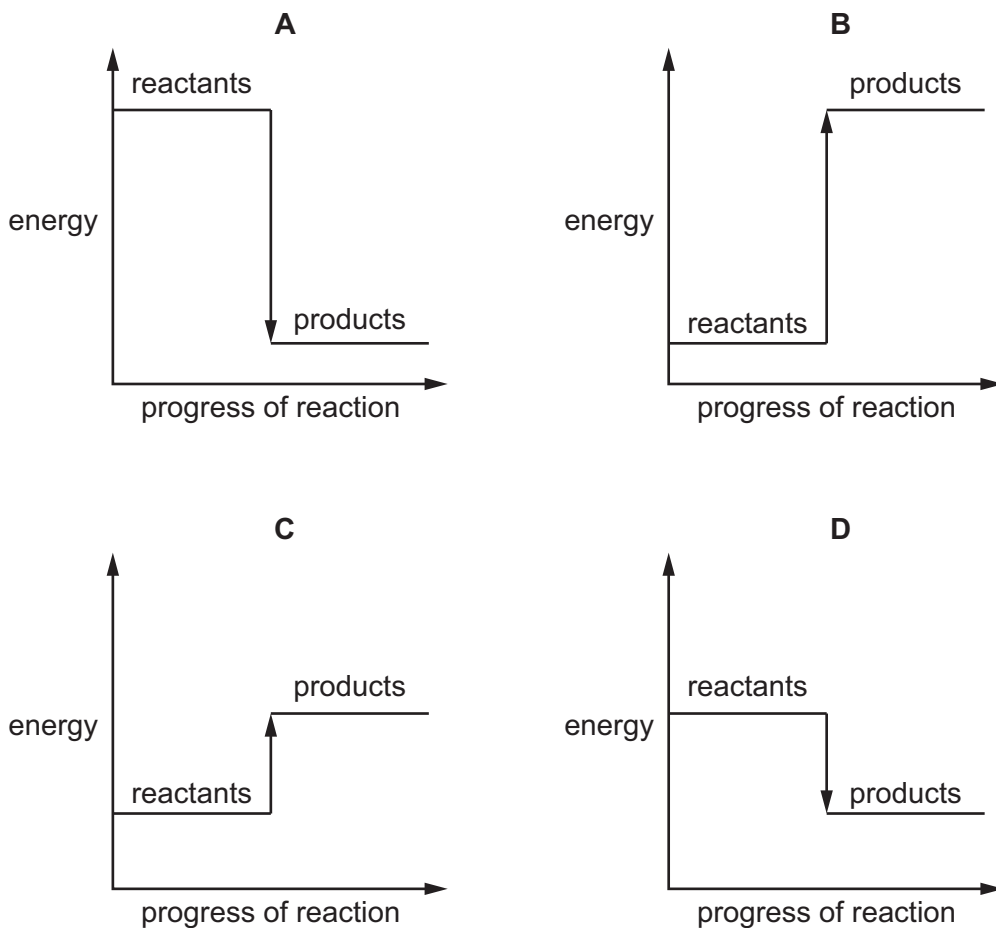


The bond energy of H–H is 436 kJ/mol and that of H–O is 464 kJ/mol.

What is the bond energy of O=O?

- A 430 kJ/mol
 B 458 kJ/mol
 C 498 kJ/mol
 D 984 kJ/mol
- 16 Which reaction pathway diagram shows the reaction that will give out the most energy?

The scale on the y-axis is the same in each diagram.



- 17 When calcium carbonate is heated strongly, a gas is given off.

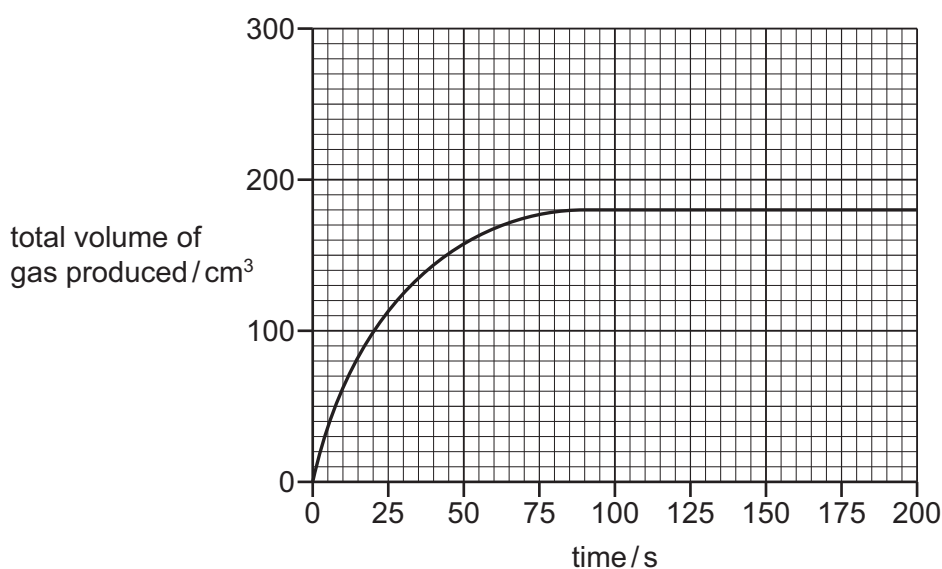
Which word describes this change?

- A chemical
- B exothermic
- C physical
- D reduction

- 18 Powdered magnesium carbonate is added to excess dilute hydrochloric acid.

The total volume of gas produced is measured over time.

A graph of the results is shown.



The experiment is repeated but the concentration of the hydrochloric acid is doubled.

All other conditions are kept the same.

Which statements about the second experiment are correct?

- 1 The final volume of gas is 360 cm^3 .
- 2 The reaction finishes before 90 seconds.
- 3 The activation energy of the reaction is lower.

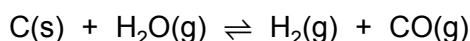
- A 1 and 2 B 1 and 3 C 2 and 3 D 2 only

19 Which statements explain why increasing the temperature changes the rate of a chemical reaction?

- 1 It increases the activation energy.
- 2 It increases the frequency of collisions between the reacting particles.
- 3 It increases the kinetic energy of the reacting particles.
- 4 It increases the number of particles per unit volume.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

20 Hydrogen is made by reacting carbon with steam. The equation for the reaction is shown.



The forward reaction is endothermic.

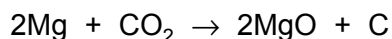
Which row describes changes in the pressure and the temperature that will **both** shift the position of equilibrium to the right?

	pressure	temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

21 Which row shows the conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process?

	pressure / atm	temperature / °C	catalyst
A	250	200	vanadium(V) oxide
B	2	450	vanadium(V) oxide
C	250	200	iron
D	2	450	iron

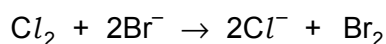
- 22 The equation for the reaction of magnesium with carbon dioxide is shown.



Which statement about this reaction is correct?

- A Magnesium is oxidised and carbon dioxide is reduced.
 - B Magnesium is reduced and carbon dioxide is oxidised.
 - C Magnesium and carbon dioxide are both oxidised.
 - D Magnesium and carbon dioxide are neither oxidised nor reduced.
- 23 Chlorine displaces bromine from aqueous potassium bromide.

The ionic equation for the reaction is shown.



Which statement about this reaction is correct?

- A Bromide ions act as an oxidising agent.
 - B Bromide ions are oxidised as electrons are lost.
 - C Chlorine acts as a reducing agent.
 - D Chlorine is reduced as electrons are lost.
- 24 Which gas is produced when ammonium chloride is warmed with aqueous sodium hydroxide?
- A ammonia
 - B chlorine
 - C hydrogen
 - D nitrogen
- 25 Which equation represents a solution of ethanoic acid in water?
- A $\text{HCOOH}(\text{aq}) \rightleftharpoons \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - B $\text{HCOOH}(\text{aq}) \rightarrow \text{HCOO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - C $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$
 - D $\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}^+(\text{aq})$

26 Four statements about the reactions of oxides with dilute hydrochloric acid and with aqueous sodium hydroxide are listed.

- 1 Aluminium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 2 Calcium oxide reacts with both dilute hydrochloric acid and aqueous sodium hydroxide.
- 3 Copper(II) oxide reacts with dilute hydrochloric acid but **not** with aqueous sodium hydroxide.
- 4 Sulfur dioxide does **not** react with either dilute hydrochloric acid or aqueous sodium hydroxide.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

27 Which statement about elements in Period 3 of the Periodic Table is correct?

- A** Aluminium is a non-metal in Group III.
B Argon is in Group VIII and has eight electrons in its outer electron shell.
C Magnesium is in Group II and has three electrons in its outer electron shell.
D Sulfur is a metal in Group VI.

28 Which row describes the structure of Group VII elements and the trend in their reactivity down the group?

	structure	reactivity down Group VII
A	diatomic	increases
B	diatomic	decreases
C	monatomic	increases
D	monatomic	decreases

29 Some information about four elements, P, Q, R and S, is shown.

	melting point in °C	density in g/cm ³	colour of chloride
P	1247	7.43	pink
Q	1410	2.33	white
R	1910	6.11	purple
S	115	2.07	red

Which elements are transition elements?

- A** P and R **B** P and S **C** Q and R **D** R and S

30 Propanoic acid is a carboxylic acid. It has similar chemical properties to ethanoic acid.

Which statements are correct?

- 1 Aqueous propanoic acid is a weaker acid than dilute hydrochloric acid.
- 2 Propanoic acid partially ionises in aqueous solution.
- 3 Propanoic acid reacts with ethanol to form propyl ethanoate.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

31 Iron rusts in the presence of oxygen and water.

Which statements about the rusting of iron are correct?

- 1 Anhydrous iron(II) oxide is produced when iron rusts.
- 2 Iron rusts more quickly when attached to a piece of zinc.
- 3 Coating the iron with plastic prevents the iron from rusting.
- 4 Iron loses electrons when it rusts.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 32** An iron nail is added to aqueous copper(II) sulfate and a different iron nail is added to aqueous magnesium sulfate.

The results are shown.

experiment	result
iron nail in aqueous copper(II) sulfate	nail is coated with a brown solid
iron nail in aqueous magnesium sulfate	no reaction

Which statement is correct?

- A** Copper atoms are oxidised more easily than magnesium atoms.
 - B** Copper atoms are reduced more easily than iron ions.
 - C** Iron atoms are oxidised more easily than copper atoms.
 - D** Iron atoms are reduced more easily than copper ions.
- 33** Which pollutant leads to the deoxygenation of water in ponds and lakes?
- A** fertilisers containing nitrates and phosphates
 - B** toxic metal compounds
 - C** combustion products of fossil fuels
 - D** acid rain
- 34** Which statement identifies a sample of water as pure?
- A** It melts at room temperature.
 - B** It turns anhydrous copper(II) sulfate blue.
 - C** It turns hydrated cobalt(II) chloride from blue to pink.
 - D** It boils at 100 °C.

35 Oxides of nitrogen are produced by car engines.

In a catalytic converter oxides of nitrogen are removed by reacting them with compound X.

Which row describes the type of reaction oxides of nitrogen undergo and identifies compound X?

	type of reaction	compound X
A	oxidation	carbon dioxide
B	oxidation	carbon monoxide
C	reduction	carbon dioxide
D	reduction	carbon monoxide

36 What is a disadvantage of producing ethanol using the catalytic addition of steam to ethene?

- A** the energy cost is low
- B** the process is continuous
- C** the process uses a non-renewable raw material
- D** the ethanol is pure

37 Which statement about the polymer PET is correct?

- A** It can be broken down into its monomers and re-polymerised.
- B** It is an addition polymer.
- C** It is a polyamide.
- D** It is made from amino acid monomers.

38 The formulae of five compounds are listed.

- 1 C_4H_{10}
- 2 $\text{C}_2\text{H}_5\text{OH}$
- 3 $\text{C}_4\text{H}_9\text{OH}$
- 4 $\text{C}_4\text{H}_9\text{COOH}$
- 5 $\text{C}_5\text{H}_{11}\text{OH}$

Which compounds are in the same homologous series?

- A** 1, 3 and 4
B 2, 3 and 5
C 3 and 4 only
D 3 and 5 only

39 Propane reacts with chlorine.

Which statements about this reaction are correct?

- 1 Ultraviolet light is used to provide the activation energy.
- 2 Propane undergoes an addition reaction.
- 3 One of the products is $\text{CH}_3\text{CH}_2\text{Cl}$.
- 4 One of the products is HCl .

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

40 Which statement about chromatography is correct?

- A** It is **not** possible for two different substances to have the same R_f value.
B It is only possible to use chromatography on substances which have a colour.
C It is possible to use chromatography on colourless substances using a locating agent.
D The R_f value of a substance = $\frac{\text{the distance travelled by the solvent}}{\text{the distance travelled by the substance}}$

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lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

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0620/22

May/June 2024

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1 Which gas has the slowest rate of diffusion?

- A** H_2 **B** NH_3 **C** CH_4 **D** CO_2

2 Which statements about the position of the elements in the Periodic Table are correct?

- 1 Elements in the same group have similar chemical properties.
- 2 Elements in the same period have similar chemical properties.
- 3 Elements in the same group have the same number of electron shells.
- 4 Elements in the same group have the same number of outer shell electrons.

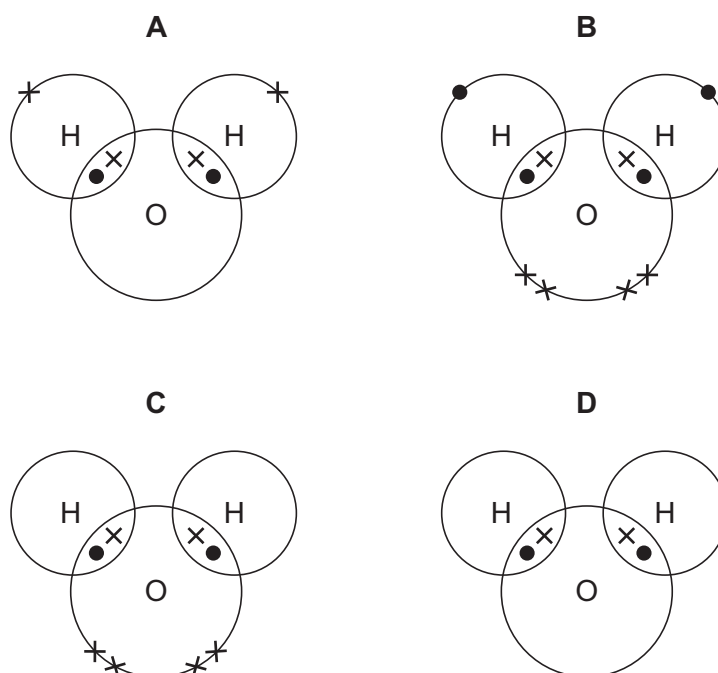
- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

3 Which statements about isotopes are correct?

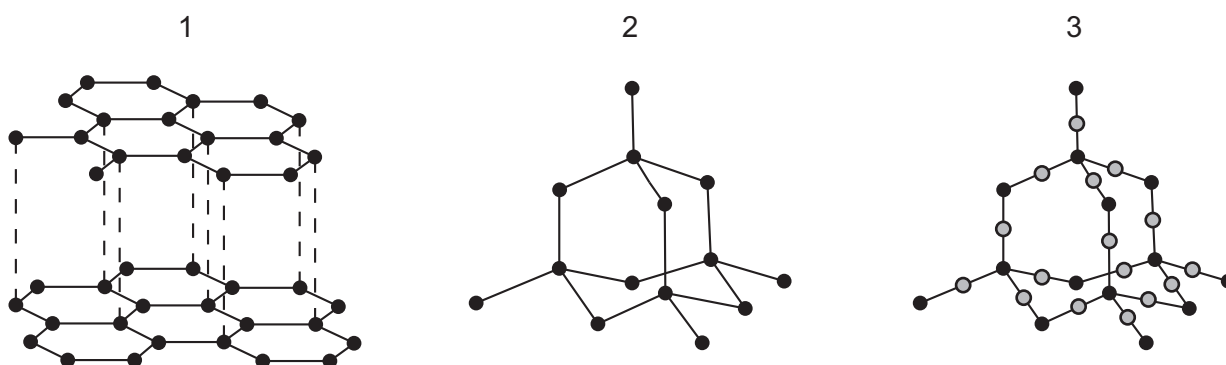
- 1 Isotopes are atoms of different elements with the same number of protons.
- 2 Isotopes of the same element have the same chemical properties.
- 3 Isotopes are atoms with the same relative atomic mass.
- 4 Isotopes of the same element have the same electronic configuration.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

4 Which diagram shows the arrangement of the outer shell electrons in a molecule of water?



5 The structures of three substances are shown.



Which substances are hard and have a high melting point?

- A** 1, 2 and 3 **B** 1 only **C** 2 only **D** 2 and 3 only

6 Information about four substances, W, X, Y and Z, is shown.

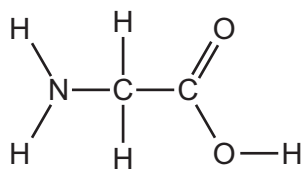
	melting point/ °C	electrical conductivity
W	1710	does not conduct when solid
X	3500	conducts when solid
Y	120	does not conduct
Z	801	conducts when molten

W, X, Y and Z are graphite, poly(ethene), sodium chloride and silicon(IV) oxide but not in that order.

What are W, X, Y and Z?

	W	X	Y	Z
A	graphite	poly(ethene)	silicon(IV) oxide	sodium chloride
B	sodium chloride	graphite	poly(ethene)	silicon(IV) oxide
C	poly(ethene)	sodium chloride	graphite	silicon(IV) oxide
D	silicon(IV) oxide	graphite	poly(ethene)	sodium chloride

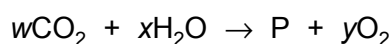
- 7 The structure of glycine is shown.



Which row is correct?

	formula of glycine	number of different elements in glycine
A	$\text{CH}_5\text{O}_2\text{N}$	10
B	$\text{C}_2\text{H}_5\text{O}_2\text{N}$	4
C	$\text{C}_2\text{H}_5\text{O}_2\text{N}$	10
D	$\text{H}_2\text{NCHCOOH}$	4

- 8 The incomplete equation for photosynthesis is shown.



Compound P is a product of the reaction.

Which row describes the values of w , x and y and gives the empirical formula of compound P?

	values of w , x and y	empirical formula of compound P
A	w , x and y are all the same	CH_2O
B	w , x and y are all the same	$\text{C}_6\text{H}_{12}\text{O}_6$
C	w and x are the same and both are greater than y	CH_2O
D	w and x are the same and both are greater than y	$\text{C}_6\text{H}_{12}\text{O}_6$

- 9 The concentration and volume of an aqueous alkali are known.

Which additional information is required to calculate the number of moles of acid needed to neutralise the aqueous alkali?

- A** the concentration of the acid
- B** the equation for the acid–alkali reaction
- C** the formula of the acid
- D** the volume of the acid required for neutralisation

10 Which statement about electrolysis is correct?

- A Electrons move through the electrolyte from the cathode to the anode.
- B Electrons move in the external circuit towards the cathode.
- C Negative ions move in the external circuit towards the anode.
- D Positive ions move through the electrolyte towards the anode.

11 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

What is the half-equation for the reaction at the cathode?

- A $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$
- C $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$
- D $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$

12 Three statements about activation energy, E_a , are listed.

- 1 Colliding particles must have at least E_a before they can react.
- 2 E_a for exothermic reactions is always greater than for endothermic reactions.
- 3 E_a is always endothermic.

Which statements are correct?

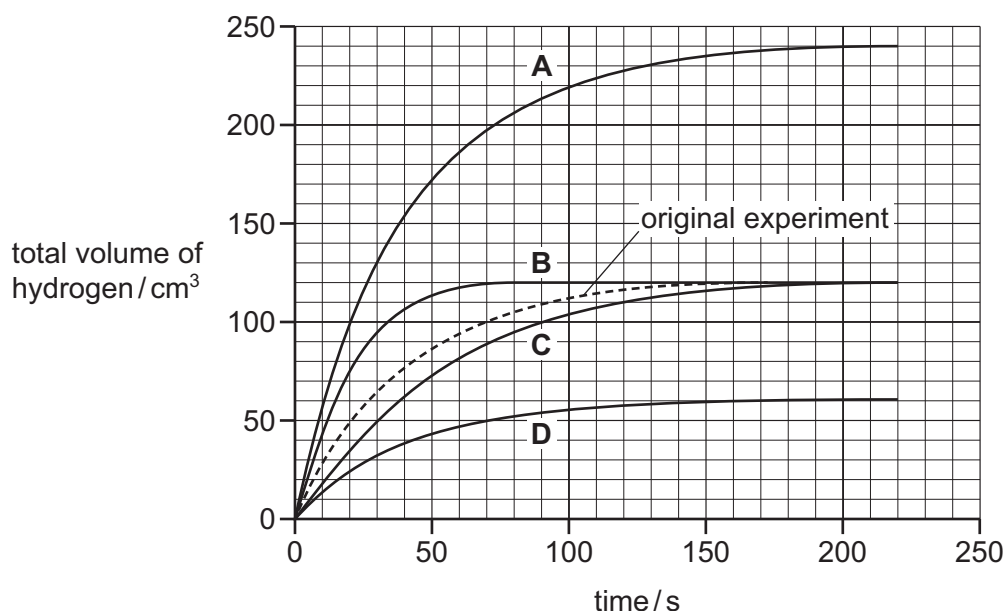
- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 13** A student adds excess magnesium ribbon to 10 cm^3 of 0.5 mol/dm^3 sulfuric acid.

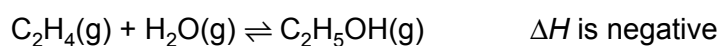
The hydrogen gas produced is collected and its total volume is measured every 10 seconds.

The experiment is repeated with 5 cm^3 of 0.5 mol/dm^3 sulfuric acid added to 5 cm^3 of water using the same mass of magnesium ribbon.

Which line on the graph shows the results of the second experiment?



- 14** The equation represents the reversible reaction between ethene and steam.



Which row describes the conditions that produce the greatest yield of ethanol?

	pressure	temperature
A	low	low
B	low	high
C	high	low
D	high	high

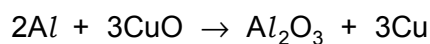
- 15 Which row identifies the pressure and the catalyst used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process?

	pressure / atm	catalyst
A	2	iron
B	2	vanadium(V) oxide
C	200	iron
D	200	vanadium(V) oxide

- 16 Which equation represents a redox reaction?

- A** $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$
B $\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$
C $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$
D $\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$

- 17 The equation for the reaction between aluminium and copper(II) oxide is shown.



Which statements about this equation are correct?

- 1 The oxidation number of the aluminium reactant is +2.
- 2 The oxidation number of the aluminium in the product is +6.
- 3 The oxidation number of the copper in the reactant is +2.
- 4 The oxidation number of the copper product is 0.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 3 and 4 **D** 4 only

- 18 What is the colour of thymolphthalein in dilute hydrochloric acid?

- A** blue
B red
C yellow
D colourless

- 19** Two acids, P and Q, with the same concentration and volume are reacted separately with the same mass of magnesium ribbon.

The reactions produce the same total volume of hydrogen gas but acid Q reacts much more slowly than acid P.

Which explanation for the difference between P and Q is correct?

- A** Acid P has a higher pH than acid Q.
 - B** Acid P has a lower concentration of hydrogen ions.
 - C** Acid Q is partially dissociated and acid P is fully dissociated.
 - D** Acid Q is a proton acceptor.
- 20** Which process is **not** used in the preparation of an insoluble salt?
- A** filtration
 - B** washing
 - C** crystallisation
 - D** drying

- 21** Lithium and potassium are metals in Group I of the Periodic Table.

Lithium has a melting point of 181 °C and a density of 0.53 g/cm³.

Which row describes the melting point and density of potassium?

	melting point in °C	density in g/cm ³
A	less than 181	less than 0.53
B	less than 181	greater than 0.53
C	greater than 181	less than 0.53
D	greater than 181	greater than 0.53

- 22** Which statements about transition elements are correct?

- 1 They have a low density.
- 2 They form ions with variable oxidation numbers.
- 3 They have a high melting point.
- 4 They form only colourless compounds.

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

23 Which statements describe the elements in Group VIII of the Periodic Table?

- 1 Their atoms have full outer electron shells.
- 2 They are unreactive metals.
- 3 They are monatomic gases.
- 4 They are diatomic gases.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

24 Four metals, Q, R, S and T, are each added to separate samples of water, steam and dilute hydrochloric acid.

The results are shown.

	observation with water	observation with steam	observation with dilute hydrochloric acid
Q	slow reaction	fast reaction	fast reaction
R	no reaction	no reaction	no reaction
S	no reaction	very slow reaction	slow reaction
T	fast reaction	explodes	explodes

Which statements are correct?

- 1 R is the least reactive metal.
- 2 T could be potassium.
- 3 S is more reactive than Q and R.
- 4 Metals generally react faster with steam than they react with water.

A 1, 2 and 4 **B** 1 and 2 only **C** 2 and 3 **D** 3 and 4

25 Metal X acts as a sacrificial metal to prevent iron from corroding.

Metal X does **not** act as a sacrificial metal to prevent aluminium from corroding.

What is X?

- A** copper
- B** magnesium
- C** silver
- D** zinc

26 Which equation represents the reduction of a compound found in hematite in the blast furnace?

- A $2\text{Al}_2\text{O}_3 \rightarrow 4\text{Al} + 3\text{O}_2$
- B $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
- C $2\text{FeO} + \text{CO} \rightarrow \text{Fe}_2\text{O}_3 + \text{C}$
- D $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

27 Which statements about the treatment of domestic water supplies are correct?

- 1 Filtration is used to remove insoluble substances from the water.
- 2 Sedimentation is used to remove soluble substances from the water.
- 3 Carbon is used to remove tastes and odours from the water.
- 4 Chlorine is used to lower the pH of the water.

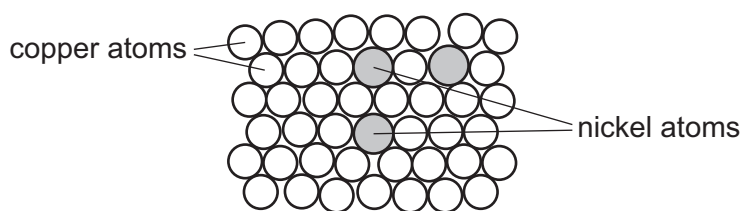
- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

28 Which statement about water is correct?

- A It turns anhydrous copper(II) sulfate from pink to blue.
- B It turns anhydrous copper(II) sulfate from white to blue.
- C It turns cobalt(II) chloride paper from blue to white.
- D It turns cobalt(II) chloride paper from pink to blue.

29 Cupronickel is used to make coins.

The arrangement of atoms in cupronickel is shown.



Which kind of substance is cupronickel?

- A an alloy
- B an isotope
- C a compound
- D a transition element

30 Which physical properties are typical of **all** metals?

- 1 good heat conductivity
- 2 low density
- 3 malleability

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only

31 Which oxide is used to neutralise acidic gases in flue gas desulfurisation?

- A** calcium oxide
B carbon dioxide
C nitrogen oxide
D sulfur dioxide

32 Some information about three gases, P, Q and R, is listed.

- Gas P forms when magnesium reacts with dilute hydrochloric acid.
- Gas Q makes up 78% of the air.
- Gas R forms when gas P reacts with gas Q.

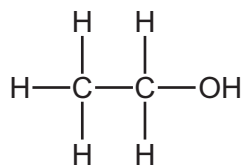
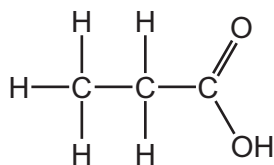
What is gas R?

- A** ammonia
B methane
C nitrogen dioxide
D water vapour

33 Which formula represents an alkene?

A CH_4 **B** C_2H_4 **C** C_2H_6 **D** $\text{C}_2\text{H}_5\text{OH}$

34 The structures of two compounds used to make an ester are shown.



What is the name of the ester?

- A ethyl propanoate
 - B propyl ethanoate
 - C ethyl ethanoate
 - D propyl propanoate
- 35 Which statement about a homologous series is correct?
- A All members have the same general formula.
 - B All members have the same molecular formula.
 - C All members have similar physical properties.
 - D Members show a trend in their chemical properties.
- 36 Which statements about aqueous ethanoic acid are correct?
- 1 It contains the functional group -COOH .
 - 2 It reacts with carbonates to produce hydrogen.
 - 3 It turns universal indicator paper blue.
 - 4 It has a pH lower than pH 7.
- A 1 and 2 B 1 and 3 C 1 and 4 D 2 and 4

37 Alkenes react with steam in an addition reaction.

Some alkenes produce only one alcohol product.

Some alkenes produce two different alcohols which are structural isomers.

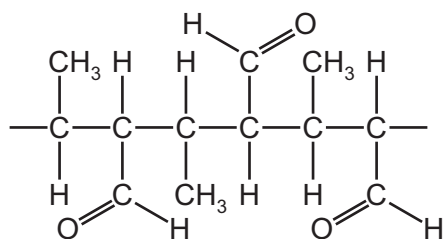
Which row gives the number of alcohol isomers formed when ethene reacts with steam and when propene reacts with steam?

	number of alcohol isomers formed	
	ethene + steam	propene + steam
A	1	1
B	1	2
C	2	1
D	2	2

38 What is an advantage of manufacturing ethanol by fermentation rather than by the addition of steam to ethene?

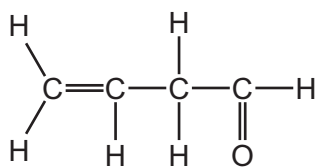
- A** No gases that cause global warming are produced.
- B** The ethanol that is produced is pure.
- C** Fermentation is a fast process.
- D** Fermentation uses renewable raw materials.

39 The diagram shows the structure of a polymer.

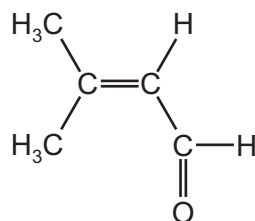


Which structure represents the monomer for this polymer?

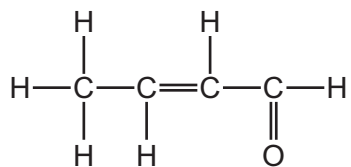
A



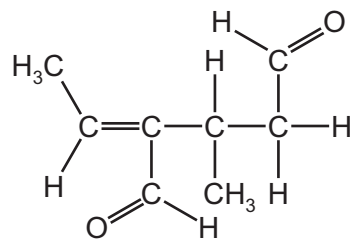
B



C



D



40 Which piece of apparatus is used to measure 24.5 cm^3 of gas produced during a reaction?

- A** beaker
- B** conical flask
- C** measuring cylinder
- D** volumetric pipette

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	

lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/23

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 Sodium chloride is a liquid at 900°C .

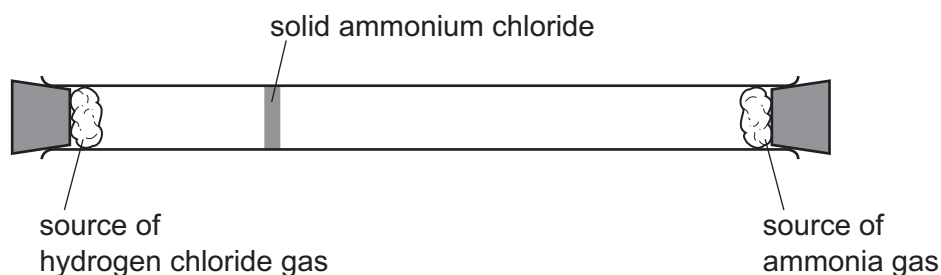
Which row describes the arrangement and the motion of the particles in sodium chloride at 900°C ?

	arrangement of particles	motion of particles
A	regular	vibrate about a fixed point
B	regular	move randomly
C	random	vibrate about a fixed point
D	random	move randomly

- 2 Hydrogen chloride gas, HCl , reacts with ammonia gas, NH_3 , to form solid ammonium chloride.

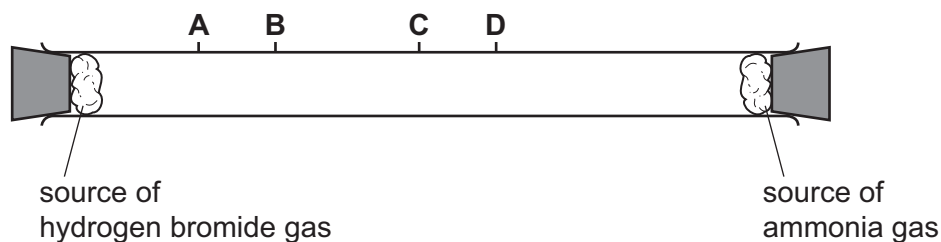
The apparatus is set up as shown.

After a few minutes, a white cloud of solid ammonium chloride forms where the two gases meet.



The experiment is repeated using hydrogen bromide gas, HBr , in place of hydrogen chloride.

How far along the tube does the white cloud of solid ammonium bromide form?



- 3 Substances P and Q both conduct electricity.

P is a mixture of two different types of atom.

Q is made of only one type of atom.

Which row describes P and Q?

	P	Q
A	alloy	element
B	alloy	compound
C	compound	alloy
D	compound	element

- 4 An atom of element R contains 15 protons, 16 neutrons and 15 electrons.

What is R?

- A** gallium
 - B** phosphorus
 - C** sulfur
 - D** zinc
- 5 Which molecule contains a double covalent bond between two atoms of the same element?
- A** carbon dioxide
 - B** ethanol
 - C** ethene
 - D** nitrogen

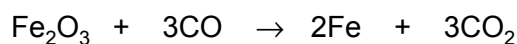
- 6 Silicon(IV) oxide is a covalently bonded compound.

Which statements are correct?

- 1 Silicon atoms form four single bonds in silicon(IV) oxide.
- 2 Oxygen atoms form two double bonds in silicon(IV) oxide.
- 3 Silicon(IV) oxide has a high melting point.
- 4 Silicon(IV) oxide contains one silicon atom and four oxygen atoms.

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 and 4

- 7 The equation for the reaction of iron(III) oxide with carbon monoxide is shown.



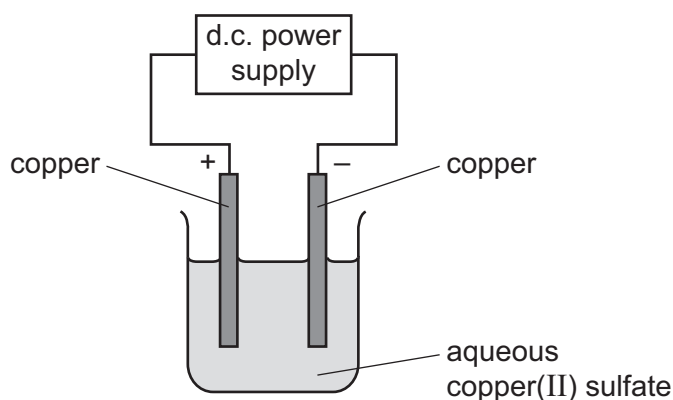
What is the percentage yield of iron when 16.8 g of carbon monoxide reacts completely with iron(III) oxide to form 8.96 g of iron?

A 26.7% **B** 40.0% **C** 53.3% **D** 80.0%

- 8 What is the volume of 14.5 g of gaseous butane, C_4H_{10} , at room temperature and pressure?

A 96.0 cm^3 **B** 6.0 cm^3 **C** 96.0 dm^3 **D** 6.0 dm^3

- 9 Aqueous copper(II) sulfate is electrolysed using copper electrodes.



Which statement describes what happens during the electrolysis?

- A** Copper atoms gain electrons at the cathode and copper(II) ions lose electrons at the anode.
- B** Electrons move in the external circuit from the positive electrode to the negative electrode.
- C** Copper(II) ions move through the electrolyte from the cathode to the anode.
- D** Copper is formed at the cathode and oxygen is formed at the anode.

- 10 Electrolysis is carried out on concentrated aqueous potassium bromide using inert electrodes.

Which products are formed at the anode and the cathode?

	anode	cathode
A	bromine	hydrogen
B	bromine	potassium
C	hydrogen	bromine
D	hydrogen	potassium

- 11 Hydrogen–oxygen fuel cells and gasoline are each used to power cars.

Which statement describes an advantage of using hydrogen–oxygen fuel cells in cars in comparison with gasoline engines?

- A** Hydrogen is a non-renewable resource.
- B** Hydrogen is produced during the fractional distillation of petroleum.
- C** Hydrogen–oxygen fuel cells do **not** release carbon dioxide.
- D** Hydrogen–oxygen fuel cells need electricity to work.

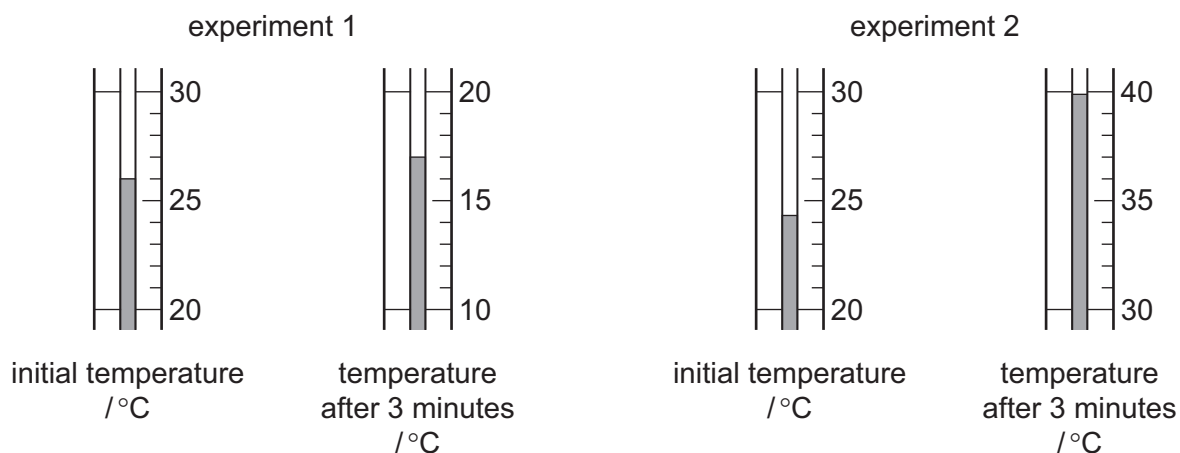
- 12 Plant cells use energy from sunlight for photosynthesis.

Which row describes and explains the energy change that occurs?

	type of energy change	explanation of energy change
A	endothermic	less energy is released making bonds than is absorbed to break bonds
B	endothermic	more energy is released making bonds than is absorbed to break bonds
C	exothermic	less energy is released making bonds than is absorbed to break bonds
D	exothermic	more energy is released making bonds than is absorbed to break bonds

- 13 Two different experiments are done to find the enthalpy change, ΔH , of each reaction.

The temperature of each reaction mixture is measured at the beginning of the reaction and after 3 minutes.

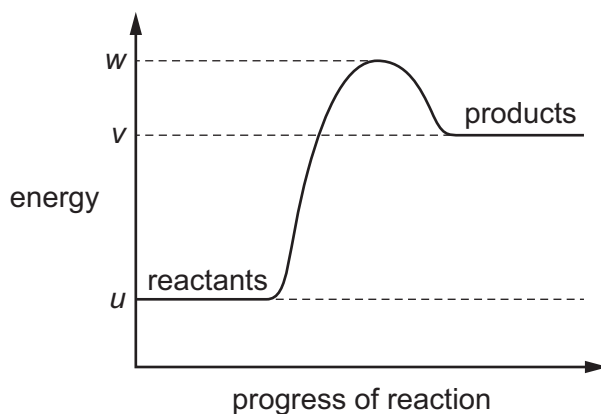


Which row gives the correct sign for the value of ΔH for each experiment and identifies if the reaction is endothermic or exothermic?

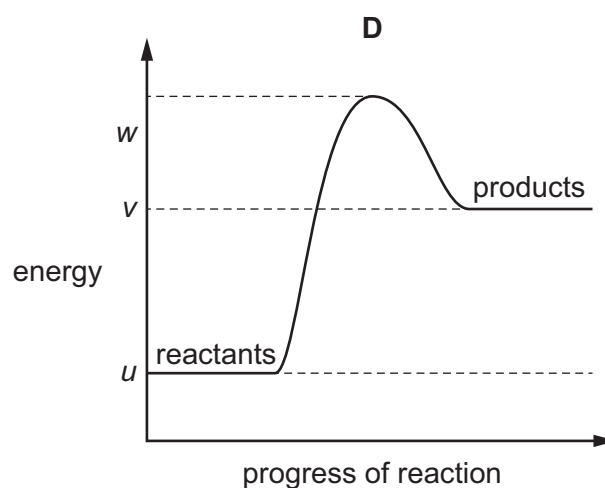
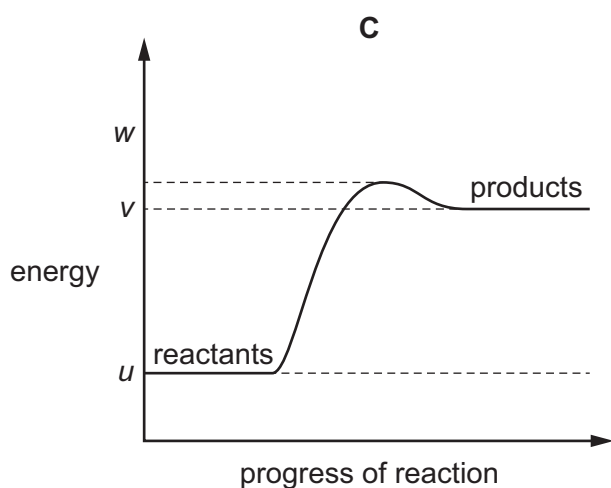
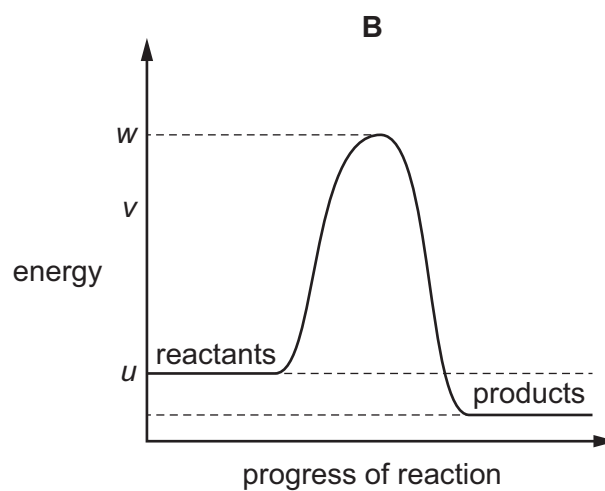
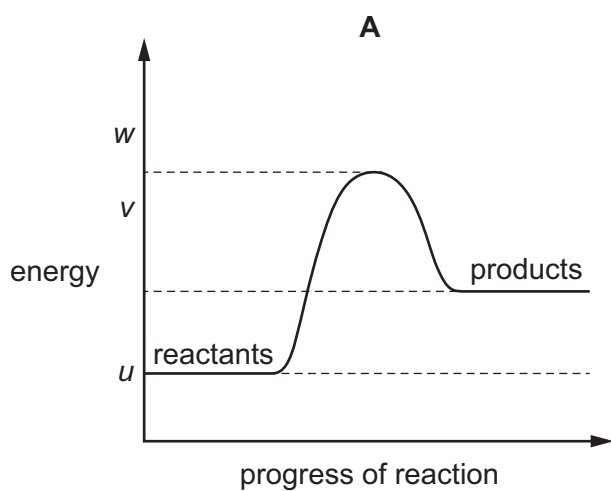
	experiment 1	experiment 2
A	negative and endothermic	positive and exothermic
B	negative and exothermic	positive and endothermic
C	positive and endothermic	negative and exothermic
D	positive and exothermic	negative and endothermic

14 The reaction pathway diagram for an endothermic reaction is shown.

u , v and w are known energy values.



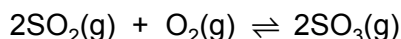
Which diagram shows the reaction pathway diagram when a catalyst is used in the reaction?



15 Which process is a physical change?

- A burning a piece of magnesium
- B reacting calcium carbonate with hydrochloric acid
- C melting an ice cube
- D the rusting of an iron nail

16 The Contact process is used to convert sulfur dioxide to sulfur trioxide. Vanadium(V) oxide is the catalyst in this process.



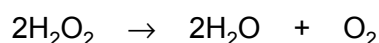
The forward reaction in this equilibrium is exothermic.

Which statements about this process are correct?

- 1 The catalyst increases the rate of both the forward and backward reactions.
- 2 A low pressure increases the yield of sulfur trioxide.
- 3 A low pressure is used to keep the costs low.
- 4 A high temperature increases the yield of sulfur trioxide.

- A 1 and 2 B 1 and 3 C 2 and 4 D 3 and 4

17 The equation for the decomposition of hydrogen peroxide, H_2O_2 , is shown.



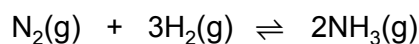
In an experiment, the total volume of oxygen produced is 100 cm^3 .

The experiment is repeated using 1.00 g of a solid catalyst. All other conditions remain the same.

Which row describes the total volume of oxygen and the mass of the catalyst at the end of the second experiment?

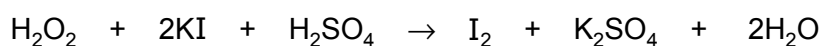
	total volume of oxygen / cm^3	mass of catalyst /g
A	more than 100	less than 1.00
B	100	less than 1.00
C	100	1.00
D	more than 100	1.00

- 18** Nitrogen reacts with hydrogen to form ammonia in the presence of an iron catalyst. The reaction is reversible.



Which statement about this reaction is correct?

- A** The iron catalyst decreases the activation energy of only the forward reaction.
B When equilibrium is reached, the forward reaction has stopped.
C When the pressure changes, the concentration of ammonia at equilibrium remains constant.
D Nitrogen and hydrogen never completely convert to ammonia.
- 19** The equation for a redox reaction is shown.



Potassium iodide is 1 agent in this reaction because iodide ions 2 electrons.

Which words complete gaps 1 and 2?

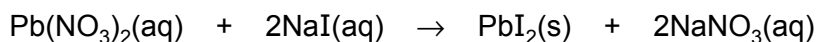
	1	2
A	an oxidising	lose
B	an oxidising	gain
C	a reducing	lose
D	a reducing	gain

- 20** Element E is in Group II of the Periodic Table.

Which row describes element E and its oxide?

	element E	oxide of E
A	metal	acidic
B	metal	basic
C	non-metal	acidic
D	non-metal	basic

- 21** Lead(II) iodide is formed as a precipitate in the reaction shown.



Which method is used to separate the lead(II) iodide from the mixture?

- A** crystallisation
 - B** distillation
 - C** evaporation
 - D** filtration
- 22** Which statement explains why sulfur, S, has similar chemical properties to selenium, Se?
- A** They both have the same number of electrons in their outer electron shell.
 - B** They are both solids at room temperature and pressure.
 - C** They are both non-metals.
 - D** They both form negative ions.
- 23** Atoms of sodium, rubidium and element Q each have one outer shell electron.

Some properties of these elements are shown.

element	melting point in °C	boiling point in °C	density in g / cm ³
sodium	98	883	0.97
rubidium	39	688	1.53
Q	28	672	1.87

What is Q?

- A** hydrogen
- B** lithium
- C** potassium
- D** caesium

24 Which statement about elements in Group VIII of the Periodic Table is correct?

- A** They all have a full outer shell of electrons.
- B** They all react with Group I elements to form ionic compounds.
- C** They are all diatomic molecules.
- D** They are all liquids at room temperature and pressure.

25 The electrical conductivity of magnesium is tested.

Magnesium is then added to dilute sulfuric acid. A gas, W, is produced.

Which row describes the electrical conductivity of magnesium and identifies W?

	electrical conductivity of magnesium	gas W
A	good	hydrogen
B	good	oxygen
C	poor	hydrogen
D	poor	oxygen

26 Aluminium metal is extracted from its purified ore by electrolysis.

Which statement about the electrolyte in this process is correct?

- A** The electrolyte is purified molten bauxite only.
- B** The electrolyte is purified bauxite dissolved in molten cryolite.
- C** The electrolyte is purified molten cryolite only.
- D** The electrolyte is purified cryolite dissolved in molten bauxite.

27 Part of the reactivity series is shown.

potassium
X
calcium
Y
aluminium

Which metals are represented by X and Y?

	X	Y
A	copper	magnesium
B	sodium	magnesium
C	sodium	silver
D	copper	silver

28 Some substances found in water extracted from a river are listed.

- 1 plastics
- 2 nitrates
- 3 oxygen

Which substances are harmful to aquatic life?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

29 Car engines which use gasoline as a fuel produce oxides of nitrogen.

Oxides of nitrogen are removed from the exhaust gases.

Which statements about the formation or removal of oxides of nitrogen are correct?

- 1 Gasoline reacts with nitrogen in the air to produce oxides of nitrogen.
- 2 Gasoline contains nitrogen.
- 3 Nitrogen and oxygen react at high temperatures to produce oxides of nitrogen.
- 4 Nitrogen monoxide, NO, reacts with carbon monoxide, CO, in a catalytic converter.

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

30 Which structures represent a pair of structural isomers?

- 1 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- 2 $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$
- 3 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$
- 4 $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

31 Alkanes are a homologous series of hydrocarbons.

The table shows the names and boiling points of the first four members of this series.

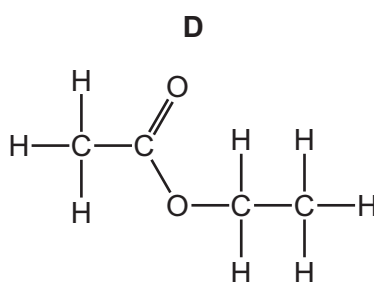
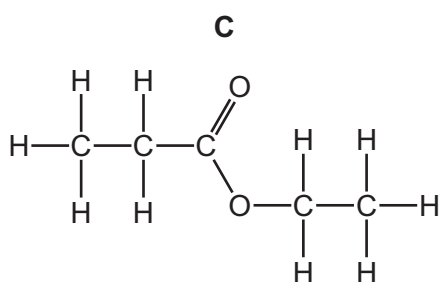
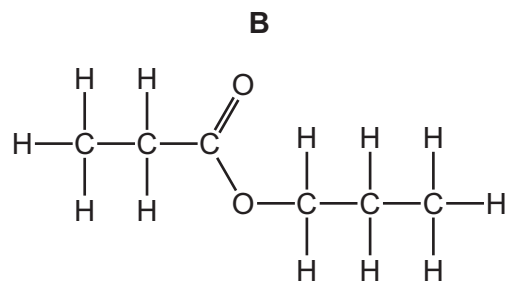
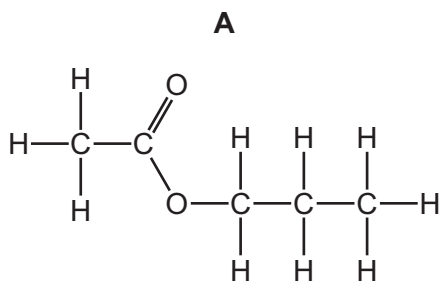
name	boiling point / °C
methane	−162
ethane	−89
propane	−42
butane	0

Pentane is the next member of the series.

Which row gives the molecular formula and the boiling point of pentane?

	molecular formula	boiling point / °C
A	C_5H_{10}	36
B	C_5H_{12}	−51
C	C_5H_{10}	−51
D	C_5H_{12}	36

- 32 Which displayed formula represents the ester formed by the reaction of propan-1-ol with ethanoic acid?



- 33 Ethanol can be manufactured by fermentation and by the catalytic addition of steam to ethene.

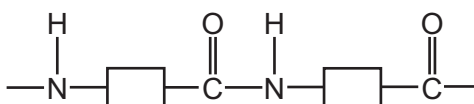
Which statement identifies an advantage of using **one** of these methods?

- A** Catalytic addition requires a high temperature and pressure.
- B** Ethanol produced by fermentation is extracted by distillation.
- C** Fermentation is a batch process.
- D** The raw material in fermentation is a renewable resource.

- 34 Which statement about carboxylic acids is correct?

- A** They react with alkalis to form a salt and water.
- B** They react with metals to form a salt and water.
- C** They react with metal carbonates to form a salt, water and hydrogen.
- D** The general formula for carboxylic acids is $C_nH_{2n+1}OH$.

35 The structure of a polymer is shown.



Which statement about this polymer is correct?

- A** Alkenes are polymerised to make the polymer.
- B** It is a polyester.
- C** It is an addition polymer.
- D** Water is produced when the polymer is made.

36 Methane undergoes substitution reactions with chlorine and complete combustion with excess oxygen.

Which row about the two reactions is correct?

	condition for reaction with chlorine	equation for the complete combustion
A	an acid catalyst	methane + oxygen → carbon dioxide + hydrogen
B	an acid catalyst	methane + oxygen → carbon dioxide + water
C	ultraviolet light	methane + oxygen → carbon dioxide + hydrogen
D	ultraviolet light	methane + oxygen → carbon dioxide + water

37 What is used to identify the end-point of an acid–base titration?

- A** balance
- B** measuring cylinder
- C** indicator
- D** volumetric pipette

- 38** Four pure substances, P, Q, R and S, are tested using chromatography. The same solvent is used each time.

The table shows the distance moved by each substance and by the solvent from the baseline.

substance	distance moved by substance / cm	distance moved by solvent / cm
P	4.5	10.0
Q	3.0	20.0
R	4.5	20.0
S	13.5	30.0

Which two substances are identical?

- A** P and R **B** P and S **C** Q and R **D** Q and S

- 39** A substance is tested with three different reagents.

Which row shows the results obtained with aqueous iron(II) nitrate?

	aqueous sodium hydroxide	acidified aqueous silver nitrate	acidified aqueous barium nitrate
A	green precipitate, insoluble in excess	no reaction	no reaction
B	green precipitate, insoluble in excess	white precipitate	white precipitate
C	white precipitate, insoluble in excess	cream precipitate	no reaction
D	white precipitate that dissolves in excess	no reaction	white precipitate

- 40** A student carries out a flame test on a sample.

The flame colour observed is light green.

Which ion is present in the sample?

- A** Ba^{2+} **B** Ca^{2+} **C** Li^+ **D** K^+

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
		11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40								
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/21

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 A gas is placed in a sealed container with a fixed volume.

The gas is heated.

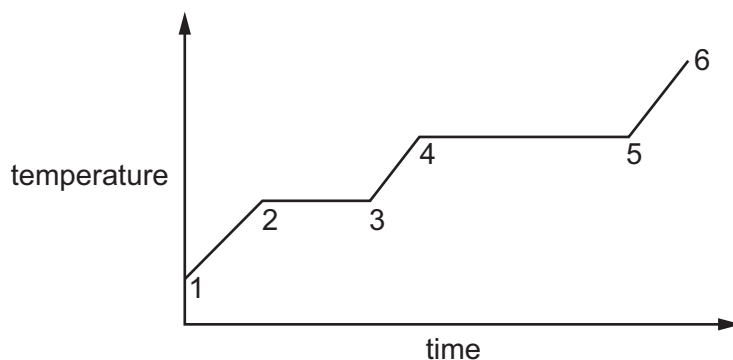
Which statements are correct?

- 1 The gas molecules move around faster.
- 2 The gas particles collide with the walls of the container more frequently.
- 3 The pressure in the container decreases.

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 2 only

- 2 A sample of ethanol is heated at a constant rate.

The heating curve for this sample is shown.



Which statement about the heating curve is correct?

- A** Ethanol changes its physical state between 3 and 4.
- B** Ethanol condenses between 2 and 3.
- C** Energy is absorbed between 3 and 4 but **not** absorbed between 4 and 5.
- D** More energy is absorbed between 4 and 5 than between 2 and 3.

- 3 Which atom has an equal number of protons, neutrons and electrons?

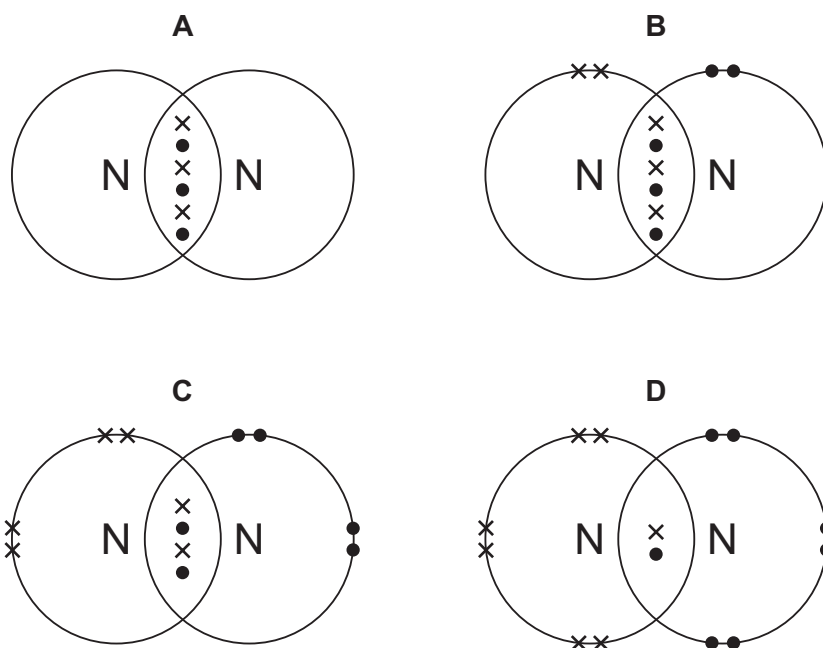
A ^{40}Ar **B** ^1H **C** ^{23}Na **D** ^{14}N

- 4 The only naturally occurring isotopes in a sample of europium are ^{151}Eu and ^{153}Eu . This sample of europium has a relative atomic mass, A_r , of 152.

What is the percentage of ^{151}Eu present in this sample to three significant figures?

A 0.987% **B** 50.0% **C** 76.0% **D** 98.7%

- 5 Which statement about potassium chloride is correct?
- A** It has a high melting point because it is a giant structure with strong covalent bonds.
- B** It conducts electricity when solid because of its oppositely charged ions.
- C** It conducts electricity when molten because its ions can move.
- D** It is insoluble in water because of its strong bonding.
- 6 Which dot-and-cross diagram shows the outer shell electron arrangement in a molecule of nitrogen?



- 7 Which statement about silicon(IV) oxide is correct?
- A** It contains silicon ions and oxide ions arranged in a giant lattice.
- B** It is used in electrical circuits because it can conduct electricity.
- C** Its bonds each contain one shared pair of electrons.
- D** Its atoms are arranged in the same way as the atoms in graphite.

8 Three organic molecules are described.

- The molecular formula of propanoic acid is $\text{CH}_3\text{CH}_2\text{COOH}$.
- The molecular formula of lactic acid is $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$.
- A glucose molecule consists of 6 carbon atoms, 6 oxygen atoms and 12 hydrogen atoms only.

Which molecules have the same empirical formula?

- A** propanoic acid, lactic acid and glucose
B propanoic acid and lactic acid only
C propanoic acid and glucose only
D lactic acid and glucose only

9 Calcium carbonate reacts with dilute hydrochloric acid.

The equation for the reaction is shown.

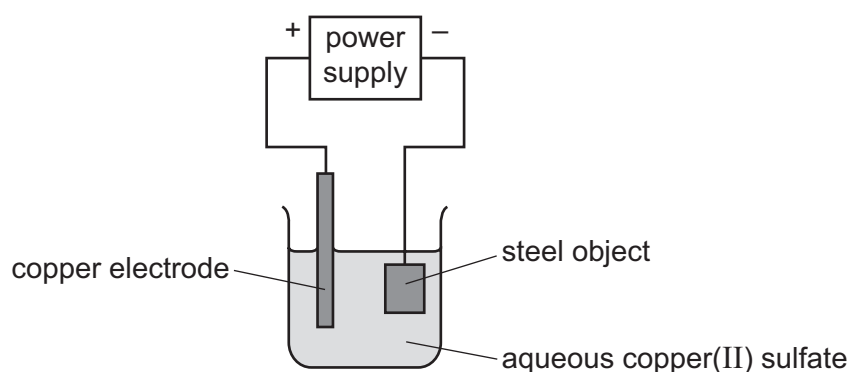


1.00 g of calcium carbonate is added to 50.0 cm^3 of 0.0500 mol/dm^3 hydrochloric acid.

Which volume of carbon dioxide is made in this reaction?

- A** 30 cm^3 **B** 60 cm^3 **C** 120 cm^3 **D** 240 cm^3

10 The diagram shows the electroplating of a steel object.



A student makes three statements.

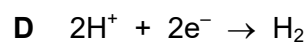
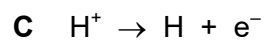
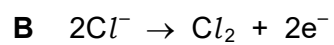
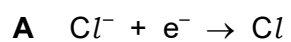
- 1 The steel object turns a reddish-brown colour.
- 2 The aqueous copper(II) sulfate changes to a paler blue colour.
- 3 The copper electrode becomes thinner.

Which statements are correct?

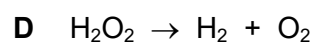
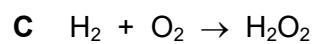
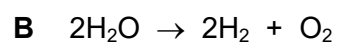
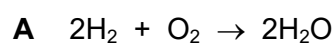
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

11 Concentrated aqueous sodium chloride is electrolysed using inert electrodes.

Which equation shows the correct ionic half-equation for the reaction at the anode?



12 Which equation represents the overall reaction in a hydrogen–oxygen fuel cell?



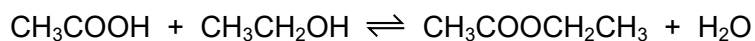
13 Water is added to anhydrous copper(II) sulfate in a test-tube.

The mixture becomes hot.

Which type of reaction and reaction pathway diagram apply to this reaction?

	type of reaction	reaction pathway diagram
A	endothermic	
B	endothermic	
C	exothermic	
D	exothermic	

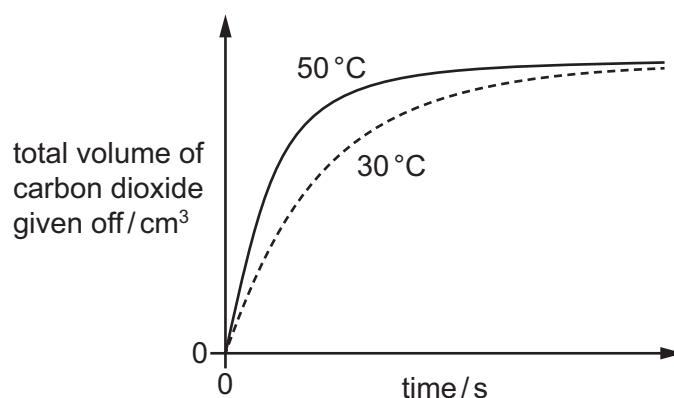
14 The equation represents the reaction between ethanoic acid and ethanol.



Which type of reaction does this show?

- A** addition
- B** neutralisation
- C** reduction
- D** reversible

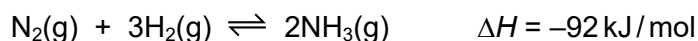
- 15 The graph shows the results of adding excess calcium carbonate to hydrochloric acid at 50 °C and at 30 °C. Only the temperature is changed.



Which row describes the reacting particles at 30 °C compared to those at 50 °C?

	frequency of collisions between particles	average kinetic energy of particles
A	higher	higher
B	higher	lower
C	lower	higher
D	lower	lower

- 16 Industrially, ammonia is made by the Haber process. The equation is shown.



The conditions used are 450 °C and 200 atm.

Which statements about the conditions are correct?

- 1 Temperatures below 450 °C give a lower rate of reaction.
- 2 Temperatures below 450 °C give a lower yield of ammonia.
- 3 Pressures above 200 atm give a lower yield of ammonia.
- 4 Pressures above 200 atm are too expensive to maintain.

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

17 Ethanol is warmed with acidified aqueous potassium manganate(VII).

Which row describes the type of reaction of the ethanol and the colour change observed in the reaction mixture?

	type of reaction	colour change
A	oxidation	colourless to purple
B	oxidation	purple to colourless
C	reduction	colourless to purple
D	reduction	purple to colourless

18 Two experiments involving acids R or S are done.

- 0.10 mol of acid R completely dissolves in 500 cm³ of water. 0.0048 mol of acid R is dissociated in the solution.
- 0.10 mol of acid S completely dissolves in 500 cm³ of water. 0.0179 mol of acid S is dissociated in the solution.

Which statement about acids R and S is correct?

- A** Acid R is a strong acid, and acid S is a weak acid.
- B** Acid R is a weak acid, and acid S is a strong acid.
- C** Acid R and acid S are both strong acids.
- D** Acid R and acid S are both weak acids.

19 Lead(II) sulfate is an insoluble salt.

Which process is **not** used to prepare a pure sample of this salt?

- A** crystallisation
- B** drying
- C** filtration
- D** precipitation

20 The formulae of two hydrated salts are shown.

- $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$
- $\text{CoCl}_2 \cdot y\text{H}_2\text{O}$

What are the values of x and y ?

	x	y
A	5	5
B	5	6
C	6	5
D	6	6

21 The properties of the elements change from left to right across the Periodic Table.

Which statement describes how the properties change across Period 3?

- A** Elements change from gases to solids.
- B** Elements change from non-conductors to electrical conductors.
- C** Elements change from metal to non-metal.
- D** Elements have a decreasing number of electron shells.

22 Which property decreases down Group VII of the Periodic Table, from chlorine to iodine?

- A** atomic number
- B** density
- C** reactivity
- D** melting point

23 Some properties of elements are listed.

- 1 They have good electrical conductivity.
- 2 They have high melting points.
- 3 They are malleable.
- 4 They have ions with variable oxidation numbers.

Which properties are correct for **both** iron and lithium?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

24 Two statements about noble gases are listed.

- 1 Noble gases are reactive, monatomic gases.
- 2 Noble gases all have a full outer shell of electrons.

Which statements about noble gases are correct?

- A Both statements are correct, and statement 2 explains statement 1.
- B Both statements are correct but statement 2 does **not** explain statement 1.
- C Statement 1 is correct but statement 2 is **not** correct.
- D Statement 2 is correct but statement 1 is **not** correct.

25 Which statement about the properties and uses of copper is correct?

- A It is used in electrical wiring, and it is ductile.
- B It is mixed with tin to make the alloy brass.
- C It is used in overhead electrical cables because it has a low density.
- D It is used in takeaway food containers because of its resistance to corrosion.

26 Three statements about solid Z are listed.

- It reacts with dilute hydrochloric acid to produce hydrogen.
- It reacts very slowly with cold water and quickly with steam.
- It is a good thermal conductor.

What is solid Z?

- A gold
- B magnesium
- C potassium
- D silver

27 Four metals J, K, L and M are added separately to aqueous solutions of their ions.

The results are shown.

	aqueous ions				
metal	J^{2+}	K^{2+}	L^{2+}	M^{2+}	
J	x	✓	✓	✓	key ✓ = reaction x = no reaction
K	x	x	x	✓	
L	x	✓	x	✓	
M	x	x	x	x	

Which statement about the metals is correct?

- A J has a greater tendency to form positive ions than K.
- B L has a greater tendency to form positive ions than J.
- C K has a greater tendency to form positive ions than L.
- D M has a greater tendency to form positive ions than K.

28 Which process is used in the extraction of iron from hematite?

- A electrolysis
- B precipitation
- C reduction
- D fractional distillation

29 Galvanising prevents iron from rusting.

Which statements about galvanising are correct?

- 1 Galvanising involves coating the iron with silver.
- 2 Galvanising prevents oxygen and water from coming into contact with the iron.
- 3 Galvanising works because the metal coating loses electrons more readily than iron.

- A 1, 2 and 3
- B 1 and 3 only
- C 2 and 3 only
- D 2 only

30 Nitrogen, phosphorus and potassium are essential elements for improved plant growth.

Which mixture provides all **three** essential elements?

	mixture	formulae
A	ammonium phosphate + potassium chloride	$(\text{NH}_4)_3\text{PO}_4$ + KCl
B	ammonium phosphate + ammonium nitrate	$(\text{NH}_4)_3\text{PO}_4$ + NH_4NO_3
C	ammonium phosphate + ammonium chloride	$(\text{NH}_4)_3\text{PO}_4$ + NH_4Cl
D	ammonium nitrate + potassium chloride	NH_4NO_3 + KCl

31 Which statements help explain why an increasing amount of carbon dioxide or methane in the atmosphere causes global warming?

- 1 Carbon dioxide absorbs thermal energy from the Earth.
- 2 Methane does **not** absorb thermal energy from the Sun.
- 3 Carbon dioxide does **not** reflect thermal energy from the Sun to the Earth.
- 4 Methane reflects thermal energy from the Earth back to the Earth.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

32 The exhaust gases from cars contain oxides of nitrogen.

How are these oxides of nitrogen formed?

- A** Nitrogen and oxygen from the air react together at the high temperatures in the engine.
- B** Nitrogen and oxygen from the petrol react together in the car exhaust.
- C** Nitrogen from the petrol reacts with oxygen at the high temperatures in the engine.
- D** Nitrogen reacts with oxygen from the air in the catalytic converter.

33 The structural formula of methyl ethanoate is $\text{CH}_3\text{COOCH}_3$.

Which compounds are structural isomers of methyl ethanoate?

- 1 $\text{HCOOCH}_2\text{CH}_3$
- 2 $\text{CH}_3\text{CH}_2\text{COOH}$
- 3 $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 3 only

34 What is the **total** number of covalent bonds in one molecule of propane?

A 8 **B** 9 **C** 10 **D** 11

35 Industrially, ethanol is produced by fermentation or by the reaction of ethene with steam.

Which row is correct?

	fermentation	ethene + steam
A	uses a temperature of 100°C	uses a temperature of 350°C
B	needs the presence of yeast	does not need a catalyst
C	slow reaction	fast reaction
D	high yield of ethanol	low yield of ethanol

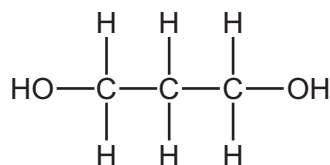
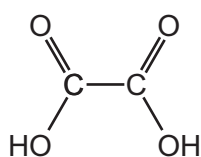
36 Some properties of an organic compound, J, are listed.

- It is a liquid at room temperature.
- It is soluble in water.
- A solution of J reacts with calcium carbonate to form carbon dioxide.
- A solution of J has a pH of 3.

In which homologous series does J belong?

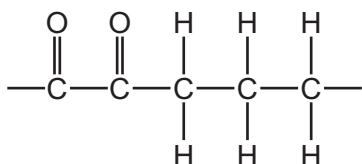
- A** alkane
- B** alkene
- C** alcohol
- D** carboxylic acid

37 The structures of two monomers that are used to make a condensation polymer are shown.

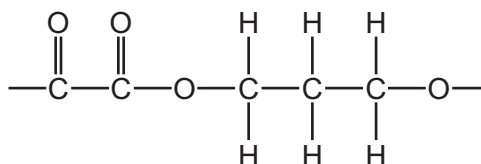


Which structure represents one repeat unit of the condensation polymer formed?

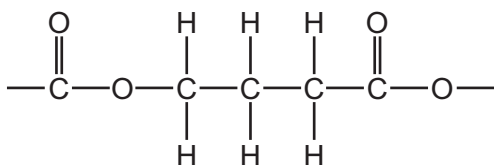
A



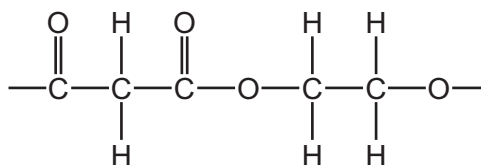
B



C



D

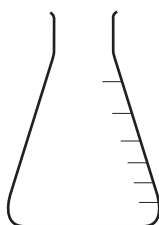


38 Which item of apparatus is used to measure 25.0 cm³ of aqueous sodium hydroxide?

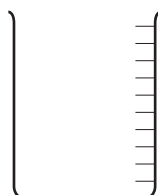
A



B



C



D

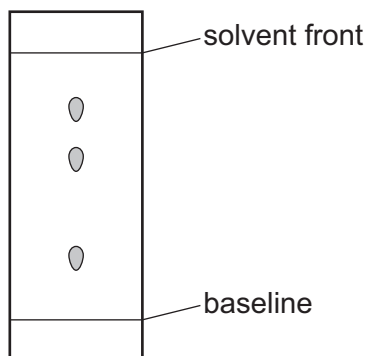


39 Which method is used to obtain copper(II) sulfate crystals from an aqueous solution of copper(II) sulfate?

- A** chromatography
- B** condensation
- C** evaporation
- D** filtration

40 A mixture of four different colourless amino acids is analysed by paper chromatography.

The final chromatogram is shown.



Why does the chromatogram only show three spots?

- A A locating agent is **not** used.
- B One of the amino acids is insoluble in the solvent.
- C The solvent front is too near the top of the paper.
- D Two of the amino acids have the same R_f value.

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The Periodic Table of Elements

Group																				
I	II											III	IV	V	VI	VII	VIII			
		1Hhydrogen1																		
		Key																		
		atomic number atomic symbol name relative atomic mass																		
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19		
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84			
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131			
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —			
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —			

lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/22

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

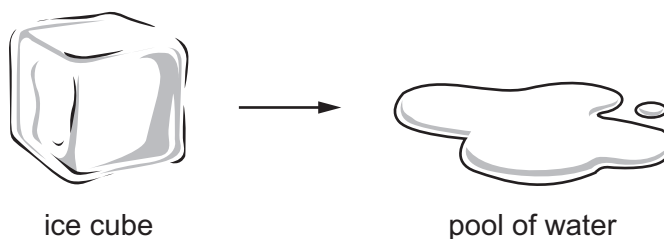
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 An ice cube melts.

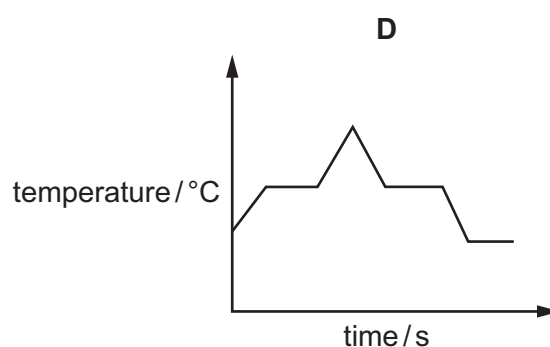
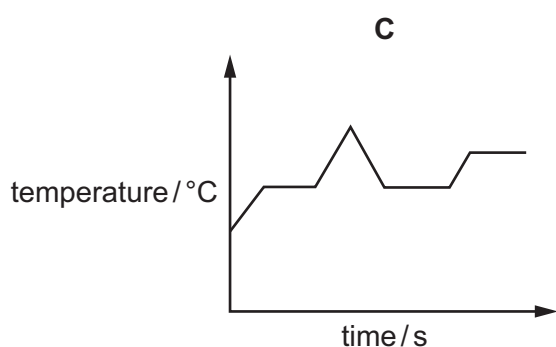
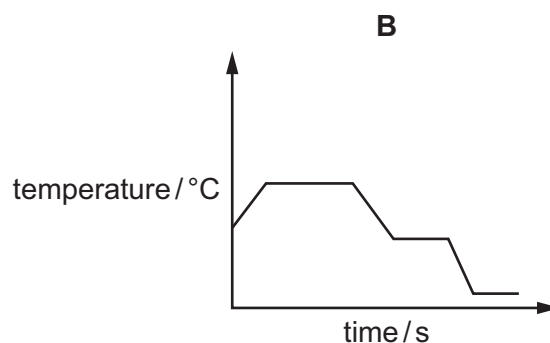
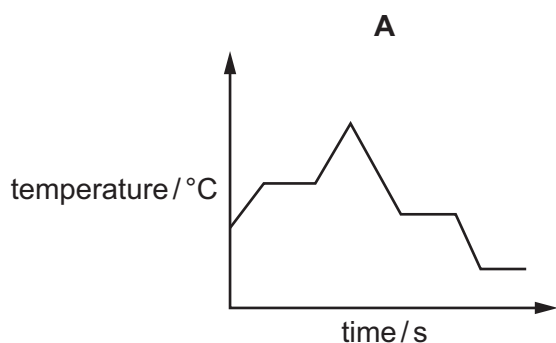


What happens to the molecules of water in the ice cube?

- A They evaporate.
 - B They dissolve.
 - C They gain energy.
 - D They lose energy.
- 2 A sample of a liquid, X, is heated to a temperature above its boiling point.

X is then cooled, so that it condenses and then freezes.

Which graph describes the change in temperature of X over time?



- 3 Which row gives the number of protons, electrons and neutrons in an atom of zinc?

	protons	electrons	neutrons
A	30	30	35
B	30	35	35
C	35	30	30
D	35	35	30

- 4 Which particles have the electronic configuration 2,8,8?

- 1 an argon atom, Ar
- 2 an aluminium ion, Al^{3+}
- 3 a sodium ion, Na^+
- 4 a chloride ion, Cl^-

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 5 A sample of element Q has two isotopes.

Their relative masses and abundances are shown.

relative mass of isotope	abundance /%
238	66
244	34

What is the relative atomic mass of this sample of Q to three significant figures?

- A** 240 **B** 241 **C** 242 **D** 243

- 6 Which statement about silicon(IV) oxide, SiO_2 , is correct?

- A** It conducts electricity because it contains delocalised electrons.
- B** It has a giant covalent structure with each silicon atom bonded to four oxygen atoms.
- C** It is a simple covalent molecule.
- D** Its structure is similar to that of graphite.

- 7 When 65 g of zinc reacts with 32 g of sulfur, 97 g of zinc sulfide is produced.

When 65 g of zinc reacts with 40 g of sulfur, the mass of zinc sulfide produced is still 97 g.

Which statement explains this observation?

- A Some of the zinc sulfide evaporates.
- B The reaction rate is slow.
- C The reaction stops before it is complete.
- D Zinc is the limiting reactant.

- 8 Aluminium reacts with iron(III) oxide to form aluminium oxide and iron.

Which chemical equation for the reaction between aluminium and iron(III) oxide is correct?

- A $Al + FeO \rightarrow AlO + Fe$
- B $2Al + Fe_2O \rightarrow Al_2O + 2Fe$
- C $Al + Fe_2O_3 \rightarrow Al_2O_3 + Fe$
- D $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$

- 9 The concentration of a solution of aqueous sodium hydroxide is 0.50 mol/dm^3 .

Which mass of sodium hydroxide is used to make 500 cm^3 of this solution?

- A 10 g B 20 g C 40 g D 160 g

- 10 Molten zinc oxide is electrolysed using inert electrodes.

Which row identifies the product at each electrode?

	anode	cathode
A	zinc	oxygen
B	hydrogen	oxygen
C	oxygen	zinc
D	oxygen	hydrogen

11 Dilute aqueous sodium chloride is electrolysed using platinum electrodes.

What is the half-equation for the reaction at the cathode?

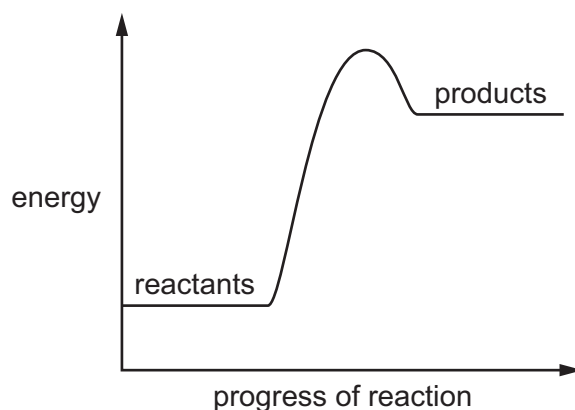
- A** $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- B** $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
- C** $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- D** $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$

12 Hydrogen–oxygen fuel cells can be used to power vehicles.

Which statement about hydrogen–oxygen fuel cells is correct?

- A** The equation for the overall reaction is $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$.
- B** The only chemical products are water and carbon dioxide.
- C** Chemical energy in the fuel is converted into electrical energy.
- D** The hydrogen used in hydrogen–oxygen fuel cells is a fossil fuel.

13 A reaction pathway diagram is shown.



Which row identifies the type of reaction and how the temperature of the surroundings changes during the reaction?

	type of reaction	temperature of the surroundings
A	endothermic	decreases
B	endothermic	increases
C	exothermic	decreases
D	exothermic	increases

- 14 The average bond energy for the C–H bond is 413 kJ/mol.

What is the enthalpy change when 1.0 mol of methane molecules is formed from carbon and hydrogen atoms?

- A –1652 kJ B –413 kJ C +413 kJ D +1652 kJ

- 15 Which statement describes the effect of adding a catalyst to a chemical reaction?

- A The activation energy, E_a , of the reaction is increased.
 B The enthalpy change, ΔH , of the reaction stays unchanged.
 C The frequency of collisions between the particles is decreased.
 D The kinetic energy of the particles is increased.

- 16 Which row describes the conditions used in the manufacture of sulfuric acid by the Contact process?

	catalyst	pressure	temperature
A	iron	high	high
B	iron	low	low
C	vanadium(V) oxide	high	low
D	vanadium(V) oxide	low	high

- 17 Which equations show the underlined species acting as a reducing agent?

- 1 Fe₂O₃ + 3CO → 2Fe + 3CO₂
 2 Fe + CuSO₄ → FeSO₄ + Cu
 3 5Fe²⁺ + MnO₄[–] + 8H⁺ → 5Fe³⁺ + Mn²⁺ + 4H₂O
 4 Fe(OH)₂ + H₂SO₄ → FeSO₄ + 2H₂O

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

- 18 A farmer incorrectly adds two substances to the soil at the same time.

They react together to form a gas which turns damp red litmus paper blue.

What are the two substances?

- A a basic oxide and a potassium salt
 B a basic oxide and an ammonium salt
 C an acidic oxide and a potassium salt
 D an acidic oxide and an ammonium salt

- 19 Which row shows the equations for the dissociation of hydrochloric acid and of ethanoic acid in aqueous solution?

	hydrochloric acid	ethanoic acid
A	$\text{HCl}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
B	$\text{HCl}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
C	$\text{HCl}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
D	$\text{HCl}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	$\text{CH}_3\text{COOH}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$

- 20 Which three oxides are **all** acidic?

- A** CaO, NO₂, SO₂
B CaO, CO₂, Na₂O
C CO₂, NO₂, SO₂
D CO₂, Na₂O, SO₂

- 21 The table shows some properties of noble gases.

noble gas	density in g/dm ³	boiling point in °C
helium	0.2	−269
neon	0.9	−246
argon	1.8	−186
krypton		
xenon	5.9	−108

Which row predicts the density and the boiling point of krypton?

	density in g/dm ³	boiling point in °C
A	1.4	−75
B	2.4	−195
C	3.8	−153
D	8.5	−213

22 Sodium is a Group I metal.

What is a physical property of sodium?

- A non-conductor of electricity
- B high melting point
- C high density
- D malleable

23 Tennessine (atomic number 117) is a manufactured element that is below astatine in Group VII of the Periodic Table.

What is the expected state of tennessine at room temperature and pressure?

- A a diatomic gas
- B a liquid
- C a monatomic gas
- D a solid

24 When chlorine is bubbled into aqueous sodium bromide, a displacement reaction occurs.

Which description of this reaction is correct?

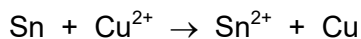
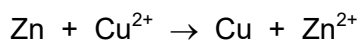
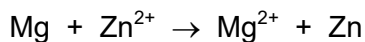
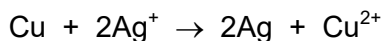
- A Chloride displaces bromide.
- B Chlorine displaces bromine.
- C Chloride displaces bromine.
- D Chlorine displaces bromide.

25 Brass is an alloy of copper.

Which statement about brass or copper is correct?

- A Brass does **not** conduct electricity because it is a compound.
- B Copper conducts electricity because its atoms are free to move.
- C Brass is harder than copper because its layers of atoms **cannot** easily slide over each other.
- D Copper is stronger than brass because it is a pure metal.

26 The ionic equations for some reactions between metals and aqueous metal ions are shown.



Which further information can be used to place copper, magnesium, silver, tin and zinc in order of their reactivity?

- A Zinc reacts with Sn^{2+} ions.
- B Copper does **not** react with Sn^{2+} ions.
- C Magnesium reacts with Cu^{2+} ions.
- D Magnesium reacts with Ag^+ ions.

27 Zinc is used to galvanise iron to prevent it from rusting.

Which statements are correct?

- 1 Galvanising is an example of a barrier method.
- 2 If the zinc is scratched, the iron will rust very quickly.
- 3 Galvanising is an example of sacrificial protection.
- 4 Zinc is more reactive than iron and so accepts electrons more readily.

- A 1, 2 and 4 B 1 and 3 C 2, 3 and 4 D 2 and 3 only

28 Which equation represents a reaction that occurs in a blast furnace during the extraction of iron from hematite?

- A $\text{Ca} + \text{CO}_2 \rightarrow \text{CaO} + \text{CO}$
- B $\text{Fe}_2\text{O}_3 + 3\text{Ca} \rightarrow 2\text{Fe} + 3\text{CaO}$
- C $\text{Ca}(\text{OH})_2 \rightarrow \text{CaO} + \text{H}_2\text{O}$
- D $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$

29 Water is added separately to anhydrous copper(II) sulfate and to anhydrous cobalt(II) chloride.

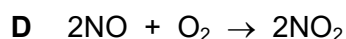
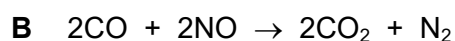
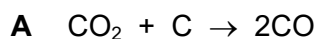
Which row shows the colour changes?

	anhydrous copper(II) sulfate	anhydrous cobalt(II) chloride
A	blue to white	blue to pink
B	blue to white	pink to blue
C	white to blue	blue to pink
D	white to blue	pink to blue

30 Gases which cause acid rain form in car engines.

These gases can be removed using a catalytic converter.

Which equation represents the removal of one of these gases using a catalytic converter?



31 Which statements about carbon dioxide are correct?

- 1 Carbon dioxide absorbs thermal energy from the Earth.
- 2 Carbon dioxide increases the thermal energy loss to space.
- 3 The level of carbon dioxide in the atmosphere has increased due to the burning of fossil fuels.
- 4 Carbon dioxide is the only gas that causes global warming.

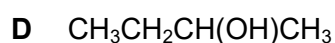
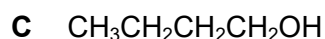
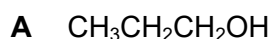
A 1 and 2

B 1 and 3

C 2 and 3

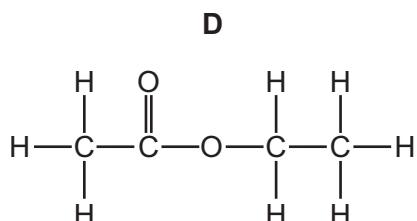
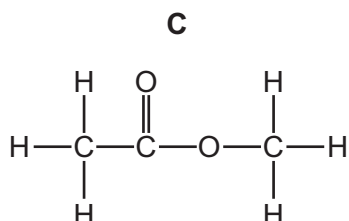
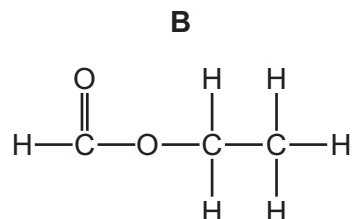
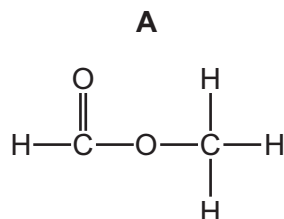
D 2 and 4

32 What is the structural formula of butan-2-ol?



- 33** Methanol reacts with ethanoic acid to form an ester.

What is the displayed formula for this ester?



- 34** Petroleum can be separated into useful fractions by fractional distillation.

Which row shows a correct use of the named fraction?

	fraction	use
A	bitumen	lubricant
B	naphtha	fuel for home heating
C	kerosene	jet fuel
D	refinery gas	making chemicals

- 35** Ethane reacts with chlorine in the presence of ultraviolet light.

Which products are formed?

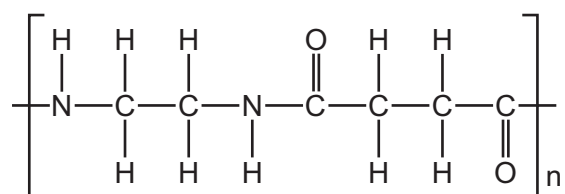
- A** $\text{ClCH}_2\text{CH}_2\text{Cl}$ and H_2
B $2\text{CH}_3\text{Cl}$ and H_2
C $\text{C}_2\text{H}_5\text{Cl}$ and HCl
D $\text{CH}_2=\text{CH}_2$ and 2HCl

- 36** Ethanol can be made by the fermentation of aqueous glucose and by the catalytic addition of steam to ethene.

What are two advantages of making ethanol by the catalytic addition of steam to ethene rather than by the fermentation of aqueous glucose?

- A** faster reaction and renewable raw materials
- B** purer product and faster reaction
- C** renewable raw materials and continuous process
- D** uses more energy and forms a purer product

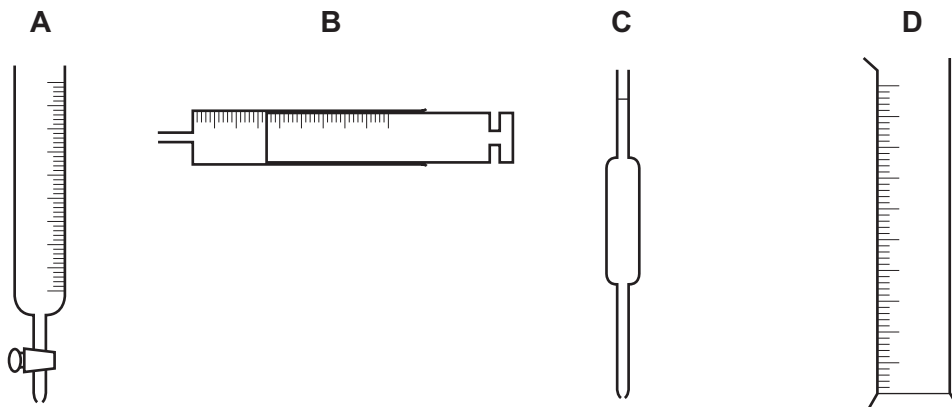
- 37** The structure of a polymer repeat unit is shown.



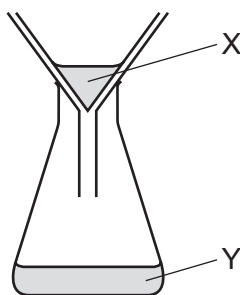
Which pair of monomers is used to make this polymer?

	monomer 1	monomer 2
A	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{N} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$	$\begin{array}{ccccc} & \text{H}-\text{O} & & \text{O}-\text{H} & \\ & & & & \\ & \text{C} & - & \text{C} & \\ & & & & \\ & \text{O} & & \text{O} & \end{array}$
B	$\begin{array}{cccc} \text{H} & \text{H} & \text{H} & \text{H} \\ & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{N} \\ & & & \\ \text{H} & \text{H} & \text{H} & \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H}-\text{O} & & \text{H} & & \text{H} & & \text{O}-\text{H} \\ & & & & & & \\ \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & \\ \text{O} & & \text{H} & & \text{H} & & \text{O} \end{array}$
C	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \\ & & & & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{C}- & \text{C}- & & \text{N} \\ & & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \end{array}$	$\begin{array}{ccccc} \text{H}-\text{O} & & & & \text{O}-\text{H} \\ & & & & \\ \text{C} & - & \text{C} & & \\ & & & & \\ \text{O} & & \text{O} & & \end{array}$
D	$\begin{array}{ccccccc} \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \\ & & & & & & \\ \text{N}- & \text{C}- & \text{C}- & \text{C}- & \text{C}- & & \text{N} \\ & & & & & & \\ \text{H} & \text{H} & \text{H} & \text{H} & \text{H} & & \text{H} \end{array}$	$\begin{array}{ccccccc} \text{H}-\text{O} & & \text{H} & & \text{H} & & \text{O}-\text{H} \\ & & & & & & \\ \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & \\ \text{O} & & \text{H} & & \text{H} & & \text{O} \end{array}$

38 Which item of apparatus is used to measure exactly 26.3 cm^3 of a liquid?



39 A mixture containing an aqueous salt and an insoluble salt is filtered.



Which row describes X and Y?

	X	Y
A	solute	pure water
B	solute	filtrate
C	residue	pure water
D	residue	filtrate

40 Pure ethanol has a melting point of -114°C and a boiling point of 78°C .

What are the melting and boiling points of a sample of ethanol with glucose dissolved in it?

	melting point/ $^\circ\text{C}$	boiling point/ $^\circ\text{C}$
A	-116	77
B	-116	79
C	-112	77
D	-112	79

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1Hhydrogen1																	
		Key																	
		atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —		

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

Paper 2 Multiple Choice (Extended)

0620/23

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.

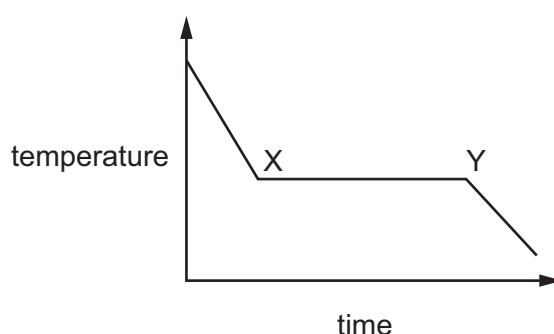


- 1 Which two changes, when applied **at the same time** to a fixed amount of gas, cause the greatest increase in volume of the gas?

	change in pressure	change in temperature
A	decrease by 10%	decrease by 10%
B	increase by 10%	decrease by 10%
C	decrease by 20%	increase by 10%
D	increase by 20%	increase by 10%

- 2 Solid G is heated until it melts. It is then left to cool to room temperature.

The graph shows the cooling curve for G.



Four statements about the cooling curve between X and Y are listed.

- 1 The speed of the molecules of G decreases.
- 2 The molecules of G become less closely packed.
- 3 Thermal energy is released by the molecules of G to the surroundings.
- 4 G changes from solid to liquid.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

- 3 Which atom has twice as many neutrons as protons?

- A** ${}^1_1\text{H}$ **B** ${}^2_1\text{H}$ **C** ${}^3_1\text{H}$ **D** ${}^4_2\text{He}$

- 4 What is the nucleon number of an atom?

- A** the number of electrons, neutrons and protons in the nucleus
B the number of neutrons and protons in the nucleus
C the number of neutrons in the nucleus
D the number of protons in the nucleus

- 5** The diagram shows the positions of elements P and Q in the Periodic Table.

A simplified periodic table grid with 18 columns and 4 rows. The first two columns are on the left, and the last six columns are on the right, with a gap in between. The element 'P' is in the second row, second column of the left block. The element 'Q' is in the second row, fourth column of the right block. There is an empty box above the gap between the two blocks.

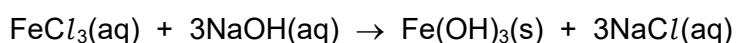
What is the formula of the compound that is formed between P and Q?

- A** PQ **B** P₂Q **C** PQ₂ **D** P₂Q₇

- 6** Which statement about ionic compounds is correct?

- A** Giant lattices are formed when ionic compounds crystallise from solution.
- B** In all ionic compounds, the metal ions have fewer electrons than the non-metal ions.
- C** In a giant lattice, ions of the same charge are closer together than ions of opposite charge.
- D** When ionic compounds melt, strong intermolecular forces are broken.

- 7 The equation for the reaction between iron(III) chloride and sodium hydroxide is shown.



What is the ionic equation for this reaction?

- A** $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{s}) + 3\text{OH}^{-}(\text{s})$
- B** $\text{Fe}^{3+}(\text{aq}) + 3\text{OH}^{-}(\text{aq}) \rightarrow \text{Fe}(\text{OH})_3(\text{s})$
- C** $3\text{Na}^{+}(\text{aq}) + 3\text{Cl}^{-}(\text{aq}) \rightarrow 3\text{Na}^{+}(\text{aq}) + 3\text{Cl}^{-}(\text{aq})$
- D** $3\text{Na}^{+}(\text{aq}) + 3\text{Cl}^{-}(\text{aq}) \rightarrow 3\text{NaCl}(\text{aq})$

- 8** A compound is analysed and found to contain 85.7% carbon and 14.3% hydrogen only.

What is its empirical formula?

- A** CH **B** CH₂ **C** C₂H₄ **D** C₆H

- 9 Aqueous barium hydroxide, $\text{Ba}(\text{OH})_2$, reacts with dilute nitric acid, HNO_3 . The products are aqueous barium nitrate, $\text{Ba}(\text{NO}_3)_2$, and water.

In a titration, 15.2 cm^3 of 0.75 mol/dm^3 aqueous barium hydroxide is required to neutralise 20.0 cm^3 of dilute nitric acid.

What is the concentration of the nitric acid?

- A 0.29 mol/dm^3
 B 0.57 mol/dm^3
 C 1.14 mol/dm^3
 D 2.03 mol/dm^3
- 10 Which particles are responsible for the transfer of charge in the external circuit during the electrolysis of molten lead(II) bromide?
- A anions
 B atoms
 C cations
 D electrons
- 11 Two statements about hydrogen are listed.

- 1 Hydrogen is used as a fuel.
- 2 When hydrogen burns in the air to form water, thermal energy is produced.

Which conclusion about these statements is correct?

- A Both statements are correct, and statement 2 explains statement 1.
 B Both statements are correct, but statement 2 does **not** explain statement 1.
 C Statement 1 is correct, but statement 2 is **not** correct.
 D Statement 2 is correct, but statement 1 is **not** correct.
- 12 Concentrated aqueous copper(II) chloride is electrolysed using carbon electrodes.

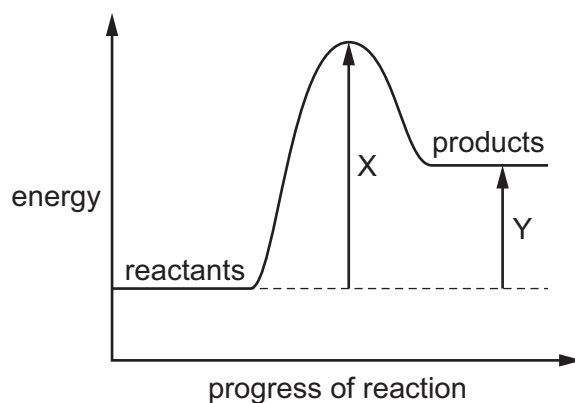
What is formed at each electrode?

	negative electrode	positive electrode
A	chlorine	copper
B	copper	chlorine
C	copper	oxygen
D	oxygen	copper

13 Which row describes an exothermic reaction?

	sign of enthalpy change, ΔH	temperature of the surroundings
A	positive	decreases
B	positive	increases
C	negative	decreases
D	negative	increases

14 A reaction pathway diagram is shown.



Which row shows the correct labels for X and Y and the type of reaction taking place?

	X	Y	type of reaction
A	enthalpy change	activation energy	exothermic
B	activation energy	enthalpy change	endothermic
C	activation energy	enthalpy change	exothermic
D	enthalpy change	activation energy	endothermic

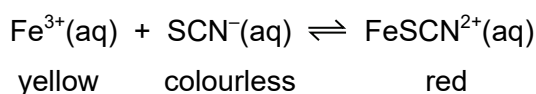
15 The rate of reaction between magnesium and dilute hydrochloric acid is increased by increasing the concentration of the acid.

How does this affect the reacting particles?

	collision rate of particles	proportion of particles with sufficient energy to react
A	increases	increases
B	increases	stays the same
C	stays the same	increases
D	stays the same	stays the same

- 16** The reaction between aqueous iron(III) ions, Fe^{3+} , and aqueous thiocyanate ions, SCN^- , is reversible.

The equation for the reaction is shown.

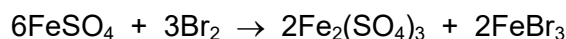


At equilibrium, the concentrations of Fe^{3+} ions, SCN^- ions, and FeSCN^{2+} ions are identical, and the mixture appears orange.

When a few drops of aqueous thiocyanate ions are added to the mixture, a new equilibrium forms.

How does the colour of the mixture change?

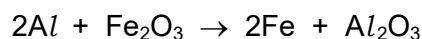
- A** It becomes colourless.
- B** It becomes a lighter orange.
- C** It becomes more yellow.
- D** It becomes more red.
- 17** The equation for the reaction between iron(II) sulfate and bromine is shown.



Which row identifies the oxidising agent and the reducing agent?

	oxidising agent	reducing agent
A	Br_2	FeSO_4
B	FeSO_4	Br_2
C	FeBr_3	$\text{Fe}_2(\text{SO}_4)_3$
D	$\text{Fe}_2(\text{SO}_4)_3$	FeBr_3

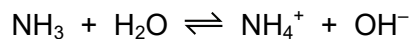
- 18** The equation represents a redox reaction.



What is the change in the oxidation number of iron in this reaction?

- A** from +2 to +3
- B** from +2 to 0
- C** from +3 to +2
- D** from +3 to 0

- 19 Aqueous ammonia forms an equilibrium with water.



In this equilibrium, which particle acts as an acid and which particle acts as a base?

	acid	base
A	NH_3	H_2O
B	NH_3	OH^-
C	H_2O	NH_3
D	H_2O	NH_4^+

- 20 Which row identifies an acidic oxide, a basic oxide and an amphoteric oxide?

	acidic	basic	amphoteric
A	ZnO	SO_2	Al_2O_3
B	SO_2	Al_2O_3	Na_2O
C	ZnO	Al_2O_3	SO_2
D	SO_2	Na_2O	Al_2O_3

- 21 A Group I metal (lithium or potassium) is reacted with a Group VII element (chlorine or iodine).

Which compound is formed when the Group I metal of highest density reacts with the Group VII element of lowest density?

- A** potassium chloride
- B** lithium chloride
- C** potassium iodide
- D** lithium iodide

- 22 Element X is in the same group of the Periodic Table as silicon.

An atom of element X has more protons than a silicon atom.

Which element could be X?

- A** aluminium
- B** carbon
- C** germanium
- D** phosphorus

23 Which statement about the elements in Group VIII is correct?

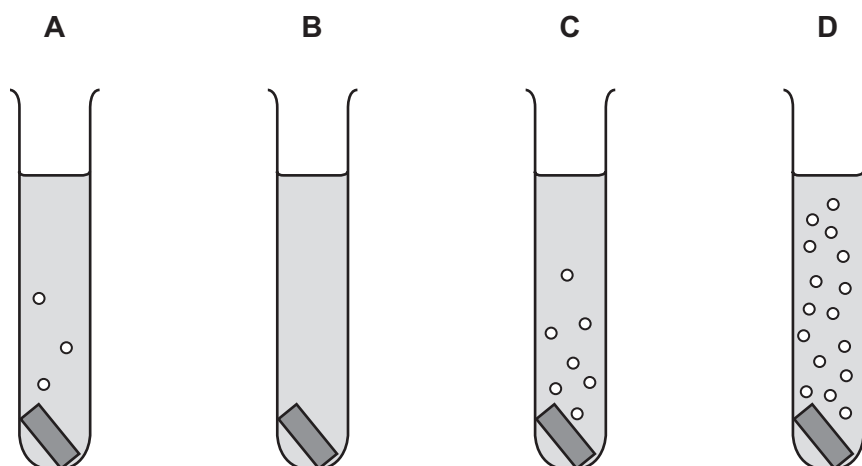
- A** They all form diatomic molecules.
- B** They all have eight electrons in their outer electron shells.
- C** They all react with oxygen to form oxides.
- D** They all are gases at room temperature.

24 Which element is a transition element?

	melting point in °C	density at r.t.p. in g/cm ³	colour of oxide at r.t.p.
A	98	1.0	white
B	328	11.3	yellow
C	651	1.7	white
D	1907	7.2	green

25 Four test-tubes contain dilute hydrochloric acid. Each test-tube has one piece of a different metal added to it. The metals are copper, iron, magnesium and zinc.

Which test-tube contains iron and dilute hydrochloric acid?



26 Brass is a mixture of copper and zinc.

Which statement describes and explains the relative strength of brass compared to copper?

- A** Brass is stronger than copper because the layers of atoms in brass **cannot** easily slide over each other.
- B** Brass is stronger than copper because the intermolecular forces are greater in brass than in copper.
- C** Brass is weaker than copper because the layers of atoms slide more easily over each other in brass than in copper.
- D** Brass is weaker than copper because the intermolecular forces are weaker in brass than in copper.

27 A sample of calcium carbonate is heated strongly, and substances U and V are formed.

Substance U is a white solid. Substance V is a colourless gas.

What are substances U and V?

	U	V
A	calcium chloride	oxygen
B	calcium hydroxide	carbon dioxide
C	calcium oxide	carbon dioxide
D	calcium sulfate	oxygen

28 A sample of water taken from a river contains harmful microbes.

What is a source of harmful microbes in the river water?

- A** sewage
- B** fertilisers
- C** detergents
- D** plastics

- 29 Anhydrous copper(II) sulfate and anhydrous cobalt(II) chloride change colour when water is added to them.

Which row shows the correct colour changes?

	copper(II) sulfate	cobalt(II) chloride
A	blue to white	blue to pink
B	blue to white	pink to blue
C	white to blue	blue to pink
D	white to blue	pink to blue

- 30 Which gas contributes to increased global warming?

- A** argon
- B** carbon dioxide
- C** nitrogen
- D** oxygen

- 31 Which statement describes how greenhouse gases increase global warming?

- A** Greenhouse gases absorb thermal energy from the Sun, reducing the Earth's temperature.
- B** Greenhouse gases reflect thermal energy back into space, cooling the Earth's surface.
- C** Greenhouse gases absorb and re-emit thermal energy radiated from the Earth's surface.
- D** Greenhouse gases release thermal energy from the Earth into space.

- 32 Nitrogen monoxide is produced in a car engine when petrol is burned.

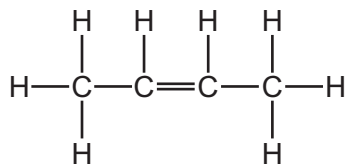
The gases from the car engine are passed through a catalytic converter.

In the catalytic converter, the nitrogen monoxide, NO, reacts with carbon monoxide, CO, to form nitrogen and carbon dioxide.

Which statement is **not** correct?

- A** Carbon monoxide is oxidised in the catalytic converter.
- B** Carbon monoxide is produced by the complete combustion of petrol.
- C** Nitrogen monoxide is formed by the reaction of nitrogen and oxygen.
- D** Nitrogen monoxide is reduced in the catalytic converter.

33 The displayed formula of but-2-ene is shown.



Which displayed formulae represent structural isomers of but-2-ene?

$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ \text{H} & - \text{C} & = & \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & & & & & & & \\ & & & & & \text{H} & & \text{H} \end{array} $	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \\ & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \\ & & & & & & & \\ & & & \text{H} & & & & \end{array} $	$ \begin{array}{ccccccc} & & & \text{H} & & & & \\ & & & & & & & \\ \text{H} & & \text{H} & - \text{C} & - & \text{H} \\ & & & & & & & \\ & & & \text{C} & = & \text{C} & & \\ & & & & & & & \\ & & & \text{H} & & \text{H} & & \\ & & & & & & & \\ & & & & & \text{H} & & \end{array} $
1	2	3

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

34 Which row describes two characteristics of members of the same homologous series?

	characteristic 1	characteristic 2
A	same functional group	trend in physical properties
B	same physical properties	same functional group
C	same general formula	different chemical properties
D	differing from one member to the next by a CH ₃ unit	similar chemical properties

35 Compound W is a hydrocarbon that contains single bonds only.

Which homologous series contains W?

- A** alkenes
B alkanes
C carboxylic acids
D alcohols

36 Methane forms chloromethane in a photochemical reaction using ultraviolet light.

Which statement about this reaction is correct?

- A** Methane is mixed with hydrogen chloride gas, and ultraviolet light prevents a combustion reaction.
- B** Methane is mixed with hydrogen chloride gas, and ultraviolet light provides the activation energy for the reaction.
- C** Methane is mixed with chlorine gas, and ultraviolet light prevents a combustion reaction.
- D** Methane is mixed with chlorine gas, and ultraviolet light provides the activation energy for the reaction.

37 Which statement about polymers is correct?

- A** Nylon contains the $\begin{array}{c} \text{O} \\ \parallel \\ \text{---C---N---} \\ | \\ \text{H} \end{array}$ linkage.
- B** Nylon is a polyester.
- C** Propane can be polymerised by addition polymerisation.
- D** The linkage in PET contains a carbon–carbon double bond.

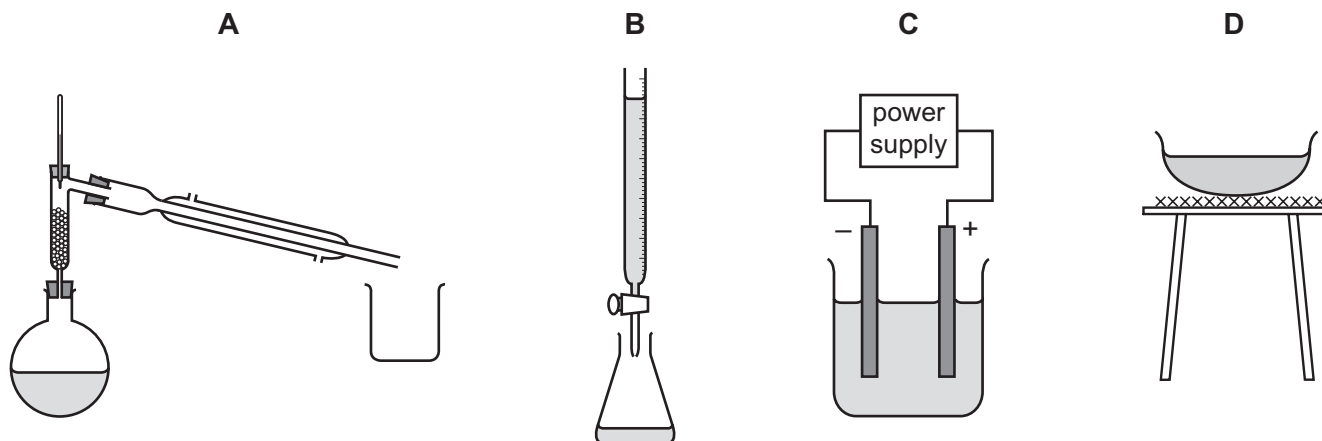
38 The steps used to separate a mixture of a soluble solid and an insoluble solid are listed.

- A solution is made by adding the mixture to a suitable1..... .
- The mixture is stirred and then poured through a filter funnel lined with filter paper.
- The insoluble solid is collected on the filter paper as the2..... .

Which words complete gaps 1 and 2?

	1	2
A	solute	filtrate
B	solute	residue
C	solvent	filtrate
D	solvent	residue

39 Which items of apparatus are used to separate a mixture of liquids with different boiling points?



40 In paper chromatography, what is the equation for the R_f value?

- A $R_f = \frac{\text{distance travelled by solvent}}{\text{distance travelled by substance}}$
- B $R_f = \frac{\text{distance travelled by locating agent}}{\text{distance travelled by substance}}$
- C $R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by locating agent}}$
- D $R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Decane has a freezing point of -30°C and a boiling point of 174°C .

A small sample of decane is placed in an open beaker in an oven at a temperature of 120°C and at atmospheric pressure for 24 hours.

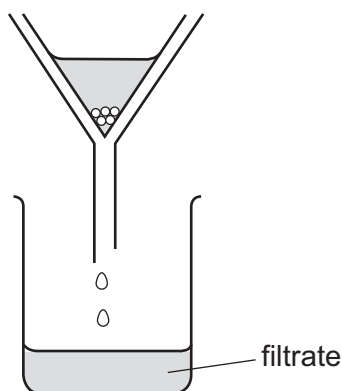
What happens to the sample of decane?

- A It boils.
 - B It evaporates.
 - C It melts.
 - D It sublimates.
- 2 A student put exactly 25.00 cm^3 of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A balance, measuring cylinder, thermometer
 - B balance, pipette, stopwatch
 - C balance, pipette, thermometer
 - D burette, pipette, thermometer
- 3 A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.



What is the filtrate?

- A broken glass only
- B broken glass and sugar solution
- C pure water
- D sugar solution

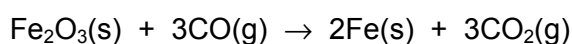
4 Which statement explains why metals conduct electricity when solid?

- A They have atoms which are free to move.
- B They have electrons which are free to move.
- C They have molecules which are free to move.
- D They have positive ions which are free to move.

5 Which description of brass is correct?

- A alloy
- B compound
- C element
- D non-metal

6 The equation for the reaction of iron(III) oxide with carbon monoxide is shown.



What is the maximum mass of iron that can be made from 480 g of iron(III) oxide?

- A 56 g B 112 g C 168 g D 336 g

7 Which statement describes the attractive forces between molecules?

- A They are strong covalent bonds which hold molecules together.
- B They are strong ionic bonds which hold molecules together.
- C They are weak forces formed between covalently-bonded molecules.
- D They are weak forces which hold ions together in a lattice.

8 Which statement about carbon is correct?

- A Diamond and graphite both have simple molecular structures.
- B Diamond and graphite are both used to make cutting tools.
- C Each carbon atom in diamond is bonded to three other carbon atoms.
- D Graphite conducts electricity and has a giant covalent structure.

9 The formula of an aluminium ion is Al^{3+} .

What is the formula of aluminium sulfate?

- A Al_2SO_4 B $\text{Al}(\text{SO}_4)_2$ C $\text{Al}_2(\text{SO}_4)_3$ D $\text{Al}_3(\text{SO}_4)_2$

10 Which statements about the products of electrolysis, using inert electrodes, are correct?

- 1 When molten lead(II) bromide is electrolysed, bromine is formed at the cathode.
- 2 When dilute sulfuric acid is electrolysed, oxygen is formed at the anode.
- 3 When concentrated aqueous sodium chloride is electrolysed, sodium is formed at the cathode.
- 4 When concentrated hydrochloric acid is electrolysed, chlorine is formed at the anode.

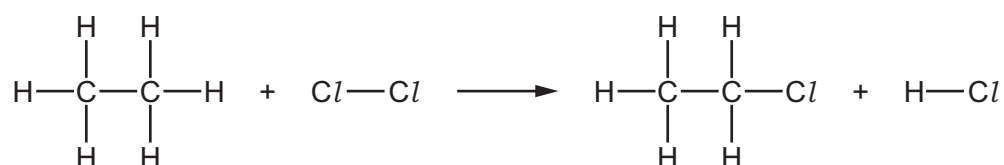
A 1 and 2

B 1 and 3

C 2 and 4

D 3 and 4

11 Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

The bond energies are shown in the table.

bond	bond energy in kJ/mol
C–Cl	+340
C–C	+350
C–H	+410
Cl–Cl	+240
H–Cl	+430

What is the energy change for the reaction?

A –1420 kJ/mol

B –120 kJ/mol

C +120 kJ/mol

D +1420 kJ/mol

12 Hydrogen is used as a fuel in rockets and is also used in hydrogen fuel cells.

Which statements are correct?

- 1 Both uses produce water vapour.
- 2 Burning hydrogen produces polluting gases.
- 3 A fuel cell produces electricity.

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

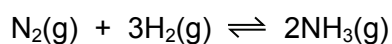
13 Which statements about the effect of increasing the temperature on the rate of a reaction are correct?

- 1 It increases the rate of a reaction.
- 2 It increases the activation energy.
- 3 It increases the frequency of collisions.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

14 Ammonia is made by reacting nitrogen with hydrogen.

The equation for the reaction is shown.



The forward reaction is exothermic.

Which changes in temperature and pressure decrease the yield of ammonia?

	temperature	pressure
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

15 X is a pink solid.

Y is a blue solid.

When X is heated, water is produced and the solid turns blue.

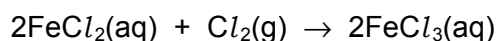
When water is added to Y, the solid turns pink.

What are X and Y?

	X	Y
A	anhydrous cobalt(II) chloride	hydrated cobalt(II) chloride
B	hydrated cobalt(II) chloride	anhydrous cobalt(II) chloride
C	anhydrous copper(II) sulfate	hydrated copper(II) sulfate
D	hydrated copper(II) sulfate	anhydrous copper(II) sulfate

16 Iron(II) chloride solution reacts with chlorine gas.

The equation is shown.



Which statements about this reaction are correct?

- 1 Fe^{2+} ions are reduced to Fe^{3+} ions.
- 2 Chlorine acts as a reducing agent.
- 3 Fe^{2+} ions each lose an electron.
- 4 Cl_2 molecules are reduced to Cl^- ions.

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

17 Which row describes the properties of an acid?

	property 1	property 2
A	proton acceptor	pH less than 7
B	proton acceptor	pH more than 7
C	proton donor	pH less than 7
D	proton donor	pH more than 7

18 Which element forms an amphoteric oxide?

- A aluminium
- B carbon
- C magnesium
- D silicon

19 Copper(II) chloride crystals are made by adding solid copper(II) carbonate to dilute hydrochloric acid until no more dissolves.

Which process is used to obtain pure copper(II) chloride crystals from the mixture?

- A distillation of the mixture
- B evaporation of the mixture
- C filtration followed by drying of the residue
- D filtration followed by evaporation of the filtrate

20 Moving from right to left across the Periodic Table the elements show increasing metallic character.

Why does metallic character increase from right to left across a period?

- A The atoms have more electrons in their outer shells.
- B The atoms more readily gain electrons to form negative ions.
- C The atoms more readily lose electrons to form positive ions.
- D The charge on the nucleus of each atom gets larger.

21 A period of the Periodic Table is shown.

group	I	II	III	IV	V	VI	VII	VIII
element	R	S	T	V	W	X	Y	Z

The letters are not their chemical symbols.

Which statement is correct?

- A Element R does not conduct electricity.
- B Elements R and Y react together to form an ionic compound.
- C Element Z exists as a diatomic molecule.
- D Element Z reacts with element T.

22 Group VII elements show trends in their physical properties going down the group.

element	X	Y	Z
chlorine	-101	-34	0.003
bromine	-7	59	3.1
iodine	114	184	4.9

Which row shows the missing headings for the properties in the table?

	X	Y	Z
A	density in g/cm ³	boiling point in °C	melting point in °C
B	melting point in °C	boiling point in °C	density in g/cm ³
C	boiling point in °C	density in g/cm ³	melting point in °C
D	boiling point in °C	melting point in °C	density in g/cm ³

23 Some properties of two metals, G and H, are shown.

metal G	metal H
the formula of the chloride is GCl reacts with cold water	high melting point has more than one oxidation state

Which row about metals G and H is correct?

	metal G	metal H
A	in Group I of the Periodic Table	in Group II of the Periodic Table
B	in Group I of the Periodic Table	transition metal
C	in Group II of the Periodic Table	in Group I of the Periodic Table
D	in Group II of the Periodic Table	transition metal

24 The noble gases are in Group VIII of the Periodic Table.

Which statement explains why noble gases are unreactive?

- A** They all have eight electrons in their outer shells.
- B** They all have full outer shells.
- C** They are all gases.
- D** They are all monoatomic.

25 Which statement is correct for **all** metals?

- A** They conduct electricity when molten.
- B** They gain electrons when they form ions.
- C** They have a low density.
- D** They have a low melting point.

26 Which statement about the extraction of metals is correct?

- A** Aluminium is extracted from the ore bauxite by electrolysis.
- B** Aluminium is extracted from the ore hematite by electrolysis.
- C** Iron is extracted from the ore bauxite by electrolysis.
- D** Iron is extracted from the ore hematite by electrolysis.

27 Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A** Aluminium does not form ions easily.
- B** Aluminium does not react with water or air.
- C** Aluminium has a protective oxide layer.
- D** Aluminium is below iron in the reactivity series.

28 Which statements about the thermal decomposition of copper(II) nitrate are correct?

- 1 A brown gas is given off.
- 2 A gas which relights a glowing splint is given off.
- 3 The solid residue is an acidic oxide.

- A** 1 only **B** 1 and 2 **C** 1 and 3 **D** 2 and 3

- 29** Covering iron with zinc prevents the iron from rusting even when the zinc is scratched.

Covering iron with tin prevents the iron from rusting, but when the tin is scratched the iron underneath starts to rust.

Which statement is correct?

- A** Both tin and zinc prevent iron from rusting by sacrificial protection.
- B** Both tin and zinc prevent iron from rusting by stopping water and carbon dioxide reaching the iron.
- C** Tin is more reactive than iron and prevents iron from rusting until it is scratched.
- D** Zinc loses electrons more easily than iron and prevents iron from rusting by corroding first.

- 30** Which statements about the Haber process are correct?

- 1 One of the raw materials is extracted from liquid air by fractional distillation.
- 2 One of the raw materials is produced by the reaction of steam and methane.
- 3 The catalyst for the Haber process is vanadium(V) oxide.

- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

- 31** Which raw material is used in the Contact process?

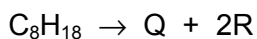
- A** air
- B** ammonia
- C** carbon
- D** nitrogen

- 32** Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A** ammonia
- B** sodium chloride
- C** sodium hydroxide
- D** sulfuric acid

- 33** An alkane molecule of molecular formula C_8H_{18} undergoes cracking. The equation for the reaction is shown.



Substance R has two carbon atoms per molecule and decolourises aqueous bromine.

What is substance Q?

- A** butane
 - B** butene
 - C** ethane
 - D** ethene
- 34** Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.

What could X and Y be?

	X	Y
A	C	H_2
B	C	C_8H_{18}
C	CH_4	H_2
D	CH_4	C_8H_{18}

- 35** Which molecule contains only single covalent bonds?

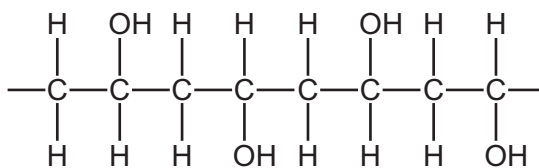
- A** propane
- B** propanoic acid
- C** propene
- D** propyl propanoate

- 36** Alkanes react with chlorine to form chloroalkanes.

Which statement about the reactions of alkanes with chlorine is correct?

- A** Alkanes react with chlorine by addition.
- B** The gaseous product turns red litmus blue.
- C** The chlorine atom in chloroethane is covalently bonded.
- D** The general formula of the chloroalkanes is $C_nH_{2n}Cl$.

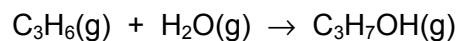
37 Part of the structure of a very large molecule is shown.



Which term describes the small unit used to make this molecule?

- A hydrocarbon
- B monomer
- C polymer
- D saturated

38 Propene reacts with steam to form propanol.



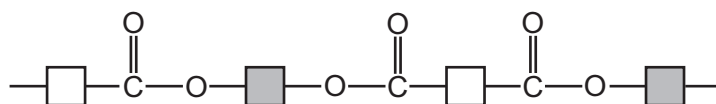
Which type of reaction takes place?

- A addition
- B condensation
- C oxidation
- D substitution

39 Which statement about aqueous ethanoic acid is correct?

- A It reacts with magnesium to produce a salt and hydrogen.
- B It reacts with sodium hydroxide to produce a salt and hydrogen.
- C It reacts with ammonium salts to produce ammonia.
- D It turns red litmus blue.

40 The diagram shows the partial structure of *Terylene*.



From which pair of compounds is it made?

- A** $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{OH}$
- B** $\text{HO}-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- C** $\text{HO}-\text{[white square]}-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- D** $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
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87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —			

lanthanoids

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



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0620/22

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October/November 2021

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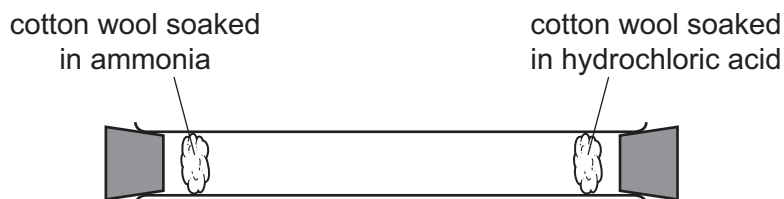
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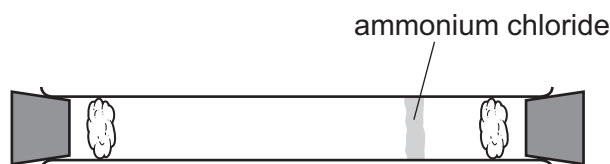
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- 1 An experiment is set up as shown.



After several minutes, a white ring of ammonium chloride appears as shown.



Which statement explains the observation after several minutes?

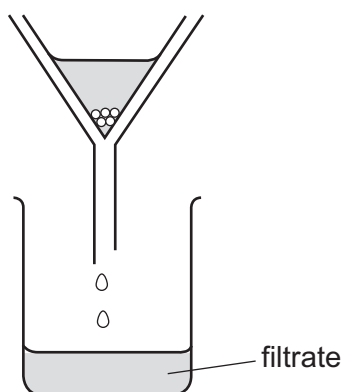
- A Ammonia gas diffuses faster than hydrogen chloride gas because its molecules have a lower molecular mass.
 - B Ammonia gas diffuses faster than hydrogen chloride gas because its molecules have a higher molecular mass.
 - C Ammonia gas diffuses slower than hydrogen chloride gas because its molecules have a lower molecular mass.
 - D Ammonia gas diffuses slower than hydrogen chloride gas because its molecules have a higher molecular mass.
- 2 A student put exactly 25.00 cm^3 of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A balance, measuring cylinder, thermometer
- B balance, pipette, stopwatch
- C balance, pipette, thermometer
- D burette, pipette, thermometer

- 3 A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.



What is the filtrate?

- A broken glass only
 - B broken glass and sugar solution
 - C pure water
 - D sugar solution
- 4 How many protons, neutrons and electrons are there in one atom of the isotope ${}_{13}^{27}\text{Al}$?

	protons	neutrons	electrons
A	13	13	13
B	13	14	13
C	14	13	13
D	14	14	13

- 5 Which description of brass is correct?

- A alloy
- B compound
- C element
- D non-metal

6 Some properties of diamond are shown.

- 1 It is very hard.
- 2 Every atom forms four bonds.
- 3 It does not conduct electricity.

Which properties are also shown by silicon(IV) oxide?

- A** 1 only **B** 1 and 2 **C** 1 and 3 **D** 2 and 3

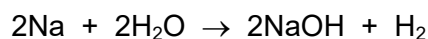
7 Which statement describes the attractive forces between molecules?

- A** They are strong covalent bonds which hold molecules together.
B They are strong ionic bonds which hold molecules together.
C They are weak forces formed between covalently-bonded molecules.
D They are weak forces which hold ions together in a lattice.

8 Which substance is described as a macromolecule?

- A** ammonia
B graphite
C iron
D sodium chloride

9 The equation for the reaction of sodium with water is shown.



What is the volume of hydrogen gas, measured at r.t.p., produced when 18.4 g of sodium reacts with excess water?

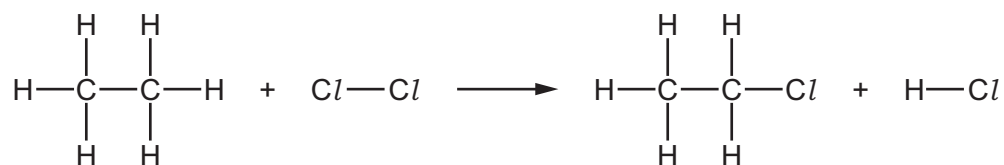
- A** 9.6 dm³ **B** 15.0 dm³ **C** 19.2 dm³ **D** 30.0 dm³

10 Iron can be electroplated with zinc to make it resistant to corrosion.

Which row about electroplating iron with zinc is correct?

	positive electrode (anode)	negative electrode (cathode)	electrolyte
A	iron	zinc	iron nitrate
B	iron	zinc	zinc nitrate
C	zinc	iron	iron nitrate
D	zinc	iron	zinc nitrate

- 11 Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

The bond energies are shown in the table.

bond	bond energy in kJ/mol
C-Cl	+340
C-C	+350
C-H	+410
Cl-Cl	+240
H-Cl	+430

What is the energy change for the reaction?

- A -1420 kJ/mol
 B -120 kJ/mol
 C +120 kJ/mol
 D +1420 kJ/mol
- 12 Chlorine gas is bubbled into aqueous potassium iodide.

What is the ionic equation for the reaction that takes place?

- A $\text{Cl} + \text{I}^- \rightarrow \text{Cl}^- + \text{I}$
 B $\text{Cl}_2 + 2\text{I}^- \rightarrow \text{Cl}_2^- + \text{I}_2$
 C $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$
 D $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + 2\text{I}$

- 13 Concentrated aqueous sodium chloride is electrolysed.

Which equation represents the reaction at the cathode?

- A $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
 B $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
 C $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
 D $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

14 Which statements about hydrogen are correct?

- 1 When hydrogen is burned, heat energy is released.
- 2 When hydrogen is used in a fuel cell, electrical energy is generated.
- 3 When hydrogen is used as a fuel, water is the only product.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 3 only

15 Solid X is heated strongly.

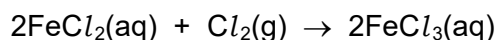
The colour of the solid changes from blue to white.

What is solid X?

- A** anhydrous cobalt(II) chloride
B calcium carbonate
C hydrated copper(II) sulfate
D lead(II) bromide

16 Iron(II) chloride solution reacts with chlorine gas.

The equation is shown.



Which statements about this reaction are correct?

- 1 Fe^{2+} ions are reduced to Fe^{3+} ions.
- 2 Chlorine acts as a reducing agent.
- 3 Fe^{2+} ions each lose an electron.
- 4 Cl_2 molecules are reduced to Cl^- ions.

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

17 Which statements about acids and bases are correct?

- 1 An acid reacts with a metal to give off hydrogen.
- 2 A base reacts with an ammonium salt to give off ammonia.
- 3 An acid reacts with a carbonate to give off carbon dioxide.
- 4 Alkaline solutions are orange in methyl orange.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

18 Oxide 1 is a solid that reacts with dilute hydrochloric acid.

Oxide 2 is a gas that reacts with sodium hydroxide solution.

What are the formulae of the oxides?

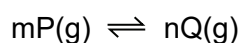
	oxide 1	oxide 2
A	CaO	MgO
B	MgO	NO ₂
C	NO ₂	SO ₂
D	SO ₂	CaO

19 Which reaction is a photochemical reaction?

- A** addition of bromine to propene
- B** esterification of ethanol and ethanoic acid
- C** oxidation of ethanol
- D** substitution of methane with chlorine

20 The equation shown represents a reaction at equilibrium.

m and n represent the balancing numbers for the reactant and product respectively.



A high temperature increases the concentration of Q.

A high pressure increases the concentration of Q.

Which statement about the reaction is correct?

- A** The forward reaction is exothermic and m is greater than n.
- B** The forward reaction is exothermic and m is less than n.
- C** The forward reaction is endothermic and m is greater than n.
- D** The forward reaction is endothermic and m is less than n.

21 A period of the Periodic Table is shown.

group	I	II	III	IV	V	VI	VII	VIII
element	R	S	T	V	W	X	Y	Z

The letters are not their chemical symbols.

Which statement is correct?

- A** Element R does not conduct electricity.
- B** Elements R and Y react together to form an ionic compound.
- C** Element Z exists as a diatomic molecule.
- D** Element Z reacts with element T.

22 All metal nitrates are soluble in water.

All metal chlorides are soluble except silver and lead.

All metal carbonates are insoluble except sodium and potassium.

Which aqueous solutions produce a precipitate when mixed together?

- 1 silver nitrate + sodium carbonate
- 2 silver nitrate + sodium chloride
- 3 barium nitrate + potassium chloride

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

23 Which row describes properties of transition elements?

	property 1	property 2	property 3
A	coloured compounds	high density	variable oxidation states
B	high density	high melting point	one oxidation state
C	high melting point	coloured compounds	one oxidation state
D	low melting point	high density	variable oxidation states

24 The noble gases are in Group VIII of the Periodic Table.

Which statement explains why noble gases are unreactive?

- A** They all have eight electrons in their outer shells.
- B** They all have full outer shells.
- C** They are all gases.
- D** They are all monoatomic.

25 Which statement is correct for **all** metals?

- A** They conduct electricity when molten.
- B** They gain electrons when they form ions.
- C** They have a low density.
- D** They have a low melting point.

26 Carbon dioxide is produced during the extraction of aluminium from bauxite.

Which statement describes how this carbon dioxide is made?

- A** Carbon monoxide reduces aluminium oxide forming carbon dioxide and aluminium.
- B** Carbon is burned in the blast furnace to release heat energy.
- C** Oxygen made in the process reacts with the carbon electrode.
- D** The ore of aluminium undergoes thermal decomposition.

27 Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A** Aluminium does not form ions easily.
- B** Aluminium does not react with water or air.
- C** Aluminium has a protective oxide layer.
- D** Aluminium is below iron in the reactivity series.

28 Which statements explain why zinc is used to protect iron from rusting?

- 1 Zinc is more reactive than iron.
- 2 Zinc is less reactive than iron.
- 3 Zinc can form alloys with iron.
- 4 Zinc acts as a sacrificial metal.

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

29 Which conditions are used in the Haber process?

	temperature / °C	pressure / atmospheres
A	100	10
B	450	10
C	450	200
D	1000	500

30 Which process does **not** produce a greenhouse gas?

- A** acid rain on limestone buildings
- B** combustion of wood
- C** digestion in cows
- D** zinc reacting with sulfuric acid

31 Which reaction involving sulfur dioxide is correct?

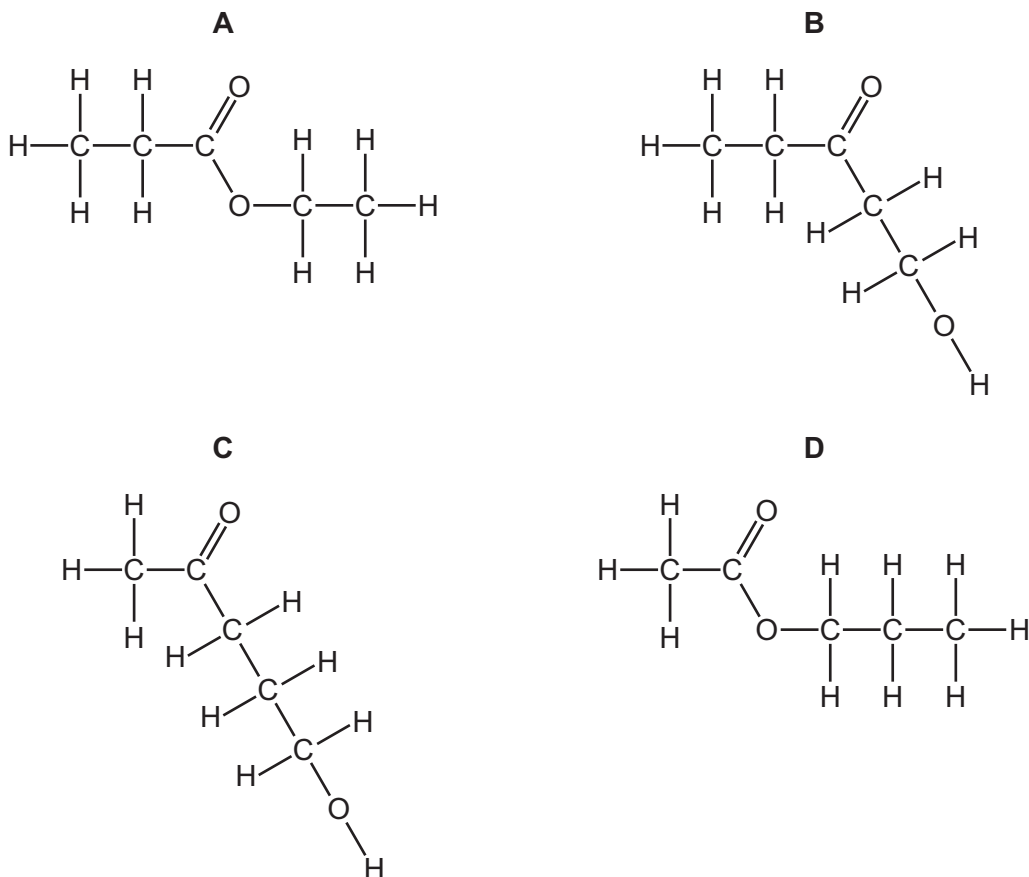
- A** It is produced during the extraction of zinc from zinc blende.
- B** It reacts with concentrated sulfuric acid to form oleum.
- C** It reacts with sulfur to form sulfur trioxide.
- D** It turns an acidified solution of potassium manganate(VII) purple.

32 Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A** ammonia
- B** sodium chloride
- C** sodium hydroxide
- D** sulfuric acid

33 What is the structure of the ester formed from ethanoic acid and propanol?

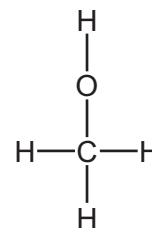
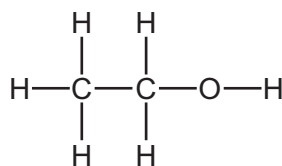
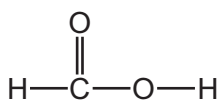
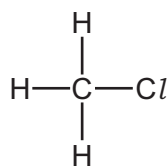


34 Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.

What could X and Y be?

	X	Y
A	C	H ₂
B	C	C ₈ H ₁₈
C	CH ₄	H ₂
D	CH ₄	C ₈ H ₁₈

35 The structures of four organic molecules are shown.



How many different homologous series are represented by these molecules?

- A** 1 **B** 2 **C** 3 **D** 4

36 Which statement about ethene is correct?

- A** It has the chemical formula C_2H_6 .
B It burns in excess oxygen producing carbon dioxide and water.
C It reacts with Br_2 to produce an orange solution.
D It reacts with oxygen to form ethanol.

37 Ethanol is manufactured by fermentation of sugars or by catalytic hydration of ethene.

Which row states an advantage of each method?

	fermentation	hydration
A	produces purer ethanol	is a batch process
B	produces purer ethanol	is a continuous process
C	uses a renewable resource	is a batch process
D	uses a renewable resource	is a continuous process

38 Which statements about unsaturated hydrocarbons are correct?

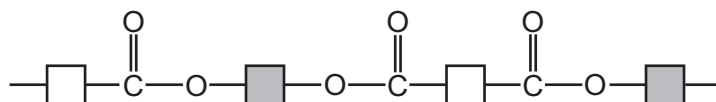
- 1 They contain both single and double bonds.
- 2 They turn aqueous bromine from colourless to brown.
- 3 They can be manufactured by cracking.

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

39 Which polymers have the same linkage between monomer units?

- A carbohydrate and polyamide
- B carbohydrate and polyester
- C protein and polyamide
- D protein and polyester

40 The diagram shows the partial structure of *Terylene*.



From which pair of compounds is it made?

- A $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{OH}$
- B $\text{HO}-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- C $\text{HO}-\text{[white square]}-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- D $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —		

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/23

Paper 2 Multiple Choice (Extended)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 Brownian motion and the diffusion of gases provide evidence for the particulate nature of matter.

Which row identifies an example of Brownian motion and how molecular mass determines the rate of diffusion of gas molecules?

	Brownian motion	diffusion
A	pollen grains in water are seen to move randomly	heavier gas molecules diffuse more quickly
B	pollen grains in water are seen to move randomly	lighter gas molecules diffuse more quickly
C	salt dissolves faster in hot water than in cold water	heavier gas molecules diffuse more quickly
D	salt dissolves faster in hot water than in cold water	lighter gas molecules diffuse more quickly

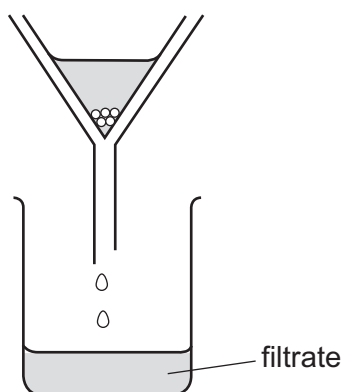
- 2 A student put exactly 25.00 cm^3 of dilute hydrochloric acid into a conical flask.

The student added 2.5 g of solid sodium carbonate and measured the change in temperature of the mixture.

Which apparatus does the student need to use?

- A** balance, measuring cylinder, thermometer
- B** balance, pipette, stopwatch
- C** balance, pipette, thermometer
- D** burette, pipette, thermometer

- 3 A student separates sugar from pieces of broken glass by dissolving the sugar in water and filtering off the broken glass.



What is the filtrate?

- A broken glass only
 - B broken glass and sugar solution
 - C pure water
 - D sugar solution
- 4 The nucleus of a particular atom consists of nineteen particles.
- Nine of them are positively charged and ten of them are uncharged.
- Which statement about this nucleus is correct?
- A The nucleus has a nucleon number of nine.
 - B The nucleus has a nucleon number of ten.
 - C The nucleus has a proton number of nine.
 - D The nucleus has a proton number of ten.
- 5 Which description of brass is correct?
- A alloy
 - B compound
 - C element
 - D non-metal

- 6 A Group I element combines with a Group VII element and forms an ionic bond.

Which row shows how the electronic structures change?

	Group I element		Group VII element	
	before bonding	after bonding	before bonding	after bonding
A	2,8,1	2,8,2	2,7	2,6
B	2,8	2,7	2,8	2,8,1
C	2,8,1	2,8	2,7	2,8
D	2,8	2,8,1	2,8	2,7

- 7 Which statement describes the attractive forces between molecules?

- A** They are strong covalent bonds which hold molecules together.
B They are strong ionic bonds which hold molecules together.
C They are weak forces formed between covalently-bonded molecules.
D They are weak forces which hold ions together in a lattice.

- 8 Which diagram shows the outer electron arrangement in a molecule of carbon dioxide?



- 9 Aluminium oxide is an ionic compound containing Al^{3+} ions and O^{2-} ions.

Aluminium hydroxide is an ionic compound containing Al^{3+} ions and OH^{-} ions.

In which row are the formulae for aluminium oxide and aluminium hydroxide correct?

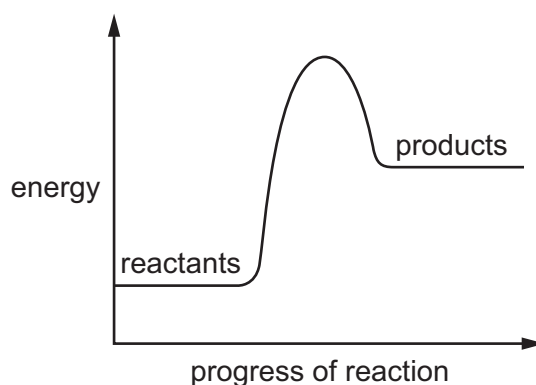
	aluminium oxide	aluminium hydroxide
A	Al_2O_3	$Al(OH)_3$
B	Al_3O_2	$AlOH_3$
C	Al_2O_3	$AlOH_3$
D	Al_3O_2	$Al(OH)_3$

- 10 Effervescence is observed at the negative electrode (cathode) during the electrolysis of concentrated aqueous sodium chloride.

Which element is produced at the negative electrode (cathode)?

- A chlorine
- B hydrogen
- C oxygen
- D sodium

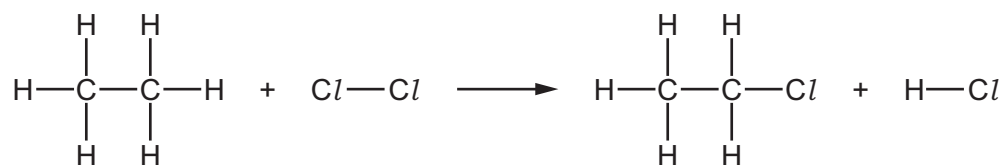
- 11 The energy level diagram for a chemical reaction is shown.



Which statement about this reaction is correct?

- A The reaction is endothermic and energy is given out to the surroundings.
- B The reaction is endothermic and energy is taken in from the surroundings.
- C The reaction is exothermic and energy is given out to the surroundings.
- D The reaction is exothermic and energy is taken in from the surroundings.

- 12 Chlorine reacts with ethane to produce chloroethane and hydrogen chloride.



The reaction is exothermic.

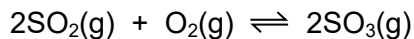
The bond energies are shown in the table.

bond	bond energy in kJ/mol
C-Cl	+340
C-C	+350
C-H	+410
Cl-Cl	+240
H-Cl	+430

What is the energy change for the reaction?

- A** -1420 kJ/mol
- B** -120 kJ/mol
- C** +120 kJ/mol
- D** +1420 kJ/mol
- 13 What is the concentration of the solution when 31.8 g of sodium carbonate, Na_2CO_3 , is dissolved in water to make a solution of 250 cm^3 ?
- A** 0.075 mol/dm^3
- B** 0.30 mol/dm^3
- C** 1.2 mol/dm^3
- D** 1.5 mol/dm^3
- 14 A fuel cell is used to generate electricity.
- Which chemicals are used in a fuel cell?
- A** hydrogen and methane
- B** hydrogen and oxygen
- C** nitrogen and methane
- D** nitrogen and oxygen

- 15** Sulfuric acid is manufactured using the Contact process. One of the reactions is shown.



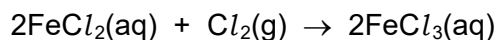
The forward reaction is exothermic.

- statement 1 The equation has more molecules on the left-hand side than on the right-hand side.
- statement 2 Using a higher pressure shifts the equilibrium to the left.
- statement 3 Higher temperatures increase the rate of reaction.
- statement 4 Increasing the temperature shifts the equilibrium to the right.

Which alternative is correct?

- A** Statement 1 is correct and explains statement 2.
- B** Statement 1 and statement 3 are correct.
- C** Statement 2 and statement 4 are correct.
- D** Statement 3 is correct and explains statement 4.
- 16** Iron(II) chloride solution reacts with chlorine gas.

The equation is shown.



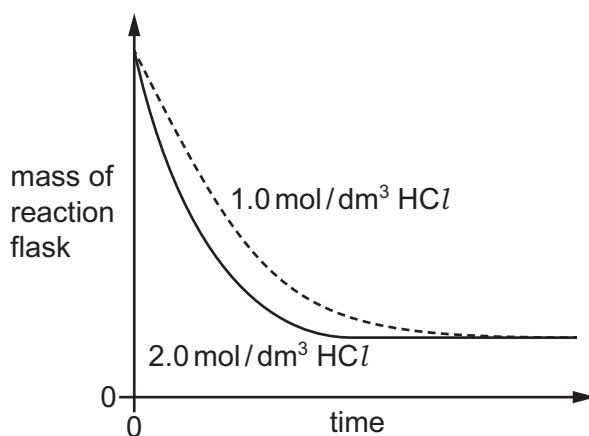
Which statements about this reaction are correct?

- 1 Fe^{2+} ions are reduced to Fe^{3+} ions.
- 2 Chlorine acts as a reducing agent.
- 3 Fe^{2+} ions each lose an electron.
- 4 Cl_2 molecules are reduced to Cl^- ions.
- A** 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

- 17** Excess dilute hydrochloric acid is added to equal masses of powdered calcium carbonate in two separate experiments.

Two different concentrations of hydrochloric acid are used. The temperature in both experiments is the same.

The results show the change in mass of the reaction flask measured over time.



Why is the rate of reaction for the 1.0 mol/dm^3 hydrochloric acid slower?

	collision energy	collision rate
A	lower	higher
B	lower	lower
C	same as for 2.0 mol/dm^3	higher
D	same as for 2.0 mol/dm^3	lower

- 18** Basic oxides are neutralised by acidic oxides.

Which element forms an oxide that neutralises calcium oxide?

- A** hydrogen
- B** magnesium
- C** sodium
- D** sulfur

19 Four solid oxides are added to dilute hydrochloric acid and aqueous sodium hydroxide.

Which row describes an amphoteric oxide?

	hydrochloric acid	sodium hydroxide
A	✓	✓
B	x	✓
C	✓	x
D	x	x

key

✓ = reacts

x = does not react

20 Which row describes an acid and an oxidising agent?

	acid	oxidising agent
A	proton acceptor	electron acceptor
B	proton acceptor	electron donor
C	proton donor	electron acceptor
D	proton donor	electron donor

21 A period of the Periodic Table is shown.

group	I	II	III	IV	V	VI	VII	VIII
element	R	S	T	V	W	X	Y	Z

The letters are not their chemical symbols.

Which statement is correct?

- A** Element R does not conduct electricity.
- B** Elements R and Y react together to form an ionic compound.
- C** Element Z exists as a diatomic molecule.
- D** Element Z reacts with element T.

26 Chromium is a more reactive metal than iron but less reactive than zinc.

Which statements are correct?

- 1 Chromium does not react with dilute hydrochloric acid.
- 2 Chromium oxide is reduced when it is heated with carbon.
- 3 Chromium reacts with zinc oxide to form zinc.
- 4 Chromium reacts with steam to form hydrogen gas.

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

27 Aluminium objects do not need protection from corrosion.

Iron objects must be protected from corrosion.

Which statement explains why aluminium resists corrosion?

- A** Aluminium does not form ions easily.
- B** Aluminium does not react with water or air.
- C** Aluminium has a protective oxide layer.
- D** Aluminium is below iron in the reactivity series.

28 Which statement describes how oxides of nitrogen are formed in a car engine?

- A** Nitrogen from the air reacts with oxygen from petrol.
- B** Nitrogen from the air reacts with oxygen from the air.
- C** Nitrogen from petrol reacts with oxygen from petrol.
- D** Nitrogen from petrol reacts with oxygen from the air.

29 Ships are made of steel, an alloy of iron.

Blocks of magnesium are attached to the underside of ships to prevent rusting.

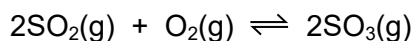
Which statement explains how the magnesium prevents rusting?

- A** Magnesium oxidises instead of iron.
- B** Magnesium stops air and water getting to the iron.
- C** The magnesium forms an alloy with iron which does not corrode.
- D** The magnesium reacts with rust as soon as it is formed.

30 Which process is used to produce hydrogen for the Haber process?

- A electrolysis of water
- B reacting aluminium with sodium hydroxide
- C reacting iron with sulfuric acid
- D reacting methane with steam

31 One of the steps in manufacturing sulfuric acid in the Contact process is shown.



Which catalyst is used to increase the rate of this reaction?

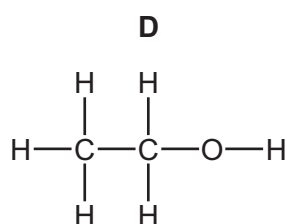
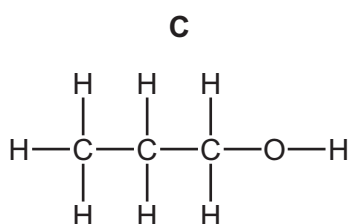
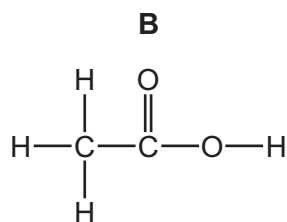
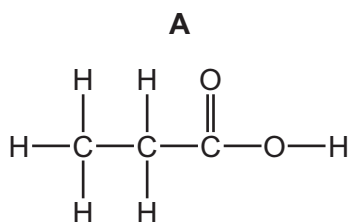
- A aluminium oxide
- B iron
- C phosphoric acid
- D vanadium(V) oxide

32 Lime (calcium oxide) is used to treat waste water from a factory.

Which substance is removed by the lime?

- A ammonia
- B sodium chloride
- C sodium hydroxide
- D sulfuric acid

33 What is the structure of propanol?



- 34 Fuel X produces carbon dioxide and water when it is burned in air. So does fuel Y.

What could X and Y be?

	X	Y
A	C	H ₂
B	C	C ₈ H ₁₈
C	CH ₄	H ₂
D	CH ₄	C ₈ H ₁₈

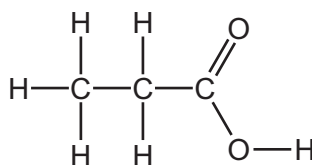
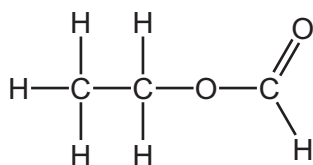
- 35 What is the main constituent of natural gas?

- A** hydrogen
- B** carbon monoxide
- C** methane
- D** nitrogen

- 36 Which statement describes the members of a homologous series?

- A** compounds with the same physical properties
- B** compounds containing the same functional group
- C** compounds containing the same number and type of bonds
- D** compounds obtained from the same raw material

- 37 The structures of two compounds are shown.

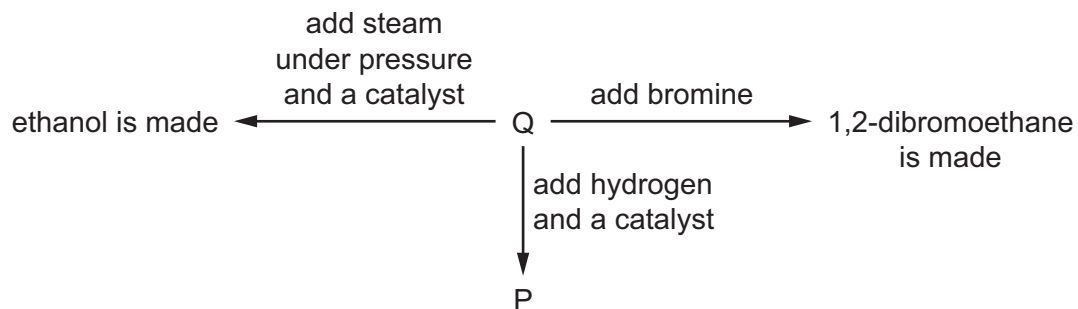


Which statements about these compounds are correct?

- 1 They have the same molecular formula.
- 2 They have similar chemical properties.
- 3 They are structural isomers.

- A** 1 only
- B** 1 and 2
- C** 2 and 3
- D** 1 and 3

38 Some reactions of substance Q are shown.



What is P?

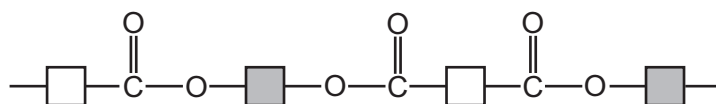
- A ethane
- B ethanoic acid
- C ethene
- D poly(ethene)

39 Proteins and starch are natural polymers.

Which row identifies the method of polymerisation of proteins and starch?

	proteins	starch
A	addition	addition
B	condensation	condensation
C	addition	condensation
D	condensation	addition

40 The diagram shows the partial structure of *Terylene*.



From which pair of compounds is it made?

- A** $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{OH}$
- B** $\text{HO}-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- C** $\text{HO}-\text{[white square]}-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$
- D** $\text{HO}-\text{C}(=\text{O})-\text{[white square]}-\text{C}(=\text{O})-\text{OH}$ + $\text{HO}-\text{C}(=\text{O})-\text{[grey square]}-\text{C}(=\text{O})-\text{OH}$

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1 H hydrogen 1																	
		Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—	—	—	—	

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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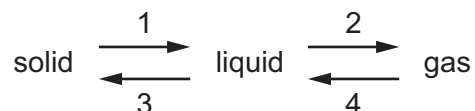
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This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows the changes of state between a solid, a liquid and a gas.



In which changes of state is energy being given out?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4
- 2 A coloured dye is separated by chromatography.

One component of the dye moves a distance of 13 cm and has an R_f value of 0.86.

Which distance did the solvent front move?

- A** 6.6 cm **B** 11.9 cm **C** 15.1 cm **D** 21.6 cm
- 3 A mixture contains salt, sand and sulfur.

Salt dissolves in water but not in xylene.

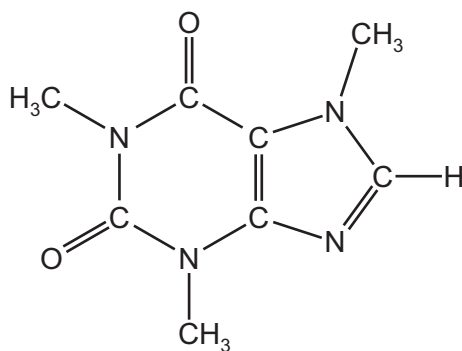
Sulfur dissolves in xylene but not in water.

Sand does not dissolve in water or xylene.

What is the order of the processes used to separate the salt, the sand and the sulfur from the mixture?

- A** add water → filter → add xylene to the filtrate → filter
- B** add water → filter → add xylene to the residue → filter
- C** add xylene → filter → add water to the filtrate → filter
- D** add xylene → filter → add xylene to the residue → filter
- 4 Which statements about isotopes of the same element are correct?
- 1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.
 - 2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.
 - 3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.
- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

- 5 Which type of structure and bonding is present in an element that is malleable and conducts electricity?
- A** covalent molecular
B ionic lattice
C covalent macromolecular
D metallic lattice
- 6 Which statements about potassium bromide are correct?
- 1 It has a high melting point.
2 It dissolves in water.
3 It conducts electricity when solid.
- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only
- 7 Which substance has a similar structure to silicon(IV) oxide?
- A** carbon dioxide
B diamond
C graphite
D sodium oxide
- 8 Caffeine is a stimulant found in coffee.



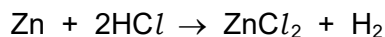
caffeine

Which formula represents caffeine?

- A** C₇H₁₀N₄O₂ **B** C₈H₁₀N₃O₂ **C** C₈H₁₀N₄O₂ **D** C₈H₁₁N₄O₂

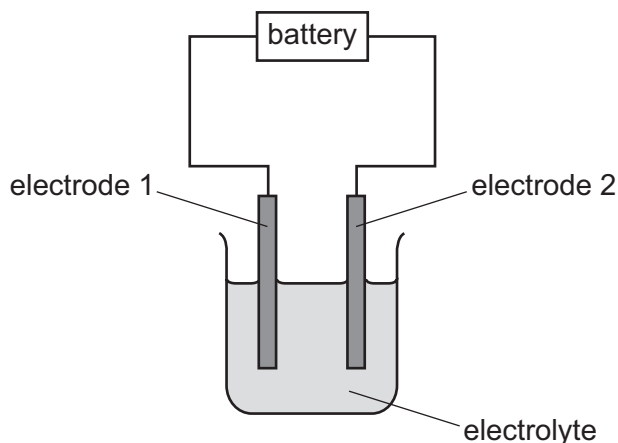
- 9 4.55 g of zinc is reacted with 50 cm³ of 2.25 mol/dm³ dilute hydrochloric acid.

The equation for the reaction is shown.



Which volume of hydrogen gas, at room temperature and pressure, is produced in the reaction?

- A** 1.35 dm³ **B** 1.67 dm³ **C** 2.70 dm³ **D** 3.34 dm³
- 10 In the electrolysis diagram, oxidation is occurring at electrode 1 and reduction at electrode 2.



Which row shows the directions of movement of the electrons in the external circuit and of the positive ions in the electrolyte?

	direction of movement of electrons in external circuit	direction of movement of positive ions in electrolyte
A	1 → 2	1 → 2
B	1 → 2	2 → 1
C	2 → 1	1 → 2
D	2 → 1	2 → 1

- 11 When an acid is added to an alkali, the temperature of the reaction mixture rises.

Which words describe this reaction?

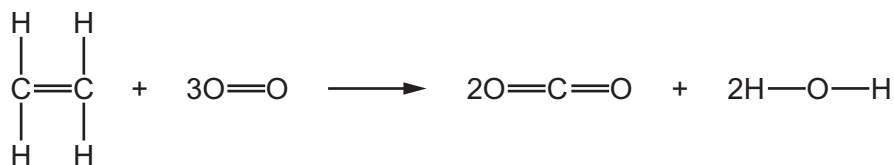
- A** decomposition and endothermic
B decomposition and exothermic
C neutralisation and endothermic
D neutralisation and exothermic

12 Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

	fuel	formula	melting point /°C	boiling point /°C
A	hydrogen	H ₂	–259	–253
B	methane	CH ₄	–182	–164
C	octane	C ₈ H ₁₈	–57	126
D	wax	C ₃₁ H ₆₄	60	400

13 Ethene can undergo complete combustion, as shown.



Some bond energies are given in the table.

bond	bond energy in kJ/mol
C=C	612
C–H	412
O–H	463
O=O	496

The energy change of the reaction is –1408 kJ/mol.

What is the bond energy of the C=O bond in CO₂?

- A** 454 kJ/mol **B** 673 kJ/mol **C** 826 kJ/mol **D** 1619 kJ/mol

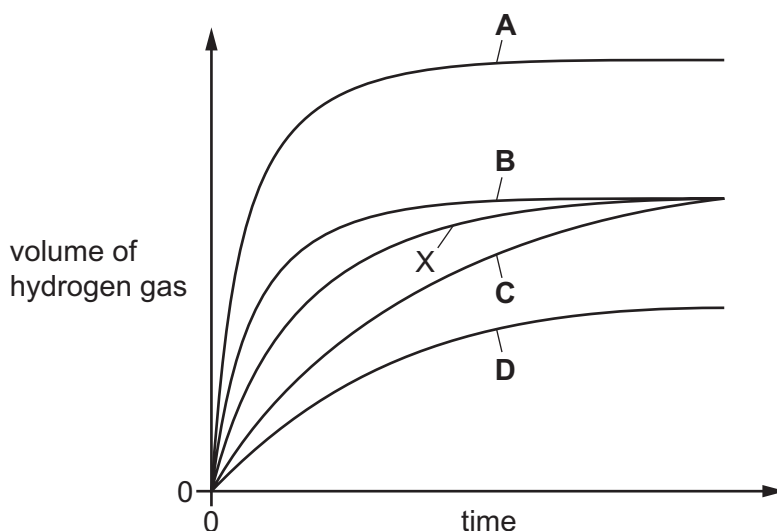
- 14** A student adds excess zinc to dilute hydrochloric acid at 25 °C.

The hydrogen gas produced is collected and measured at room temperature and pressure.

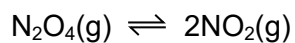
The results are plotted and labelled as curve X on the graph.

The experiment is repeated at 50 °C with all other conditions remaining the same.

Which graph shows the results at 50 °C?



- 15** Dinitrogen tetroxide, N_2O_4 , is converted into nitrogen dioxide, NO_2 , in a reversible reaction.



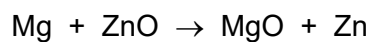
The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

	pressure /atmospheres	temperature
A	2	high
B	2	low
C	50	high
D	50	low

16 When magnesium is heated with zinc oxide a reaction occurs.

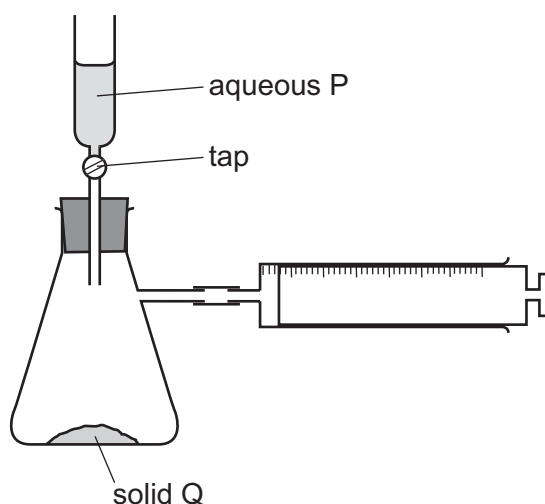
The equation is shown.



Which substance is oxidised?

- A magnesium
- B magnesium oxide
- C zinc
- D zinc oxide

17 The diagram shows an experiment.



A small volume of aqueous P is poured on to solid Q and the tap of the funnel closed.

Which pairs of substances cause the syringe to fill with gas?

	HNO_3 and Mg	HCl and Cu	H_2SO_4 and Na_2CO_3
A	✓	✓	✓
B	✓	✓	x
C	✓	x	✓
D	x	✓	✓

- 18** Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

	ethanoic acid	water
A	accepts a proton	donates a proton
B	accepts an electron	donates an electron
C	donates a proton	accepts a proton
D	donates an electron	accepts an electron

- 19** Aqueous ammonium sulfate is made by reacting aqueous ammonia with dilute sulfuric acid.

How is solid ammonium sulfate obtained from the resulting solution?

- A** crystallisation
- B** distillation
- C** filtration
- D** solvent extraction

- 20** Carbon forms two oxides: carbon monoxide, CO, and carbon dioxide, CO₂.

Which row describes these two oxides?

	CO	CO ₂
A	acidic	acidic
B	acidic	neutral
C	neutral	acidic
D	neutral	neutral

- 21** Group II elements show the same trends as Group I elements.

Which statement about elements in Group II is correct?

- A** The melting point of barium is higher than the melting point of calcium.
- B** Barium is more reactive than beryllium.
- C** Strontium would not react with oxygen.
- D** Magnesium is more dense than barium.

22 Some information about properties of Group I elements is shown.

element	melting point / °C	density in g/cm ³
lithium	181	0.53
sodium	98	0.97
potassium	X	
rubidium	Y	Z

What are the values for X, Y and Z?

	X	Y	Z
A	63	252	0.26
B	63	39	0.26
C	39	63	1.53
D	63	39	1.53

23 Which statements describe properties of transition elements?

- 1 They form coloured compounds.
- 2 They have variable oxidation states.
- 3 They have low densities.
- 4 They are volatile.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

24 Which statement about the extraction of aluminium by electrolysis is correct?

- A** Aluminium is extracted from its ore, cryolite.
- B** Aluminium is formed at the positive electrode.
- C** Bauxite is used to lower the temperature of the extraction process.
- D** Graphite is used for both the positive and negative electrodes.

25 Copper(II) nitrate and zinc carbonate are heated strongly in separate test-tubes.

Which row identifies the gases produced?

	copper(II) nitrate	zinc carbonate
A	oxygen and nitrogen dioxide	carbon dioxide only
B	oxygen and nitrogen dioxide	carbon dioxide and oxygen
C	nitrogen dioxide only	carbon dioxide and oxygen
D	nitrogen dioxide only	carbon dioxide only

26 Iron from a blast furnace can be converted to steel.

Which statements about steel are correct?

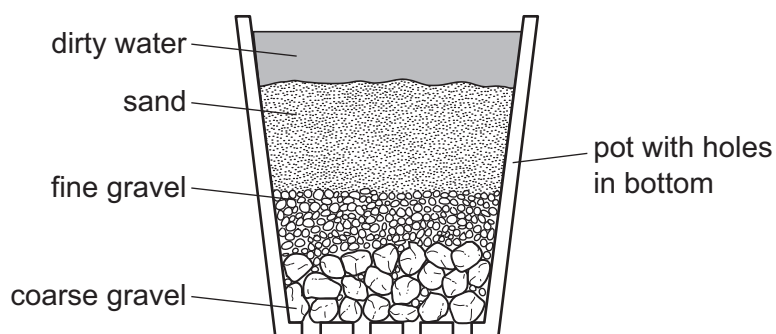
- 1 Steel contains more carbon than the iron obtained from the blast furnace.
- 2 Steel is produced by blowing oxygen through the iron.
- 3 Calcium oxide is added to molten iron to remove basic oxides.

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 only

27 Which metal is used to galvanise steel?

- A** copper
B lead
C tin
D zinc

28 The diagram shows a stage in the purification of dirty water.



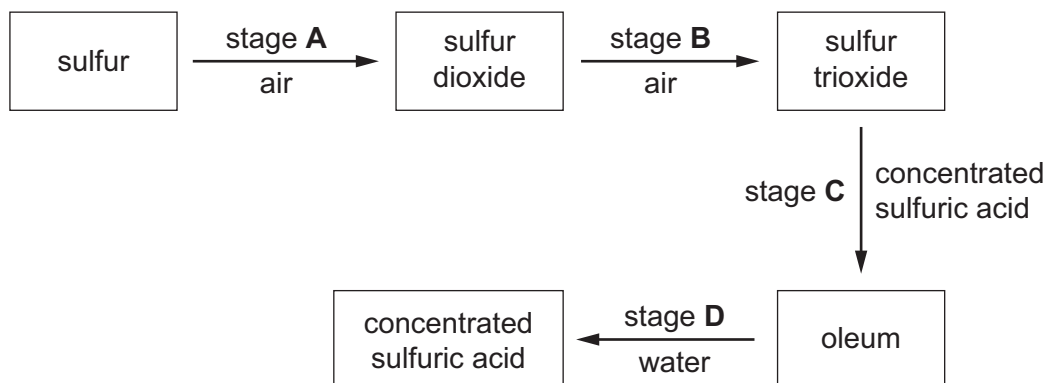
Which process does this apparatus show?

- A chlorination
 - B condensation
 - C distillation
 - D filtration
- 29 Which substance in polluted air damages stonework and kills trees?
- A carbon dioxide
 - B carbon monoxide
 - C lead compounds
 - D sulfur dioxide
- 30 Which row explains why a high temperature and an iron catalyst are used in the manufacture of ammonia by the Haber process?

	high temperature	iron catalyst
A	increases the rate of the reaction	increases the equilibrium yield of ammonia
B	increases the rate of the reaction	increases the rate of the reaction
C	increases the equilibrium yield of ammonia	increases the equilibrium yield of ammonia
D	increases the equilibrium yield of ammonia	increases the rate of the reaction

31 The scheme shows four stages in the conversion of sulfur to sulfuric acid.

In which stage is a catalyst used?



32 Which element has an oxide that is used as a food preservative?

- A helium
- B hydrogen
- C iron
- D sulfur

33 Which substance gives off carbon dioxide on heating?

- A lime
- B limestone
- C limewater
- D slaked lime

34 Which formula represents ethanol?

- A CH_3CH_3 B CH_2CH_2 C $\text{CH}_3\text{CH}_2\text{OH}$ D CH_3COOH

35 Which statement about structural isomers is correct?

- A They have the same structure but different reactivity.
- B They have the same general formula but a different number of carbon atoms in their molecules.
- C They have the same structure but different relative molecular masses.
- D They have different structures but the same numbers of each type of atom.

36 Which formula is the same in methanol, ethanol and propanol?

- A** empirical formula
- B** general formula
- C** molecular formula
- D** structural formula

37 Ethene reacts with water under suitable conditions.

Which statement about this reaction is correct?

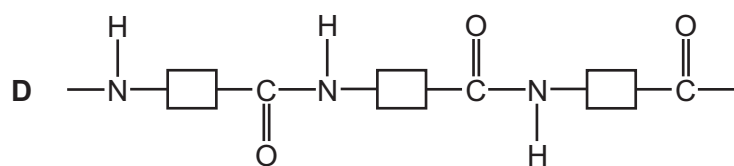
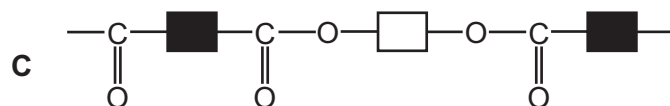
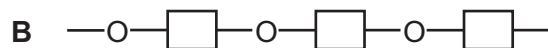
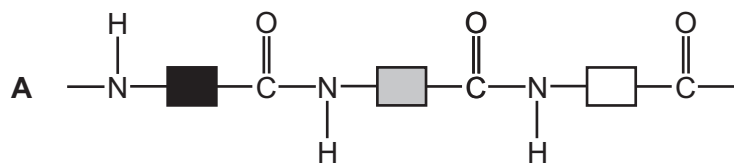
- A** The product of this reaction has an M_r of 46.
- B** The reaction produces two different products.
- C** The reaction occurs when ethene gas is bubbled into cold water in the presence of an acid catalyst.
- D** The reaction is a redox reaction.

38 Ethanoic acid is made by reacting ethanol with acidified potassium manganate(VII).

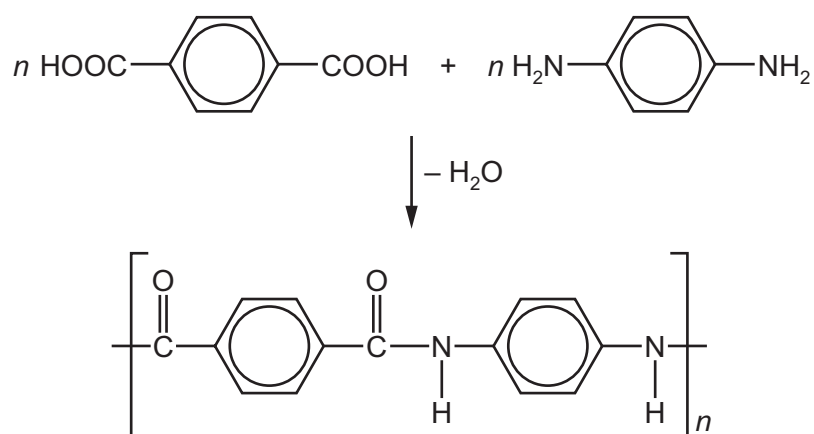
Which type of reaction occurs when ethanol reacts with acidified potassium manganate(VII)?

- A** displacement
- B** fermentation
- C** oxidation
- D** neutralisation

39 Which structure represents *Terylene*?



40 The equation shows the formation of a polymer called *Kevlar*.



Which row describes *Kevlar*?

	how the polymer is formed	type of polymer
A	addition polymerisation	polyamide
B	addition polymerisation	polyester
C	condensation polymerisation	polyamide
D	condensation polymerisation	polyester

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lanthanoids

actinoids

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89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

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0620/22

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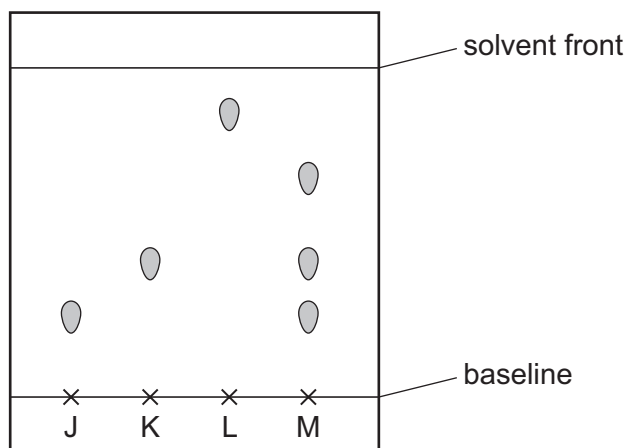
This document has **16** pages.



- 1 The rate of diffusion of three gases, ammonia, carbon dioxide and methane, is measured.

What is the order of the rate of diffusion of the gases from slowest to fastest?

- A** $\text{CO}_2 \rightarrow \text{NH}_3 \rightarrow \text{CH}_4$
B $\text{CO}_2 \rightarrow \text{CH}_4 \rightarrow \text{NH}_3$
C $\text{CH}_4 \rightarrow \text{NH}_3 \rightarrow \text{CO}_2$
D $\text{NH}_3 \rightarrow \text{CH}_4 \rightarrow \text{CO}_2$
- 2 Which description of Brownian motion is correct?
- A** random movement of particles due to bombardment by larger particles
B random movement of particles due to bombardment by smaller particles
C random movement of particles from a high concentration to a low concentration
D random movement of particles from a low concentration to a high concentration
- 3 The chromatogram obtained using four substances, J, K, L and M, is shown.



Which statement about M is correct?

- A** It is a mixture of J and K only.
B It is a pure substance.
C It is a mixture of J, K and L.
D It is a mixture of J, K and an unknown substance.

4 Which statements about isotopes of the same element are correct?

- 1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.
- 2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.
- 3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

5 Which statement about solid magnesium oxide is correct?

- A** It is a giant structure made up of magnesium and oxygen atoms bonded covalently.
- B** It is an electrical conductor with mobile magnesium ions and oxygen ions.
- C** Magnesium loses electrons and these electrons move freely through a lattice.
- D** Oxygen ions and magnesium ions are attracted to each other in a giant lattice.

6 Which molecule contains only three shared pairs of electrons?

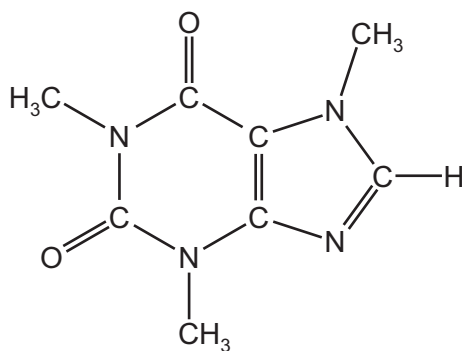
A CH_3OH **B** Cl_2 **C** H_2O **D** N_2

7 Which particles are present in the structure of metals?

- 1 positive ions
- 2 negative ions
- 3 shared pairs of electrons
- 4 mobile electrons

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 8 Caffeine is a stimulant found in coffee.

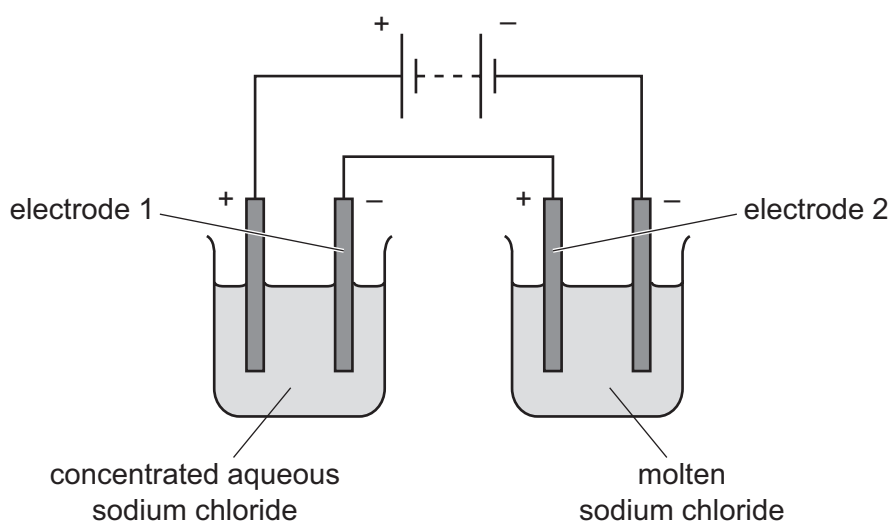


caffeine

Which formula represents caffeine?

- A** $C_7H_{10}N_4O_2$ **B** $C_8H_{10}N_3O_2$ **C** $C_8H_{10}N_4O_2$ **D** $C_8H_{11}N_4O_2$
- 9 Which sample does **not** contain a number of atoms equal to the Avogadro constant?
- A** 14 g of nitrogen, N_2
B 6 g of water, H_2O
C 4 g of helium, He
D 28 g of carbon monoxide, CO

- 10 The electrolysis of concentrated aqueous sodium chloride and molten sodium chloride is shown.



What are the products at electrodes 1 and 2?

	electrode 1	electrode 2
A	chlorine	chlorine
B	hydrogen	chlorine
C	hydrogen	sodium
D	sodium	sodium

- 11 When an acid is added to an alkali, the temperature of the reaction mixture rises.

Which words describe this reaction?

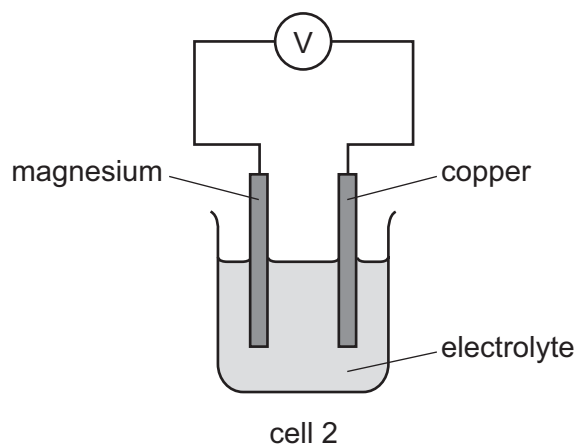
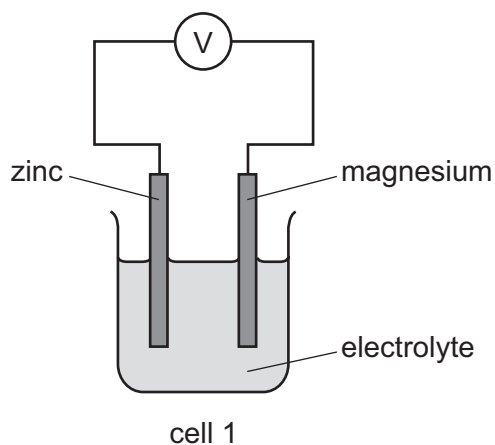
- A** decomposition and endothermic
- B** decomposition and exothermic
- C** neutralisation and endothermic
- D** neutralisation and exothermic

12 Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

	fuel	formula	melting point / °C	boiling point / °C
A	hydrogen	H ₂	–259	–253
B	methane	CH ₄	–182	–164
C	octane	C ₈ H ₁₈	–57	126
D	wax	C ₃₁ H ₆₄	60	400

13 The electrical energy, or voltage, of two simple cells is measured.



- statement 1 The voltage of cell 1 is greater than cell 2.
- statement 2 Zinc is more reactive than copper.
- statement 3 Magnesium is oxidised in both cells.
- statement 4 Magnesium atoms lose electrons to form magnesium ions.

Which option is correct?

- A** All the statements are correct.
- B** Only statements 1 and 3 are correct.
- C** Statement 2 is correct and explains statement 1.
- D** Statement 4 is correct and explains statement 3.

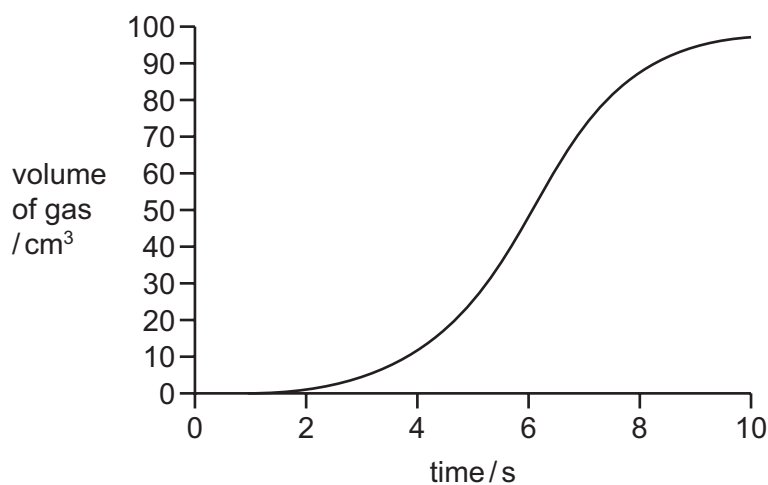
- 14 Dilute aqueous sodium chloride is electrolysed using carbon electrodes.

What is the product at the anode?

- A carbon dioxide
- B hydrogen
- C oxygen
- D sodium

- 15 The volume of gas given off in a chemical reaction is measured over time.

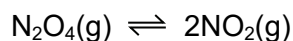
The results are shown.



At which time is the rate of reaction greatest?

- A 0 s
- B 4 s
- C 6 s
- D 10 s

- 16 Dinitrogen tetroxide, N_2O_4 , is converted into nitrogen dioxide, NO_2 , in a reversible reaction.



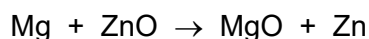
The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

	pressure / atmospheres	temperature
A	2	high
B	2	low
C	50	high
D	50	low

17 When magnesium is heated with zinc oxide a reaction occurs.

The equation is shown.



Which substance is oxidised?

- A** magnesium
- B** magnesium oxide
- C** zinc
- D** zinc oxide

18 X and Y are oxides of two different elements.

- X reacts with water to produce aqueous solution Z.
- Z turns universal indicator paper blue.
- An aqueous solution of Y reacts with sodium carbonate to produce carbon dioxide gas.

Which statement is correct?

- A** X and Y are both the oxides of metals.
- B** X and Y are both the oxides of non-metals.
- C** X is the oxide of a metal and Y is the oxide of a non-metal.
- D** X is the oxide of a non-metal and Y is the oxide of a metal.

19 Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

	ethanoic acid	water
A	accepts a proton	donates a proton
B	accepts an electron	donates an electron
C	donates a proton	accepts a proton
D	donates an electron	accepts an electron

20 Copper(II) sulfate is a soluble salt.

Calcium sulfate is an insoluble salt.

Which row shows suitable reactants for preparing a pure sample of the named salt?

	salt	reactants
A	calcium sulfate	calcium carbonate + dilute sulfuric acid
B	calcium sulfate	aqueous calcium chloride and aqueous sodium sulfate
C	copper(II) sulfate	copper + dilute sulfuric acid
D	copper(II) sulfate	aqueous copper(II) chloride and aqueous sodium sulfate

21 Strontium displaces magnesium from molten magnesium chloride.

Bromine displaces iodine from aqueous potassium iodide.

Which row describes the change in reactivity down both Group II and Group VII of the Periodic Table?

	reactivity down the group	
	Group II	Group VII
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

22 Elements J and K are in the same period in the Periodic Table.

J reacts with acids to produce a salt and hydrogen.

K reacts with sodium to form an ionic compound.

Which statement about J and K is correct?

- A** An atom of J has more electrons than an atom of K.
- B** J and K are both metals.
- C** J and K are both non-metals.
- D** J is to the left of K in the Periodic Table.

26 Silver is below copper in the reactivity series.

Which row describes the reactions of silver?

	reaction with steam	reaction with dilute hydrochloric acid
A	no reaction	no reaction
B	no reaction	reacts to produce hydrogen gas
C	reacts to produce hydrogen gas	no reaction
D	reacts to produce hydrogen gas	reacts to produce hydrogen gas

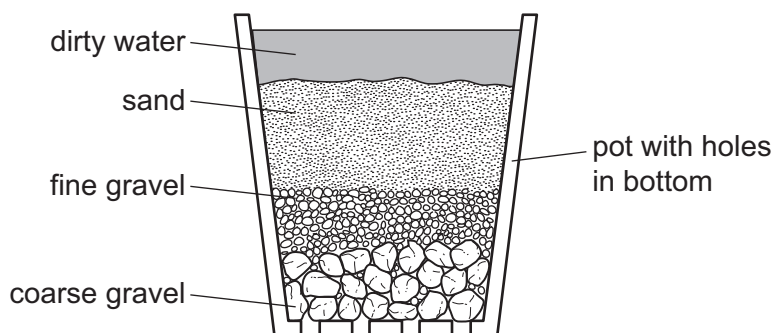
27 Iron is galvanised by coating it in zinc.

Brass is made by mixing copper with zinc.

Which row gives the reasons for each of these uses of zinc?

	reason for galvanising iron	reason for making brass
A	prevents corrosion	produces a softer metal
B	prevents corrosion	produces a harder metal
C	produces a harder metal	produces a softer metal
D	produces a harder metal	produces a harder metal

28 The diagram shows a stage in the purification of dirty water.



Which process does this apparatus show?

- A** chlorination
- B** condensation
- C** distillation
- D** filtration

29 Which substance in polluted air damages stonework and kills trees?

- A carbon dioxide
- B carbon monoxide
- C lead compounds
- D sulfur dioxide

30 Ammonium nitrate, NH_4NO_3 , is a fertiliser and is added to fields to help crops grow.

Slaked lime, $\text{Ca}(\text{OH})_2$, is an alkali and is added to fields to reduce the acidity of the soil.

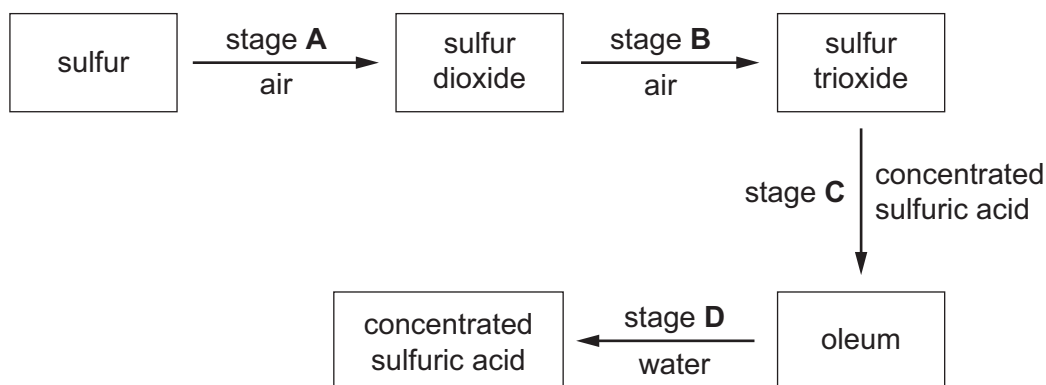
Ammonium nitrate and slaked lime should not be added to a field at the same time because they react with each other to form a gas, Z.

What is Z?

- A ammonia
- B hydrogen
- C nitrogen
- D oxygen

31 The scheme shows four stages in the conversion of sulfur to sulfuric acid.

In which stage is a catalyst used?



32 Which element has an oxide that is used as a food preservative?

- A helium
- B hydrogen
- C iron
- D sulfur

33 Which substance gives off carbon dioxide on heating?

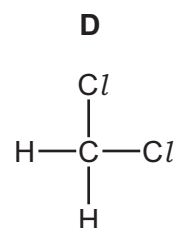
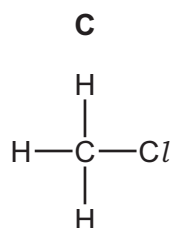
- A lime
- B limestone
- C limewater
- D slaked lime

34 Which compound has the most $\text{-CH}_2\text{-}$ groups in one molecule?

- A butane
- B butanoic acid
- C butan-1-ol
- D but-1-ene

35 Methane reacts with chlorine in the presence of ultraviolet light.

Which substance is **not** produced in this reaction?



36 Ethene reacts with both hydrogen and steam.

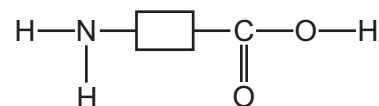
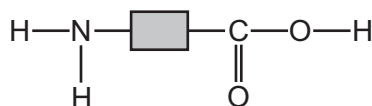
Which row about these reactions is correct?

	reactant with ethene	type of reaction	catalyst used
A	hydrogen	substitution	phosphoric acid
B	hydrogen	addition	nickel
C	steam	substitution	phosphoric acid
D	steam	addition	nickel

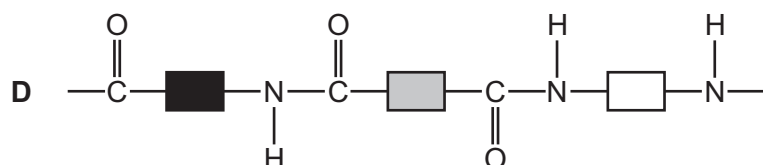
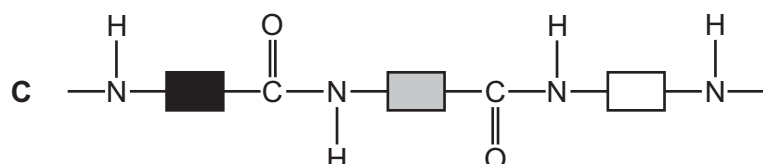
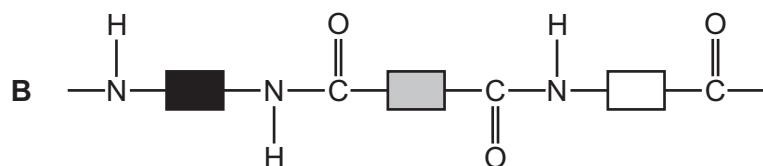
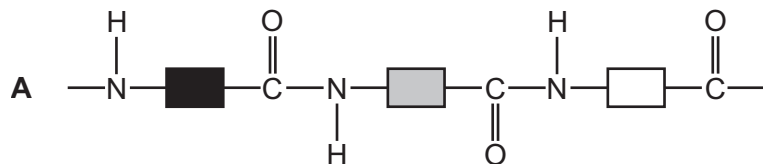
37 Which type of reaction occurs when ethanol is converted to ethanoic acid?

- A combustion
- B decomposition
- C neutralisation
- D oxidation

38 Hydrolysis of polymer P produces the three compounds shown.



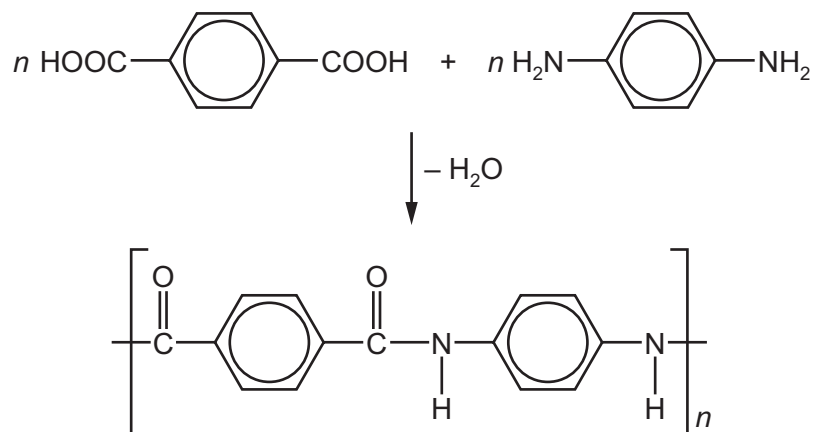
What is the structure of polymer P?



39 Which statement about unsaturated hydrocarbons is correct?

- A** $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3$ is an unsaturated hydrocarbon.
- B** Ethene has more hydrogen atoms per molecule than ethane.
- C** Unsaturated hydrocarbons have double bonds between carbon and hydrogen atoms.
- D** Unsaturated hydrocarbons turn aqueous bromine from colourless to brown.

40 The equation shows the formation of a polymer called *Kevlar*.



Which row describes *Kevlar*?

	how the polymer is formed	type of polymer
A	addition polymerisation	polyamide
B	addition polymerisation	polyester
C	condensation polymerisation	polyamide
D	condensation polymerisation	polyester

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1 H hydrogen 1																	
		Key atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	114 Fl flerovium —	116 Lv livermorium —	—	—	—	—	

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/23

Paper 2 Multiple Choice (Extended)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



1 Which gas diffuses the most slowly?

A CH_4

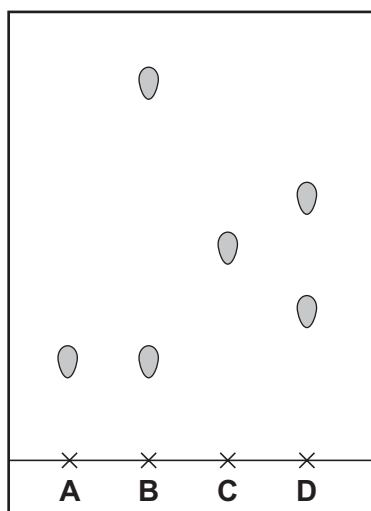
B CO_2

C H_2

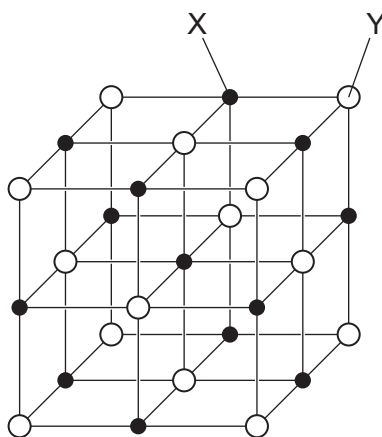
D NH_3

2 The chromatogram from four different substances is shown.

Which pure substance has the largest R_f value?



3 The structure of sodium chloride can be represented as shown.



What are X and Y?

	X	Y
A	metal atom	non-metal atom
B	negative ion	electron
C	positive ion	negative ion
D	positive ion	electron

4 Which two particles have the same electronic structure?

- A C and O^{2-}
- B F^{-} and Na
- C K^{+} and S^{2-}
- D Mg and Na^{+}

5 Which statements about isotopes of the same element are correct?

- 1 They are atoms which have the same chemical properties because they have the same number of electrons in their outer shell.
- 2 They are atoms which have the same number of electrons and neutrons but different numbers of protons.
- 3 They are atoms which have the same number of electrons and protons but different numbers of neutrons.

- A 1 and 2 B 1 and 3 C 2 only D 3 only

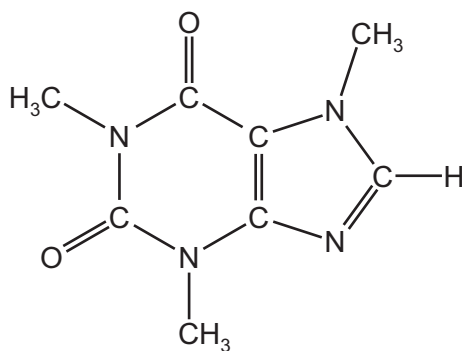
6 What is the total number of shared electrons in a molecule of methanol, CH_3OH ?

- A 4 B 5 C 8 D 10

7 Which row about the structures and uses of diamond and graphite is correct?

	structure	use
A	diamond has a giant covalent structure	diamond is used to make electrodes
B	diamond has a simple covalent structure	diamond is used to make cutting tools
C	graphite has a giant covalent structure	graphite is used as a lubricant
D	graphite has a simple covalent structure	graphite is used to make cutting tools

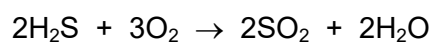
- 8 Caffeine is a stimulant found in coffee.



caffeine

Which formula represents caffeine?

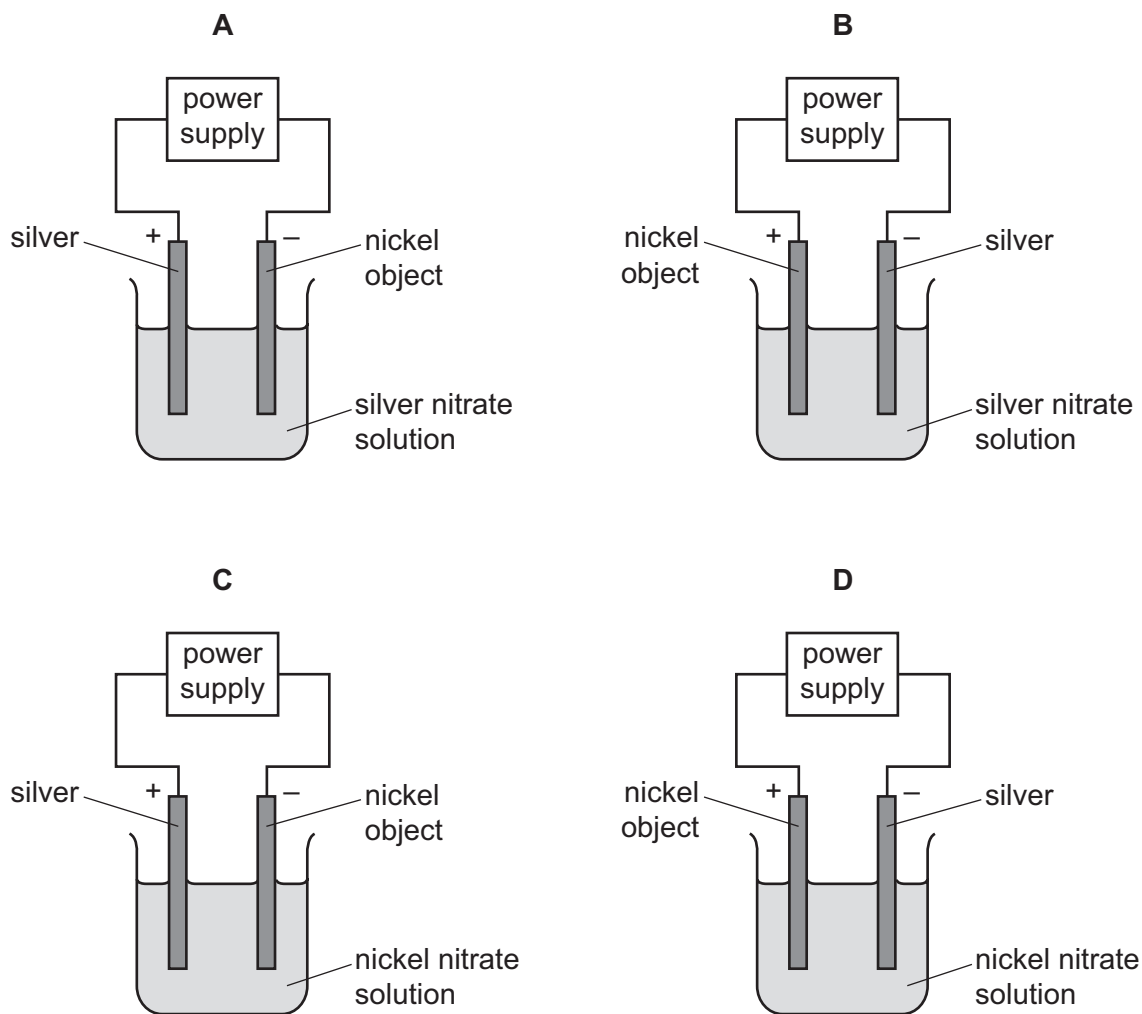
- A** $C_7H_{10}N_4O_2$ **B** $C_8H_{10}N_3O_2$ **C** $C_8H_{10}N_4O_2$ **D** $C_8H_{11}N_4O_2$
- 9 The equation for the reaction between hydrogen sulfide, H_2S , and oxygen is shown.



Which mass of oxygen is required to react with 5.1 g of hydrogen sulfide?

- A** 2.4 g **B** 4.8 g **C** 7.2 g **D** 14.4 g

10 Which apparatus is used to plate a nickel object with silver?



11 When an acid is added to an alkali, the temperature of the reaction mixture rises.

Which words describe this reaction?

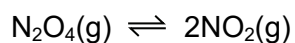
- A** decomposition and endothermic
- B** decomposition and exothermic
- C** neutralisation and endothermic
- D** neutralisation and exothermic

12 Some properties of four fuels are shown.

Which fuel is a gas at room temperature and makes two products when it burns in a plentiful supply of air?

	fuel	formula	melting point /°C	boiling point /°C
A	hydrogen	H ₂	–259	–253
B	methane	CH ₄	–182	–164
C	octane	C ₈ H ₁₈	–57	126
D	wax	C ₃₁ H ₆₄	60	400

13 Dinitrogen tetroxide, N₂O₄, is converted into nitrogen dioxide, NO₂, in a reversible reaction.



The forward reaction is endothermic.

Which conditions give the highest equilibrium yield of nitrogen dioxide?

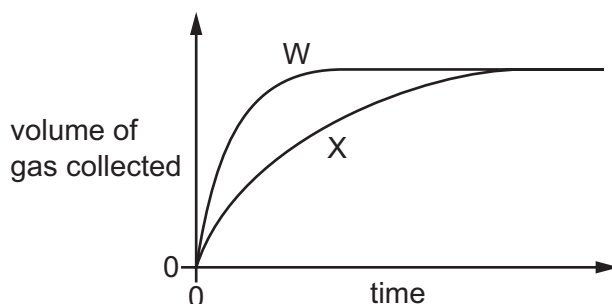
	pressure /atmospheres	temperature
A	2	high
B	2	low
C	50	high
D	50	low

- 14** Dilute hydrochloric acid is reacted with excess calcium carbonate and the total volume of gas is measured at regular intervals.

The results are shown by line W on the graph.

The experiment is repeated but with one change.

The results of the second experiment are shown by line X on the graph.



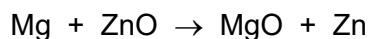
Which change is made in the second experiment?

- A** A catalyst is added.
 - B** The calcium carbonate is broken into smaller pieces.
 - C** The concentration of the dilute hydrochloric acid is increased.
 - D** The temperature of the dilute hydrochloric acid is decreased.
- 15** When hydrated copper(II) sulfate is heated, it produces white copper(II) sulfate. When water is added, the white copper(II) sulfate turns blue.

Which type of reaction is shown by these observations?

- A** decomposition
 - B** displacement
 - C** redox
 - D** reversible
- 16** When magnesium is heated with zinc oxide a reaction occurs.

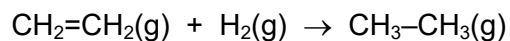
The equation is shown.



Which substance is oxidised?

- A** magnesium
- B** magnesium oxide
- C** zinc
- D** zinc oxide

- 17 The equation for the reaction between ethene and hydrogen is shown.



The bond energies are shown.

bond	bond energy in kJ/mol
C=C	612
H-H	436
C-C	348
C-H	416

What is the overall energy change during this reaction?

- A** -284 kJ/mol
B -132 kJ/mol
C +132 kJ/mol
D +284 kJ/mol
- 18 Ethanoic acid reacts with water to produce an acidic solution.

Which row describes the roles of ethanoic acid and water in this reaction?

	ethanoic acid	water
A	accepts a proton	donates a proton
B	accepts an electron	donates an electron
C	donates a proton	accepts a proton
D	donates an electron	accepts an electron

19 Tests are done on an aqueous solution.

test	a few drops of aqueous sodium hydroxide are added	aqueous sodium hydroxide is added in excess
observation	white precipitate	precipitate dissolves to give a colourless solution

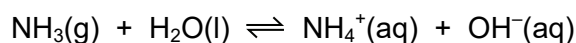
Which cations produce these observations?

- 1 aluminium, Al^{3+}
- 2 calcium, Ca^{2+}
- 3 zinc, Zn^{2+}

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 and 3

20 Ammonia, NH_3 , dissolves in water to form a dilute solution of ammonium hydroxide, NH_4OH .

The reaction is reversible and exists as an equilibrium mixture.



Which statement about the mixture is correct?

- A** All of the ammonia and water molecules have turned into ions.
- B** The ammonia and water molecules have stopped changing into ions.
- C** The concentrations of the ammonia molecules and ammonium ions are always equal.
- D** The rate of the formation of ammonia molecules is equal to the rate of formation of the ammonium ions.

21 Elements E and F are in Group I of the Periodic Table.

E has a higher melting point than F.

Elements J and L are in Group VII of the Periodic Table.

J has a higher density than L.

Which elements have the highest atomic numbers in each group?

A E and J **B** E and L **C** F and J **D** F and L

22 Which metal forms ions with one oxidation state?

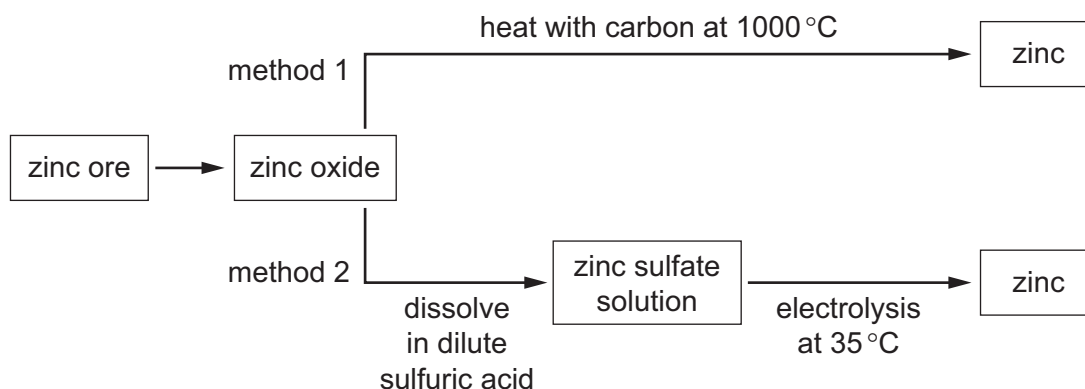
- A aluminium
- B chromium
- C copper
- D iron

23 How does the nature of the oxides change across Period 3 from sodium to chlorine?

- A basic → amphoteric → acidic
- B basic → acidic → amphoteric
- C amphoteric → basic → acidic
- D acidic → amphoteric → basic

24 Zinc is a metal with a boiling point of 907 °C.

Two methods of making zinc are shown.



Which statement is correct?

- A Carbon oxidises zinc oxide in method 1.
- B Zinc vapour is produced in both methods.
- C Zinc is produced at the anode in method 2.
- D Zinc compounds are reduced in both methods.

25 Which statement about the reactions of metals is correct?

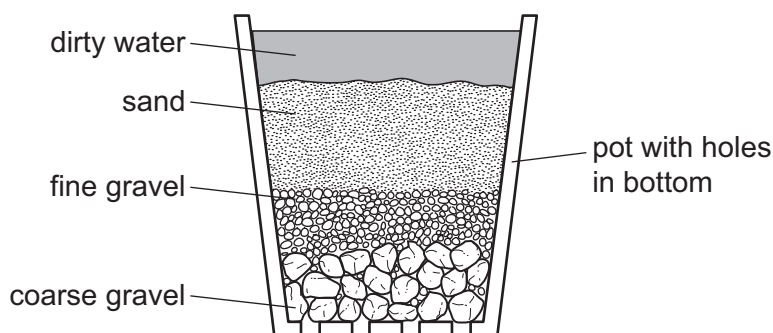
- A Iron and carbon dioxide are produced when iron(III) oxide is heated with carbon.
- B Magnesium reacts with dilute hydrochloric acid producing hydrogen and chlorine.
- C Potassium reacts vigorously with water producing hydrogen and an acidic solution.
- D Zinc reacts with dilute sulfuric acid producing sulfur dioxide.

- 26 12.4 g of copper(II) carbonate is heated in a test-tube. Only 50% is decomposed.

[M_r : CuCO_3 , 124; CuO , 80]

What will be the final mass of the substances in the test-tube?

- A 9.4 g B 9.8 g C 10.2 g D 10.6 g
- 27 Which statement about the manufacture of ammonia is correct?
- A Ammonia is manufactured by heating hydrogen and nitrogen at 50 °C and 1.0 atm.
B Ammonia is obtained by heating hydrogen and nitrogen in the Contact process.
C Hydrogen for the manufacture of ammonia is extracted from air.
D The reaction between hydrogen and nitrogen to form ammonia is reversible.
- 28 The diagram shows a stage in the purification of dirty water.



Which process does this apparatus show?

- A chlorination
B condensation
C distillation
D filtration
- 29 Which substance in polluted air damages stonework and kills trees?
- A carbon dioxide
B carbon monoxide
C lead compounds
D sulfur dioxide

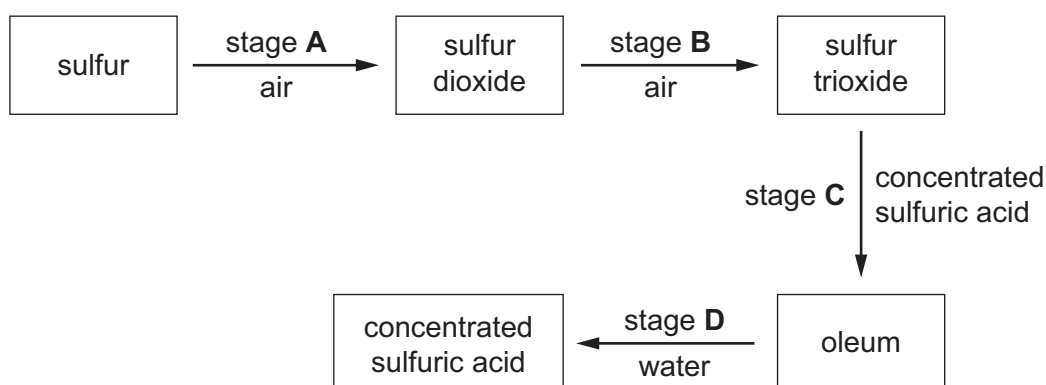
30 Petrol-fuelled cars produce oxides of nitrogen.

Which statement explains how oxides of nitrogen are formed?

- A In the catalytic converter, the elements nitrogen and oxygen combine.
- B Oxygen and nitrogen compounds in petrol combine in the car engine.
- C The high temperatures in the engine provide oxygen and nitrogen with the activation energy needed to react.
- D In the car engine, nitrogen compounds in petrol combine with oxygen.

31 The scheme shows four stages in the conversion of sulfur to sulfuric acid.

In which stage is a catalyst used?



32 Which element has an oxide that is used as a food preservative?

- A helium
- B hydrogen
- C iron
- D sulfur

33 Which substance gives off carbon dioxide on heating?

- A lime
- B limestone
- C limewater
- D slaked lime

34 Which formula represents ethyl butanoate?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$
- B $\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$
- D $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

35 Methanol, CH_3OH , is a member of the homologous series of alcohols.

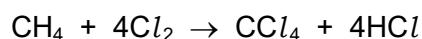
What is the formula of the alcohol in the same homologous series which contains three carbon atoms?

- A $\text{C}_3\text{H}_5\text{OH}$ B $\text{C}_3\text{H}_6\text{OH}$ C $\text{C}_3\text{H}_7\text{OH}$ D $\text{C}_3\text{H}_8\text{OH}$

36 Which type of compound reacts with hydrogen in an addition reaction?

- A alkanes
- B alkenes
- C alcohols
- D carboxylic acids

37 The equation for the reaction between methane and chlorine is shown.



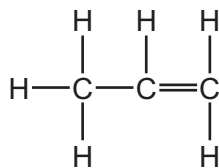
Which type of reaction does methane undergo?

- A substitution
- B reduction
- C condensation
- D addition

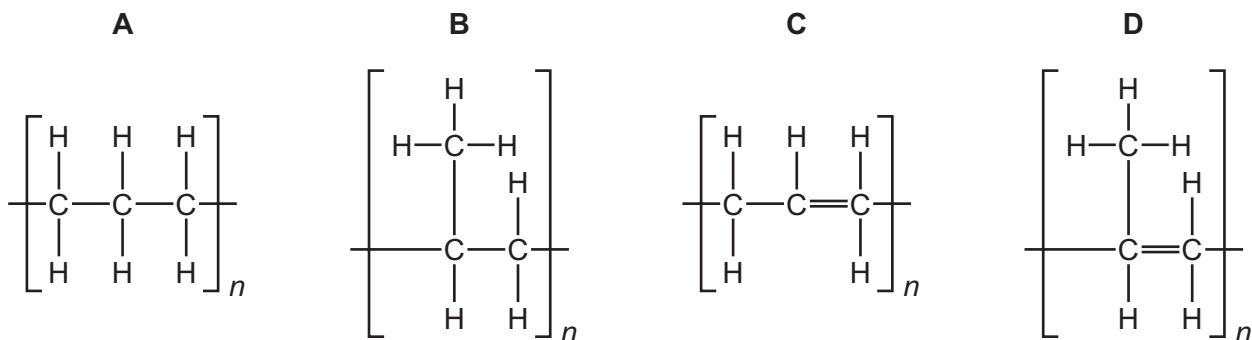
38 Which functional groups form an amide linkage?

- A $\text{H}_2\text{N}-$ and $-\text{COOH}$
- B $\text{H}_2\text{N}-$ and $\text{H}_2\text{N}-$
- C $-\text{OH}$ and $-\text{COOH}$
- D $-\text{OH}$ and $\text{H}_2\text{N}-$

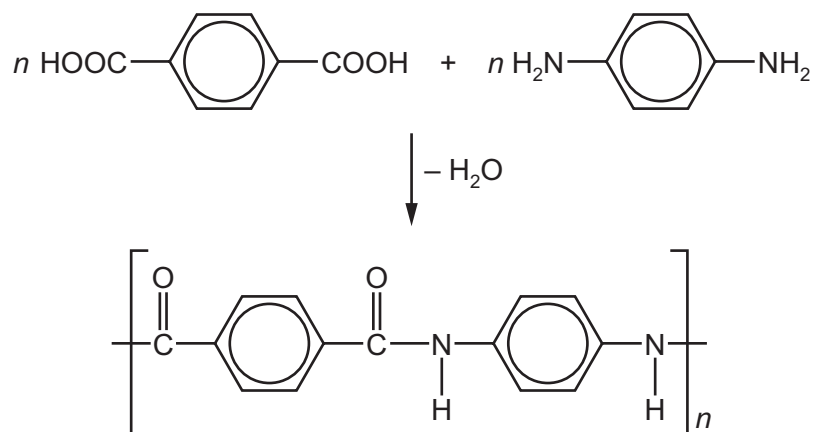
39 The structure of propene is shown.



Which diagram represents poly(propene)?



40 The equation shows the formation of a polymer called *Kevlar*.



Which row describes *Kevlar*?

	how the polymer is formed	type of polymer
A	addition polymerisation	polyamide
B	addition polymerisation	polyester
C	condensation polymerisation	polyamide
D	condensation polymerisation	polyester

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —		114 Fl flerovium —		116 Lv livermorium —		

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 A gas is placed in a sealed container. The gas has a pressure of one atmosphere and a temperature of 50 °C.

It is heated to 100 °C.

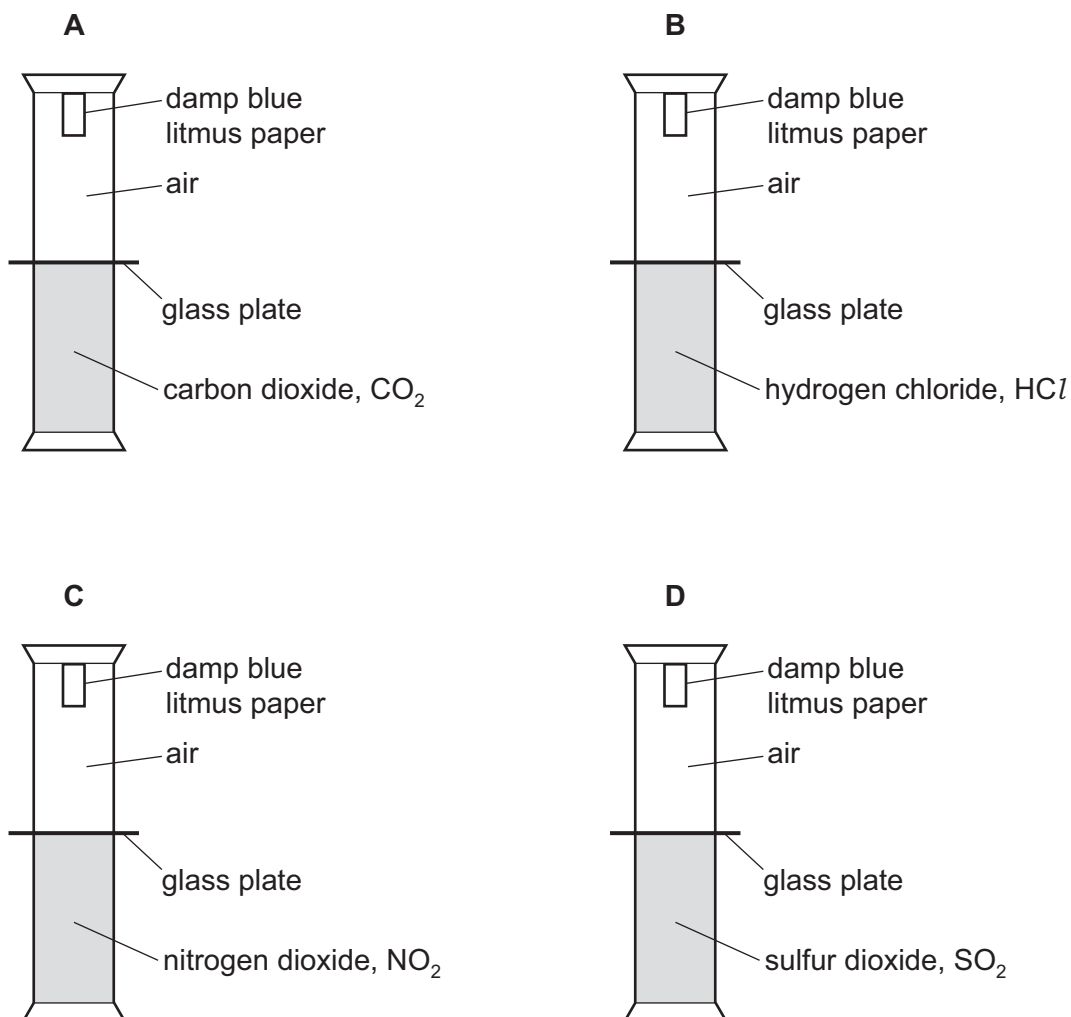
Which row describes the cause of the pressure of the gas and the effect of increasing the temperature of the gas?

	cause of gas pressure	the effect of increased temperature of the gas
A	collisions between gas particles	collisions become less frequent
B	collisions between gas particles	the average speed of the gas particles increases
C	collisions between gas particles and the container	collisions become less frequent
D	collisions between gas particles and the container	the average speed of the gas particles increases

- 2 Four experiments, each containing a different acidic gas, are set up as shown.

The dividing glass plates are removed at the same time.

In which set of apparatus does the litmus turn red first?



- 3 The Group I element potassium forms an ionic bond with the Group VII element fluorine.

Which two ions are produced?

- A** K^+ and F^+ **B** K^+ and F^- **C** K^- and F^- **D** K^- and F^+

4 X and Y are atoms.

- X and Y have the same number of electron shells.
- X and Y have the same number of outer electrons.
- X and Y have different mass numbers.

Which statements about X and Y are correct?

- 1 X and Y are isotopes.
- 2 X and Y have the same total number of electrons.
- 3 X and Y have the same chemical properties.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

5 Lithium chloride is an ionic compound and silicon(IV) oxide is a covalent compound.

Which statement about **both** compounds is correct?

- A** They are not soluble in water.
- B** They conduct electricity when melted.
- C** They do not conduct electricity in solid form.
- D** They have low melting points.

6 Which equations are balanced?

- 1 $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- 2 $\text{ZnCO}_3 + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{CO}_2 + 2\text{H}_2\text{O}$
- 3 $\text{Mg}(\text{NO}_3)_2 + \text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 + 2\text{NaNO}_3$
- 4 $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

7 Which row shows the formulae of sodium carbonate, zinc nitrate and ammonium sulfate?

	sodium carbonate	zinc nitrate	ammonium sulfate
A	Na_2CO_3	ZnNO_3	$(\text{NH}_4)_2\text{SO}_4$
B	Na_2CO_3	$\text{Zn}(\text{NO}_3)_2$	$(\text{NH}_4)_2\text{SO}_4$
C	NaCO_3	ZnNO_3	$(\text{NH}_3)_2\text{SO}_4$
D	NaCO_3	$\text{Zn}(\text{NO}_3)_2$	$(\text{NH}_3)_2\text{SO}_4$

8 Which statements about hydrogen and oxygen are correct?

	hydrogen and oxygen can react to produce electrical energy	hydrogen and oxygen can be made by the electrolysis of dilute aqueous sodium chloride
A	x	x
B	x	✓
C	✓	x
D	✓	✓

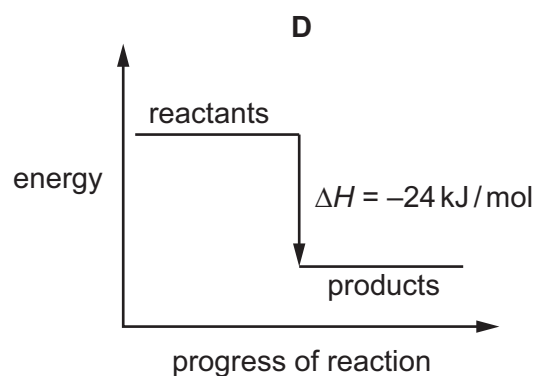
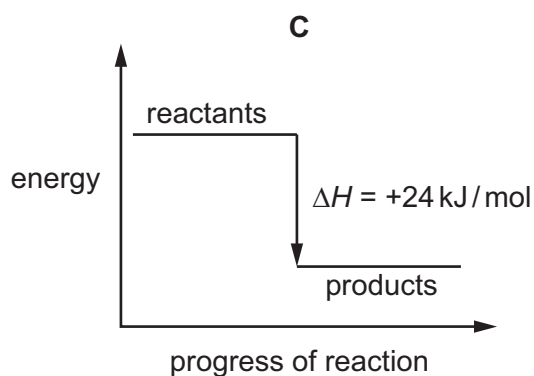
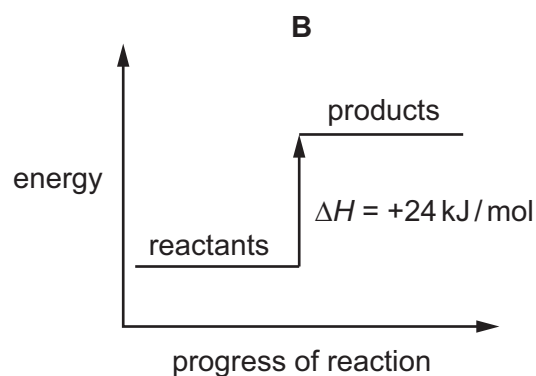
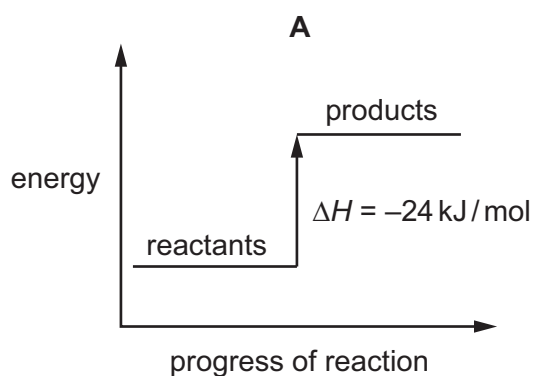
9 Graphite has a giant covalent structure.

Which statements about graphite are correct?

- 1 Carbon atoms form four covalent bonds with neighbouring atoms.
- 2 There are delocalised electrons between layers of carbon atoms.
- 3 Graphite is a useful lubricant.
- 4 Graphite is a good conductor of electricity.

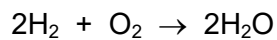
A 1 and 2 **B** 1, 3 and 4 **C** 2, 3 and 4 **D** 3 and 4 only

10 Which reaction pathway diagram represents an endothermic reaction?



11 Hydrogen burns in oxygen.

The equation for the reaction is shown.



The table shows the bond energies involved.

bond	bond energy in kJ/mol
H–H	436
O=O	498
O–H	464

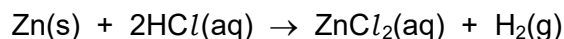
What is the energy given out during the reaction?

- A** –3226 kJ/mol
- B** –884 kJ/mol
- C** –486 kJ/mol
- D** –442 kJ/mol

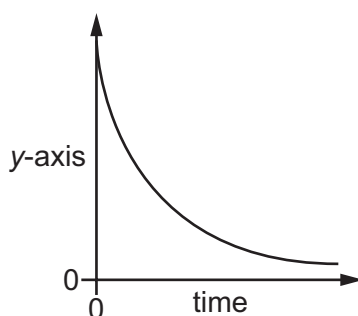
12 Which process involves a chemical change?

- A** adding sodium to water
- B** boiling water
- C** dissolving sodium chloride in water
- D** producing water from aqueous sodium chloride

- 13 An experiment is carried out to find the rate of reaction between hydrochloric acid and zinc.

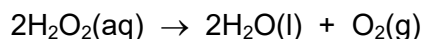


The results of the experiment are shown.



What is the label on the y-axis?

- A amount of ZnCl_2 produced
 - B concentration of HCl
 - C mass of Zn reacted
 - D volume of H_2 produced
- 14 Hydrogen peroxide, H_2O_2 , decomposes to form water and oxygen.



Manganese(IV) oxide catalyses the decomposition reaction.

The reaction is investigated in four experiments.

experiment	volume and concentration of hydrogen peroxide	conditions
1	12.5 cm ³ of 1.0 mol/dm ³	25 °C with manganese(IV) oxide powder added
2	12.5 cm ³ of 2.0 mol/dm ³	40 °C with manganese(IV) oxide powder added
3	25 cm ³ of 1.0 mol/dm ³	40 °C without manganese(IV) oxide powder
4	25 cm ³ of 1.0 mol/dm ³	40 °C with manganese(IV) oxide powder added

All reactions go to completion and all measurements of gas volumes are at room temperature and pressure.

Which statement is correct?

- A Experiment 1 produces less gas than experiment 4, but at the same rate.
- B Experiment 2 produces more gas than experiment 1, but at the same rate.
- C Experiment 2 and experiment 4 each produce the same volume of gas, but at different rates.
- D Experiment 3 and experiment 4 each produce the same volume of gas and at the same rate.

15 Sulfuric acid is produced by the Contact process.

Which row shows the typical conditions used in the process?

	catalyst	pressure / kPa	temperature / °C
A	iron	200	300
B	iron	20 000	450
C	vanadium(V) oxide	200	450
D	vanadium(V) oxide	20 000	300

16 Which equation shows the reduction of copper?

- A** $\text{CuO} + \text{C} \rightarrow \text{Cu} + \text{CO}$
- B** $2\text{CuS} + 3\text{O}_2 \rightarrow 2\text{CuO} + 2\text{SO}_2$
- C** $\text{Cu(g)} \rightarrow \text{Cu(l)}$
- D** $\text{Cu(l)} \rightarrow \text{Cu(s)}$

17 Which statement about acids is correct?

- A** A weak acid partially dissociates in aqueous solution.
- B** An acid accepts protons when added to water.
- C** Ethanoic acid acts as a strong acid when added to water.
- D** Hydrochloric acid is a strong acid that ionises in water to form H^+ ions.

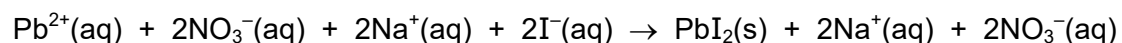
18 Copper(II) sulfate is formed by reacting excess solid copper(II) carbonate with dilute sulfuric acid.

Which processes are part of the preparation of solid copper(II) sulfate?

- 1 crystallisation
- 2 distillation
- 3 filtration
- 4 titration

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

19 Which type of reaction is represented by the equation shown?



- A addition
- B redox
- C neutralisation
- D precipitation

20 Which compound is likely to be coloured?

- A KMnO_4 B KNO_3 C K_2CO_3 D K_2SO_4

21 Which statements about the metal zinc are correct?

- 1 It is extracted from the ore bauxite.
- 2 It is used to galvanise steel.
- 3 It is used to make the alloy brass.
- 4 It reacts with dilute hydrochloric acid to produce hydrogen gas.

- A 1, 2 and 4 B 1, 3 and 4 C 2, 3 and 4 D 2 and 3 only

22 The electronic configurations of four elements, P, Q, R and S, are shown.

element	electronic configuration
P	2
Q	2,2
R	2,6
S	2,8

Which elements are unreactive monatomic gases?

- A P and Q B P and S C Q and R D S only

- 23** Which row compares the strength of alloys with pure metals and explains the difference in strength?

	strength of an alloy compared to a pure metal	explanation
A	weaker	larger atoms slide more easily over smaller atoms
B	weaker	larger atoms make it harder for layers to slide over one another
C	stronger	larger atoms slide more easily over smaller atoms
D	stronger	larger atoms make it harder for layers to slide over one another

- 24** Zinc oxide reacts with carbon to produce zinc.

Which equation represents this reaction?

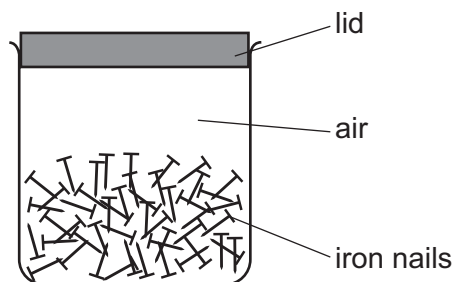
- A** $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}$
- B** $2\text{ZnO} + 2\text{C} \rightarrow 2\text{Zn} + 2\text{CO}_2$
- C** $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
- D** $\text{ZnO} + 2\text{C} \rightarrow \text{Zn} + 2\text{CO}_2$

- 25** When a piece of aluminium foil is added to dilute hydrochloric acid, no effervescence is seen.

Which statement explains why no effervescence is seen?

- A** Aluminium does not make a gas when it reacts with an acid.
- B** Aluminium has a surface layer of aluminium oxide.
- C** Aluminium is less reactive than hydrogen.
- D** Aluminium only reacts with concentrated acid.

26 Iron nails are stored in an airtight container.



The nails begin to rust after a few days.

How can the rusting of the nails be prevented?

- A Leave the lid off.
- B Replace the air with argon.
- C Put the container in a warm place.
- D Seal the container in a bag.

27 Four substances present in the blast furnace during iron extraction are listed.

- 1 calcium carbonate
- 2 carbon dioxide
- 3 carbon monoxide
- 4 iron(III) oxide

Which substances are both a reactant and a product during the reactions occurring in the blast furnace?

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

28 Aluminium is extracted from purified bauxite by electrolysis.

Which row shows the ionic half-equations for the reaction at each electrode?

	anode	cathode
A	$Al \rightarrow Al^{3+} + 3e^{-}$	$2O^{2-} + 4e^{-} \rightarrow O_2$
B	$Al^{3+} + 3e^{-} \rightarrow Al$	$2O^{2-} \rightarrow O_2 + 4e^{-}$
C	$2O^{2-} + 4e^{-} \rightarrow O_2$	$Al \rightarrow Al^{3+} + 3e^{-}$
D	$2O^{2-} \rightarrow O_2 + 4e^{-}$	$Al^{3+} + 3e^{-} \rightarrow Al$

29 Which test is used to show that a sample of water is pure?

- A Evaporate the water to see if any solids remain.
- B Heat the water to check its boiling point.
- C Test with anhydrous cobalt(II) chloride.
- D Use universal indicator paper to check its pH.

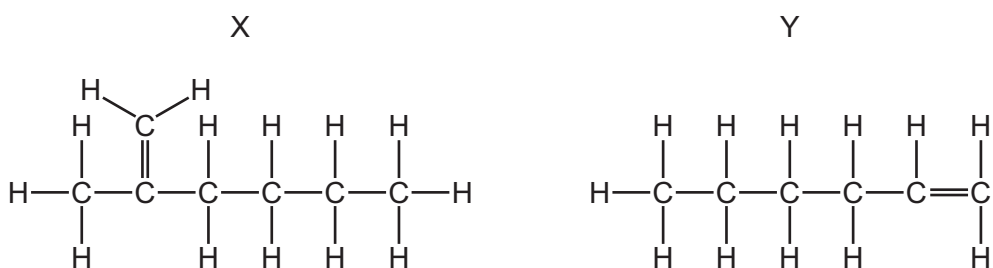
30 Catalytic converters in car exhausts change polluting gases into non-polluting gases.

Which statements about oxides of nitrogen and car engines are correct?

- 1 The nitrogen in oxides of nitrogen comes from compounds in gasoline.
- 2 The oxygen in oxides of nitrogen comes from the air in the car engine.
- 3 Catalytic converters convert oxides of nitrogen into nitrogen.

- A 1 and 2 B 2 and 3 C 2 only D 3 only

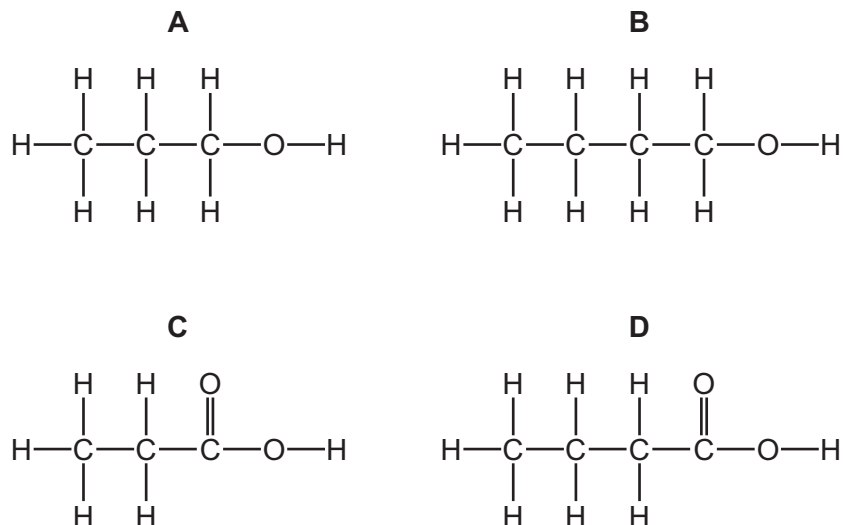
31 The structures of two molecules, X and Y, are shown.



Which row describes X and Y?

	structural isomers	belong to same homologous series
A	no	no
B	no	yes
C	yes	no
D	yes	yes

32 What is the structure of butanoic acid?



33 When a mixture of methane and chlorine is exposed to ultraviolet light, a reaction takes place.

Which statements about this reaction are correct?

- 1 It is an addition reaction.
- 2 The ultraviolet light provides the activation energy.
- 3 An equation for the reaction is $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{H}_2$.
- 4 CH_3Cl is made in the reaction.

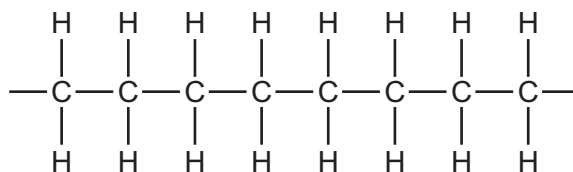
A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

34 Esters are formed when a carboxylic acid reacts with an alcohol.

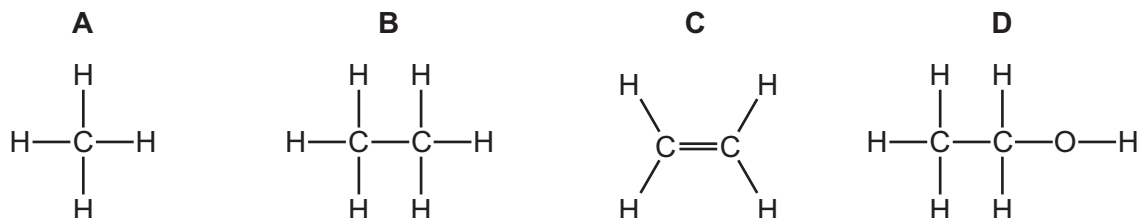
What is the catalyst for this reaction?

- A** aqueous potassium manganate(VII)
B iron
C sulfuric acid
D vanadium(V) oxide

35 The diagram shows part of a polymer.



Which diagram shows the monomer from which this polymer is made?



36 Nylon and PET are polymers.

Which statements about these polymers are correct?

- 1 They are both condensation polymers.
- 2 $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$ could be a monomer for both polymers.
- 3 The complete combustion of both polymers gives two products only.

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 and 3

37 Ethane is used as a fuel.

Which equation shows the complete combustion of ethane?

- A** $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$
- B** $2\text{C}_2\text{H}_6 + 5\text{O}_2 \rightarrow 4\text{CO} + 6\text{H}_2\text{O}$
- C** $\text{C}_2\text{H}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$
- D** $\text{C}_2\text{H}_4 + 2\text{O}_2 \rightarrow 2\text{CO} + 2\text{H}_2\text{O}$

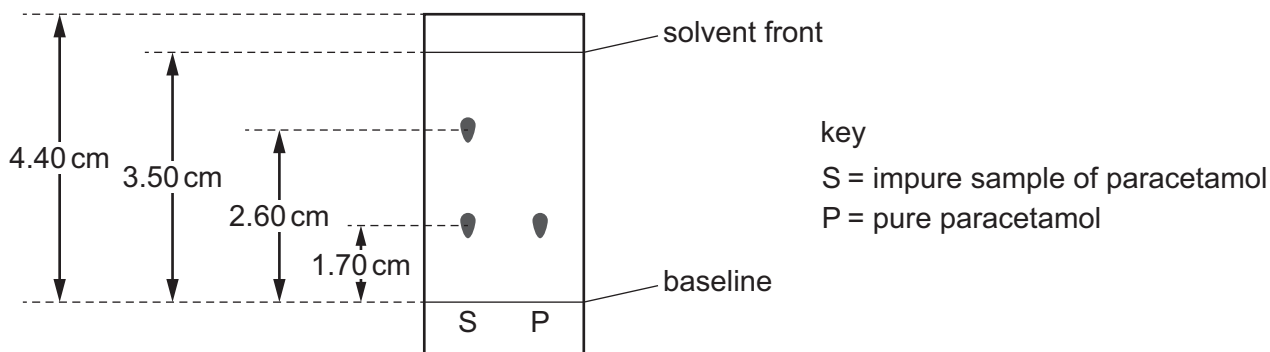
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- 38 The painkiller paracetamol is synthesised from 4-aminophenol.

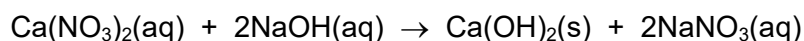
Chromatography is done on an impure sample of paracetamol. The results are shown. The diagram is not drawn to scale.



The sample of paracetamol is contaminated with 4-aminophenol only.

What is the R_f value of 4-aminophenol?

- A** 0.49 **B** 0.65 **C** 0.74 **D** 1.35
- 39 The equation for the reaction of aqueous calcium nitrate and aqueous sodium hydroxide is shown.



Which process is used to remove calcium hydroxide from the mixture?

- A** chromatography
B crystallisation
C distillation
D filtration
- 40 The results of two tests on aqueous compound X are given.

test	result
warm with aluminium foil and aqueous sodium hydroxide	ammonia is produced
aqueous sodium hydroxide	brown precipitate

What is X?

- A** iron(III) nitrate
B iron(II) nitrate
C iron(III) sulfate
D iron(II) sulfate

The Periodic Table of Elements

[illegible]

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/22

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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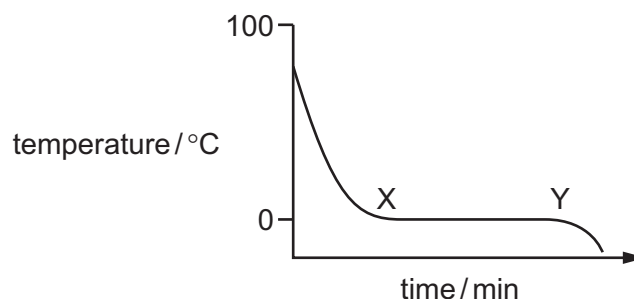
INFORMATION

- The total mark for this paper is 40.
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This document has **16** pages.

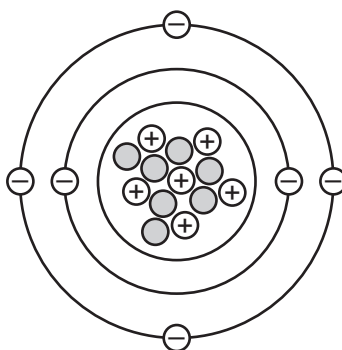


- 1 Part of a cooling curve for water is shown.



What is occurring between points X and Y?

- A** Steam is condensing into water.
 - B** The temperature of the water is decreasing.
 - C** Ice is melting.
 - D** Particles are losing heat to the surroundings.
- 2 Which statements about clean, dry air are correct?
- 1 It is a mixture of elements only.
 - 2 It is a mixture of elements and compounds.
 - 3 It contains only non-metals.
- A** 1 and 3 **B** 1 only **C** 2 and 3 **D** 2 only
- 3 A representation of an atom is shown.



What is the nucleon number of this atom?

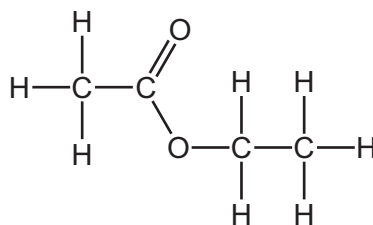
- A** 6 **B** 7 **C** 12 **D** 13

- 4 The percentage abundances of three isotopes in a sample of neon are shown.

isotope	percentage abundance / %
$^{20}_{10}\text{Ne}$	90.48
$^{21}_{10}\text{Ne}$	0.27
$^{22}_{10}\text{Ne}$	9.25

What is the relative atomic mass, A_r , of this sample of neon?

- A** 10.19 **B** 20.19 **C** 21.00 **D** 30.19
- 5 Potassium reacts with iodine to form potassium iodide.
- Which statement about potassium iodide is correct?
- A** Each potassium atom shares a pair of electrons with an iodine atom.
- B** In potassium iodide, the particles of potassium have more protons than electrons.
- C** Potassium iodide has a high melting point because it is a covalent compound.
- D** Potassium iodide has a low melting point because it is an ionic compound.
- 6 Which substance has the lowest melting point?
- A** graphite
- B** methanol
- C** silicon(IV) oxide
- D** sodium chloride
- 7 The diagram shows the structure of a molecule of ethyl ethanoate.



What is the molecular formula of a molecule of ethyl ethanoate?

- A** CHO **B** $\text{C}_4\text{H}_8\text{O}_2$ **C** $\text{C}_4(\text{H}_2)_2(\text{O}_2)$ **D** $\text{C}_2\text{H}_4\text{O}$

- 8 A hydrocarbon contains 85.7% of carbon by mass.

What is the empirical formula of the hydrocarbon?

A CH₂ **B** CH₄ **C** C₂H₅ **D** C₃H₆

- 9 The formula of a compound containing element X is Na₂X₂O₃.

The relative formula mass of the compound is 158.

What is the relative atomic mass of X?

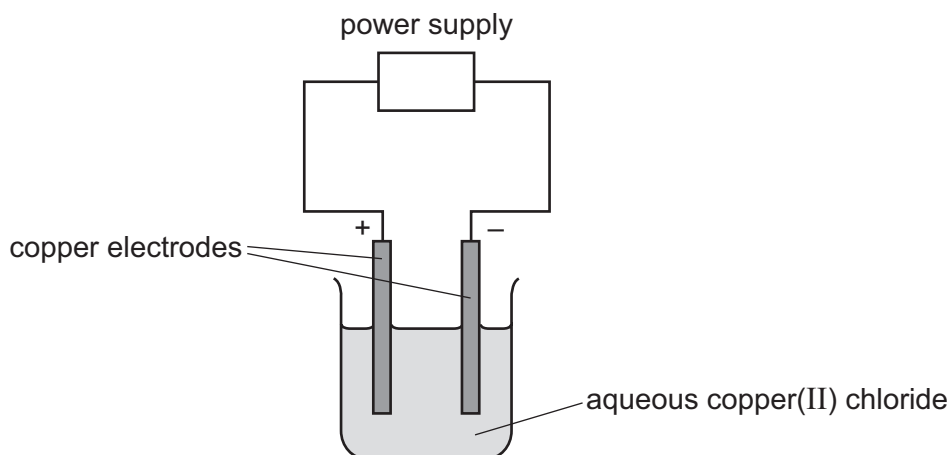
A 32 **B** 59.5 **C** 64 **D** 119

- 10 Dilute aqueous potassium chloride is electrolysed using platinum electrodes.

Which row identifies the product at each electrode?

	anode	cathode
A	chlorine	hydrogen
B	chlorine	potassium
C	oxygen	hydrogen
D	oxygen	potassium

- 11 Concentrated aqueous copper(II) chloride is electrolysed using copper electrodes, as shown.



What happens to the mass of each electrode during this process?

	positive electrode	negative electrode
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 12 The initial and final temperatures of four different reactions are measured.

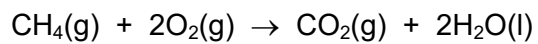
Which reaction is the **least** exothermic?

	initial temperature / °C	final temperature / °C
A	19	25
B	21	18
C	22	17
D	22	26

- 13 Which equation represents an endothermic reaction?

- A** $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$
B $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$
C $\text{H}(\text{g}) + \text{H}(\text{g}) \rightarrow \text{H}_2(\text{g})$
D $2\text{K}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{KOH}(\text{aq}) + \text{H}_2(\text{g})$

14 Methane burns in oxygen to form carbon dioxide and water.



The bond energies are shown.

bond	bond energy in kJ/mol
C–H	410
C–O	360
C=O	805
O–H	460
O–O	146
O=O	496

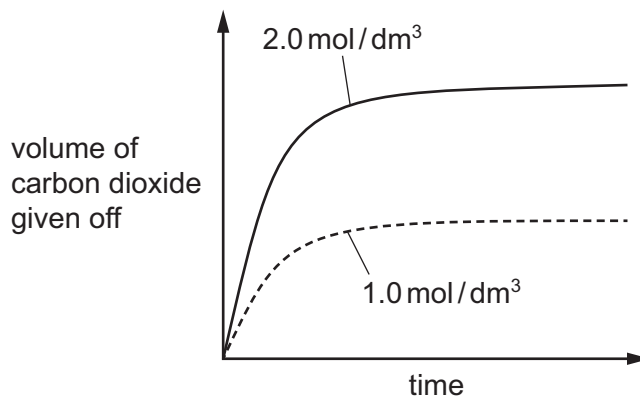
What is the energy change for this reaction?

A –818 kJ/mol **B** –102 kJ/mol **C** +102 kJ/mol **D** +818 kJ/mol

- 15** Hydrochloric acid is added to excess calcium carbonate in two separate experiments.

Two different concentrations of hydrochloric acid are used but the temperature is the same in both experiments.

The graph of the results shows the volume of carbon dioxide gas given off over time.

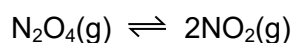


Which row is correct?

	particles in 2.0 mol/dm ³ compared to 1.0 mol/dm ³	
	collision rate	collision energy
A	higher	no change
B	higher	higher
C	lower	no change
D	lower	higher

- 16** The decomposition of dinitrogen tetroxide, N₂O₄, into nitrogen dioxide, NO₂, is a reversible reaction.

The equation for the reaction is shown.

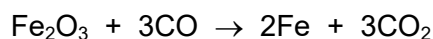


The forward reaction is endothermic.

Which row shows the effect on the position of equilibrium and the rate of the reverse reaction when the temperature is increased?

	position of equilibrium	rate of the reverse reaction
A	shifts to the left	decreases
B	shifts to the left	increases
C	shifts to the right	decreases
D	shifts to the right	increases

- 17 In a blast furnace, iron(III) oxide is converted to iron and carbon monoxide is converted to carbon dioxide.



What happens to each of these reactants?

- A Both iron(III) oxide and carbon monoxide are oxidised.
- B Both iron(III) oxide and carbon monoxide are reduced.
- C Iron(III) oxide is oxidised and carbon monoxide is reduced.
- D Iron(III) oxide is reduced and carbon monoxide is oxidised.

- 18 Which row describes what happens to Fe^{2+} ions when they are oxidised?

	electron movement	oxidation number of iron
A	they gain electrons	decreases
B	they gain electrons	increases
C	they lose electrons	decreases
D	they lose electrons	increases

- 19 In which reaction does an acid react with a base?

- A Dilute sulfuric acid is added to a piece of magnesium ribbon producing hydrogen.
- B Dilute sulfuric acid is added to aqueous barium chloride producing a white precipitate of barium sulfate.
- C Aqueous sodium hydroxide is added to aqueous copper(II) sulfate producing a blue precipitate of copper(II) hydroxide.
- D Aqueous sodium hydroxide is added to solid ammonium sulfate producing gaseous ammonia.

- 20 Which element forms an oxide that reacts with an aqueous solution of a base?

- A argon
- B sulfur
- C magnesium
- D copper

21 Which method is used to produce insoluble salts?

- A addition of excess insoluble base to an acid
- B addition of excess metal to an acid
- C precipitation using two aqueous solutions
- D titration using an acid and an alkali

22 The noble gases are in Group VIII of the Periodic Table.

Some properties of the first four noble gases are shown.

noble gas	boiling point in °C	density in g/dm ³
helium	−267	0.179
neon	−246	0.900
argon	−186	1.782
krypton	−152	3.708

Which row identifies the trends in boiling point and in density as Group VIII is descended?

	boiling point	density
A	decreasing	increasing
B	increasing	increasing
C	decreasing	decreasing
D	increasing	decreasing

23 Some properties of element R are shown.

melting point in °C	98
boiling point in °C	883
reaction with cold water	gives off H ₂ gas
reaction when heated with oxygen	burns to give a white solid

In which part of the Periodic Table is R found?

- A Group I
- B Group VII
- C Group VIII
- D transition elements

24 Which pair of compounds shows that transition elements have variable oxidation states?

- A** Cr_2O_3 and CrBr_3
- B** CuSO_4 and CuCl_2
- C** Fe_2O_3 and FeCl_2
- D** NiO and NiCl_2

25 The list gives the order of some metals and hydrogen in the reactivity series.

Metal X is also included.

most reactive	K
	Mg
	Zn
	H
	X
least reactive	Cu

Which row shows the properties of metal X?

	reacts with dilute acids	oxide reduced by carbon
A	no	no
B	no	yes
C	yes	no
D	yes	yes

26 When zinc is added to an aqueous solution containing magnesium ions, there is no reaction.

Which species has the greatest tendency to lose electrons?

- A** Mg
- B** Mg^{2+}
- C** Zn
- D** Zn^{2+}

27 Which gas in the air is needed for iron to rust?

- A** argon
- B** carbon dioxide
- C** nitrogen
- D** oxygen

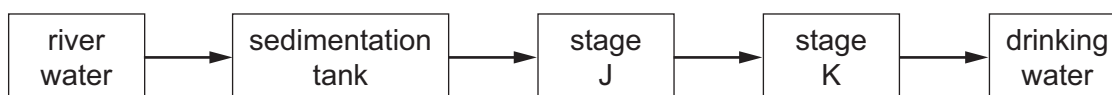
28 Which coating prevents iron from rusting even when the coating is damaged?

- A grease
- B paint
- C plastic
- D zinc

29 Why is limestone added to the blast furnace?

- A It neutralises the molten slag produced.
- B It reacts with impurities to form slag.
- C It releases carbon dioxide which reduces the iron(III) oxide.
- D It removes acidic gases such as carbon dioxide.

30 The flow chart shows stages in the treatment of river water to produce drinking water.



What occurs at stages J and K?

	J	K
A	distillation	chlorination
B	distillation	filtration
C	filtration	chlorination
D	filtration	distillation

- 31** Carbon dioxide acts as a greenhouse gas by interacting with a particular type of energy that radiates from the Earth's surface into the atmosphere.

Which type of energy is involved and what happens when this energy interacts with carbon dioxide molecules?

	type of energy involved	what happens
A	thermal	carbon dioxide molecules increase the Earth's energy loss to space
B	thermal	carbon dioxide molecules absorb the energy
C	light	carbon dioxide molecules increase the Earth's energy loss to space
D	light	carbon dioxide molecules absorb the energy

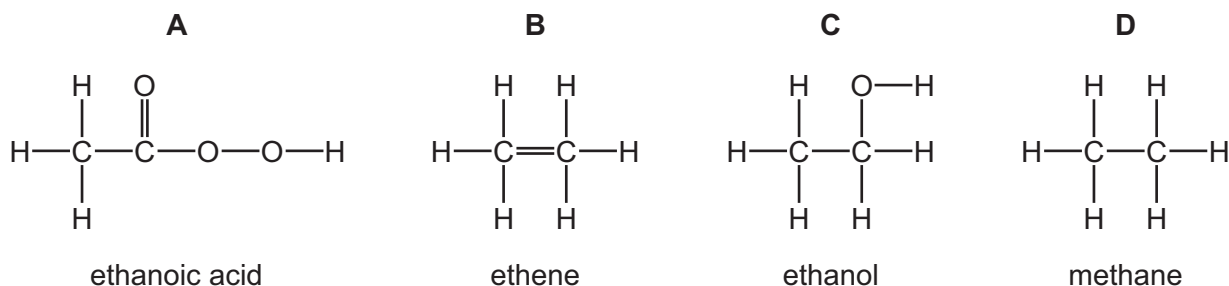
- 32** Oxides of nitrogen, such as NO and NO₂, are formed in the petrol engines of cars.

They are removed from the exhaust gases by reactions in the car's catalytic converter.

Which row describes how oxides of nitrogen are formed in a petrol engine and a reaction that happens in the catalytic converter?

	how oxides of nitrogen are formed	a reaction that happens in the catalytic converter
A	by the reaction between nitrogen and oxygen from the air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
B	by the reaction between nitrogen and oxygen from the air	$2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$
C	by the reaction between nitrogen compounds in petrol and oxygen from the air	$2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$
D	by the reaction between nitrogen compounds in petrol and oxygen from the air	$2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$

- 33** Which diagram shows the displayed formula for the named organic compound?



34 What is the total number of covalent bonds in a molecule of butane, C_4H_{10} ?

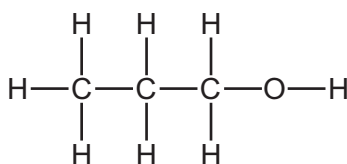
- A 3 B 10 C 13 D 14

35 Propane reacts with chlorine in a substitution reaction.

Which reaction condition is required for the reaction to occur?

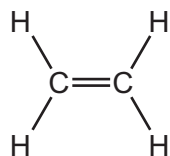
- A acid catalyst
B iron catalyst
C temperature of 400°C
D ultraviolet light

36 The structure of an organic compound is shown.

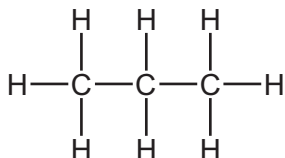


Which structure represents a molecule that reacts with steam to produce this product?

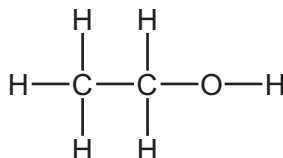
A



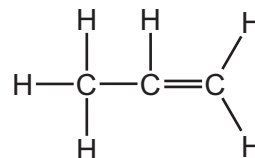
B



C



D

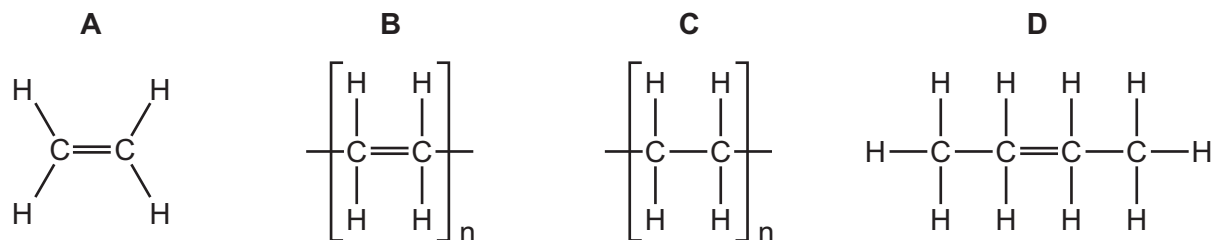


37 Which term describes nylon?

- A addition polymer
B natural polymer
C polyamide
D polyester

38 Ethene can be polymerised.

Which diagram represents the structure of the product formed?



39 An acid–base titration is described.

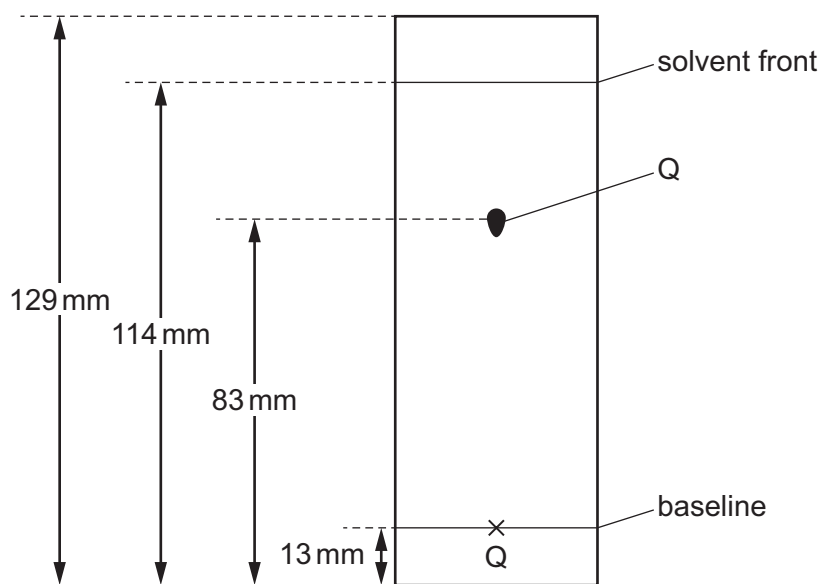
- 25.0 cm³ of dilute aqueous alkali is put into a conical flask.
- Indicator is added to the flask.
- Dilute acid is added to the aqueous alkali until the indicator changes colour.
- The volume of acid used is then recorded.

Which use of apparatus is correct?

- A** The 25.0 cm³ of aqueous alkali is measured using a volumetric pipette.
- B** The 25.0 cm³ of aqueous alkali is measured using the lines on the conical flask.
- C** The volume of acid is measured using a measuring cylinder.
- D** The volume of acid is measured using a volumetric pipette.

40 Substance Q is investigated using chromatography.

The chromatogram is shown. The diagram is not drawn to scale.



What is the R_f value of Q?

- A** 0.60 **B** 0.64 **C** 0.69 **D** 0.72

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The Periodic Table of Elements

Group																																																					
I	II	Key										III	IV	V	VI	VII	VIII																																				
3 Li lithium 7	4 Be beryllium 9	atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	2 He helium 4																																				
		1 H hydrogen 1																																																			
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84																												
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesion —

lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	ytterbium	173	71	Lu	lutetium	175	
	actinoids	89	Ac	actinium	—	90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium	—	94	Pu	plutonium	—	95	Am	americium	—	96	Cm	curium	—	97	Bk	berkelium	—	98	Cf	californium	—	99	Es	einsteinium	—	100	Fm	fermium	—	101	Md	mendelevium	—	102	No	nobelium	—	103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/23

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

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- The Periodic Table is printed in the question paper.

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- 1 A sample of a gas occupies 340 cm^3 at room temperature and pressure.

The temperature and pressure are both increased, but the volume occupied by the gas remains 340 cm^3 .

Which row describes what happens to the particle speed and the average distance between the particles in the gas when the temperature and pressure are both increased?

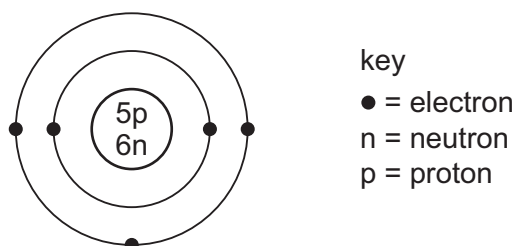
	particle speed	average distance between particles
A	unchanged	unchanged
B	unchanged	increased
C	increased	unchanged
D	increased	increased

- 2 Which statements about the rate of diffusion of the gases ammonia, carbon monoxide, nitrogen and oxygen are correct?

- 1 Nitrogen and carbon monoxide will diffuse at the same rate.
- 2 Oxygen will diffuse slowest because it is an element, whereas the others are compounds.
- 3 Ammonia will diffuse fastest.

A 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 and 3

- 3 The structure of an atom of element X is shown.



What is element X?

- A** boron
B carbon
C sodium
D sulfur

- 4 Which statement explains why isotopes of an element have the same chemical reactions?
- A** They have different numbers of neutrons.
- B** They have ions with different numbers of electrons.
- C** They have the same number of outer shell electrons.
- D** They have the same number of protons.

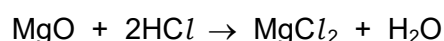
- 5 Magnesium reacts with oxygen to form magnesium oxide.

What happens to magnesium atoms and oxygen atoms during this reaction?

- A** Magnesium and oxygen share two electrons.
- B** Magnesium gains two electrons and oxygen loses two electrons.
- C** Magnesium loses one electron and oxygen gains one electron.
- D** Magnesium loses two electrons and oxygen gains two electrons.
- 6 Which row about the properties of both diamond and silicon(IV) oxide is correct?

	conductor of electricity	type of molecule
A	yes	giant covalent
B	yes	simple covalent
C	no	giant covalent
D	no	simple covalent

- 7 The equation represents the reaction between solid magnesium oxide and dilute hydrochloric acid to form magnesium chloride and water.



Which row shows the state symbols for hydrochloric acid, magnesium chloride and water?

	HCl	MgCl ₂	H ₂ O
A	(aq)	(aq)	(l)
B	(aq)	(l)	(l)
C	(l)	(aq)	(aq)
D	(l)	(l)	(aq)

8 Which substance is a mixture?

- A air
- B graphite
- C oxygen
- D water

9 The number of moles of atoms X, Y and Z, in a compound, are shown.

atom	moles
X	0.6
Y	1.2
Z	0.3

What is the formula of the compound?

- A XY_2Z_4 B XY_4Z_2 C X_2YZ_4 D $\text{X}_2\text{Y}_4\text{Z}$

10 1.0 mol of silver nitrate, AgNO_3 , contains 1.2×10^{24} ions.

How many ions are there in 0.25 mol of iron(III) oxide, Fe_2O_3 ?

- A 1.5×10^{23} B 3.0×10^{23} C 7.5×10^{23} D 3.0×10^{24}

11 Concentrated aqueous magnesium bromide is electrolysed using carbon electrodes.

Which equations represent the reactions occurring at each electrode?

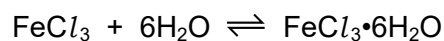
	positive electrode	negative electrode
A	$2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{e}^-$	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$
B	$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$	$2\text{O}^{2-}(\text{aq}) \rightarrow \text{O}_2(\text{aq}) + 4\text{e}^-$
C	$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	$2\text{Br}^-(\text{aq}) \rightarrow \text{Br}_2(\text{aq}) + 2\text{e}^-$
D	$2\text{O}^{2-}(\text{aq}) \rightarrow \text{O}_2(\text{aq}) + 4\text{e}^-$	$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$

12 Aqueous copper(II) sulfate is electrolysed using carbon electrodes.

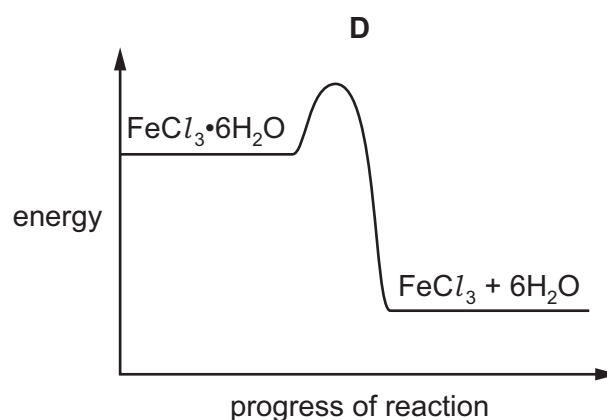
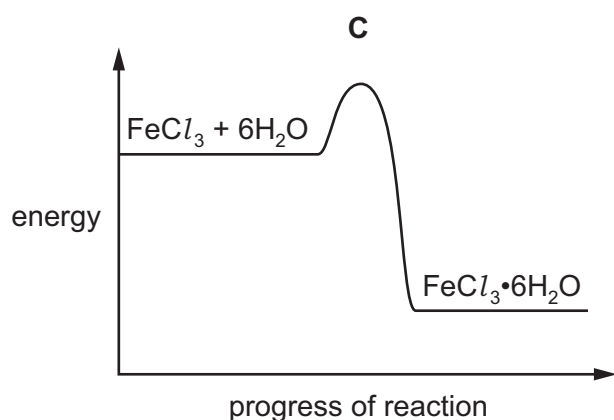
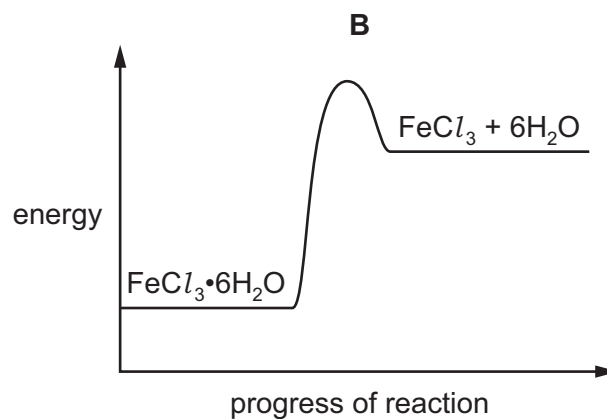
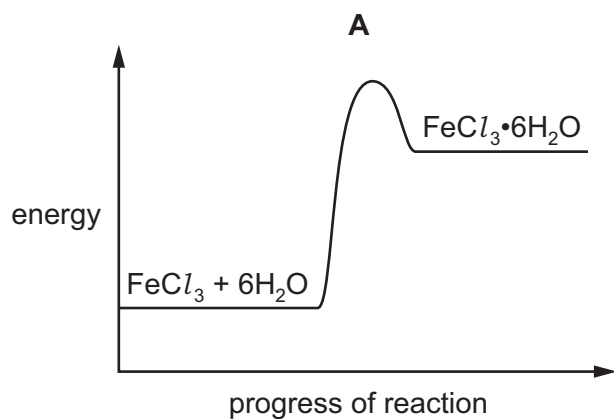
Which statement is correct?

- A Bubbles of hydrogen gas are formed at the anode.
- B Bubbles of oxygen gas are formed at the cathode.
- C Copper is deposited at the anode.
- D The blue colour of the solution fades.

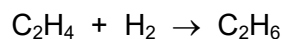
- 13 When water is added to anhydrous iron(III) chloride, FeCl_3 , hydrated iron(III) chloride, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, is formed and energy is given out.



Which reaction pathway diagram represents the formation of anhydrous iron(III) chloride in the **reverse** reaction?



14 Ethene reacts with hydrogen. The equation is shown.



The bond energies are shown.

bond	bond energy in kJ/mol
C–C	+350
C=C	+610
C–H	+410
H–H	+436

What is the energy change for the reaction?

A –560 kJ/mol **B** –124 kJ/mol **C** +486 kJ/mol **D** +5496 kJ/mol

15 Statements about four different acids are listed.

- A 0.0100 mol/dm³ solution of hydrochloric acid has a pH of 2.
- A 0.0100 mol/dm³ solution of ethanoic acid has a pH of 3.4.
- Hydrobromic acid, HBr, is a strong acid.
- Ethanoic acid is a slightly stronger acid than trimethylethanoic acid.

What are the pH values of 0.0100 mol/dm³ HBr and 0.0100 mol/dm³ trimethylethanoic acid?

	pH of 0.0100 mol/dm ³ HBr	pH of 0.0100 mol/dm ³ trimethylethanoic acid
A	2	3.3
B	2	3.5
C	3.4	3.3
D	3.4	3.5

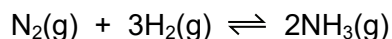
16 Anhydrous cobalt(II) chloride is blue and turns pink when water is added.

How is this reaction reversed?

- A** adding dilute acid
- B** filtering
- C** heating
- D** cooling

- 17 The reaction between hydrogen and nitrogen is reversible.

The forward reaction is exothermic.



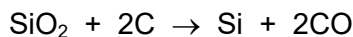
Which change to the conditions would increase the yield of ammonia?

- A add a catalyst
 - B increase the pressure
 - C increase the temperature
 - D reduce the concentration of nitrogen
- 18 Ethanol can be turned into ethanoic acid by passing it over hot copper(II) oxide.



What is this type of reaction?

- A precipitation
 - B redox
 - C thermal decomposition
 - D neutralisation
- 19 When heated strongly, silicon(IV) oxide reacts with carbon.



Which term describes what happens to silicon(IV) oxide?

- A thermal decomposition
 - B neutralisation
 - C oxidation
 - D reduction
- 20 Which statement about aqueous weak acids is correct?
- A Weak acids are always dilute aqueous solutions.
 - B Weak acids dissociate fully in aqueous solution.
 - C When a weak acid is added to blue litmus paper, it stays blue.
 - D When a weak acid is added to solid magnesium, effervescence is seen.

21 Which oxides are basic?

- 1 calcium oxide
- 2 sodium oxide
- 3 iron(II) oxide

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 3 only

22 Zinc oxide is an amphoteric oxide.

Zinc oxide is added to excess dilute hydrochloric acid.

Zinc oxide is added to excess aqueous sodium hydroxide.

Which row describes the observations made in these reactions?

	excess dilute hydrochloric acid	excess aqueous sodium hydroxide
A	colourless solution forms	colourless solution forms
B	colourless solution forms	no visible change
C	fizzing	colourless solution forms
D	fizzing	no visible change

23 Which row shows properties of an element that is in the same group of the Periodic Table as lithium?

	electrical conductivity	density in g/cm ³
A	high	0.97
B	high	8.93
C	low	0.07
D	low	3.12

24 The elements in Group VII include chlorine, bromine and iodine.

Which statements are correct?

- 1 Iodine is more dense than chlorine.
- 2 Iodine displaces chlorine from a solution containing chloride ions.
- 3 Bromine is a diatomic non-metal.
- 4 Chlorine gas is darker in colour than bromine vapour.

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

25 Cobalt is a transition element.

What is a property of cobalt?

- A** It can form coloured compounds.
- B** It is a poor electrical conductor.
- C** It has a low density.
- D** It has a low melting point.

26 Which metal has variable oxidation numbers?

- A** aluminium
- B** calcium
- C** copper
- D** sodium

27 Which statement about alloys is correct?

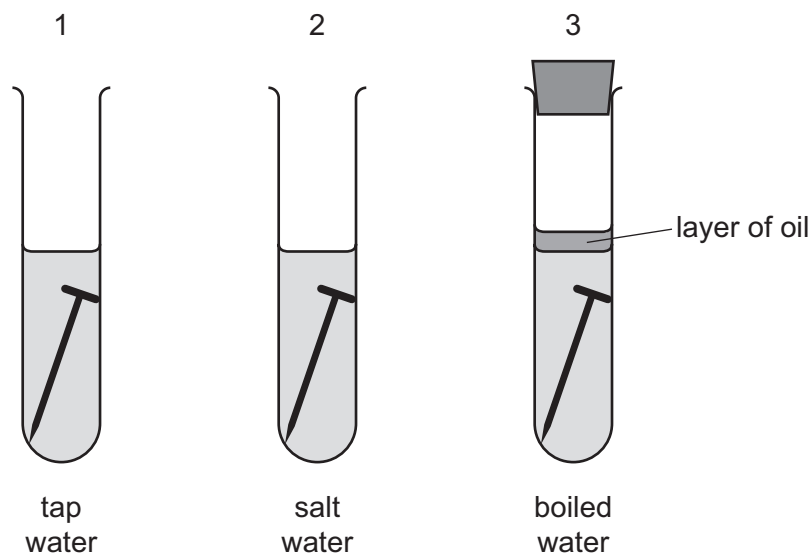
- A** Alloys are pure metal elements.
- B** At least two or more metals react together to make alloys.
- C** Alloys can be harder and stronger than a pure metal.
- D** Steel is **not** an alloy because it can contain the non-metal carbon.

28 A metal M is between sodium and magnesium in the reactivity series.

Which reactions occur with M and its oxide?

	M reacts with steam	M can be extracted by heating its oxide with carbon
A	no	no
B	no	yes
C	yes	no
D	yes	yes

29 The diagrams show experiments to investigate rusting of iron nails.



In which test-tubes do the nails rust?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 1 only

30 Which equation represents a reaction that takes place when iron is extracted from its ore in the blast furnace?

- A** $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
B $\text{CaO} + \text{CO}_2 \rightarrow \text{CaCO}_3$
C $2\text{CO} \rightarrow \text{C} + \text{CO}_2$
D $2\text{Fe} + 3\text{CO}_2 \rightarrow \text{Fe}_2\text{CO}_3 + 3\text{CO}$

31 Some uses of water are listed.

- 1 for drinking
- 2 in chemical reactions
- 3 in swimming pools
- 4 in washing

For which uses is it necessary to chlorinate the water?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

- 32** Oxides of nitrogen are formed in car engines and are a source of air pollution.

To decrease this pollution, catalytic converters are fitted to car exhausts.

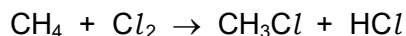
What happens to the oxides of nitrogen in the catalytic converter?

- A** combustion
- B** cracking
- C** oxidation
- D** reduction

- 33** Which pair of compounds are structural isomers of each other?

- A** $\text{CH}_3\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- B** $\text{CH}_2=\text{CHCH}_3$ and $\text{CH}_3\text{CH}=\text{CH}_2$
- C** $\text{CH}_2(\text{OH})\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ and $\text{CH}_3\text{COOCH}_2\text{CH}_3$

- 34** Methane reacts with chlorine in sunlight.



Which statements about this reaction are correct?

- 1 It is a substitution reaction.
- 2 It is an addition reaction.
- 3 It is a photochemical reaction.
- 4 It is catalysed by nickel.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 35** Propene reacts with bromine to give one product only.

What is the formula of the product?

- A** $\text{CH}_3\text{CH}_2\text{CHBr}_2$
- B** $\text{CH}_2\text{BrCH}_2\text{CH}_2\text{Br}$
- C** $\text{CH}_3\text{CHBrCH}_2\text{Br}$
- D** $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$

36 Ethanol can be manufactured by fermentation or by the catalytic addition of steam to ethene.

Which statements describe an advantage of manufacturing ethanol by fermentation?

- 1 The yield of ethanol is low.
- 2 The method uses a batch process.
- 3 The process takes place at a lower temperature.
- 4 The ethanol is made from a renewable source.

A 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

37 A compound with the formula $\text{CH}_3\text{COOC}_2\text{H}_5$ is formed from ethanol in two separate reactions.

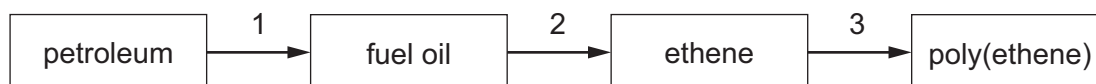
reaction 1 Ethanol reacts to form ethanoic acid.

reaction 2 Ethanoic acid and ethanol react together to form $\text{CH}_3\text{COOC}_2\text{H}_5$.

Which row describes reaction 1 and reaction 2?

	reaction 1	reaction 2
A	oxidation	ester formation
B	oxidation	addition
C	reduction	ester formation
D	reduction	addition

38 The flow diagram shows how poly(ethene) may be made from petroleum.



What are stages 1, 2 and 3?

	1	2	3
A	cracking	polymerisation	fractional distillation
B	cracking	fractional distillation	polymerisation
C	fractional distillation	cracking	polymerisation
D	fractional distillation	polymerisation	cracking

39 R_f values are used to identify unknown substances using paper chromatography.

Which statements about R_f values are correct?

- 1 R_f values are always less than 1.0.
- 2 R_f value = distance travelled by solvent $\div$ distance travelled by unknown substance.
- 3 The higher the R_f value, the further the unknown substance travels.
- 4 R_f values are **not** affected by the solubility of the unknown substance.

A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 and 4

40 The results of some tests on an aqueous solution of substance X are listed.

- 1 A cream precipitate is produced when adding aqueous silver nitrate.
- 2 Adding aqueous sodium hydroxide produces a green precipitate which dissolves in excess alkali.
- 3 Adding aqueous ammonia produces a green precipitate which is insoluble in excess ammonia.

What is substance X?

- A** chromium(III) bromide
B chromium(III) chloride
C iron(II) bromide
D iron(II) chloride

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The Periodic Table of Elements

Group																	
I	II	1 H hydrogen 1										III	IV	V	VI	VII	VIII
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
	11 Na sodium 23																
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —

lanthanoids

actinoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



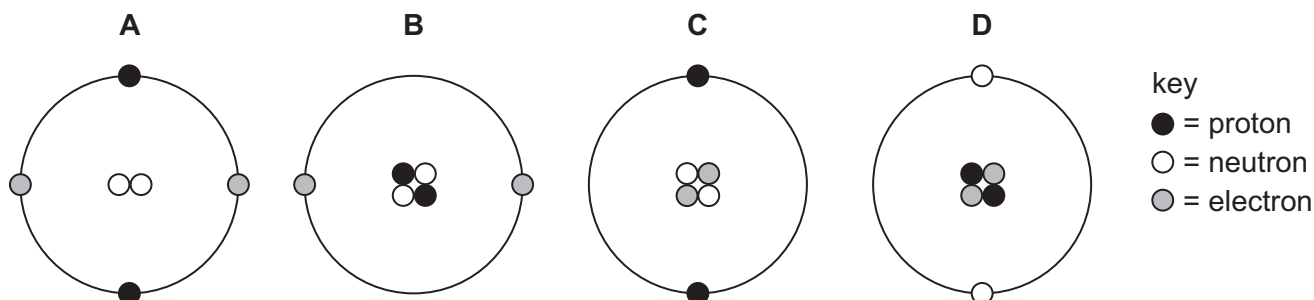
1 Which row describes the arrangement and motion of the particles in a liquid?

	arrangement	motion
A	random and particles are touching	moving slowly
B	random with space between all particles	moving slowly
C	an ordered lattice with all particles touching	moving slowly
D	an ordered lattice with space between all particles	moving quickly

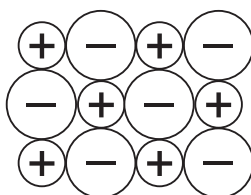
2 Which gas has the lowest rate of diffusion at room temperature and pressure?

- A** the gas produced when ammonium chloride is heated with aqueous sodium hydroxide
- B** the gas which makes up approximately 78% of clean, dry air
- C** the gas produced when sodium carbonate is added to dilute hydrochloric acid
- D** the gas produced when zinc is added to dilute sulfuric acid

3 Which diagram represents one helium atom?



4 The diagram shows part of an ionic lattice structure.



Which compound does the diagram represent?

- A** potassium bromide
- B** sodium oxide
- C** magnesium chloride
- D** carbon monoxide

- 5 Which statement about nitrogen molecules and ethene molecules is correct?
- A** A nitrogen molecule has 2 more shared electrons than an ethene molecule.
- B** An ethene molecule has 3 more shared electrons than a nitrogen molecule.
- C** A nitrogen molecule has 4 more shared electrons than an ethene molecule.
- D** An ethene molecule has 6 more shared electrons than a nitrogen molecule.

- 6 Sulfur is a simple molecule with the formula S₈.

Which row describes and explains the melting point of sulfur?

	melting point	explanation
A	high	the covalent bonds between sulfur atoms are strong
B	high	the covalent bonds between sulfur molecules are strong
C	low	the forces of attraction between sulfur atoms are weak
D	low	the forces of attraction between sulfur molecules are weak

- 7 Which row identifies a property and an explanation of the property for both diamond and silicon(IV) oxide?

	property	explanation of property
A	very hard	diamond has a giant covalent structure and silicon(IV) oxide has a giant ionic structure
B	high melting point	both have giant covalent structures with many strong bonds between the atoms
C	good lubricant	both have layers of atoms, which can slide over each other
D	poor conductor	both contain only non-metal elements and are simple molecules

- 8 Which statement about the structure of metals explains why metals are malleable?
- A** The electrons can move freely throughout the lattice.
- B** The layers of metal ions can slide over each other.
- C** The metal ions are positively charged.
- D** There is a strong force of attraction between the metal ions and the electrons.

9 What is the formula of iron(III) oxide?

- A FeO B Fe₃O₄ C FeO₂ D Fe₂O₃

10 Calcium carbonate is heated. Calcium oxide and carbon dioxide gas are formed.

The equation for the reaction is shown.



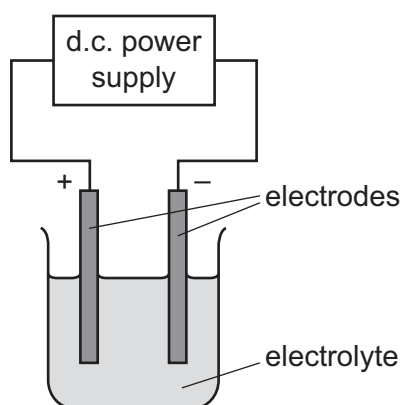
225 kg of calcium carbonate is heated until there is no further change in mass.

The yield of calcium oxide is 85 kg.

What is the percentage yield?

- A 37.8% B 47.2% C 67.5% D 85.0%

11 The apparatus used for electrolysis is shown.



Which statement is correct?

- A Copper forms at the anode in some electrolysis reactions.
B Hydrogen forms at the cathode in some electrolysis reactions.
C Oxygen forms at the cathode in some electrolysis reactions.
D Sodium forms at the anode in some electrolysis reactions.

12 Which statement about the electrolysis of aqueous copper(II) sulfate is correct?

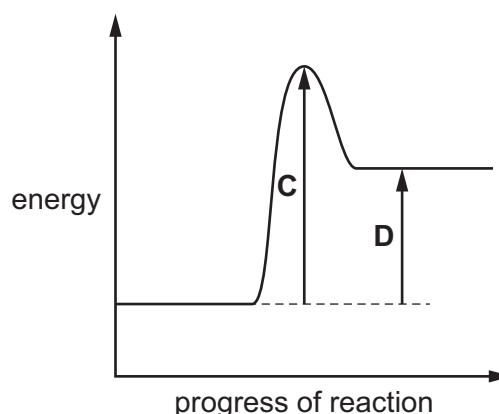
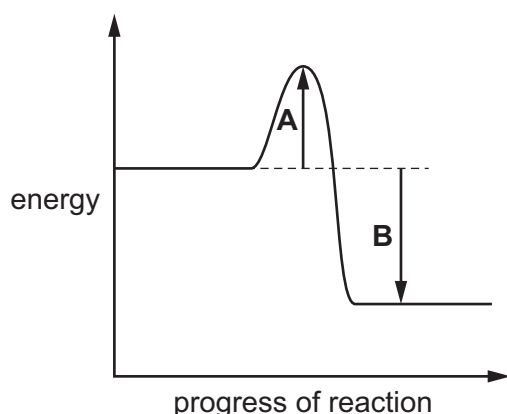
- A When copper electrodes are used, the solution turns from blue to colourless.
B When graphite electrodes are used, bubbles of gas are formed at the cathode.
C When copper electrodes are used, the anode gets smaller.
D When graphite electrodes are used, the colour of the solution does **not** change.

13 Which statement describes an advantage of using a hydrogen–oxygen fuel cell in a car compared to a gasoline engine?

- A The hydrogen is difficult to store.
- B The hydrogen is highly flammable.
- C The hydrogen used is made from hydrocarbons.
- D The only chemical product is water.

14 Two reaction pathway diagrams are shown.

Which arrow represents the activation energy for a reaction which releases thermal energy?



15 Which statements about the Haber process are correct?

- 1 A high temperature is used because the reaction is slow at room temperature.
- 2 A high pressure is used because there are more moles of gaseous reactants than moles of gaseous product.
- 3 A nickel catalyst is used to increase the rate of reaction.
- 4 An iron catalyst is used to increase the equilibrium yield of ammonia.

- A 1 and 2 B 1 and 4 C 2 and 3 D 4 only

16 Which substance is a raw material used to manufacture sulfuric acid?

- A vanadium(V) oxide
- B sulfur
- C sulfur dioxide
- D sulfur trioxide

- 17 Which colours are seen when litmus and methyl orange are added to separate samples of aqueous sodium hydroxide?

	litmus	methyl orange
A	blue	orange
B	blue	yellow
C	purple	orange
D	purple	yellow

- 18 Information about the solubility in water of four oxides is shown.

Which oxide, when added to water, gives a solution with a pH less than pH 7?

	name of oxide	solubility in water
A	nitrogen dioxide	soluble
B	copper(II) oxide	insoluble
C	silicon(IV) oxide	insoluble
D	barium oxide	soluble

- 19 Copper(II) sulfate is made when copper(II) carbonate reacts with dilute sulfuric acid.



Pure copper(II) sulfate crystals are obtained.

Which reagent is in excess and how are the crystals obtained?

	reagent in excess	how the crystals are obtained
A	copper(II) carbonate	filter and evaporate the solution to dryness
B	copper(II) carbonate	filter, evaporate the solution to crystallising point and then cool
C	dilute sulfuric acid	evaporate the solution to dryness
D	dilute sulfuric acid	evaporate the solution to crystallising point and then cool

- 20 Which statement about elements in Group I or Group VII of the Periodic Table is correct?

- A** Bromine reacts with potassium chloride to produce chlorine.
- B** Iodine is a monatomic non-metal.
- C** Lithium has a higher melting point than potassium.
- D** Sodium is more reactive with water than potassium.

21 Some information about an element from Group VII of the Periodic Table is shown.

melting point/°C	–7
boiling point/°C	59

What is the element?

- A fluorine
- B chlorine
- C bromine
- D iodine

22 Manganese(IV) oxide, MnO_2 , is a black solid.

The equation for the reaction between manganese(IV) oxide and dilute hydrochloric acid is shown.



The reaction produces a pale pink solution.

Which properties of transition elements does this reaction show?

- 1 They can act as catalysts.
- 2 They form coloured compounds.
- 3 They have high melting points.
- 4 They have variable oxidation numbers.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

23 Part of a steel ship is protected from rusting using a sacrificial metal.

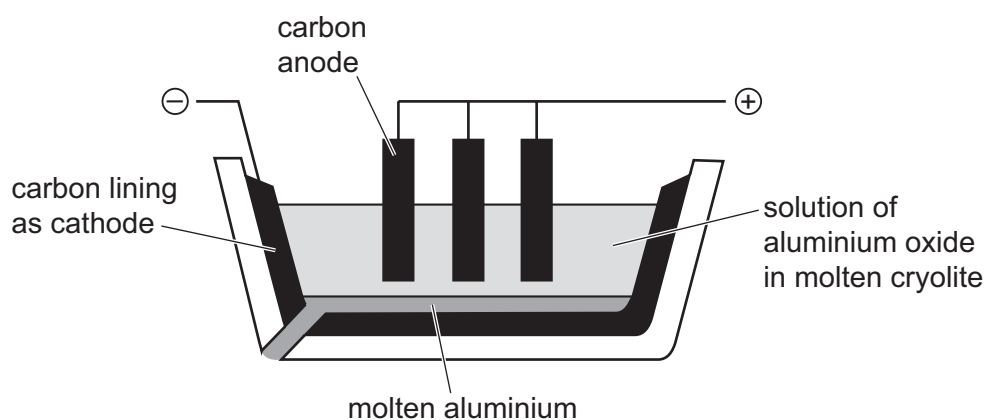
What is a suitable sacrificial metal?

- A copper
- B zinc
- C silver
- D potassium

24 Which row gives a use for the named metal and two properties which **both** explain this use?

	metal	use	property 1	property 2
A	aluminium	aircraft construction	high density	resistant to corrosion
B	copper	electrical wiring	good electrical conductivity	ductile
C	aluminium	food containers	resistant to corrosion	not malleable
D	copper	aircraft construction	malleable	low density

25 The apparatus used for the extraction of aluminium by electrolysis is shown.



Which equation represents the reaction at the anode?

- A** $\text{O} + 2\text{e}^- \rightarrow \text{O}^{2-}$
- B** $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
- C** $\text{Al}^{3-} \rightarrow \text{Al} + 3\text{e}^-$
- D** $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

26 Which gas is both an element and present in clean, dry air?

- A** argon
- B** carbon dioxide
- C** chlorine
- D** water vapour

27 Oxides of nitrogen formed in a car's engine are removed using a catalytic converter.

What happens to the oxides of nitrogen in the catalytic converter?

- A** They are hydrated.
- B** They are neutralised.
- C** They are oxidised.
- D** They are reduced.

28 What is the equation for photosynthesis?

- A** $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- B** $2\text{CO}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2$
- C** $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{CO}_2 + 2\text{C}_2\text{H}_5\text{OH}$
- D** $\text{C}_2\text{H}_5\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$

29 Four statements about members of the same homologous series are listed.

- 1 They have the same volatility.
- 2 They have the same molecular formula.
- 3 They have the same functional group.
- 4 They have the same general formula.

Which statements are correct?

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

30 Ethene reacts with steam to produce ethanol.

Which row describes each compound?

	ethene	ethanol
A	saturated	saturated
B	saturated	unsaturated
C	unsaturated	saturated
D	unsaturated	unsaturated

31 Which process is used to make an alkene from a long-chain alkane?

- A combustion
- B condensation
- C cracking
- D polymerisation

32 Which fraction obtained from petroleum has the lowest boiling point?

- A diesel oil
- B fuel oil
- C kerosene
- D naphtha

33 Alkanes undergo substitution reactions with chlorine in the presence of ultraviolet light.

Which equation shows a reaction of this type?

- A $\text{C}_3\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_6\text{Cl}_2$
- B $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_6\text{Cl}_2 + \text{H}_2$
- C $\text{C}_3\text{H}_8 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{HCl}$
- D $\text{C}_3\text{H}_6 + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_5\text{Cl} + \text{HCl}$

34 Information about two reactions of ethene is listed.

- Reaction 1 requires a nickel catalyst.
- Reaction 2 requires an acid catalyst.

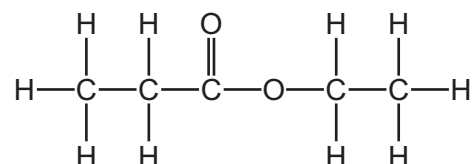
Which substance reacts with ethene in each reaction?

	reaction 1	reaction 2
A	bromine	steam
B	bromine	hydrogen
C	hydrogen	bromine
D	hydrogen	steam

35 Which process converts $\text{CH}_3\text{CH}_2\text{OH}$ to CH_3COOH ?

- A bacterial oxidation
- B fermentation
- C catalytic addition of steam
- D catalytic addition of hydrogen

36 The structure of an ester is shown.



Which row identifies the name of the ester and the two compounds from which it is made?

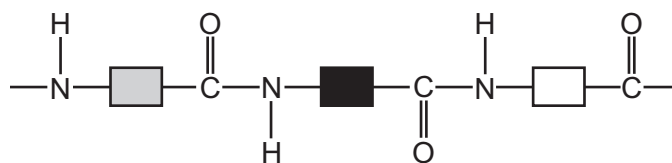
	name	compound 1	compound 2
A	ethyl propanoate	ethanol	propanoic acid
B	ethyl propanoate	propanol	ethanoic acid
C	propyl ethanoate	ethanol	propanoic acid
D	propyl ethanoate	propanol	ethanoic acid

37 Which statements about monomers or polymers are correct?

- 1 Monomers are **always** joined together by addition reactions.
- 2 A polymer can be formed from a single type of monomer.
- 3 A polymer can be formed by joining two different types of monomer.
- 4 Water is **always** produced when monomer molecules join together.

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

38 The diagram shows the structure of a naturally occurring polymer, Q.



What is Q?

- A an amino acid
- B nylon
- C a protein
- D PET

39 Which row shows how the boiling point and the melting point of water change when a soluble impurity is added to the water?

	boiling point	melting point
A	increases	increases
B	decreases	decreases
C	increases	decreases
D	decreases	increases

40 X is a white powder. The following tests are done on X.

- When a few drops of aqueous sodium hydroxide are added to a solution of X, no precipitate is seen.
- When X is heated with aqueous sodium hydroxide, no gas is formed.
- X gives a lilac colour when put into a flame.
- When acidified aqueous silver nitrate is added to a solution of X, a yellow precipitate is seen.

What is X?

- A ammonium bromide
- B ammonium iodide
- C potassium bromide
- D potassium iodide

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
		1Hhydrogen1																	
		Key																	
		atomic number atomic symbol name relative atomic mass																	
3 Li lithium 7	4 Be beryllium 9													5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24													13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids		72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids		104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —		

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/22

Paper 2 Multiple Choice (Extended)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Oxygen melts at -219°C and boils at -183°C .

At which temperature is oxygen a liquid?

- A** -225°C **B** -189°C **C** -175°C **D** 25°C

- 2 The pressure of a sample of gas is decreased. The temperature is kept constant.

Which row describes the effects on the particles?

	movement of particles	collisions between particles
A	slower	occur less often
B	slower	occur with more force
C	no change in speed	occur less often
D	no change in speed	occur with more force

- 3 Rubidium has two isotopes, $^{85}_{37}\text{Rb}$ and $^{87}_{37}\text{Rb}$.

Which statement explains why both isotopes have the same chemical properties?

- A** They have the same number of protons.
B They have the same electronic configuration.
C They have different numbers of neutrons.
D They have different mass numbers.

- 4 Which pair of elements react to form a compound with a strong attraction between oppositely charged ions?

- A** carbon and bromine
B carbon and nitrogen
C sodium and oxygen
D sodium and potassium

5 Four substances, P, Q, R and S, are described.

- P is diatomic.
- Q is a good conductor of electricity when solid and when molten.
- R is a silver solid with a very high melting point.
- S reacts with oxygen to form a brown gas.

Which substances are metals?

- A P and Q B P and S C Q and R D R and S

6 Which diagram shows the covalent bonding in a molecule of carbon dioxide?

- A O—C—O B O=C—O C O=C=O D O≡C≡O

7 The bonding, structure and melting point of sodium chloride and sulfur dichloride are shown.

compound	bonding	structure	melting point / °C
sodium chloride	ionic	giant lattice	801
sulfur dichloride	covalent	simple molecular	–121

Why does sulfur dichloride have a lower melting point than sodium chloride?

- A The covalent bonds in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- B The covalent bonds in sulfur dichloride are weaker than the ionic bonds in sodium chloride.
- C The attractive forces between molecules in sulfur dichloride are weaker than the attractive forces between molecules in sodium chloride.
- D The attractive forces between molecules in sulfur dichloride are weaker than the ionic bonds in sodium chloride.

8 Diamond and graphite have giant covalent structures of carbon atoms.

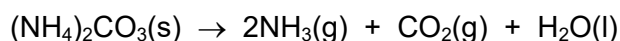
Which statement describes graphite?

- A It has a strong, rigid three-dimensional structure.
- B It has four strong covalent bonds between each carbon atom.
- C It has layers, which can slide over each other.
- D It has no delocalised electrons so does **not** conduct electricity.

- 9 Which row explains the malleability and electrical conductivity of a solid metal?

	malleability	electrical conductivity
A	Delocalised electrons can move freely through the structure.	Delocalised electrons can move freely through the structure.
B	Delocalised electrons can move freely through the structure.	Positive ions can move freely through the structure.
C	Rows of positive ions can slide over each other.	Delocalised electrons can move freely through the structure.
D	Rows of positive ions can slide over each other.	Positive ions can move freely through the structure.

- 10 The equation for the decomposition of ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is shown.



[M_r : $(\text{NH}_4)_2\text{CO}_3$, 96]

The **total** volume of gas produced is 360 cm^3 at r.t.p.

Which mass of ammonium carbonate, $(\text{NH}_4)_2\text{CO}_3$, is decomposed?

- A** 0.24 g **B** 0.48 g **C** 0.96 g **D** 1.44 g
- 11 What is the empirical formula of a compound that contains 3.66 g of hydrogen, 37.8 g of phosphorus and 58.5 g of oxygen?
- A** $\text{H}_6\text{P}_2\text{O}_6$ **B** H_4PO_4 **C** H_3PO_3 **D** HPO
- 12 Aqueous copper(II) sulfate is electrolysed using graphite electrodes.

Which row identifies the product and observations at each electrode during the electrolysis?

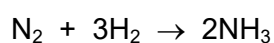
	anode		cathode	
	product	observation	product	observation
A	oxygen	bubbles of gas	copper	electrode turns pink
B	copper	electrode turns pink	oxygen	bubbles of gas
C	none	electrode dissolves	copper	electrode turns pink
D	oxygen	bubbles of gas and electrode dissolves	hydrogen	bubbles of gas

13 Molten sodium chloride is electrolysed using inert electrodes.

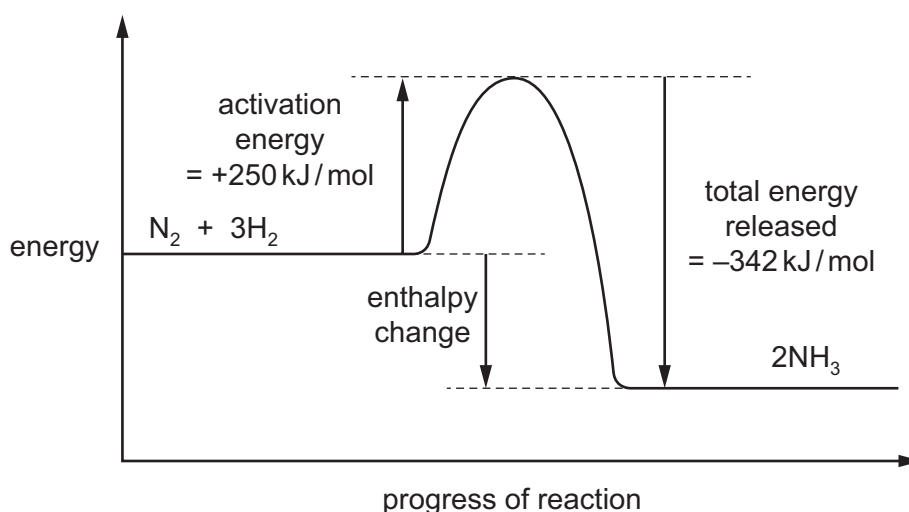
Which row shows the products formed at the cathode and anode?

	cathode	anode
A	chlorine	hydrogen
B	chlorine	sodium
C	hydrogen	chlorine
D	sodium	chlorine

14 The equation for the formation of ammonia is shown.



The reaction pathway diagram for the reaction is shown.



What is the enthalpy change for the reaction?

- A** -592 kJ/mol
- B** -92 kJ/mol
- C** +92 kJ/mol
- D** +592 kJ/mol

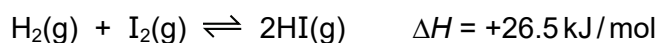
15 Sulfur dioxide is converted to sulfur trioxide in the Contact process.

The conditions used are 450 °C and 200 kPa with a vanadium(V) oxide catalyst.

Which row describes and explains the effect of changing conditions on the rate of reaction?

	change in conditions	effect on rate	explanation
A	no catalyst	lower	the activation energy is higher
B	higher pressure	higher	the particles have more kinetic energy
C	lower temperature	lower	the particles collide more frequently
D	lower pressure	higher	there are more particles per unit volume

16 Hydrogen gas reacts with iodine gas to form hydrogen iodide gas in an equilibrium reaction.

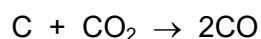


Which changes increase the yield of HI at equilibrium?

- 1 adding a catalyst
- 2 adding more hydrogen gas
- 3 increasing the pressure
- 4 increasing the temperature

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

17 The equation for the reaction of carbon with carbon dioxide is shown.



Which row identifies the carbon atom that is reduced and its change in oxidation number?

	atom that is reduced	change in oxidation number
A	carbon in CO ₂	+2 → +4
B	carbon in CO ₂	+4 → +2
C	elemental carbon, C	0 → +2
D	elemental carbon, C	+2 → 0

- 18** Aqueous iron(II) sulfate is added to acidified potassium manganate(VII). The purple colour of the potassium manganate(VII) disappears.

Aqueous potassium iodide is added to acidified potassium dichromate(VI). A dark brown solution forms.

Which row identifies the role of the iron(II) sulfate and the potassium dichromate(VI) in these reactions?

	iron(II) sulfate	potassium dichromate(VI)
A	oxidising agent	oxidising agent
B	oxidising agent	reducing agent
C	reducing agent	reducing agent
D	reducing agent	oxidising agent

- 19** Which row shows the difference between a weak acid and a strong acid?

	weak acid	strong acid
A	fully dissociated	partially dissociated
B	concentrated	dilute
C	dilute	concentrated
D	partially dissociated	fully dissociated

- 20** Which substance turns methyl orange red?

- A** aqueous ammonia
- B** dilute hydrochloric acid
- C** aqueous sodium hydroxide
- D** distilled water

- 21** Which row describes zinc oxide and calcium oxide?

	zinc oxide	calcium oxide
A	basic	acidic
B	acidic	basic
C	amphoteric	acidic
D	amphoteric	basic

22 Which row shows the properties of a transition element?

	catalyst	colour of oxide	electrical conductivity
A	yes	red	good
B	yes	green	poor
C	no	yellow	good
D	no	white	poor

23 Fluorine is the element at the top of Group VII of the Periodic Table.

Which statement describes fluorine?

- A** It is inert.
- B** It is monatomic.
- C** It is non-metallic.
- D** It is a solid at room temperature.

24 When aluminium is placed in dilute hydrochloric acid, there is no reaction.

When zinc is placed in dilute hydrochloric acid, bubbles of gas are immediately given off.

Which statement correctly explains these observations?

- A** Aluminium is coated with a layer of aluminium oxide.
- B** Aluminium is more reactive than hydrogen.
- C** Aluminium is less reactive than zinc.
- D** Zinc is less reactive than hydrogen.

25 Which statements about the use of sacrificial protection to prevent iron from rusting are correct?

- 1 A more reactive metal than iron is used as a sacrificial protector because it undergoes reduction before iron.
- 2 Zinc is used as a sacrificial protector because it gains electrons more readily than iron.
- 3 Copper is **not** used as a sacrificial protector because it is less reactive than iron.
- 4 Magnesium is used as a sacrificial protector because it loses electrons more readily than iron.

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

- 26 Aluminium is extracted from its ore by electrolysis.

What is the role of cryolite in this process?

- A to lower the operating temperature
- B to lower the boiling point of bauxite
- C to raise the melting point of bauxite
- D to act as a catalyst

- 27 Which row identifies two greenhouse gases and three processes by which they contribute to global warming?

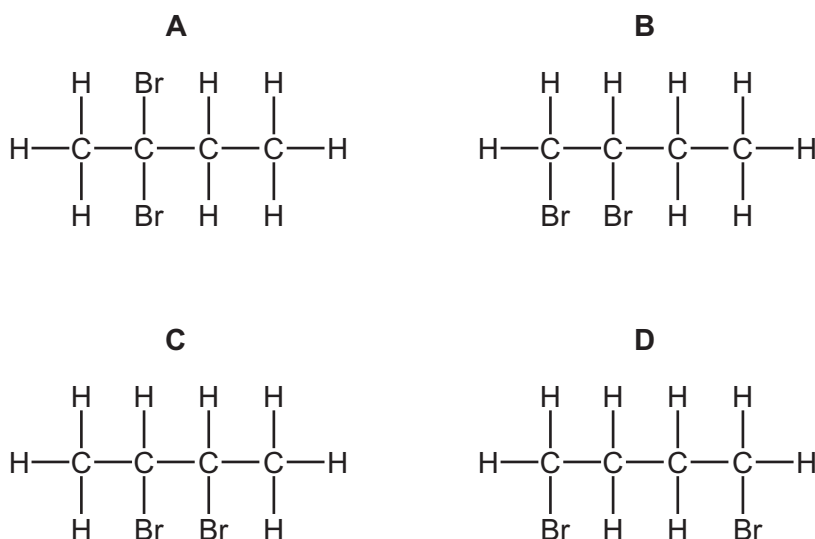
	two greenhouse gases	three processes
A	carbon dioxide and methane	absorption, creation and reflection of thermal energy
B	carbon dioxide and oxygen	absorption, creation and reflection of thermal energy
C	carbon dioxide and methane	absorption, emission and reflection of thermal energy
D	methane and oxygen	absorption, emission and reflection of thermal energy

- 28 Which mixture contains all of the elements in a typical NPK fertiliser?

- A ammonium nitrate and calcium phosphate
- B ammonium phosphate and potassium chloride
- C potassium nitrate and ammonium chloride
- D potassium carbonate and ammonium nitrate

- 29 Bromine reacts with but-2-ene.

What is the displayed formula of the product of this reaction?



30 Which statement is correct?

- A Bitumen is used as a fuel for ships.
- B Coal, natural gas and oxygen are all fuels.
- C Hydrogen is the main constituent of natural gas.
- D Petroleum is separated into useful substances by fractional distillation.

31 Which statement explains why ethanoic acid is saturated?

- A The molecule dissociates completely in water.
- B There is a carbon–oxygen double bond in the molecule.
- C The carbon–carbon bond in the molecule is a single bond.
- D All the carbon–hydrogen bonds in the molecule are single bonds.

32 Which statement about compounds in the same homologous series is correct?

- A They have the same chemical properties because they have the same number of carbon atoms.
- B They have the same physical properties because they have the same number of carbon atoms.
- C They have different chemical properties because they have different numbers of carbon atoms.
- D They have different physical properties because they have different numbers of carbon atoms.

33 Which row shows the properties of methane?

	soluble in water	state at room temperature	gives a positive test with aqueous bromine
A	no	gas	no
B	no	gas	yes
C	yes	liquid	no
D	yes	liquid	yes

34 The table shows two methods used to make ethanol.

method	type of process	conditions			source of raw material
		temperature / °C	pressure / atm	catalyst	
fermentation	batch	35	1	yeast	sugar cane
adding steam to ethene	continuous	300	60	acid	petroleum

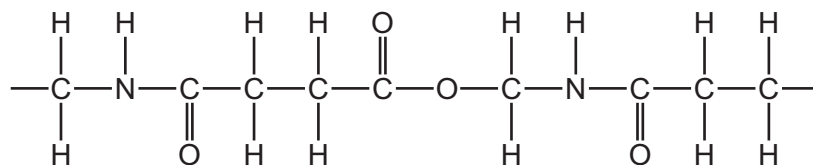
Which statement gives an advantage of preparing ethanol by fermentation rather than by adding steam to ethene?

- A Fermentation takes several days to complete.
- B Little energy is used in the fermentation process.
- C The fermentation of glucose from sugar cane produces pure ethanol.
- D Fermentation uses a non-renewable raw material.

35 Which equation represents an addition reaction?

- A $\text{CH}_3\text{CHO} + \text{HCN} \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CN}$
- B $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$
- C $\text{NH}_4\text{Br} \rightarrow \text{NH}_3 + \text{HBr}$
- D $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_2\text{H}_4 + \text{C}_8\text{H}_{18} + \text{C}_4\text{H}_8$

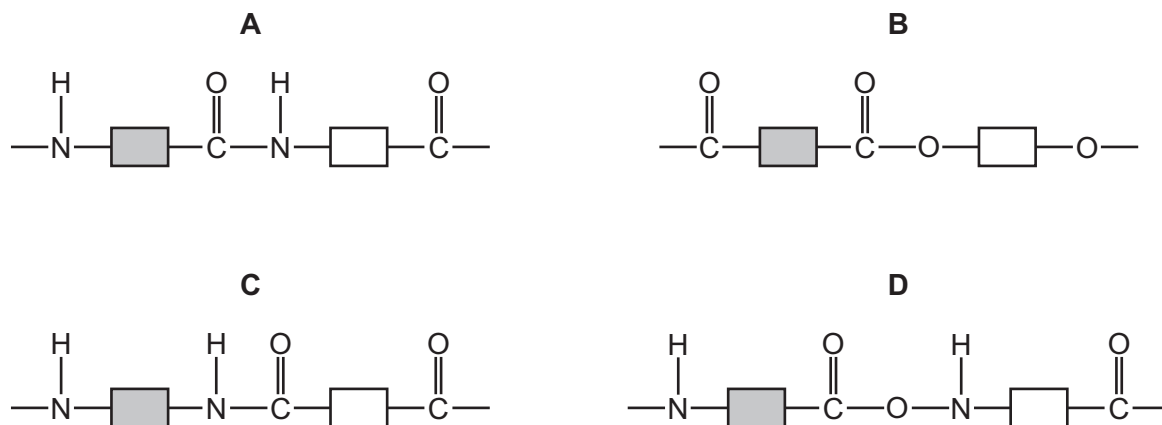
36 The structure of part of a polymer is shown.



How many amide and ester linkages are included in the structure shown?

	amide linkages	ester linkages
A	1	0
B	1	1
C	2	1
D	2	2

37 Which structure represents part of a protein?



38 Which piece of apparatus can only measure a single fixed volume?

- A** a 250 cm³ beaker
- B** a 50 cm³ burette
- C** a 100 cm³ measuring cylinder
- D** a 25 cm³ volumetric pipette

39 Pure solid copper(II) nitrate can be obtained from a mixture of copper(II) nitrate and copper powder.

Three stages in the method are listed.

- X add water and stir
- Y crystallise
- Z filter

After the three stages, the copper(II) nitrate is washed and dried.

What is the correct order of stages X, Y and Z to obtain pure solid copper(II) nitrate from the mixture?

- A** X → Y → Z
- B** X → Z → Y
- C** Y → X → Z
- D** Z → X → Y

40 Which row describes a test and the observation for aqueous sulfate ions?

	test	observation
A	add dilute nitric acid	a gas is produced which turns limewater cloudy
B	add dilute nitric acid and aqueous barium nitrate	white precipitate forms
C	add dilute nitric acid and aqueous potassium manganate(VII)	solution decolourises
D	add dilute nitric acid and aqueous silver nitrate	white precipitate forms

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
1Hhydrogen1 Key atomic number atomic symbol name relative atomic mass																		
3Li lithium 7	4Be beryllium 9																	
11Na sodium 23	12Mg magnesium 24	13Al aluminium 27	14Si silicon 28	15P phosphorus 31	16S sulfur 32	17Cl chlorine 35.5	18Ar argon 40											
19K potassium 39	20Ca calcium 40	21Sc scandium 45	22Ti titanium 48	23V vanadium 51	24Cr chromium 52	25Mn manganese 55	26Fe iron 56	27Co cobalt 59	28Ni nickel 59	29Cu copper 64	30Zn zinc 65	31Ga gallium 70	32Ge germanium 73	33As arsenic 75	34Se selenium 79	35Br bromine 80	36Kr krypton 84	
37Rb rubidium 85	38Sr strontium 88	39Y yttrium 89	40Zr zirconium 91	41Nb niobium 93	42Mo molybdenum 96	43Tc technetium —	44Ru ruthenium 101	45Rh rhodium 103	46Pd palladium 106	47Ag silver 108	48Cd cadmium 112	49In indium 115	50Sn tin 119	51Sb antimony 122	52Te tellurium 128	53I iodine 127	54Xe xenon 131	
55Cs caesium 133	56Ba barium 137	57–71 lanthanoids		72Hf hafnium 178	73Ta tantalum 181	74W tungsten 184	75Re rhenium 186	76Os osmium 190	77Ir iridium 192	78Pt platinum 195	79Au gold 197	80Hg mercury 201	81Tl thallium 204	82Pb lead 207	83Bi bismuth 209	84Po polonium —	85At astatine —	86Rn radon —
87Fr francium —	88Ra radium —	89–103 actinoids		104Rf rutherfordium —	105Db dubnium —	106Sg seaborgium —	107Bh bohrium —	108Hs hassium —	109Mt meitnerium —	110Ds darmstadtium —	111Rg roentgenium —	112Cn copernicium —	113Nh nihonium —	114Fl flerovium —	115Mc moscovium —	116Lv livermorium —	117Ts tennessine —	118Og oganesson —

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/23

Paper 2 Multiple Choice (Extended)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A sample of ethanol is left in an open beaker at room temperature.

After 24 hours, no ethanol remains in the beaker.

What has happened to the ethanol?

- A It has boiled.
- B It has condensed.
- C It has evaporated.
- D It has frozen.

- 2 A gas is in a sealed container with a fixed volume.

Which statements describe what happens to the molecules in the gas when the temperature is increased?

- 1 They move more slowly.
- 2 They collide with the walls of the container more frequently.
- 3 They collide with the walls of the container with less force.
- 4 They have greater kinetic energy.

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

- 3 What happens when sodium atoms combine with chlorine atoms to form sodium chloride?

- A Sodium atoms each gain one electron, and chlorine atoms each lose one electron.
- B Sodium atoms each lose one electron, and chlorine atoms each gain one electron.
- C Sodium atoms and chlorine atoms share one electron with each other.
- D Sodium atoms and chlorine atoms share two electrons with each other.

- 4 The table shows some properties of four substances.

substance	melting point	electrical conductivity when solid	electrical conductivity when molten
1	high	poor	poor
2	high	poor	good
3	low	poor	poor
4	high	good	good

Which substances are ionic?

- A 1, 3 and 4 B 1 and 3 only C 2 and 4 D 2 only

5 Which statement about methane is correct?

- A In methane, positive hydrogen ions are attracted to negative carbon ions.
- B In methane, electrons are shared between carbon atoms and hydrogen atoms.
- C Methane has a high boiling point.
- D Methane is a good conductor of electricity.

6 A sample of iridium has a relative atomic mass of 192.29.

The sample contains two isotopes only.

64.50% of the sample is ^{193}Ir .

What is the other isotope in the sample?

- A ^{189}Ir B ^{190}Ir C ^{191}Ir D ^{192}Ir

7 Ammonium iron(III) citrate contains in its formula:

- more than one ammonium ion
- one iron ion
- two $\text{C}_6\text{H}_4\text{O}_7^{4-}$ ions.

What is the formula of ammonium iron(III) citrate?

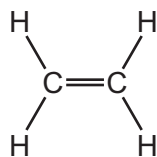
- A $(\text{NH}_4)_4\text{Fe}(\text{C}_6\text{H}_4\text{O}_7)_2$
- B $(\text{NH}_4)_5\text{Fe}(\text{C}_6\text{H}_4\text{O}_7)_2$
- C $(\text{NH}_4)_6\text{Fe}(\text{C}_6\text{H}_4\text{O}_7)_2$
- D $(\text{NH}_4)_7\text{Fe}(\text{C}_6\text{H}_4\text{O}_7)_2$

8 Silicon(IV) oxide reacts with chlorine and carbon to form liquid silicon(IV) chloride, SiCl_4 , and carbon dioxide gas.

If the reaction is carried out at r.t.p., which symbol equation represents this reaction?

- A $\text{SiO}_2(\text{l}) + 2\text{Cl}_2(\text{g}) + \text{C}(\text{s}) \rightarrow \text{SiCl}_4(\text{l}) + \text{CO}_2(\text{g})$
- B $\text{SiO}_2(\text{l}) + 2\text{Cl}_2(\text{g}) + \text{C}(\text{g}) \rightarrow \text{SiCl}_4(\text{l}) + \text{CO}_2(\text{g})$
- C $\text{SiO}_2(\text{s}) + 2\text{Cl}_2(\text{g}) + \text{C}(\text{s}) \rightarrow \text{SiCl}_4(\text{g}) + \text{CO}_2(\text{g})$
- D $\text{SiO}_2(\text{s}) + 2\text{Cl}_2(\text{g}) + \text{C}(\text{s}) \rightarrow \text{SiCl}_4(\text{l}) + \text{CO}_2(\text{g})$

- 9 The structure of ethene is shown.



How many hydrogen atoms and how many carbon atoms are in one mole of ethene?

	hydrogen atoms	carbon atoms
A	2.4×10^{24}	1.2×10^{24}
B	2.4×10^{24}	6.0×10^{23}
C	6.0×10^{23}	1.2×10^{22}
D	6.0×10^{23}	6.0×10^{23}

- 10 A known volume and concentration of aqueous sodium hydroxide is titrated against dilute hydrochloric acid.

The volume of dilute hydrochloric acid needed to exactly neutralise the sodium hydroxide is measured.

Five calculation steps are shown.

- 1 Calculate the amount of hydrochloric acid in moles.
- 2 Calculate the relative formula mass of hydrochloric acid.
- 3 Calculate the concentration of hydrochloric acid in g/dm^3 .
- 4 Calculate the amount of sodium hydroxide in moles.
- 5 Calculate the concentration of hydrochloric acid in mol/dm^3 .

What is the order of these steps to calculate the concentration of the hydrochloric acid in g/dm^3 ?

- A** 1 → 4 → 3 → 5 → 2
- B** 1 → 2 → 4 → 5 → 3
- C** 4 → 1 → 5 → 2 → 3
- D** 4 → 2 → 1 → 3 → 5

- 11 Two different substances are electrolysed using inert electrodes in two separate experiments.

Hydrogen is produced in both experiments.

Which row identifies the two substances and the electrode at which hydrogen is produced?

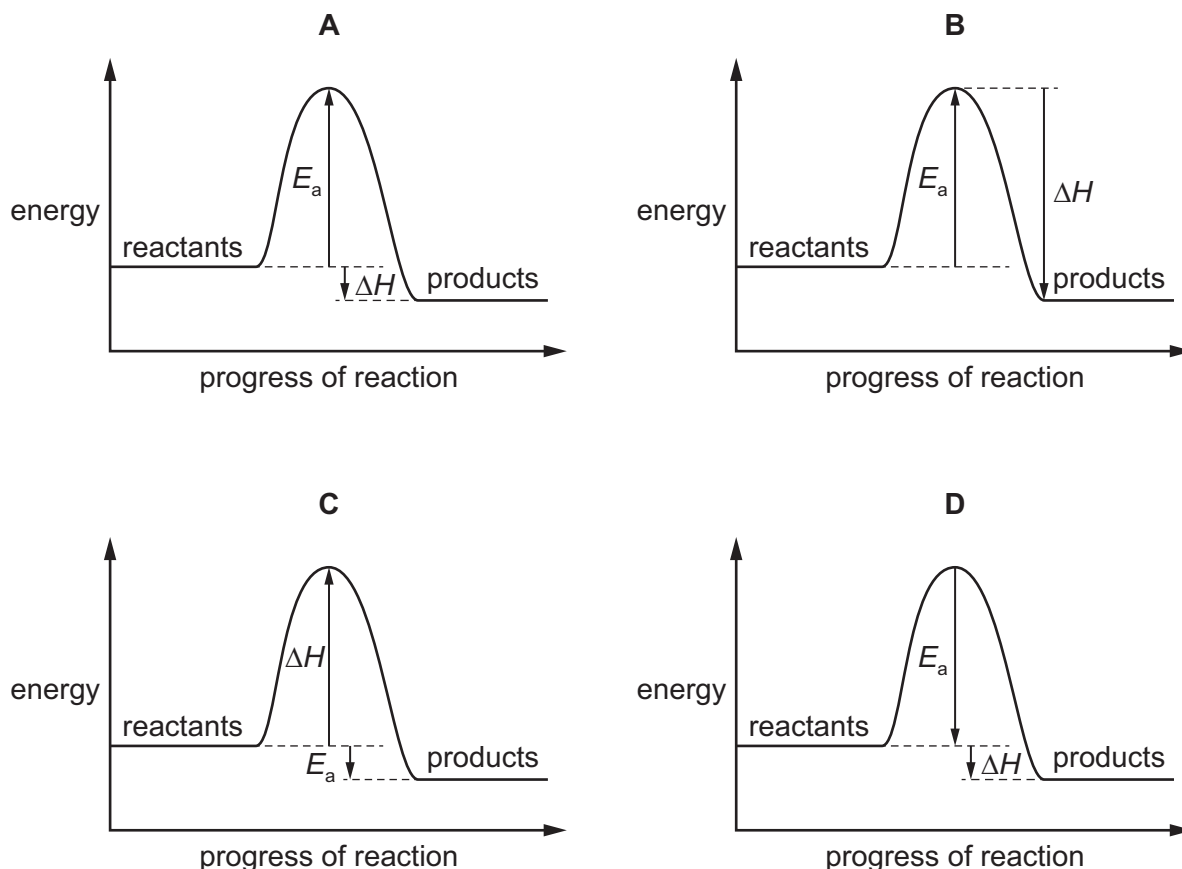
	substance 1	substance 2	electrode
A	molten sodium chloride	aqueous sodium chloride	anode
B	molten sodium chloride	aqueous sodium chloride	cathode
C	dilute sulfuric acid	concentrated hydrochloric acid	anode
D	dilute sulfuric acid	concentrated hydrochloric acid	cathode

- 12 Aqueous copper(II) sulfate can be electrolysed using either carbon electrodes or copper electrodes.

Which statement describes what happens at the positive electrode?

- A** Copper is deposited if the electrode is made from carbon.
- B** Copper is deposited if the electrode is made from copper.
- C** Oxygen gas is produced if the electrode is made from carbon.
- D** Oxygen gas is produced if the electrode is made from copper.
- 13 Which statement about a hydrogen–oxygen fuel cell is **not** correct?
- A** Chemical energy is converted into electrical energy.
- B** Hydrogen is oxidised.
- C** The reaction that takes place is endothermic.
- D** Water is the only chemical product.

14 Which reaction pathway diagram is correctly labelled?



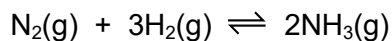
15 Which row describes a reaction where the overall energy change is exothermic?

	energy needed for breaking bonds / kJ	energy released by forming bonds / kJ	temperature of the surroundings
A	600	300	decreases
B	600	1200	decreases
C	900	300	increases
D	900	1200	increases

16 Which process involves a physical change only?

- A** heating calcium carbonate strongly
- B** burning wood
- C** melting an ice cube
- D** mixing an acid and a base

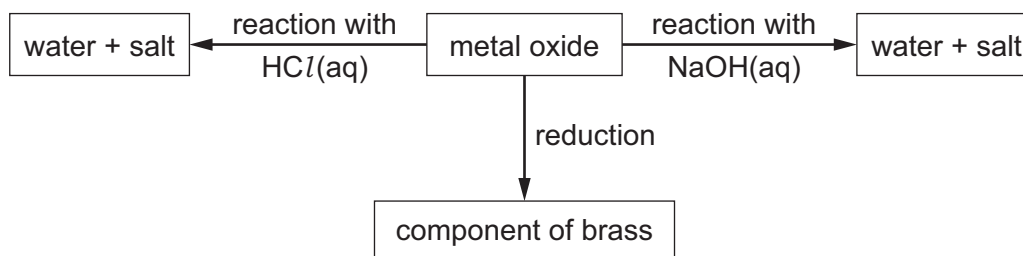
- 17 In the Haber process, an equilibrium is established.



The forward reaction is exothermic.

Which change to the reaction conditions will move the position of equilibrium to the left?

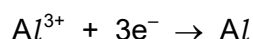
- A decreasing the pressure by 100 atm
 - B decreasing the temperature by 100 °C
 - C adding more nitrogen gas to the mixture
 - D removing the iron catalyst
- 18 The flow chart shows some properties of a metal oxide.



What is the metal oxide?

- A aluminium oxide
 - B copper(II) oxide
 - C iron(III) oxide
 - D zinc oxide
- 19 Which statement about reactants in redox reactions is correct?
- A An oxidising agent donates electrons, and a reducing agent accepts electrons.
 - B When one element gains electrons, the oxidation number of a different element increases.
 - C When the oxidation number of one element increases, a different element gains oxygen.
 - D When the oxidation number of one element increases, a different element loses electrons.

- 20 Aluminium is extracted from aluminium oxide by electrolysis. The ionic half-equation for the reaction at one of the electrodes is shown.



Which row describes the change in oxidation number of the aluminium and the type of reaction at this electrode?

	change in oxidation number of aluminium	type of reaction
A	decrease	reduction
B	decrease	oxidation
C	increase	reduction
D	increase	oxidation

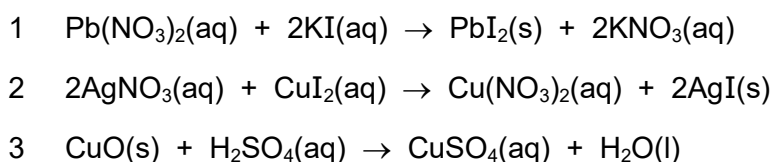
- 21 Which statement about dilute hydrochloric acid is correct?

- A** It is a strong acid as it fully dissociates.
B It is a strong acid as it partially dissociates.
C It is a weak acid as it fully dissociates.
D It is a weak acid as it partially dissociates.

- 22 Which row describes and gives the formula of hydrated copper(II) sulfate?

	description of hydrated copper(II) sulfate	formula of hydrated copper(II) sulfate
A	aqueous copper(II) sulfate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
B	aqueous copper(II) sulfate	$\text{CuSO}_4(\text{aq})$
C	copper(II) sulfate chemically combined with water molecules	$\text{CuSO}_4(\text{aq})$
D	copper(II) sulfate chemically combined with water molecules	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

- 23 The equations for three reactions are shown.

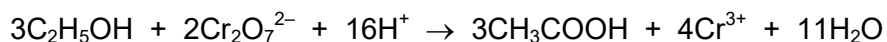


Which reactions are suitable for making a salt by precipitation?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

24 Acidified potassium dichromate(VI), $\text{K}_2\text{Cr}_2\text{O}_7$, is used to oxidise ethanol, $\text{C}_2\text{H}_5\text{OH}$.

The ionic equation for the reaction is shown.



Which properties of transition elements are shown by chromium in this reaction?

	acts as a catalyst	variable oxidation number
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

25 Which statements describe the Periodic Table?

- 1 The elements are arranged in order of their nucleon number.
- 2 The elements are arranged in order of their proton number.
- 3 It is used to predict the properties of elements.

A 1 and 3 **B** 1 only **C** 2 and 3 **D** 2 only

26 Which row shows the correct order of reactivity of the four named metals?

	most reactive	—————→			least reactive
A	magnesium	copper	zinc		silver
B	magnesium	zinc	copper		silver
C	silver	copper	zinc		magnesium
D	silver	zinc	copper		magnesium

27 Four iron nails are added to four different metal sulfate solutions.

In which solution does a displacement reaction occur?

- A** copper(II) sulfate
B magnesium sulfate
C sodium sulfate
D zinc sulfate

- 28 A fertiliser contains ammonium nitrate and potassium phosphate.

Why is the fertiliser described as an NPK fertiliser?

- A It provides nitrogen, which is an essential element for improved plant growth.
- B It contains the element oxygen, which neutralises acidic soil.
- C It contains the elements nitrogen and phosphorus.
- D It provides the three main elements needed for improved plant growth.

- 29 What are the approximate percentages of oxygen and nitrogen in clean, dry air?

	percentage of oxygen	percentage of nitrogen
A	19	80
B	21	78
C	80	19
D	78	21

- 30 Which compounds have similar chemical properties?

- A butanol and butanoic acid
- B ethane and ethene
- C methane and butane
- D propene and propanol

- 31 Four statements about organic compounds P, Q, R and S are listed.

P is a saturated hydrocarbon.

The formula of Q is CH_3CH_3 .

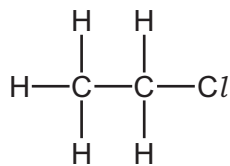
A molecule of R contains only one oxygen atom.

Compound S is a carboxylic acid.

Which statement about these compounds is correct?

- A P and Q are members of different homologous series.
- B P and S are members of the same homologous series.
- C Q and S are members of the same homologous series.
- D Q, R and S are all members of different homologous series.

32 The structure of an organic compound is shown.



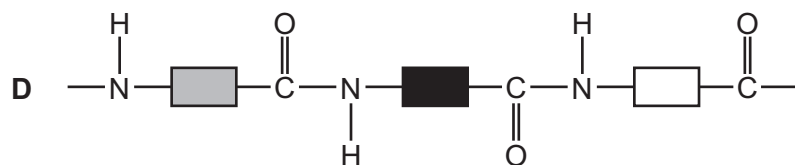
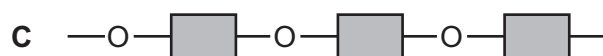
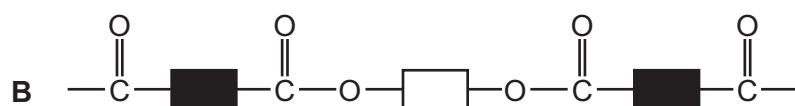
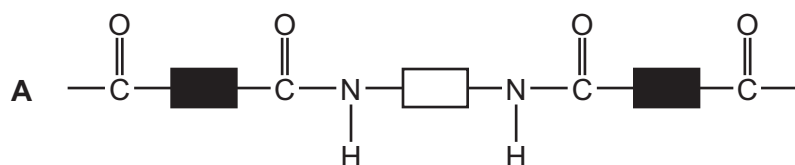
What is the name of the compound?

- A chloroethane
 - B chloroethene
 - C chloroethanol
 - D chloroethanoic acid
- 33 Which statement about the manufacture of ethene from larger alkane molecules is correct?
- A A low temperature is required.
 - B The process is called cracking.
 - C The process requires an excess of oxygen.
 - D Water is also a product.
- 34 Which processes are used to make ethanoic acid?
- 1 heating ethanol with acidified aqueous potassium manganate(VII)
 - 2 bacterial oxidation of ethanol
 - 3 distilling ethanol using a fractionating column
- A 1 and 2 B 1 only C 2 and 3 D 3 only
- 35 Which statement about propene, C₃H₆, is correct?
- A Propene reacts with bromine in the dark in a substitution reaction.
 - B Propene reacts with steam in the presence of an alkaline catalyst, forming an alcohol.
 - C Propene undergoes addition polymerisation, forming poly(ethene).
 - D Propene undergoes an addition reaction to form an alkane.

36 How many of each type of bond are present in ethanoic acid, CH_3COOH ?

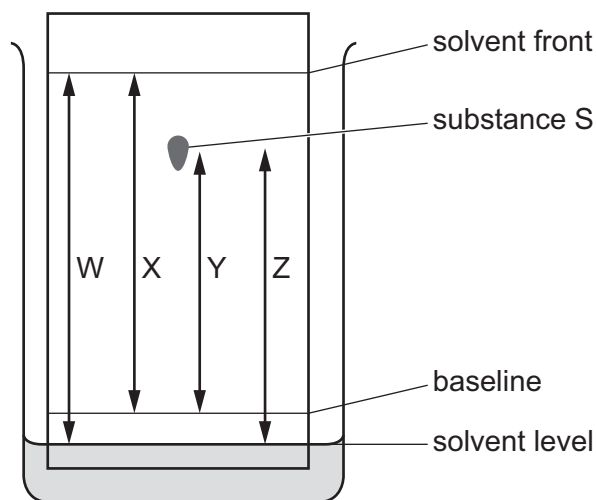
	type of bond		
	C-H	C-C	C=O
A	3	1	1
B	3	0	2
C	4	0	2
D	4	1	2

37 Which diagram represents the structure of a protein?



38 The chromatogram of substance S is shown.

Some distances, W, X, Y and Z, are labelled on the diagram.



How is the R_f value of substance S calculated?

A $\frac{X}{Y}$

B $\frac{W}{Z}$

C $\frac{Y}{X}$

D $\frac{Y}{W}$

39 Some information about solid silver chloride and solid sodium chloride is shown.

- Silver chloride and sodium chloride do **not** dissolve in kerosene.
- Silver chloride is insoluble in water, but sodium chloride is soluble in water.
- The boiling point of silver chloride is 1547°C and the boiling point of sodium chloride is 1413°C .

Which processes are used to separate a mixture of solid silver chloride and solid sodium chloride?

- A add kerosene, stir and then filter
- B add water, stir and then filter
- C add water, stir and then leave to crystallise
- D add water, stir and then perform fractional distillation

40 Which statement describes how a flame test is done?

- A The tip of a clean wire is dipped into the substance and the wire is placed in a blue Bunsen burner flame.
- B The tip of a clean wire is dipped into the substance and the wire is placed in a yellow Bunsen burner flame.
- C A wooden splint is lit and is placed above a test-tube containing the gas being tested.
- D A wooden splint is lit, blown out and the glowing splint put into a test-tube of the gas being tested.

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The Periodic Table of Elements

Group																			
I	II											III	IV	V	VI	VII	VIII		
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20	18 Ar argon 40
11 Na sodium 23	12 Mg magnesium 24	13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5													
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84		
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131		
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —		
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —		

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actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).